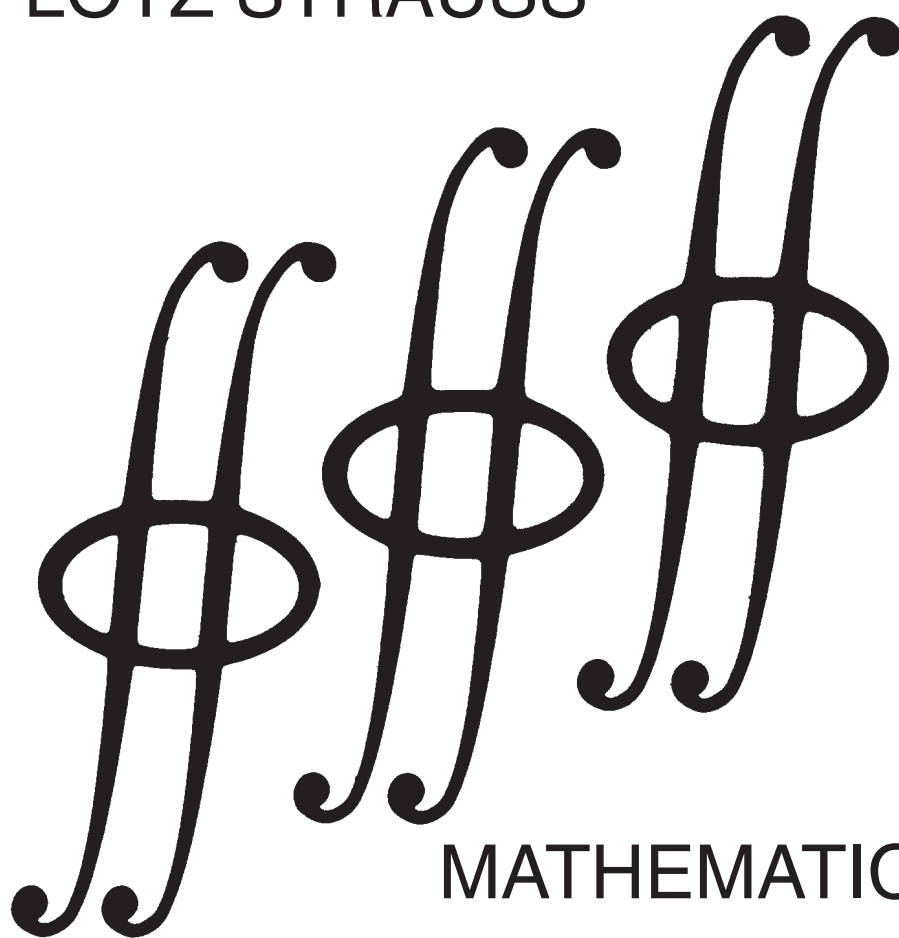


L. STRAUSS - MATHEMATICS FOR PHYSICAL SCIENCE

LÖTZ STRAUSS



MATHEMATICS
FOR
PHYSICAL SCIENCE

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Preface

Mathematics for Physical Science contains all the mathematical tools required for the study of introductory physical science at first year university or college level. It is intended mainly for students who require a practical knowledge of elementary differential and integral calculus and also elementary vector algebra and vector analysis. In writing the book, the requirements for courses in biological sciences were also taken into account.

Although the mathematical concepts are dealt with fairly systematically, the accent is on the development of the student's skill in the use of mathematical techniques in the description of theories in physical science and the solution of problems. In order to attain this goal, the text contains a large number of problems which are worked out in full detail to aid weaker students. At the end of each chapter is a variety of problems of which the answers are supplied at the end of the book.

The prerequisite for the use of this textbook is the successful completion of a mathematics course at high school level. Most of chapter 1 and much of chapter 2 is a revision of work which is normally included in school courses and is intended to stress those topics which are most important for understanding the rest of the course.

Most of this text is presented as an introductory course to Physics 1 at the University of Pretoria. This introductory course is usually covered in about 18 lectures. The students are not required to be able to prove the mathematical deductions which are treated, but they are required to understand the meaning of the concepts and apply them skilfully. In the physics courses for which this textbook is intended, differential and integral calculus and also vector algebra and simple vector analysis are used. The student is required to know definitions and understand physical principles and then use the mathematical tools to solve problems. The final examinations for these courses consist of problems only. Remembering formulas and their derivation do not form a part of the final examination.

Some aspects are treated at a later stage in the physics courses when the need arises. Line integrals are introduced when the concept of work is treated. Experience has shown that students should encounter the concept of a **differential equation** as soon as possible. This enables one to use the techniques of differential and integral calculus at a more elegant level in the solution of problems.

My heartfelt thanks to my dear wife for proofreading at level one (correcting her husband's appallingly poor English spelling and grammar). The most difficult task was the detailed checking of the manuscript which is a translation of the original version in Afrikaans. This immense task was performed by my dear and respected friend Etienne Malherbe. For the elimination of errors missed by the other readers, the final draft was scanned by Danie Steyn, another dear friend who was deeply involved in the preparation of the original Afrikaans version. Sareta van Tonder made the index. My sincere thanks to her for this task. (Now the reader knows who to blame if a concept cannot be traced through the index!)

My thanks to my Maker for the privilege, opportunity and ability to plan and accomplish this task. *Soli Deo Gloria!*

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January 1994*

Lötz Strauss

Chapter 1

INTRODUCTION

1.1 Important symbols and constants

$=$	is equal to	$\neq$	is not equal to
$\approx$	is approximately equal to	$\equiv$	is identical to, definition of
$\sim$	is of same order of magnitude as	$\propto$	is proportional to
$>$	is greater than	$\gg$	is much greater than
$<$	is less than	$\ll$	is much less than
$\geq$	is greater than or equal to	$\leq$	is less than or equal to
$\therefore$	because	$\therefore$	therefore
$\Rightarrow$	this implies, from this follows	$\rightarrow$	tends to
∞	infinitely large		

Δx or δx a change in x . $\Delta x = (\text{final value of } x) - (\text{initial value of } x)$

$ a $	the absolute value (modulus) of a $ a = a$ if $a > 0$ (a real, positive) $ a = 0$ if $a = 0$ $ a = -a$ if $a < 0$ (a real, negative)
$n!$	n -factorial, $n! = n(n-1)(n-2)(n-3)\dots$ 5.4.3.2.1 $0! = 1$ by definition
π	(circumference of a circle)/(diameter of the circle) = 3,14159265...
e	base of natural (Napierian) logarithms = 2,71828183...

$$\sum_i^n x_i \quad \text{sum of all the } x_i \text{ values} = x_1 + x_2 + x_3 + \dots + x_n$$

1.2 The Greek alphabet

A	α	alpha	N	ν	nu
B	β	beta	Ξ	ξ	xi
Γ	γ	gamma	O	o	omicron
Δ	δ	delta	Π	π	pi
E	ϵ	epsilon	P	ρ	rho
Z	ζ	zeta	Σ	σ	sigma
H	η	eta	T	τ	tau
Θ	θ	theta	Υ	υ	upsilon
I	ι	iota	Φ	ϕ	phi
K	κ	kappa	X	χ	chi
Λ	λ	lambda	Ψ	ψ	psi
M	μ	mu	Ω	ω	omega

1.3 Exponents and logarithms

The following rules are valid for real numbers:

$$\begin{aligned}
 a^n &= a \times a \times a \times a \dots n \text{ times. } n \text{ is a positive integer.} \\
 a^{-n} &= 1/a^n \\
 a^{1/n} &= \sqrt[n]{a} \\
 a^0 &= 1 \\
 (ab)^p &= a^p \times b^p \\
 a^p \times a^q &= a^{p+q} \\
 a^p \div a^q &= a^{p-q} \\
 a^{p/q} &= \sqrt[q]{a^p} = (\sqrt[q]{a})^p \\
 (a^p)^q &= a^{pq}
 \end{aligned}$$

If $a > 0$ and $a^n = b$, then $b > 0$ for all real values of n . In this relationship a is called the **base**, b is called a power (the n -th power in this case) of a and n is the **exponent** or **logarithm** to which a has to be raised to give b . If n is to be made the subject of the formula, it is written as follows: $n = \log_a b$. Another way in which b may be written as the subject of the formula is $b = \text{antilog}_a n$ which is equivalent to $b = a^n$ in all respects. By writing the relationship $a = b^{1/n}$ the roles of the base and the power are exchanged. Since $1/n = \log_b a$, it follows directly that $\log_a b = 1/(\log_b a)$.

The following are different ways of expressing one and the same relationship between the numbers a (the base), n (the exponent or logarithm) and b (the power or antilogarithm):

$$\begin{array}{lll}
b = a^n = \text{antilog}_a n & a = b^{1/n} = \text{antilog}_b (1/n) & n = \log_a b \\
= n\text{-th power of } a & = n\text{-th root of } b & = (\log_b a)^{-1}
\end{array}$$

Before calculators were available, one had to make use of tables of logarithms and antilogarithms to perform difficult calculations involving multiplication, division, exponentiation and the extraction of roots. For this purpose logarithms to the base 10 were used. To save time, the base was omitted.

If $r = \log s$, it means that $s = 10^r$

Logarithms to the base $e = 2,71828\dots$ are called **natural logarithms** or **Napierian logarithms**. If $y = \log_e b = \ln b$, then $b = e^y$. The notation $\ln b$ (pronounced “lin- b ”) is the conventional way for writing natural logarithms.

The rules which express calculations involving logarithms, are just a different way to express those which apply to exponents. With P , Q and a positive numbers and $a \neq 1$, the following rules are valid:

$$\begin{array}{ll}
\log_a P^n &= n \log_a P \\
\log_a \sqrt[n]{P} &= \frac{1}{n} \log_a P \\
\log_a 1 &= 0 \\
\log_a 1/P &= -\log_a P \\
\log_a (P \times Q) &= \log_a P + \log_a Q \\
\log_a P/Q &= \log_a P - \log_a Q \\
\log_a P &= (\log_b P) \times (\log_a b) = (\log_b P) \div (\log_b a)
\end{array}$$

in which also $b > 0$ and $b \neq 1$.

1.4 The binomial theorem

For small positive integer values of n , the expansion of $(a + b)^n$ may be done directly by the rules which apply to the multiplication of algebraic expressions, e.g.

$$\begin{array}{ll}
(a + b)^0 &= 1 \\
(a + b)^1 &= a + b \\
(a + b)^2 &= a^2 + 2ab + b^2 \\
(a + b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3 \\
(a + b)^4 &= a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4 \quad \text{etc.}
\end{array}$$

For n not a positive integer (or a large positive integer) the expansion is calculated by using the **binomial theorem**.

For n a **positive integer** we have

$$(a+b)^n = a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3 + \frac{n(n-1)(n-2)(n-3)}{4!}a^{n-4}b^4 + \dots + b^n \quad 1.4(1)$$

which has $n+1$ terms. Equation 1.4(1) may be written in a shorter way as

$$(a+b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r \quad 1.4(2)$$

in which

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} \quad 1.4(3)$$

is known as the **binomial coefficient** of term number $r+1$.

Equation 1.4(1) also holds for negative and fractional values of n , provided that $|b| < |a|$ or $|b/a| < 1$ in which case the expansion has an infinite number of terms which form a convergent series.

In physics the binomial expansion is often used for the case where $a = 1$ and n not necessarily an integer or positive. For the condition that $|b| < 1$, it is an infinite series as follows:

$$(1+b)^n = 1 + nb + \frac{n(n-1)}{2!}b^2 + \dots$$

If b is much less than unity, the first two or three terms give such a good approximation that the higher powers may be disregarded.

1.5 Factors

The reader should be familiar with the following identities which occur in the study of physics.

$$\begin{aligned} a^2 - b^2 &= (a-b)(a+b) \\ a^4 - b^4 &= (a^2 - b^2)(a^2 + b^2) = (a-b)(a+b)(a^2 + b^2) \\ a^3 \pm b^3 &= (a \pm b)(a^2 \mp ab + b^2) \\ ax^2 + bx + c &= a(x - x_-)(x - x_+) \quad \text{in which} \quad x_{\pm} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \end{aligned}$$

1.6 Radian measure

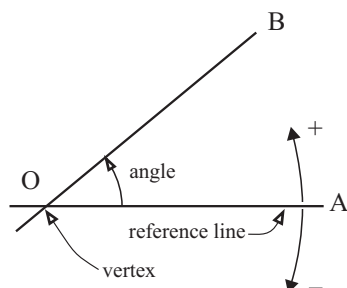


Figure 1.6-1

An angle measures rotation. Consider straight lines OA and OB in Figure 1.6-1. If OA is a fixed line, the angle AOB measures the amount of rotation of OB from the position of OA about the fixed point O to the position indicated in the sketch. The rotation occurs in the plane which contains OA and OB. If the rotation is counter-clockwise, it is taken as positive and if it is clockwise, it is negative.

Observers viewing the rotation from opposite sides, will assign different signs to one and the same angle. The possibility of ambiguities is not excluded.

An angle thus specifies the direction of one straight line with respect to another. The two lines are called the **arms of the angle** and their point of intersection, the **vertex**.

The simplest unit in which an angle may be measured, is the revolution. A popular unit is the **degree**. In one revolution there are 360 degrees (360°). A quarter of a revolution (90°) is called a **right angle**. The degree is subdivided into 60 **minutes** ($60'$) and each minute into 60 seconds ($60''$). Most pocket calculators have a function for the conversion of degrees, minutes and seconds to degrees expressed in decimals and vice versa. (*Comment: Since the same factors apply, this function may also be used for the conversion of time units.*)

$$1 \text{ revolution} = 360^\circ$$

$$1 \text{ straight angle} = \frac{1}{2} \text{ revolution} = 180^\circ$$

$$1 \text{ right angle} = \frac{1}{4} \text{ revolution} = 90^\circ$$

For scientific and mathematical work, these units are often not suitable. The preferred unit is the **radian**.

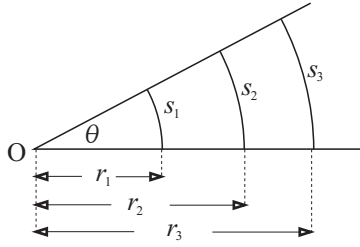


Figure 1.6-2

Consider the sketch in Figure 1.6-2. If r_1 , r_2 and r_3 are the radii of three circles with centre O and s_1 , s_2 and s_3 the corresponding arcs between the arms of the angle, then

$$s_1/r_1 = s_2/r_2 = s_3/r_3$$

The ratio (arc length)/(radius) depends on the magnitude of the angle only. This allows a new unit for the measurement of angles to be defined.

$$\theta = \frac{\text{arc length}}{\text{radius}} = \frac{s}{r} \quad 1.6(1)$$

If s and r are measured in the same units, the answer is in **radians** (abbreviation **rad**).

It follows from Equation 1.6(1) that the length of the arc s is given by

$$s = \text{radius} \times \text{angle} = r\theta \quad \theta \text{ in radians} \quad 1.6(2)$$

$$\begin{aligned} 1 \text{ revolution} &= \frac{\text{circumference of a circle}}{\text{radius of the circle}} = \frac{2\pi r}{r} = 2\pi \text{ rad} \\ 1 \text{ right angle} &= \frac{\frac{1}{4} \text{ circumference of a circle}}{\text{radius of the circle}} = \frac{\frac{1}{4} \times 2\pi r}{r} = \frac{\pi}{2} \text{ rad} \end{aligned}$$

It is recommended that the reader remember the relationship $180^\circ = \pi \text{ rad}$ and that this be used to convert degrees to radians and vice versa. From this it follows that

$$\begin{aligned} 1 \text{ rad} &= 180/\pi = 57,29577\dots \approx 57,3^\circ \\ 1^\circ &= \pi/180 = 0,01745\dots \text{rad} \end{aligned}$$

Since a calculator will be used for conversions, it will not be necessary to remember these two values. Some calculators can do the conversions directly by the use of one or two keys only. If such a facility does not exist, the following procedure may be used:

$$\begin{aligned} \text{angle in radians} &= (\text{angle in degrees}) \times \pi/180 \\ \text{angle in degrees} &= (\text{angle in radians}) \times 180/\pi \end{aligned}$$

Some calculators make provision for yet another unit which is called **grad**. This is an attempt to decimalise angles. There are 100 grad in a right angle but these units will not be used in this book.

1.7 Useful goniometric relationships

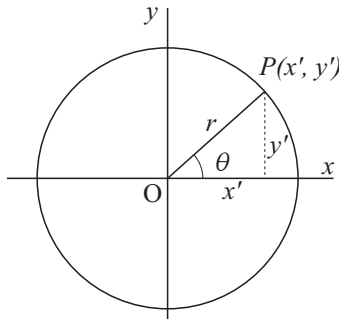


Figure 1.7-1

To indicate the position of a point on a plane surface, a **Cartesian** frame of reference, also referred to as a rectangular frame of reference, is used. Two straight lines which intersect at right angles, are called the axes. It is convenient to think of the axes as rulers. Their point of intersection is called the origin which is usually the common zero on both scales. It is conventional (though not necessary) to refer to the axes as the x -axis and y -axis respectively. On the x -axis (see Figure 1.7-1) the values to the left of the origin are

taken as negative and those to the right as positive. On the y -axis the values below the origin are taken as negative, and those above as positive. The position of any point on the plane surface is given unambiguously by the specification of two ordered real numbers, (x', y') , which are known as its **co-ordinates**. In chapter 4, frames of reference will be dealt with in greater depth.

Consider the Cartesian frame of reference in Figure 1.7-1. Describe a circle, radius r and centre at the origin. θ is the angle between the positive x -axis and radius vector OP . If the angle is measured counter-clockwise, it is taken as positive and when it is measured clockwise, it is negative. In the following goniometric ratios (also called trigonometric ratios), r is always a positive number but the signs of x' and y' are in accordance with the quadrants in which they lie.

$$\begin{aligned}
 \sin \theta &= y'/r & \csc \theta &= (\sin \theta)^{-1} = r/y' \\
 \cos \theta &= x'/r & \sec \theta &= (\cos \theta)^{-1} = r/x' \\
 \tan \theta &= y'/x' & \cot \theta &= (\tan \theta)^{-1} = x'/y'
 \end{aligned}
 \tag{1.7(1)}$$

These definitions are valid for all values of θ , excluding those for which the denominator which appears in the definition is equal to zero. At such values of θ the corresponding trigonometric ratio is undefined. The problem occurs with all the ratios excepting $\sin \theta$ and $\cos \theta$. The graphs of $\sin \theta$ and $\cos \theta$ are shown in Figure 1.7-2.

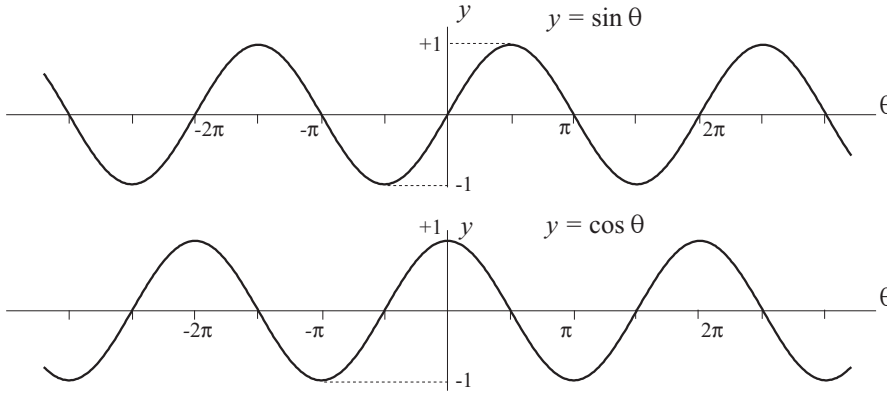


Figure 1.7-2

From the definitions of the goniometric ratios the following identities may be derived. They are often utilised in the mathematics which is used in physics and the reader should be quite familiar with them.

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \quad 1.7(2)$$

$$\sin^2 \theta + \cos^2 \theta = 1 \quad 1.7(3)$$

$$\sin(\theta \pm \phi) = \sin \theta \cos \phi \pm \cos \theta \sin \phi \quad 1.7(4)$$

$$\cos(\theta \pm \phi) = \cos \theta \cos \phi \mp \sin \theta \sin \phi \quad 1.7(5)$$

From Equations 1.7(3) to 1.7(4) the following may be deduced quite easily:

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta \quad 1.7(6)$$

$$\sin 2\theta = 2 \cos \theta \sin \theta \quad 1.7(7)$$

From Equation 1.7(6) we have

$$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta) \quad 1.7(8)$$

$$\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta) \quad 1.7(9)$$

The following identities also follow from 1.7(4) and 1.7(5):

$$\cos \theta \cos \phi = \frac{1}{2}[\cos(\theta + \phi) + \cos(\theta - \phi)] \quad 1.7(10)$$

$$\sin \theta \sin \phi = \frac{1}{2}[\cos(\theta - \phi) - \cos(\theta + \phi)] \quad 1.7(11)$$

$$\sin \theta \cos \phi = \frac{1}{2}[\sin(\theta + \phi) + \sin(\theta - \phi)] \quad 1.7(12)$$

$$\cos \theta \sin \phi = \frac{1}{2}[\sin(\theta + \phi) - \sin(\theta - \phi)] \quad 1.7(13)$$

$$\cos \theta + \cos \phi = 2 \cos \frac{1}{2}(\theta + \phi) \cos \frac{1}{2}(\theta - \phi) \quad 1.7(14)$$

$$\cos \theta - \cos \phi = -2 \sin \frac{1}{2}(\theta + \phi) \sin \frac{1}{2}(\theta - \phi) \quad 1.7(15)$$

$$\sin \theta + \sin \phi = 2 \sin \frac{1}{2}(\theta + \phi) \cos \frac{1}{2}(\theta - \phi) \quad 1.7(16)$$

$$\sin \theta - \sin \phi = 2 \cos \frac{1}{2}(\theta + \phi) \sin \frac{1}{2}(\theta - \phi) \quad 1.7(17)$$

By using differential calculus, the following infinite series may be derived:

$$\sin \theta = \frac{\theta}{1!} - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \dots (\theta \text{ in rad}) \quad 1.7(18)$$

$$\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} + \dots (\theta \text{ in rad}) \quad 1.7(19)$$

The higher-order terms in these infinite series are small compared to those which occur early in the series. It will often be sufficient to make an approximation in which a number of the lower-order terms only are used. Calculators make use of these series when calculating the numerical values of trigonometric ratios of different angles. From the series it should be apparent that $\cos \theta$ is symmetrical about $\theta = 0$ and $\sin \theta$, antisymmetrical. For these reasons, $\cos \theta$ is known as an **even function** and $\sin \theta$ as an **uneven function**.

By means of Figure 1.7-1 $\sin \theta$ was defined as $\sin \theta = y'/r$. The **inverse** goniometric function may be written as

$$\theta = \arcsin(y'/r) \quad \text{or} \quad \theta = \sin^{-1} y'/r \quad 1.7(20)$$

which is equivalent to the statement

$$\theta = \text{the angle of which the sine is equal to } y'/r$$

This is known as the **arc sine** of the given quantity. The second way of writing this function often appears on calculators and should not be confused with the reciprocal of the sine function. Similarly, with reference to Figure 1.7-1, we have

$$\theta = \arctan(y'/x') \quad \text{and} \quad \theta = \arccos(x'/r)$$

By means of a calculator the numerical values of $\arcsin$, $\arccos$ and $\arctan$ may be determined directly. For reasons which will become apparent when the

relevant mathematics is studied in depth, these functions exist only between the limits as indicated.

For	arcsin:	$-\pi/2$	and	$\pi/2$
	arccos:	0	and	π
	arctan:	$-\pi/2$	and	$\pi/2$

Answers which are read from a calculator will lie within these limits. The reader is advised to read the instruction booklet supplied with his/her pocket calculator and become familiar with the use of these functions. An important point to be aware of, is whether an answer is required in degrees or radians. The calculator has to be adjusted beforehand and it is indicated on the read-out panel. Verify the following answers as an exercise to become familiar with your calculator.

$\arcsin 0,1234$	$=$	$7,088^\circ$	$\arcsin 0,8660$	$=$	$60,00^\circ$
$\arcsin 0,5000$	$=$	$30,00^\circ$	$\arcsin 1,1234$	does not exist	
$\arcsin 0,5000$	$=$	$0,524 \text{ rad}$	$\arccos 0,5236$	$=$	$1,020 \text{ rad}$
$\arcsin(-0,5000)$	$=$	$-30,00^\circ$	$\arccos(-0,5000)$	$=$	$120,0^\circ$
$\arctan(-0,5000)$	$=$	$-26,57^\circ$	$\arctan 123,0$	$=$	$89,53^\circ$
$\arctan 0$	$=$	0	$\arctan 1,000$	$=$	$45,00^\circ$
$\arccos 0$	$=$	$90,00^\circ$	$\arccos 1,000$	$=$	0

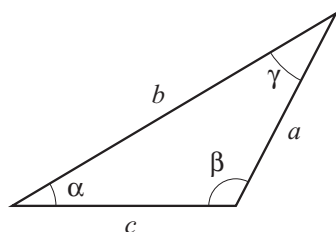


Figure 1.7-3

When a calculator is switched to degrees, numerical answers of the inverse functions will be in degrees and decimal parts of degrees, not degrees, minutes and seconds. Most calculators will perform this conversion by the use of one or two keys. If a trigonometric ratio of an angle which is specified in degrees, minutes and seconds is to be found by means of a calculator, it must first be converted to degrees and decimal parts of a degree. This can also be accomplished directly by means of a calculator.

If certain dimensions of a triangle (sides and angles) are specified, they may define its shape and size unambiguously. If that is the case, the unspecified dimensions may be calculated by using the cosine rule or the sine rule. The cosine rule is of prime importance in the study of physics since it forms the base of that branch of mathematics which is known as vector algebra.

Consider the triangle which is shown in Figure 1.7-3.

The sides opposite to angles α , β and γ are a , b and c respectively. If any two sides and the included angle are known, the third side may be calculated by means of the cosine rule.

$$\begin{aligned} a^2 &= b^2 + c^2 - 2bc \cos \alpha \\ b^2 &= c^2 + a^2 - 2ca \cos \beta \\ c^2 &= a^2 + b^2 - 2ab \cos \gamma \end{aligned} \quad 1.7(21)$$

For the special case where $\gamma = 90^\circ$, the cosine rule for the calculation of c may be rewritten as

$$c^2 = a^2 + b^2 \quad 1.7(22)$$

which is known as the theorem of Pythagoras.

The sine rule supplies two more relationships for the calculation of unknown quantities. For the triangle which is shown in Figure 1.7-3 the following is valid:

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma} \quad 1.7(23)$$

The sides of the triangles in Figure 1.7-4, may be calculated by means of the theorem of Pythagoras. If one of the angles of a triangle is equal to zero, the two adjacent sides will coincide and the opposite side will be equal to zero. By means of these facts, the goniometric ratios of the angles given in Table 1.7-1 may be calculated.

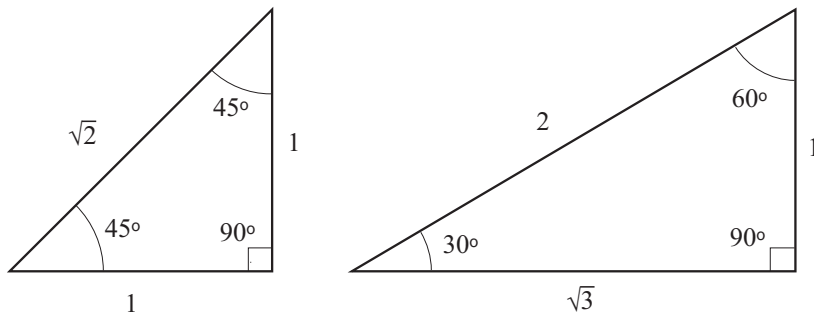


Figure 1.7-4

θ	$\sin \theta$	$\cos \theta$	$\tan \theta$
0°	0	1	0
30°	$1/2 = 0,5000$	$\sqrt{3}/2 = 0,8660$	$1/\sqrt{3} = 0,5774$
45°	$1/\sqrt{2} = 0,7071$	$1/\sqrt{2} = 0,7071$	1
60°	$\sqrt{3}/2 = 0,8660$	$1/2 = 0,5000$	$\sqrt{3} = 1,7920$
90°	1	0	undefined

Table 1.7-1

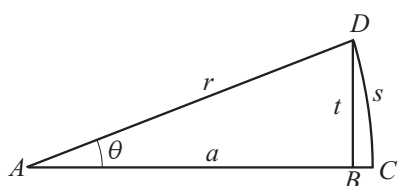


Figure 1.7-5

Figure 1.7-5 shows a triangle of which angle θ is very small. Angle $ABD = 90^\circ$. $AD = r (= AC)$ is the hypotenuse of the triangle. $AB = a$ is the adjacent arm of the angle θ . $BD = t$ is the side opposite angle θ . $DC = s$ is the arc length which the angle θ subtends at a distance of r . From the sketch it can be seen that $s \approx t$ and $a \approx r$ when θ is small. With these approximations the following are valid:

$$\begin{aligned}\sin \theta &= \frac{t}{r} \approx \frac{s}{r} = \theta & (\theta \text{ in radians}) \\ \tan \theta &= \frac{t}{a} \approx \frac{s}{r} = \theta & (\theta \text{ in radians}) \\ \cos \theta &= \frac{a}{r} \approx \frac{r}{r} = 1\end{aligned}$$

Summary: For small θ : $\sin \theta \approx \tan \theta \approx \theta$ (θ in radians) and $\cos \theta \approx 1$ 1.7(24)

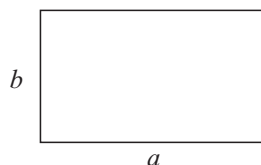
The approximations for $\sin \theta$ and $\cos \theta$ also follow directly from the series given in Equations 1.7(18) and 1.7(19). The approximations are often used in physics and they represent one of the advantages of using radians as units for measuring angles.

1.8 Geometric formulae

■ Rectangle, length a and width b

Perimeter $= 2a + 2b$

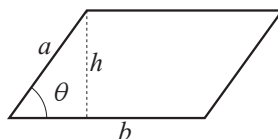
Area $= ab$



■ Parallelogram, base b and height h

$$\text{Perimeter} = 2a + 2b$$

$$\text{Area} = bh = ab \sin \theta$$

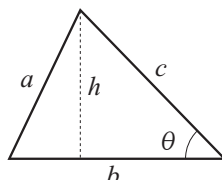


■ Triangle, base b and height h

$$\text{Perimeter} = a + b + c$$

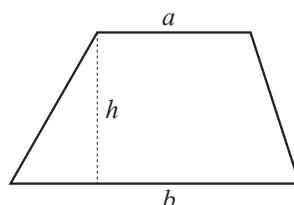
$$\text{Semi-perimeter} = s = \frac{1}{2}(a + b + c)$$

$$\begin{aligned} \text{Area} &= \frac{1}{2}bh = \frac{1}{2}bc \sin \theta \\ &= \sqrt{s(s-a)(s-b)(s-c)} \end{aligned}$$



■ Trapezium with parallel sides a and b and height h

$$\text{Area} = \frac{1}{2}(a + b)h$$

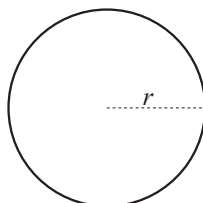


■ Circle, radius r

$$\text{Diameter } D = 2r$$

$$\text{Circumference} = \pi D = 2\pi r$$

$$\text{Area} = \frac{1}{4}\pi D^2 = \pi r^2$$

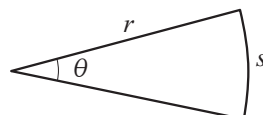


■ Sector of a circle with radius r

$$\text{Arc length} = s = r\theta \quad (\theta \text{ in rad})$$

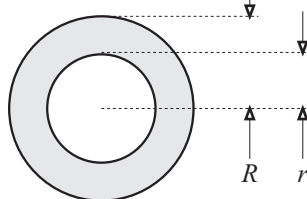
$$\text{Perimeter} = 2r + r\theta \quad (\theta \text{ in rad})$$

$$\text{Area} = \frac{1}{2}rs = \frac{1}{2}r^2\theta \quad (\theta \text{ in rad})$$



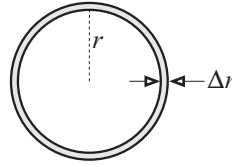
■ Annulus, inner radius r , outer radius R

$$\begin{aligned} \text{Area} &= \pi(R^2 - r^2) \\ &= \pi(R - r)(R + r) \end{aligned}$$



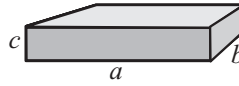
- Narrow annulus, radius r and width Δr

$$\begin{aligned} \text{Area} &\approx \text{circumference} \times \text{width} \\ &= 2\pi r \Delta r \end{aligned}$$



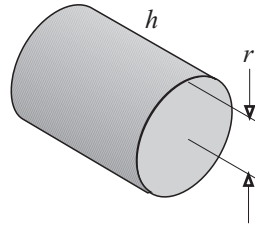
- Rectangular parallelepiped, sides a , b and c

$$\begin{aligned} \text{Area} &= 2(ab + bc + ca) \\ \text{Volume} &= abc \end{aligned}$$



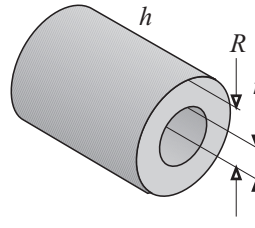
- Solid right circular cylinder with radius r and length h

$$\begin{aligned} \text{Area} &= 2\pi r^2 + 2\pi r h \\ &= 2\pi r(r + h) \\ \text{Volume} &= \pi r^2 h \end{aligned}$$



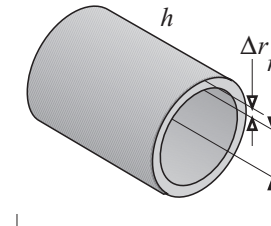
- Hollow right circular cylinder, inner radius r , outer radius R , length h

$$\begin{aligned} \text{Area} &= 2\pi(R^2 - r^2) + 2\pi h(R + r) \\ &= 2\pi(R + r)(R - r + h) \\ \text{Material volume} &= \pi(R^2 - r^2)h \end{aligned}$$



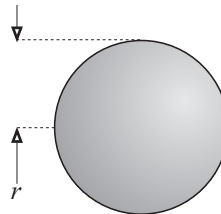
- Hollow right circular cylinder, radius r , length h and thin wall Δr

$$\begin{aligned} \text{Area} &\approx 4\pi r h \\ \text{Volume of material} &\approx 2\pi r h \Delta r \end{aligned}$$



- Sphere, radius r

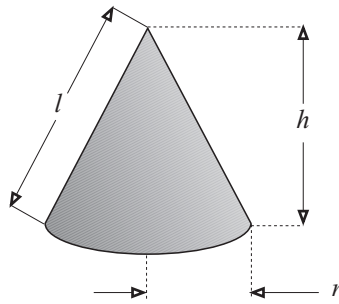
$$\begin{aligned} \text{Area} &= 4\pi r^2 \\ \text{Volume} &= \frac{4}{3}\pi r^3 \end{aligned}$$



■ Right circular cone, radius r , height h

$$\text{Area} = \pi r l + \pi r^2$$

$$\text{Volume} = \frac{1}{3} \pi r^2 h$$



1.9 PROBLEMS CHAPTER 1

1. Calculate the following without the use of calculating aids:

- | | | |
|----------------------|---|--|
| (a) $64^{1/6}$ | (b) $8^{2/3}$ | (c) $27^{-2/3}$ |
| (d) $2^3 \times 3^2$ | (e) $\sqrt[7]{2 \times 8^2}$ | (f) $\sqrt[3]{8^{-2}}$ |
| (g) $6^4/3^3$ | (h) $10^6 \times 10^4$ | (i) $10^6/10^4$ |
| (j) $10^6/10^{-4}$ | (k) $9^{2,5}$ | (l) $16^{1,5}$ |
| (m) $8^{-0,6667}$ | (n) $\left(\frac{32 \times 10^5}{2^3 \times 10^3}\right)^{\frac{1}{2}}$ | (o) $\left[10 \times \left(\frac{1,08 \times 10^4}{300}\right)^{\frac{1}{2}} + 4\right]^{\frac{1}{2}}$ |

2. Calculate the following without the use of calculating aids:

- | | | |
|-----------------|-------------------------------|---------------------|
| (a) $\log_2 32$ | (b) $\log_3 27^{\frac{2}{3}}$ | (c) $\log_2 (1/32)$ |
| (d) $\log_5 1$ | (e) $\log_5 (-1)$ | (f) $\ln (1/e^2)$ |
| (g) $\log 100$ | (h) $\log 0$ | (i) $\log 10^{-5}$ |

3. Use a calculator to calculate the following:

- | | | |
|-----------------------------|--------------------------------|--------------------------------|
| (a) $12, 34^2$ | (b) $12, 34^3$ | (c) $12, 34^{2,56}$ |
| (d) $12, 34^{-2}$ | (e) $12, 34^{\frac{1}{2}}$ | (f) $12, 34^{\frac{1}{3}}$ |
| (g) $12, 34^{-\frac{1}{3}}$ | (h) $(12, 34^3)^{\frac{1}{5}}$ | (i) $1/(12, 34)^{\frac{1}{3}}$ |

4. Use a calculator to calculate the following:

- | | | |
|-------------------------|--|--------------------------|
| (a) $\log (12, 34^2)$ | (b) $(\log 12, 34)^2$ | (c) $\log \sqrt{12, 34}$ |
| (d) $\log_8 12, 34$ | (e) $\log_2 12, 34$ | (f) $\log_5 12, 34$ |
| (g) $\log_{12,34} 10$ | (h) $\log (12, 34/10)$ | (i) $(\log 12, 34)/10$ |
| (j) $\log (12, 34/100)$ | (k) $(\log 12, 34)/(\log \sqrt{12, 34})$ | |

5. Use the binomial theorem to calculate the following:

- | | | |
|------------------------------------|-----------------------------|---------------------------------|
| (a) $(a + b)^5$ | (b) $(x + 2)^4$ | (c) $(2y - 3)^4$ |
| (d) $(1 - v^2/c^2)^{-\frac{1}{2}}$ | (e) $(1 + 0,002)^2$ | (f) $(1 - 0,005)^{\frac{1}{2}}$ |
| (g) $(1,003)^{\frac{1}{2}}$ | (h) $(0,995)^{\frac{1}{2}}$ | (i) $(0,995)^{-\frac{1}{2}}$ |

6. Calculate the following angles in radians without the use of a calculating aid.

Give the answers in terms of π .

- | | | |
|-----------------|-----------------|-----------------|
| (a) 30° | (b) 60° | (c) 45° |
| (d) 270° | (e) 216° | (f) 720° |
| (g) 450° | (h) 630° | (i) 72° |

7. Calculate the following angles in radians:

- | | | |
|-------------------------|--------------------------|-------------------------|
| (a) $85,34^\circ$ | (b) $123,4^\circ$ | (c) $815,6^\circ$ |
| (d) $89^\circ 39' 42''$ | (e) $219^\circ 54' 29''$ | (f) $356^\circ 0' 56''$ |

8. The following angles are in radians. Express them in degrees without making use of a calculator.

- | | | |
|--------------|--------------|--------------|
| (a) $3\pi/2$ | (b) $\pi/2$ | (c) $2\pi/7$ |
| (d) $4\pi/5$ | (e) $2,5\pi$ | (f) 4π |
| (g) 3π | (h) $\pi/6$ | (i) $4\pi/3$ |

9. Use a calculator to express the following angles in degrees:

- | | | |
|----------------|----------------------|---------------------|
| (a) 7,345 rad | (b) 0,1234 rad | (c) 0,5678 rad |
| (d) -5,568 rad | (e) 0,5678 π rad | (f) 14,56 π rad |

10. Use a calculator to calculate the following:

- | | | |
|--------------------------------|-----------------------------------|-----------------------------------|
| (a) $\sin 56,78^\circ$ | (b) $\cos 56,78^\circ$ | (c) $\tan 56,78^\circ$ |
| (d) $\cot 56,78^\circ$ | (e) $\csc 56,78^\circ$ | (f) $\sec 56,78^\circ$ |
| (g) $\sin 98^\circ 34' 45''$ | (h) $\cos 125^\circ 28' 48''$ | (i) $\tan 135^\circ 45' 38''$ |
| (j) $\tan (2,65 \text{ rad})$ | (k) $\csc (0,567 \text{ rad})$ | (l) $\cos (2,345\pi \text{ rad})$ |
| (m) $\cot (-27^\circ 5' 27'')$ | (n) $\sin (-1,23\pi \text{ rad})$ | (o) $\sec (-0,56\pi \text{ rad})$ |

11. Use a calculator to calculate the following angles, first in degrees and then in radians:

- | | | |
|------------------------|-----------------------------------|--------------------------------------|
| (a) $\arctan 1,234$ | (b) $\arctan (-1,234)$ | (c) $\arccos (-0,8660)$ |
| (d) $\arcsin 1,234$ | (e) $\operatorname{arccsc} 1,234$ | (f) $\operatorname{arccot} (-1,234)$ |
| (g) $\arcsin 0,0012$ | (h) $\arctan 0,0012$ | (i) $\operatorname{arcsec} 2,3456$ |
| (j) $\arccos (-1,543)$ | (k) $\operatorname{arccos} 1,543$ | (l) $\arcsin 0,8660$ |

12. Write down, without the use of a calculator, the sine, cosine and tangent of the following angles. Check your answers by means of Table 1.7-1.
 0° , 30° , 45° , 60° , 90° and 180° .

13. The following apply to the triangle in Figure 1.7-3. Calculate in each case the unspecified quantities if the following are known:

- | | |
|---|--|
| (a) $a = 20 \text{ mm}$, $c = 30 \text{ mm}$, $\beta = 120^\circ$ | (b) $c = 12,5 \text{ mm}$, $\alpha = 40^\circ$, $\beta = 60^\circ$ |
| (c) $c = 63 \text{ m}$, $\alpha = 62^\circ$, $\gamma = 55^\circ$ | (d) $a = 25 \text{ m}$, $b = 35 \text{ m}$, $c = 45 \text{ m}$ |
| (e) $c = 50 \text{ m}$, $a = 30 \text{ m}$, $\alpha = 20^\circ$ | |

14. Calculate the areas of the triangles specified in Question 13. If possible, avoid using quantities which had to be calculated.
15. Calculate the area of an equilateral triangle with side length 2 m by (i) using the magnitude of one of its angles, (ii) using only the lengths of the sides.
16. A circular race track has an internal circumference of 2 km and a width of 6 m. Calculate the area of the paving on the track in square metres.
17. The sides of a trapezium are 1 m, 1 m, 3 m and 4 m respectively. Calculate its area.
18. Calculate the area of a circle with diameter 12,34 m.
19. Calculate the diameter of a circle with area 4 m^2 .
20. Calculate the area of a right circular cylinder with radius 20 mm and length 40 mm.
21. An empty cylindrical pipe has an outer diameter of 30 mm and an inner diameter of 25 mm. Calculate the material volume of 1 m of the pipe and also the area exposed to the atmosphere. Calculate the volume of water that would fill 2 m of this pipe.

Chapter 2

DIFFERENTIAL CALCULUS

2.1 Functions

2.1.1 Introduction

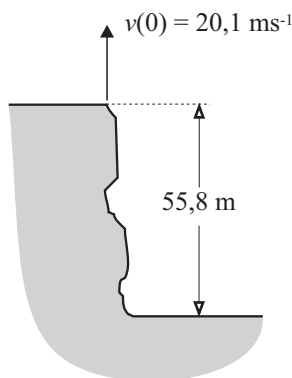


Figure 2.1-1

Figure 2.1-1 shows a precipice which is 55,8 m deep. Standing on the edge, a person throws an object vertically upwards at an initial speed of $20,1 \text{ ms}^{-1}$. The height of the object is measured from the edge and it is indicated by x . Values of x which are measured upwards are taken as positive and those in the opposite direction, negative. As time progresses, x will first increase and then decrease until the object hits the bottom of the precipice.

At each instant, t , there exists one value of x . It should be clear that the object cannot be at two different positions at one instant and also that x may only assume values which lie between two limits, the maximum height which the object reaches and the bottom of the cliff. The complete motion of the object lasts a finite interval of time, and therefore the possible values of t

also lie between two limits. It is said that x is a **function** of t . Since one and the same value of x may correspond to different values of t , t is not a function of x in this time interval.

This relationship between x and t which is called a **function**, may be represented by

$$x = x(t) \quad 2.1(1)$$

In this representation, t is known as the **independent variable** and x the **dependent variable**.

The quantitative information which is gathered during an experiment is known as **data**. The data is used to determine the **functional relationship** between the variables which describes a process. There are three ways in which a function may be represented.

- (i) a table of numbers
- (ii) a graph
- (iii) an equation

For the example of the object which is thrown vertically upwards from the edge of a cliff, Table 2.1-1 gives the relationship between x and t for the duration of the motion. The value of x is given at the beginning and thereafter at the end

Time, t , in seconds	Position, x , in metres
0,0	0,0
1,0	15,2
2,0	20,6
3,0	16,2
4,0	2,0
5,0	-22,0
6,0	-55,8

Table 2.1-1

of each second while the motion exists. At instant $t = 6$ s, the motion stops. Although this table gives valuable information about the motion, it has certain shortcomings. The first is that it can never be complete, which means that values of x at unspecified times, cannot be known. A superficial inspection might lead to the conclusion that the maximum height is 20,6 m. This is incorrect and more information is required before the correct maximum value can be determined.

A graphic representation of this data makes it more informative. The independent variable is usually represented by a suitable scale along a straight line from left to right. In this case it will be t and the line is called the **abscissa axis**. The dependent variable, x in this case, is represented on a line perpendicular to the first and which is known as the **ordinate axis**. The pairs of corresponding numbers are called co-ordinates. Each pair of co-ordinates gives one point on the graphic representation. Figure 2.1-2(a) gives a graphic representation of the data in Table 2.1-1.

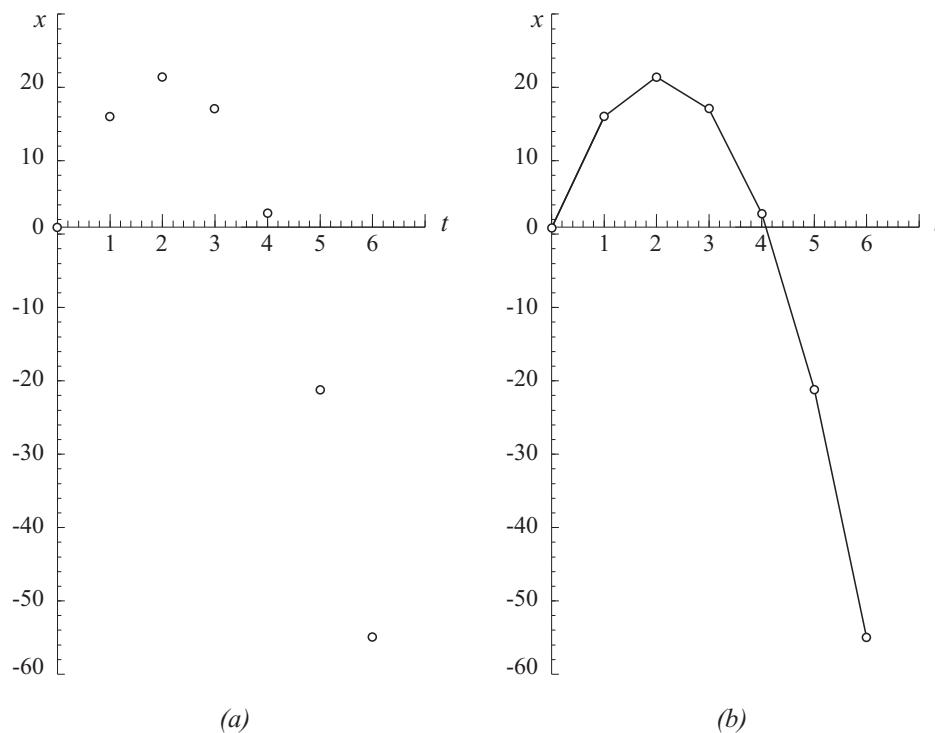


Figure 2.1-2

If this graph is to be used to obtain information about values of x at values of t which lie between the points which are known, the technique of **interpolation** is used. This term applies to any reasonable method by which points that lie between those which are known, are determined. If the known points are joined with straight lines as shown in Figure 2.1-2(b), they do not describe the function correctly, but are an aid to facilitate interpolation. By means of drawing aids such as curve stencils or flexible drawing guides, a smooth curve may be drawn through the points. Such a curve gives more accurate values when interpolating. Figure 2.1-3 was drawn from the data in Table 2.1-1 with the aid of a computer program. The best technique to use is that of curve fitting by means of the method of **least squares of deviations**. That was the technique used by the computer program which was used to produce the graph in Figure 2.1-3. The technique of curve fitting will not be treated in this book but readers are advised to gain access to computer programs which may be used for this purpose.

Known data may be used in some cases to predict the behaviour of a system in regions which lie outside the known domain. This technique is known as

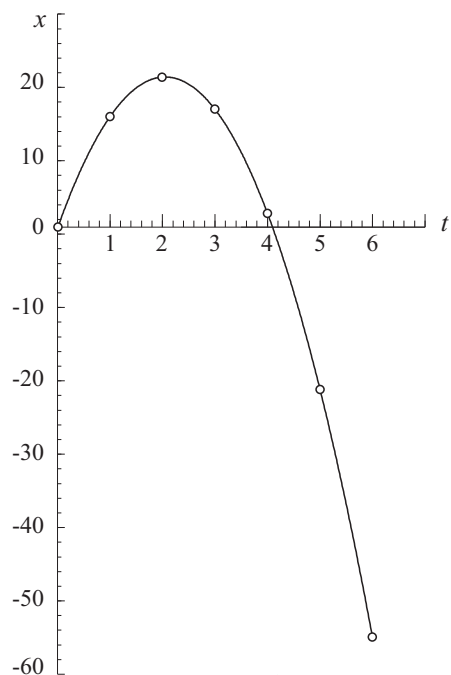


Figure 2.1-3

extrapolation. Graphic extrapolation implies the extension of a curve to a region where no known data points exist. The shape of the portion of the curve which is known to be correct is used as a guide. In some cases graphic extrapolation is fairly easy but in others it might be very difficult or impossible.

In the study of physics, the aim is mostly to describe a function by means of an equation. With the techniques at our disposal this may be done quite readily in most cases. The equation which describes the motion of the object in question is

$$x = 20t - 4.9t^2 \quad 2.1(2)$$

in which the position, x , is measured in metres and the time, t , in seconds. By means of this equation, the value of x

at any admissible value of t , may be calculated and it describes the behaviour of the function for the complete motion.

Having considered a practical example, it is now possible to give a more formal definition of a function. If two variables x and y are related in such a way that for each value of x in a set of admissible values which is called the *domain* of x , there exists only one value of y from another set of admissible values which is called the **range** of y , then y is said to be a function of x . In physics the following notation is used to indicate a function:

$$y = y(x) \quad 2.1(3)$$

It is convenient to interpret this notation as follows: A variable y depends on x of which it is a function. In the notation, x is called the independent variable (abscissa) and y the dependent variable (ordinate). Each co-ordinate pair, (x, y) , represents one point on the graph which represents it. In Figure 2.1-2, x was used to represent the ordinate whereas in a graph of Equation 2.1(3) it would be the abscissa. This was done on purpose to impress upon the reader that a symbol may be used for any purpose.

The symbolic notation $y = y(x)$ contains no information for the calculation

of the pairs of co-ordinates. That is possible only when the equation for the function is known as is shown in the example below.

$$\begin{array}{ll}
 \text{Consider} & y = y(x) = 3x^2 - 5x + 2 \\
 \text{Then} & y(0) = 3(0)^2 - 5(0) + 2 = 2 \\
 & y(1) = 3(1)^2 - 5(1) + 2 = 0 \\
 & y(-1) = 3(-1)^2 - 5(-1) + 2 = 10 \\
 & y(2) = 3(2)^2 - 5(2) + 2 = 4 \quad \text{etc.}
 \end{array}$$

There are also functions of more than one variable as shown in the following examples:

(i) The length, c , of the hypotenuse of a variable right angled triangle, depends on both the other two sides, a and b .

$$c = (a^2 + b^2)^{\frac{1}{2}}$$

The symbolic notation $c = c(a, b)$ indicates that c is a function of both variables a and b .

(ii) The volume, V , of an ideal gas, depends on the number of moles, n , of the gas, its pressure, p , and the temperature, T .

$$V = V(n, p, T) = RnT/p \quad \text{in which } R \text{ is constant}$$

In this book, mostly functions of a single variable will be considered.

2.1.2 A number of important functions of a single variable and their graphic representations

(1) $y \propto x$ or $y = mx$ in which m is a constant. In this relationship y is said to be **directly proportional** to x and m is known as the **proportionality constant**. The graph is a straight line which always passes through the origin. The graph in Figure 2.1-4 represents a direct proportionality for which $m > 0$. In this chapter the meaning of m will be of prime importance.

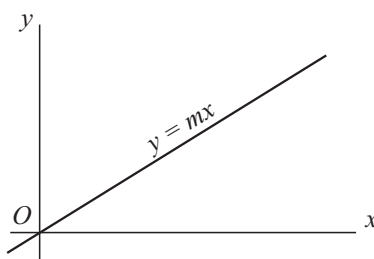


Figure 2.1-4

(2) $\Delta y \propto \Delta x$ or $y = mx + c$ in which m and c are constants. In this case the *change* in y is directly proportional to the *increase* in x . The graph is also a straight line but it does not generally pass through the origin. In the equation c is the **intercept** on the ordinate axis. The sketch in Figure 2.1-5(a) shows the graphic representation of such a proportionality for which $m < 0$ and $c > 0$.

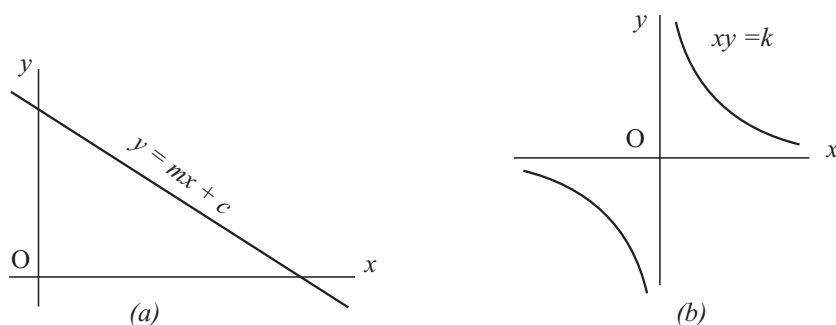


Figure 2.1-5

(3) $y \propto 1/x$ or $x \propto 1/y$ which may also be written as $y = k/x$ or $xy = k$. The relationship is an **inverse proportionality** and its graphic representation is called a **rectangular hyperbola**. The sketch in Figure 2.1-5(b) represents an inversely proportional relationship with $k > 0$. In the region visible in the sketch, the graph, which consists of two branches, is curved. When x becomes very small or very large, it is difficult to distinguish the graph from a straight line and the co-ordinate axes become tangents to it. This behaviour is known as **asymptotic behaviour** and the axes are called its **asymptotes**. Not all graphs have asymptotes.

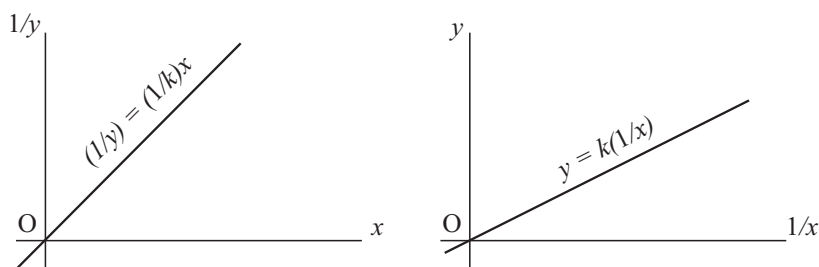


Figure 2.1-6

An inverse proportionality may be represented by a straight line by plotting the value of $1/y$ along the ordinate axis instead of y , or $1/x$ along the abscissa axis instead of x . Figure 2.1-6 shows these two possibilities. The straight-line representation is often used in physics since the graphic extrapolation of a hyperbola is difficult.

(4) $y = ax^2 + bx + c$. The graph of this function is called a **parabola** and its **axis of symmetry**, which is given by $x = -b/2a$, is parallel to the y -axis. If $b^2 > 4ac$, the parabola intersects the x -axis at two positions given by

$x = (-b \pm \sqrt{b^2 - 4ac})/2a$. If $b^2 = 4ac$, the x -axis is a tangent to it at position $x = -b/2a$. If $b^2 < 4ac$, the graph does not intersect or touch the x -axis. The graph shown in Figure 2.1-7(a) represents a parabola for which a , b and c are positive and $b^2 > 4ac$. c is the intercept on the ordinate axis.

(5) $y^2 + x^2 = r^2$ in which r is constant, is the equation for a **circle** with radius r and centre at the origin. The relationship $y^2 = r^2 - x^2$ is not a function because two values of y (or x) correspond to each value of x (or y). The circle consists of two functions namely $y = +\sqrt{r^2 - x^2}$ and $y = -\sqrt{r^2 - x^2}$ in which the domain of x is given by $-r \leq x \leq r$. The circle may also be described by means of the two **parametric equations** $x = r \cos \theta$, $y = r \sin \theta$. The **parameter** θ is shown in figure 2.1-7(b).

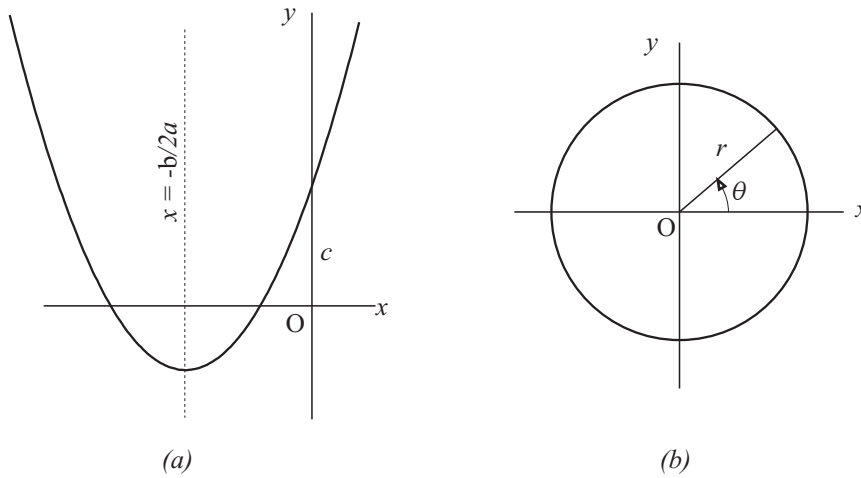


Figure 2.1-7

(6) $x^2/a^2 + y^2/b^2 = 1$ is the equation of an **ellipse** of which the axes of symmetry have lengths of $2a$ and $2b$ respectively and which coincide with the co-ordinate axes. The ellipse consists of the two functions $y = (b/a)\sqrt{a^2 - x^2}$ and $y = -(b/a)\sqrt{a^2 - x^2}$ in which the domain of x is given by $-a \leq x \leq a$. The parametric equations of an ellipse are $x = a \cos \theta$ and $y = b \sin \theta$. It should be clear that an ellipse becomes a circle if $a = b$. Figure 2.1-8(a) shows an ellipse for which $a > b$. a is the **semi major axis** of the ellipse and b the **semi minor axis**.

The functions which have been discussed up to this stage are related in a wonderful way. They are generated by the intersection of a plane surface with a right circular cone. For this reason they are known as **conic sections**. They are of prime importance in physics because they describe the orbits of electri-

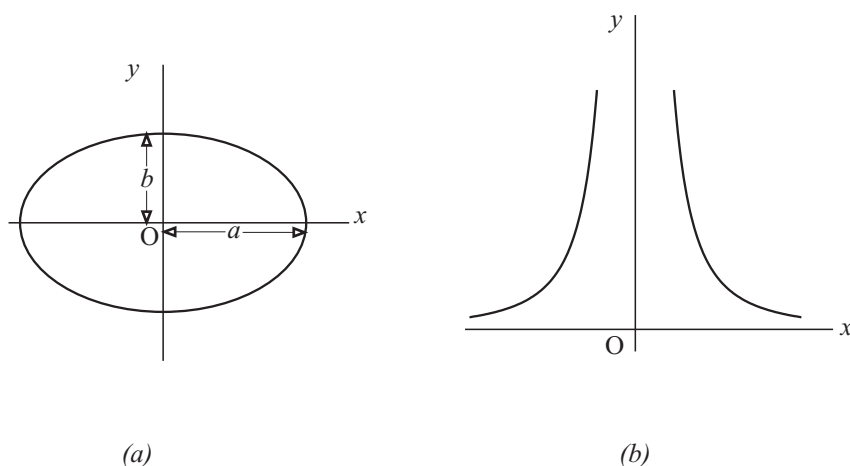


Figure 2.1-8

cally charged particles, planets, comets, etc., under the influence of an electrical and/or a gravitational force.

(7) $y \propto 1/x^2$ or $y = k/x^2$. The graph of this function consists of two branches and has no special name. The law which governs the magnitude of y , is known as the **inverse square law**. This law describes an important property of the three-dimensional space in which we live: It determines the way in which electric point charges and point masses exert forces on one another respectively. Figure 2.1-8(b) shows a graph of this function in which the constant k is positive.

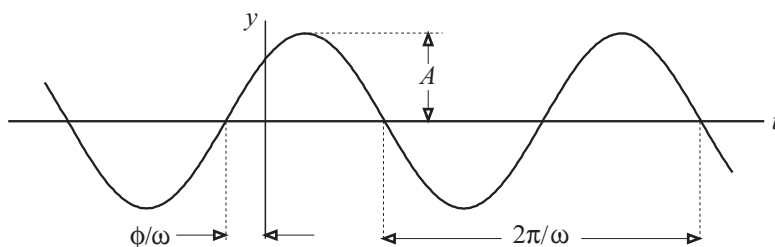


Figure 2.1-9

(8) $y = A \sin(\omega t + \phi)$, in which A , ω and ϕ are constants. When this function occurs in physics, it usually involves time and for this reason the time, t is used as independent variable. In this function A is called the **amplitude** and $(\omega t + \phi)$ the **phase angle** or **phase** in short. ω is the **angular frequency** and ϕ the **phase constant** or **initial phase**. The function $y = A \cos(\omega t + \phi)$ has exactly the same shape as that shown in Figure 2.1-9 but as if this curve has been moved a distance $\pi/2\omega$ to the left.

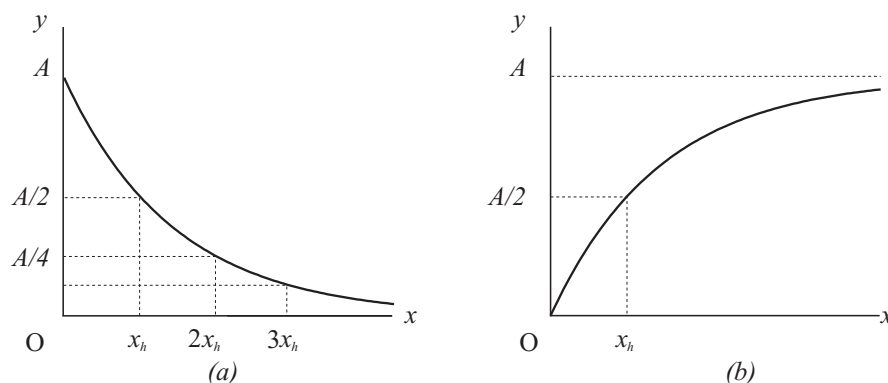


Figure 2.1-10

(9) $y = Ae^{-kx}$ in which A and k are positive constants, is known as an **exponential decay curve** with **decay constant** k . It has the interesting property that its value is A when $x = 0$ and after interval $\Delta x = x_h = (\ln 2)/k$, it has half this value, namely $A/2$. After every increment of $\Delta x = x_h$ the value of the function is half of what it is at the beginning of the interval. When $x \rightarrow \infty$, y tends to zero. The graph is shown in Figure 2.1-10(a).

(10) $y = A(1 - e^{-kx})$ in which A and k are positive constants, is known as **bounded or limited exponential growth** with **growth constant** k . The way in which this function grows, is a mirror image of the decay of that in (9). The graph of this function is shown in Figure 2.1-10(b).

In studying physics, the reader will also encounter other functions. There are, however, few phenomena in nature in which one or more of the functions mentioned here, do not play a part.

2.2 Calculations with zero and infinity. Limits

2.2.1 Calculations involving zero and numbers tending to infinity

For any finite number a we have

$$a + 0 = a, \quad a - 0 = a, \quad a \times 0 = 0 \quad 2.2(1)$$

“Division by zero” illustrates the fact that the arithmetic operations are *defined*, and that the definitions are chosen so as to be consistent with previously stated

rules of arithmetic.

Assume that $a/0 = b$, a finite number. It follows from the definition of a quotient and 2.2(1) that $a = b \times 0 = 0$. But this last equation cannot be valid, because $b \times 0 = 0$, no matter what number we choose for b , and we have assumed specifically that $a \neq 0$. Since we cannot assign any number b to the symbol $a/0$ that will be consistent with the method of assigning numbers to other quotients, we simply do not define the symbol $a/0$. Therefore $a/0$ is meaningless, that is, no meaning has been assigned to division of a finite number by zero.

A somewhat different problem is the case of the quotient $0/0$. Assume that $0/0 = b$. It follows from the definition of a quotient that $0 = b \times 0$. But this last equation is true for any number b , and there is no particular reason to prefer one number over any other. Again we avoid the difficulty by leaving $0/0$ undefined.

Consider the function $y = y(x) = 1/x$. As x tends to 0 from the positive side (written $x \rightarrow 0^+$), $1/x$ increases without bound. That is, $1/x$ becomes greater than any finite number we may think of. In mathematical notation we write $1/x \rightarrow \infty$ as $x \rightarrow 0^+$.

If x tends to zero from the negative side (written $x \rightarrow 0^-$), $1/x$ decreases without bound and we may write $1/x \rightarrow -\infty$ as $x \rightarrow 0^-$.

If x increases or decreases without bound, $|1/x|$ becomes **infinitesimally** small, that is, it tends to zero. We may write $1/x \rightarrow 0$ as $x \rightarrow \pm\infty$. In this case a limit value exists (see section 2.2.2) which is expressed as

$$\lim_{x \rightarrow \pm\infty} \frac{1}{x} = 0$$

When we say that n “tends to ∞ ”, we simply mean that n is supposed to assume a series of values which increase beyond all limit. **There is no number “infinity”**. An equation such as

$$n = \infty$$

as it stands, is *meaningless*. The reader will always have to bear in mind that ∞ *by itself* means nothing, although *phases containing it*, sometimes mean something, e.g. the phrase “tends to ∞ ”.

Problems in which factors tend to zero and infinity will be understood well once the techniques discussed in the theory of *limits* have been mastered. It is of prime importance that $a/0$, $0/0$, ∞/∞ and $0 \times \infty$ are not defined and therefore have no meaning. Calculators do not have a button for infinity and cannot give it as a read-out, but the reader might try some undefined calculations involving zero to observe the calculator’s response.

2.2.2 Limits

Consider the function $y = 3x/(x + 1)$, $x \neq -1$.

In order to illustrate an interesting property, the value of y is calculated for different values of $x \geq 0$.

$x = 0$	$y = 0/1$	$= 0$
$x = 1$	$y = 3/2$	$= 1,500000000$
$x = 2$	$y = 6/3$	$= 2,000000000$
$x = 3$	$y = 9/4$	$= 2,250000000$
$x = 5$	$y = 15/6$	$= 2,500000000$
$x = 10$	$y = 30/11$	$= 2,727272727$
$x = 50$	$y = 150/51$	$= 2,941176470$
$x = 100$	$y = 300/101$	$= 2,970297029$
$x = 10^3$	$y = 3000/1001$	$= 2,997002997$
$x = 10^6$	$y = 3000000/1000001$	$= 2,999997000$

When $x \rightarrow \infty$, y takes on the form ∞/∞ which has no meaning since it is undefined. From the calculation it would seem to be a reasonable conclusion that if x becomes very much larger than any of the values used in the calculations, the function tends towards a **limit value** of 3.

This result may also be obtained by changing the procedure of the calculation. The numerator and denominator of the fraction which constitute the function may be divided by any non-zero number without affecting a change in its value. Dividing numerator and denominator by x ($x \neq 0$), the function becomes

$$y = \frac{3x}{x+1} = \frac{3}{1+1/x}$$

From this follows that the contribution of the term $1/x$ in the denominator becomes less significant as the magnitude of x increases and the following statement becomes valid:

The limiting value of y when $x \rightarrow \infty$, is $\frac{3}{1+0}$. In short this may be written as

$$\lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} \frac{3x}{x+1} = \lim_{x \rightarrow \infty} \frac{3}{1+1/x} = \frac{3}{1+0} = 3$$

Consider also the following function:

$$y = \frac{x^2 - 9}{x - 3}, \quad x \neq 3$$

If x is assigned the value 3, we obtain the form $0/0$ which is meaningless. This means that the function $(x^2 - 9)/(x - 3)$ is not defined at $x = 3$. That is why $x = 3$ has been excluded from the definition of the function under consideration. It would, however, be of interest to investigate the behaviour of the function when $x \rightarrow 3$. Since $x \neq 3$, $x - 3 \neq 0$ and we may divide numerator and denominator by $x - 3$ and investigate the result when $x \rightarrow 3$ as shown below.

$$\begin{aligned}\lim_{x \rightarrow 3} y &= \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} \frac{(x + 3)(x - 3)}{x - 3} \\ &= \lim_{x \rightarrow 3} (x + 3) = 6\end{aligned}$$

For the purpose of this study, the following example is the most important since it will be needed to develop a fundamental concept in calculus.

Consider the function $y = 2x^2 - 3x + 4$. Calculate the value of

$$\lim_{h \rightarrow 0} \frac{y(x + h) - y(x)}{h}$$

Direct substitution of $h = 0$ raises the problem of zero divided by zero which is undefined. It is, however, permissible to calculate the value of the expression for a very small value of h and investigate the behaviour of the function when h tends to zero.

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{y(x + h) - y(x)}{h} &= \lim_{h \rightarrow 0} \frac{[2(x + h)^2 - 3(x + h) + 4] - [2x^2 - 3x + 4]}{h} \\ &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - 3h}{h} = \lim_{h \rightarrow 0} (4x + 2h - 3) \\ &= 4x - 3\end{aligned}$$

In this section the concept of a limit was treated very superficially. It is assumed that the reader will also follow a course in mathematics in which the topic will be treated in a more satisfactory way. The present treatment is sufficient to deal with the scope of this work.

2.3 The gradient of a straight line

In section 2.1.2(2) it was shown that the equation of a straight line is $y = mx + c$ in which m and c are constants. The **gradient** or **slope** of the straight line which represents this function, is the **rate** at which y changes with increasing x and

may be defined as follows: The gradient (slope) of the straight line $y = mx + c$ is the *change* in y between two points on it divided by the corresponding *increase* in x .

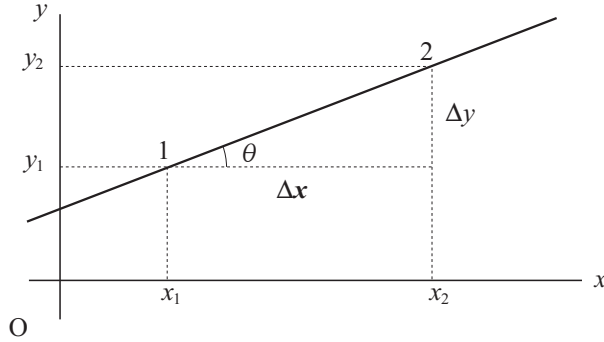


Figure 2.3-1

Consider the graph of $y = mx + c$ which is shown in Figure 2.3-1. Points 1 and 2 have co-ordinates (x_1, y_1) and (x_2, y_2) respectively and $x_2 > x_1$. According to the definition of the gradient, it follows that

$$\begin{aligned}
 \text{gradient} &= \frac{\text{change in } y}{\text{increase in } x} = \frac{\Delta y}{\Delta x} \quad (\Delta x > 0) \\
 &= \frac{y_2 - y_1}{x_2 - x_1} \quad (x_2 > x_1 \text{ because of increase}) \\
 &= \frac{y(x_2) - y(x_1)}{x_2 - x_1} \\
 &= \frac{(mx_2 + c) - (mx_1 + c)}{x_2 - x_1} \\
 &= \frac{m(x_2 - x_1)}{x_2 - x_1} \\
 &= m
 \end{aligned}$$

In Figure 2.3-1 the angle θ between the x -axis and the graph is called the **angle of inclination** or **slope angle**. In the solution of physics problems, it is often helpful to make use of the fact that the tangent of the angle of inclination is equal to the gradient.

$$\boxed{m = \text{gradient} = \Delta y / \Delta x = \tan \theta} \qquad 2.3(1)$$

Consider the straight lines in Figure 2.3-2.

In sketch (a) the function increases if x increases ($y_2 > y_1$; $\Delta y = y_2 - y_1 > 0$),

and the gradient is a positive number. For the function $y = mx + c$ it means that $m > 0$.

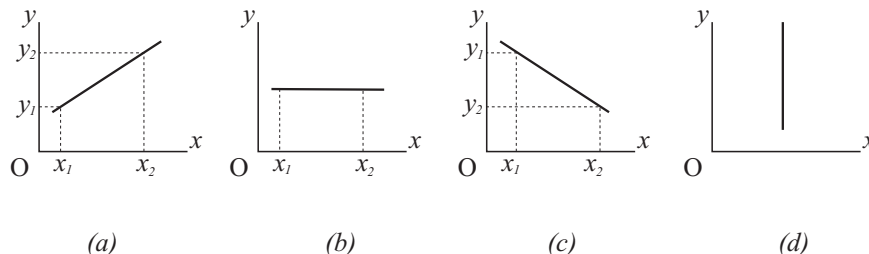


Figure 2.3-2

In sketch (b), y remains unchanged if x increases ($y_2 = y_1$; $\Delta y = 0$) and the gradient is zero. For the function $y = mx + c$ it means that $m = 0$ so that $y = c$ (constant).

In sketch (c) y decreases with increase in x ($y_2 < y_1$; $\Delta y < 0$) and the gradient is negative ($m < 0$).

In sketch (d) the straight line is *not* the representation of a function since an infinite number of y -values correspond to one and the same value of x . Strictly, the gradient of this line is undefined since it would imply dividing by zero. For some problems in physics it is useful to refer to the “gradient” of the line as $+\infty$ or $-\infty$.

From the preceding material it should be clear that the gradient or slope of a straight line is a measure of its *inclination*. If the direction of the x -axis is taken as horizontal and that of the y -axis as vertical and the inclination of the line is examined in the direction in which x increases, the following will be valid:

$m > 0$ means that the slope of the line is “uphill”. If $m_1 > m_2$, it means that the line with gradient m_1 is steeper uphill than one with a gradient of m_2 .

$m = 0$ means that the line is horizontal.

$m < 0$ means that the slope of the line is “downhill”.

$m \rightarrow \pm\infty$ refers to a vertical line.

Spoornet uses the notation 1:200 to indicate the incline of a stretch of rail. This notation refers to the sine instead of the tangent of the angle of incline. Can you think of a reason why this is used? Is the difference in definitions of any practical importance?

2.4 The gradient of a curve

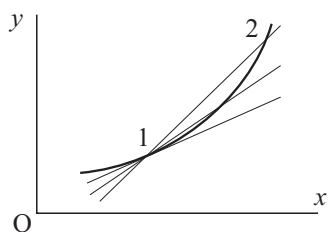


Figure 2.4-1

If a graph is said to be curved in a given region, it simply means that its gradient is not constant and then we refer to the graph as a **curve**. The procedure described for the determination of the gradient of a straight line, is not directly applicable to a curve for two simple reasons. Firstly, the gradient differs from one position to the next and secondly, the answer for a given position (x_1, y_1) will be influenced by the magnitude of the interval Δx .

A new definition is required to specify the gradient of the curve at a given position unambiguously. The problem is solved if the gradient of a curve at a given position is defined as the gradient of the tangent to the curve at that position. Strictly, this definition will be valid only if a single tangent exists at the specified point.

The gradient of the tangent can be calculated by using the procedure for a straight line. Although the definition is correct, it does not contain information about a method to calculate the gradient of the curve at the required position. A beginner might try a graphic method by drawing a tangent at the given position and base the calculation on measurement. The accuracy of this method is not very reliable.

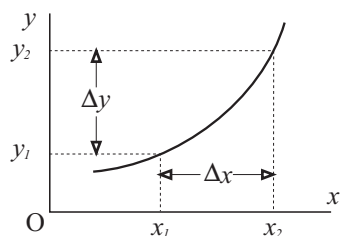


Figure 2.4-2

Figure 2.4-1 not only gives a slightly different approach to the definition, but also supplies information for the calculation of the gradient of the curve at a given position. Suppose that the gradient is required at position 1 of which the co-ordinates are (x_1, y_1) . Position 2 is a second point on the curve to the right of 1 ($x_2 > x_1$). The gradient of the cord (the straight line joining points 1 and 2) will generally differ from the gradient of the tangent at point 1. From the

sketch it should be clear that the gradient of the cord will differ less from the gradient of the tangent if position 2 is chosen nearer to 1. It may also be stated

in the following way: As position 2 tends to position 1, the gradient of the cord tends to the gradient of the tangent, which is the gradient of the curve at the required position.

To state that point 2 tends to point 1, is the same as to state that $x_2 \rightarrow x_1$, or $\Delta x \rightarrow 0$. If $\Delta x \rightarrow 0$, then $\Delta y \rightarrow 0$ and the gradient apparently becomes $\Delta y / \Delta x = 0/0$ which is undefined. The sketch in Figure 2.4-1 shows that the gradient of the cord tends to a limit (that of the tangent) when $\Delta x \rightarrow 0$, and the technique discussed in section 2.2.2 may be applied to calculate it. This allows for a better definition for the gradient of a curve which also incorporates a method for its calculation.

$$\text{gradient} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \quad 2.4(1)$$

Other notations which are preferred by some, are as follows:

$$\text{gradient} = \lim_{h \rightarrow 0} \frac{y(x_1 + h) - y(x_1)}{h} = \lim_{h \rightarrow 0} \frac{f(x_1 + h) - f(x_1)}{h} \quad 2.4(2)$$

Example: Calculate the gradient of the function $y = x^2 - 6x + 10$ at the positions where (a) $x = 4$, (b) $x = 1, 5$. The graph is shown in Figure 2.4-3.

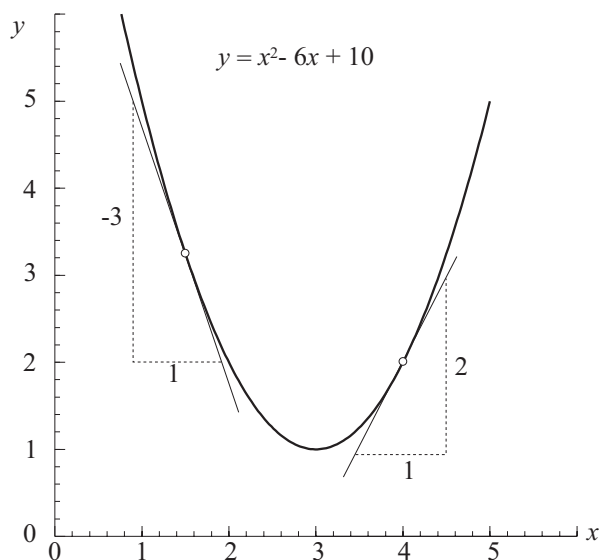


Figure 2.4-3

$$\begin{aligned}
\text{(a) At } x_1 = 4, \quad y_1 &= y(4) = 4^2 - 6(4) + 10 = 2 \\
\text{At } x_2 = 4 + \Delta x, \quad y_2 &= y(4 + \Delta x) \\
&= (4 + \Delta x)^2 - 6(4 + \Delta x) + 10 \\
&= (16 + 8\Delta x + [\Delta x]^2) - (24 + 6\Delta x) + 10 \\
&= 2 + 2\Delta x + [\Delta x]^2
\end{aligned}$$

From the definition, Equation 2.4(1), it follows

$$\begin{aligned}
\text{gradient} &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{y_2 - y_1}{x_2 - x_1} \\
&= \lim_{\Delta x \rightarrow 0} \frac{(2 + 2\Delta x + [\Delta x]^2) - (2)}{(4 + \Delta x) - (4)} = \lim_{\Delta x \rightarrow 0} \frac{2\Delta x + [\Delta x]^2}{\Delta x} \\
&= \lim_{\Delta x \rightarrow 0} (2 + \Delta x) = 2
\end{aligned}$$

(b) In the same way as in (a) above, the gradient may be calculated at $x = 1, 5$.

$$\begin{aligned}
\text{gradient} &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{y(1, 5 + \Delta x) - y(1, 5)}{\Delta x} \\
&= \lim_{\Delta x \rightarrow 0} \frac{([1, 5 + \Delta x]^2 - 6[1, 5 + \Delta x] + 10) - (1, 5^2 - 6[1, 5] + 10)}{\Delta x} \\
&= \lim_{\Delta x \rightarrow 0} \frac{(2, 25 + 3\Delta x + [\Delta x]^2 - 9 - 6\Delta x + 10) - (2, 25 - 9 + 10)}{\Delta x} \\
&= \lim_{\Delta x \rightarrow 0} \frac{-3\Delta x + [\Delta x]^2}{\Delta x} = \lim_{\Delta x \rightarrow 0} (-3 + \Delta x) = -3
\end{aligned}$$

Excluding cases where the behaviour of the function is such that its gradient is undefined and does not exist at a given position, the method may be used in principle to calculate the gradient of a function at any position.

2.5 Derived functions or derivatives

From the work done in the previous section, it should be obvious that the gradient of a function changes if the abscissa changes. The gradient is indeed also a function of the independent variable. If it is possible to determine the function which describes the gradient (i.e. an equation or a formula by means of which the gradient may be calculated at any position), it will not be necessary to follow the procedure which was illustrated in section 2.4 for the calculation of the gradient at a given position.

The function which expresses the gradient in terms of x is called the **gradient function**, the **derived function**, or **derivative** and it will allow the calculation of the gradient at each position where it exists.

For the calculation of the derivative a similar procedure is followed as that in section 2.4. The difference is that instead of using a fixed value x_1 at which to calculate the gradient, an unspecified value x is used. We use x instead of x_1 and $x + \Delta x$ instead of x_2 . The derivative may be defined as follows:

$$\text{derivative} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{y(x + \Delta x) - y(x)}{\Delta x}$$

Unfortunately there are a large number of different notations to indicate the derivatives of $y = y(x)$. Although each notation has advantages, this variety may initially confuse one who is new to the study of calculus. A number of different notations are shown.

$\frac{dy}{dx}$ or dy/dx : This is read *d-y-d-x* and *not* *dy* divided by *dx*. It is a symbolic notation for the derivative, i.e. a function by means of which the gradient of $y = y(x)$ may be calculated. In this book preference will be given to this notation

$\frac{d}{dx}(y)$: This notation is read *d-d-x of y*. It is seldom used symbolically since the position of y is usually occupied by the expression for y in terms of x . The portion d/dx is known as an **operator** which may be interpreted as a procedure to be applied to y in order to determine its derivative.

$D(y)$ or $D_x y$: The same as above with operators D and D_x instead of d/dx . They are read *d-of-y* and *d-x-of-y* respectively.

$y', f'(x), \dot{y}$: They are read *y-prime*, *f-prime-of-x* and *y-dot* respectively. They are all symbolic notations for the derivative of y .

By means of the notation which is preferred in this book, the definition for the derivative of the function $y = y(x)$ may be written as follows:

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{y(x + \Delta x) - y(x)}{\Delta x} \quad 2.5(1)$$

This definition is now applied to determine the derivative of the function of which the graph is shown in Figure 2.4-3, namely $y = x^2 - 6x + 10$.

$$\begin{aligned} \frac{dy}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{y(x + \Delta x) - y(x)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{([x + \Delta x]^2 - 6[x + \Delta x] + 10) - (x^2 - 6x + 10)}{\Delta x} \end{aligned}$$

$$\begin{aligned}
&= \lim_{\Delta x \rightarrow 0} \frac{(x^2 + 2x[\Delta x] + [\Delta x]^2 - 6x - 6\Delta x + 10) - (x^2 - 6x + 10)}{\Delta x} \\
&= \lim_{\Delta x \rightarrow 0} \frac{2x[\Delta x] - 6\Delta x + [\Delta x]^2}{\Delta x} \\
&= \lim_{\Delta x \rightarrow 0} (2x - 6 + \Delta x) \\
&= 2x - 6
\end{aligned}$$

The formula $dy/dx = 2x - 6$ enables one to calculate the gradient of the function $y = x^2 - 6x + 10$ at any value of x .

$$\left. \frac{dy}{dx} \right|_{x=-1} = 2(-1) - 6 = -8 \qquad \left. \frac{dy}{dx} \right|_{x=1,5} = 2(1,5) - 6 = -3$$

$$\left. \frac{dy}{dx} \right|_{x=0} = 2(0) - 6 = -6 \qquad \left. \frac{dy}{dx} \right|_{x=4} = 2(4) - 6 = 2$$

The two values on the right agree with those calculated previously with much more effort. It should be clear that the determination of the derivative of a function and then using it to calculate the gradient of the function at any desired position, is much easier than calculating the gradient at each position by using the definition.

The procedure by which the derivative is determined is called **differentiation** and the verb is **differentiate**. An assignment to *differentiate a given function*, means that its derivative is to be determined. The only means at our disposal at this stage is the procedure which has been followed in the above example.

2.6 Useful differentiation formulas

A number of functions often occur in the study of physics. For this reason it has definite advantages to differentiate them by using the definition and then remembering the results. The results are then simply used as a formulae to write down the answers when such functions are to be differentiated.

2.6.1 $y = x^n$ in which n is any real finite constant

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{y(x + \Delta x) - y(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{(x + \Delta x)^n - x^n}{\Delta x}$$

But $(x + \Delta x)^n = x^n + nx^{n-1}\Delta x + [n(n-1)/2!]\Delta x^2 + \text{higher powers of } \Delta x$. This result follows directly from the binomial theorem which is explained in section 1.4. Using this result, the derivative becomes

$$\begin{aligned}\frac{dy}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{(x^n + nx^{n-1}\Delta x + \text{higher powers of } \Delta x) - x^n}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} (nx^{n-1} + \text{powers of } \Delta x) = nx^{n-1}\end{aligned}\quad 2.6(1)$$

2.6.2 $y = ax^n$ in which a and n are real finite numbers

The derivative is calculated in exactly the same way as that in 2.6.1. It is left to the reader as an exercise.

$$\frac{d}{dx}(ax^n) = anx^{n-1} \quad 2.6(2)$$

Equation 2.6(1) is the special case of 2.6(2) where $a = 1$. Since the latter represents a more general case it is the one that needs to be remembered.

2.6.3 $y = \sin ax$ in which a is any real finite number

$$\begin{aligned}\frac{dy}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{\sin a(x + \Delta x) - \sin ax}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\sin ax \cos a\Delta x + \cos ax \sin a\Delta x - \sin ax}{\Delta x}\end{aligned}$$

In the last step, Equation 1.7(4), namely $\sin(\theta + \phi) = \sin \theta \cos \phi + \cos \theta \sin \phi$, was used. Since Δx and also $a\Delta x$ may be chosen to be as small as desired, the following approximations may be applied: $\sin a\Delta x \approx a\Delta x$ (x in radians) and $\cos a\Delta x \approx 1$. The derivative becomes

$$\begin{aligned}\frac{dy}{dx} &= \lim_{\Delta x \rightarrow 0} \frac{\sin ax + a\Delta x \cos ax - \sin ax}{\Delta x} \\ &= a \cos ax\end{aligned}\quad 2.6(3)$$

2.6.4 $y = \cos ax$ in which a is any real finite number

The procedure is the same as that used in 2.6.3 and it is left to the reader as an exercise.

$$\frac{d}{dx}(\cos ax) = -a \sin ax \quad 2.6(4)$$

Examples:

Problems (a) to (f) are applications of the result in 2.6.2

$$(a) \quad d/dx(x^3) = 3x^{3-1} = 3x^2$$

$$(b) \quad d/dx(7x^3) = 21x^2$$

$$(c) \quad d/dx(7) = 0 \quad \text{The graph is horizontal. See Fig. 2.3-2(b).}$$

$$(d) \quad d/dx(1/x^3) = d/dx(x^{-3}) = -3x^{-3-1} = -3x^{-4}$$

$$(e) \quad d/dx(x^{1/3}) = (1/3)x^{1/3-1} = (1/3)x^{-2/3}$$

$$(f) \quad d/dx(\sqrt{x}) = d/dx(x^{1/2}) = (1/2)x^{-1/2} = 1/(2\sqrt{x})$$

Problems (g) to (i) are applications of the results in 2.6.3 and 2.6.4.

$$(g) \quad d/dx(\sin 5x) = 5 \cos 5x$$

$$(h) \quad d/dx(\sin x) = \cos x$$

$$(i) \quad d/dt(\cos 7t) = -7 \sin 7t$$

2.7 Useful differentiation rules

2.7.1 The sum rule

The functions for which the derivatives were calculated in 2.6, all consist of a single term. The sum rule shows how a function consisting of the sum of a number of terms may be differentiated.

Let $u = u(x)$ and $v = v(x)$, i.e. both u and v are functions of x . Form a new function by the addition of $u(x)$ and $v(x)$: $y = u(x) + v(x)$.

If x increases by Δx , then $u(x)$ and $v(x)$ will generally change and that will also be the case for $y = y(x)$ so that

$$\begin{aligned} u(x + \Delta x) &= u + \Delta u & \text{and} & & v(x + \Delta x) &= v + \Delta v \\ \text{and} \quad y(x + \Delta x) &= y + \Delta y \\ \text{so that} \quad y + \Delta y &= (u + \Delta u) + (v + \Delta v) \\ \Rightarrow \quad \Delta y &= \Delta u + \Delta v \end{aligned}$$

For $\Delta x \neq 0$, the left and right-hand sides may be divided by Δx as follows:

$$\begin{aligned} \frac{\Delta y}{\Delta x} &= \frac{\Delta u}{\Delta x} + \frac{\Delta v}{\Delta x} \\ \text{so that } \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta u}{\Delta x} + \frac{\Delta v}{\Delta x} \right) \end{aligned}$$

There is a theorem about the limit of a sum which will not be proved here. It is as follows: *The limit of the sum of a number of expressions is equal to the sum of the limits of the individual expressions.* Applied to the above, it follows that

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} + \lim_{\Delta x \rightarrow 0} \frac{\Delta v}{\Delta x} \\ \text{which, by definition, is } \frac{dy}{dx} &= \frac{du}{dx} + \frac{dv}{dx} \end{aligned} \quad 2.7(1)$$

This relationship is known as the sum rule and may be formulated in words as follows: *The derivative of the sum of a number of expressions is equal to the sum of the derivatives of the individual expressions.* Since u and v may each consist of more terms than one, the sum rule applies to the sum of any number of terms.

Examples:

- (a) $d/dx(3x^2 - 4x^5) = 6x - 20x^4$
- (b) $d/dx(5x^4 - 3 \sin 4x) = 20x^3 - 12 \cos 4x$
- (c) $d/dx(2x^{-3} + 3 \cos 2x - 7) = -6x^{-4} - 6 \sin 2x$
- (d) $d/dr(\sqrt{r^3} - 4 \cos 2r - 4r) = 1.5r^{1/2} + 8 \sin 2r - 4$

2.7.2 The product rule

The product rule is used to differentiate the product of two functions. If $u = u(x)$ and $v = v(x)$ and a new function is formed by their product $y = y(x) = u(x) \times v(x)$, then

$$\begin{aligned} y(x + \Delta x) &= u(x + \Delta x) \times v(x + \Delta x) \\ \text{i.e. } y + \Delta y &= (u + \Delta u) \times (v + \Delta v) \\ &= uv + u\Delta v + v\Delta u + \Delta u\Delta v \\ \Rightarrow \Delta y &= u\Delta v + v\Delta u + \Delta u\Delta v \end{aligned}$$

$$\begin{aligned}
\text{so that} \quad \frac{\Delta y}{\Delta x} &= u \frac{\Delta v}{\Delta x} + v \frac{\Delta u}{\Delta x} + \Delta u \frac{\Delta v}{\Delta x} \\
\text{and} \quad \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \left(u \frac{\Delta v}{\Delta x} + v \frac{\Delta u}{\Delta x} + \Delta u \frac{\Delta v}{\Delta x} \right) \\
&= \lim_{\Delta x \rightarrow 0} u \frac{\Delta v}{\Delta x} + \lim_{\Delta x \rightarrow 0} v \frac{\Delta u}{\Delta x} + \lim_{\Delta x \rightarrow 0} \Delta u \frac{\Delta v}{\Delta x}
\end{aligned}$$

Another theorem about limits which is needed and which will not be proved here, is as follows: *The limit of the product of two expressions is equal to the product of the limits of the two individual expressions.* The last term on the right-hand side of the above equation thus becomes

$$\lim_{\Delta x \rightarrow 0} \Delta u \frac{\Delta v}{\Delta x} = \left(\lim_{\Delta x \rightarrow 0} \Delta u \right) \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta v}{\Delta u} \right) = 0 \times \frac{dv}{dx} = 0$$

Finally it follows that

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx} \quad 2.7(2)$$

which is known as the product rule.

Examples:

(a) $y = (3x^2 - 5)(2x^3 + 7x)$. Calculate dy/dx .

$$\begin{aligned}
\text{Let } u &= 3x^2 - 5. & \text{Then } du/dx &= 6x \\
\text{and } v &= 2x^3 + 7x. & \text{Then } dv/dx &= 6x^2 + 7
\end{aligned}$$

$$\begin{aligned}
\text{But } \frac{dy}{dx} &= u \frac{dv}{dx} + v \frac{du}{dx} \\
&= (3x^2 - 5)(6x^2 + 7) + (2x^3 + 7x)(6x) \\
&= 18x^4 - 9x^2 - 35 + 12x^4 + 42x^2 \\
&= 30x^4 + 33x^2 - 35
\end{aligned}$$

The answer may be calculated directly by the sum rule after the expansion of the product of the two factors.

$$y = (3x^2 - 5)(2x^3 + 7x) = 6x^5 + 11x^3 - 35x, \text{ so that } dy/dx = 30x^4 + 33x^2 - 35$$

$$\begin{aligned}
\text{(b)} \quad y &= (3x^4 - 7x^2) \sin 3x \\
\frac{dy}{dx} &= (3x^4 - 7x^2) \frac{d}{dx} (\sin 3x) + (\sin 3x) \frac{d}{dx} (3x^4 - 7x^2) \\
&= (3x^4 - 7x^2)(3 \cos 3x) + (12x^3 - 14x)(\sin 3x)
\end{aligned}$$

$$\begin{aligned}
(c) \quad y &= \sin^2 x = (\sin x)(\sin x) \\
\frac{dy}{dx} &= (\sin x)(\cos x) + (\sin x)(\cos x) \\
&= 2 \sin x \cos x = \sin 2x
\end{aligned}$$

2.7.3 The quotient rule

If a function consists of the quotient of two expressions, it may be differentiated by means of the quotient rule. If $u = u(x)$ and $v = v(x)$ and a new function is formed by their quotient as follows: $y = y(x) = u(x)/v(x)$, then

$$\begin{aligned}
y + \Delta y &= \frac{u + \Delta u}{v + \Delta v} \\
\Delta y &= \frac{u + \Delta u}{v + \Delta v} - y = \frac{u + \Delta u}{v + \Delta v} - \frac{u}{v} \\
&= \frac{v(u + \Delta u) - u(v + \Delta v)}{v(v + \Delta v)} = \frac{v\Delta u - u\Delta v}{v^2 + v\Delta v} \\
\text{so that } \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \frac{v(\Delta u/\Delta x) - u(\Delta v/\Delta x)}{v^2 + v\Delta v}
\end{aligned}$$

The following theorem regarding the limit of a quotient is stated without proof: *The limit of the quotient of two expressions is equal to the quotient of the limits of the two expressions.* From this follows that

$$\begin{aligned}
\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} &= \frac{\lim_{\Delta x \rightarrow 0} v(\Delta u/\Delta x) - \lim_{\Delta x \rightarrow 0} u(\Delta v/\Delta x)}{\lim_{\Delta x \rightarrow 0} (v^2 + v\Delta v)} \\
\text{so that } \frac{dy}{dx} &= \frac{v(du/dx) - u(dv/dx)}{v^2} \quad 2.7(3)
\end{aligned}$$

which is known as the quotient rule.

Examples:

$$\begin{aligned}
(a) \quad y &= \frac{x^3 + 8}{x + 2}, \quad x \neq -2 \\
\text{Let } u &= x^3 + 8. & \text{Then } du/dx = 3x^2 \\
\text{and } v &= x + 2. & \text{Then } dv/dx = 1 \\
\text{but } \frac{dy}{dx} &= \frac{v(du/dx) - u(dv/dx)}{v^2} \\
&= \frac{(x + 2)(3x^2) - (x^3 + 8)(1)}{(x + 2)^2} \\
&= \frac{3x^3 + 6x^2 - x^3 - 8}{(x + 2)^2} = \frac{2x^3 + 6x^2 - 8}{(x + 2)^2} \\
&= 2x - 2
\end{aligned}$$

The final answer may be obtained by factorising the numerator or using long division.

For the calculation of this derivative the use of the quotient rule could be avoided since it could be written directly that

$$y = \frac{x^3 + 8}{x + 2} = \frac{(x + 2)(x^2 - 2x + 4)}{x + 2} = x^2 - 2x + 4$$

Application of the sum rule gives $dy/dx = 2x - 2$. This control shows that the quotient rule was used correctly.

$$(b) \quad y = \frac{3x - 5}{7x - 2}, \quad x \neq 2/7$$

Let $u = 3x - 5 \Rightarrow du/dx = 3$ and $v = 7x - 2 \Rightarrow dv/dx = 7$, so that

$$\frac{dy}{dx} = \frac{(7x - 2)(3) - (3x - 5)(7)}{(7x - 2)^2} = \frac{29}{(7x - 2)^2}$$

(c) $y = \tan x$. Since $\tan x = (\sin x)/(\cos x)$, the quotient rule may be used.

$$\begin{aligned} \frac{dy}{dx} &= \frac{(\cos x)(\cos x) - (\sin x)(-\sin x)}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x \end{aligned}$$

$$(d) \quad y = (2x^3 - 3)^{-1} = 1/(2x^3 - 3)$$

Let $u = 1 \Rightarrow du/dx = 0$ and $v = 2x^3 - 3 \Rightarrow dv/dx = 6x^2$

$$\frac{dy}{dx} = \frac{(2x^3 - 3)(0) - (1)(6x^2)}{(2x^3 - 3)^2} = -\frac{6x^2}{(2x^3 - 3)^2}$$

By means of the chain rule which is treated in section 2.7.4, this differentiation may be done in a much simpler way.

$$(e) \quad y = \csc x = 1/\sin x.$$

$$\frac{dy}{dx} = \frac{(\sin x)(0) - (1)(\cos x)}{\sin^2 x} = -\frac{\cos x}{\sin^2 x} = -\csc x \cot x$$

In this calculation the chain rule would also have supplied the result by a simpler method.

2.7.4 The chain rule

The chain rule is a most powerful tool in the calculation of derivatives and is used mostly in differentiating **a function of a function**. We call $\cos x$ a

function of x and although $\cos(2x^2 - 1)$ is also a function of x , it may be taken to be a function of $(2x^2 - 1)$ which in itself is also a function of x . $\cos 3x$ may be taken to be a function of $3x$, but its derivative, $(d/dx)(\cos 3x)$ may be calculated directly by the result in Equation 2.6(4). That is not the case for $(d/dx)(\cos [2x^2 - 1])$. If, however, the substitution $u = 2x^2 - 1$ is made, the function becomes $y = \cos u$ for which Equation 2.6(4) is applicable if dy/du is to be calculated. For the determination of dy/dx a different approach will have to be found.

If $y = y(u)$ and $u = u(x)$, it means that y is a function of u , which is a function of x and it may be said that if x increases by Δx , u will change by Δu . This change will cause y to change by an amount equal to Δy . For any *finite* values of Δx , Δu and Δy , we may write that

$$\frac{\Delta y}{\Delta x} = \frac{\Delta y}{\Delta u} \frac{\Delta u}{\Delta x}$$

If $\Delta x \rightarrow 0$, both Δu and Δy also tend to zero and from this follows that

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} &= \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta u} \frac{\Delta u}{\Delta x} \right) \\ &= \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta u} \right) \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} \right) \end{aligned}$$

$$\text{so that} \quad \frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} \quad 2.7(4)$$

This rule is known as the chain rule and may be extended for a function of a function of a function, etc. to contain as many factors ("links") as may be necessary. Such an extension will be demonstrated in the examples.

Examples:

(a) $y = (2x^2 - 3)^2$. Calculate dy/dx .

$$\begin{aligned} \text{Let } u &= 2x^2 - 3 \quad \Rightarrow \quad du/dx = 4x \\ \text{Then } y &= u^2 \quad \Rightarrow \quad dy/du = 2u = 2(2x^2 - 3) \\ \text{and } \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} = 2(2x^2 - 3)(4x) = 16x^3 - 24x \end{aligned}$$

This answer may be verified in the following way:

$$y = (2x^2 - 3)^2 = 4x^4 - 12x^2 + 9 \quad \Rightarrow \quad dy/dx = 16x^3 - 24x$$

(b) $y = \sqrt{3x^5 - 2x^2}$. Calculate dy/dx .

$$\text{Let } p = 3x^5 - 2x^2 \quad \Rightarrow \quad dp/dx = 15x^4 - 4x$$

$$\begin{aligned}
\text{Then } y &= \sqrt{p} = p^{1/2} \Rightarrow dy/dp = \frac{1}{2}p^{-1/2} = \frac{1}{2}(3x^5 - 2x^2)^{-1/2} \\
\text{and } \frac{dy}{dx} &= \frac{dy}{dp} \frac{dp}{dx} = (15x^4 - 4x) \times \frac{1}{2}(3x^5 - 2x^2)^{-1/2} \\
&= (15x^4 - 4x)/(2\sqrt{3x^5 - 2x^2})
\end{aligned}$$

(c) $y = \sin(3x^2 - 2x)$. Calculate dy/dx .

$$\begin{aligned}
\text{Let } s &= 3x^2 - 2x \Rightarrow ds/dx = 6x - 2 \\
\text{Then } y &= \sin s \Rightarrow dy/ds = \cos s = \cos(3x^2 - 2x) \\
\text{and } \frac{dy}{dx} &= \frac{dy}{ds} \frac{ds}{dx} = (6x - 2) \cos(3x^2 - 2x)
\end{aligned}$$

Comment: With practice this answer may be written down without showing the substitution in writing. The reader should work towards attaining this goal. In the preceding problem

dy/db = the derivative of the sine function with respect to the entire angle
= the cosine of the same angle as shown in example (h) on page 39.

db/dx = the derivative of the angle $(3x^2 - 2x)$ with respect to x and that is equal to $6x - 2$ which follows from the application of the sum rule.

The final answer is the product of these two factors.

(d) $y = \cos^3(3x^3 - 2x)$. Calculate dy/dx .

In this calculation it will be necessary to make more than one substitution and the “chain” will consist of more than two “links”.

$$\begin{aligned}
\text{Let } v &= 3x^3 - 2x \Rightarrow dv/dx = 9x^2 - 2. \\
\text{Let } u &= \cos(3x^3 - 2x) = \cos v \Rightarrow du/dv = -\sin v = -\sin(3x^3 - 2x) \\
\text{then } y &= u^3 \Rightarrow dy/du = 3u^2 = 3\cos^2(3x^3 - 2x) \\
\text{and } \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dv} \frac{dv}{dx} = (9x^2 - 2) \times [-\sin(3x^3 - 2x)] \times [3\cos^2(3x^3 - 2x)] \\
&= -3(9x^2 - 2) \sin(3x^3 - 2x) \cos^2(3x^3 - 2x)
\end{aligned}$$

2.8 Expansion in series and exponential functions

2.8.1 Expansion in series

Subject to a few conditions which will not be mentioned or treated here, a large variety of functions of one variable may be expressed by a **power series** of the independent variable as follows:

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 \dots\dots \quad 2.8(1)$$

in which the coefficients a_0, a_1, a_2 , etc. are constants which may be positive, negative or equal to zero. They may be calculated by a method which will be illustrated later in this book.

This phenomenon is known as the **theorem of Maclaurin**. The expansions of $\sin x$ and $\cos x$ which are given in Equations 1.7(18) and 1.7(19) respectively, and also the binomial theorem in section 1.4, are examples of such power series.

Expansion in a series is important in the study of physics. One of its many applications, is curve-fitting when experimental data is to be described by means of a smooth curve.

2.8.2 The derivative of an exponential function

The function $y = a^x$ in which the base, a , is a positive constant number, is an **exponential function**. In an exponential function the exponent is variable. It must not be confused with a power function such as $y = x^n$ in which the base is variable and the exponent a constant number. To calculate its derivative, the definition of a derivative is used.

$$\begin{aligned} dy/dx &= \lim_{h \rightarrow 0} \frac{y(x+h) - y(x)}{h} = \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} \\ &= \lim_{h \rightarrow 0} \frac{a^x a^h - a^x}{h} = \lim_{h \rightarrow 0} \frac{a^x (a^h - 1)}{h} \\ &= a^x \lim_{h \rightarrow 0} \frac{a^h - 1}{h} \end{aligned}$$

In this result the factor a^x is independent of h and may be written outside the limit. Since the rest of the expression contains no x , it must be a constant which can depend on a only (h tends to zero). Provisionally we represent this constant

about which very little is known, by $k(a)$. The derivative of an exponential function can thus be written as follows:

$$\frac{d}{dx}(a^x) = [k(a)]a^x \quad 2.8(2)$$

It is possible to gain information about the magnitude of $k(a)$ by means of an experiment. A number of different values of a are chosen and a graph of $y = a^x$ is drawn in each case. By means of construction and measurement, the gradient of each graph is determined at one position. By substituting these values in equation 2.8(2), it will be possible to calculate the value of $k(a)$ for each value of a .

When $x = 1$, the value of the function, $y = 1^x$, is always equal to 1. The graph is a straight line parallel to the x -axis of which the gradient is zero at all positions as shown in Figure 2.8-1. Substitution in Equation 2.8(2) gives

$$0 = [k(1)] \times 1^x = [k(1)] \times 1$$

so that

$$k(1) = 0$$

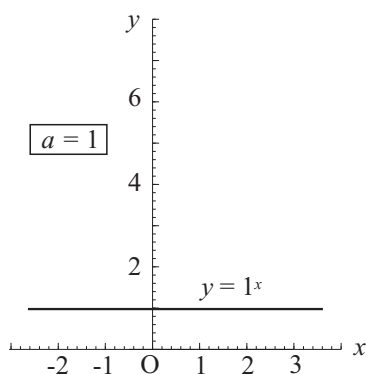


Figure 2.8-1

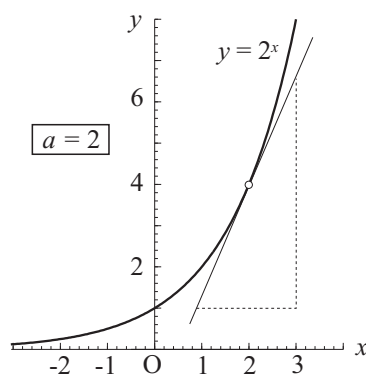


Figure 2.8-2

The next function is that for which $a = 2$ so that $y = 2^x$. The graph is shown in Figure 2.8-2. Its gradient is different at each position. The tangent at $x = 2$ is drawn as well as it is possible by means of a ruler and its gradient determined by measurement and calculation. According to the construction shown in the sketch, the gradient is approximately equal to 2.8. By substituting the relevant values in Equation 2.8(2), it follows that

$$2.8 \approx [k(2)] \times 2^2 = 4k(2)$$

so that $k(2) \approx 0,7$

Since it is based on a construction of questionable accuracy, this result is only an approximation. It is, however, quite clear that $k(2) < 1$ and about this result there can be little doubt. It may be verified by using other points on the graph in Figure 2.8-2.

If the same procedure is followed for $y = 3^x$, it will follow that $k(3) \approx 1,1$. Similarly $k(4) \approx 1,4$. It is quite clear that a value of a exists between 2 and 3 for which $k(a) = 1$. This value of a is indicated by the symbol e . At this stage it is known that $k(e) = 1$ and $2 < e < 3$. From this and Equation 2.8(2) follows that

$$\frac{d}{dx}(e^x) = [k(e)] \times e^x = 1 \times e^x = e^x \quad 2.8(3)$$

Experimentally we have discovered a number which has the strange property that if it is used as the base of an exponential function, the function is the same as its derivative. The expansion of this function as a power series, allows one to calculate the numerical value of e . From 2.8(1) it follows that

$$e^x = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots$$

The coefficients a_0, a_1, a_2 , etc. may be calculated in the way which is illustrated below.

$$\begin{aligned} e^x &= a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots \\ \Rightarrow e^0 &= 1 = a_0 + 0 + 0 + 0 + \dots \\ \text{so that } a_0 &= 1 \\ \text{Furthermore } \frac{d}{dx}(e^x) &= \frac{d}{dx}(a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + \dots) \\ e^x &= 0 + a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots \\ \Rightarrow e^0 &= 1 = a_1 + 0 + 0 + 0 + \dots \\ \text{so that } a_1 &= 1 \\ \text{Also } \frac{d}{dx}(e^x) &= \frac{d}{dx}(a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots) \\ e^x &= 0 + 2a_2 + 6a_3x + 12a_4x^2 + \dots \\ \Rightarrow e^0 &= 1 = 2a_2 + 0 + 0 + 0 + \dots \\ \text{so that } a_2 &= 1/2 = 1/2! \end{aligned}$$

In the same way it may be shown by successive differentiation that the other coefficients are $1/3!, 1/4!, 1/5!, 1/6!$, etc. so that

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots \quad 2.8(4)$$

When $x = 1$, the value of e may be calculated.

$$\begin{aligned} e^1 = e &= 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \dots \\ \Rightarrow e &= 2,71828183\dots \end{aligned} \quad 2.8(5)$$

As π , e is an irrational number which is used as the base for natural or Napierian logarithms with the following notation:

$$\text{If } b = e^c, \quad \text{then } c = \log_e b = \ln b$$

2.8.3 The derivative of a logarithmic function

Consider the expression $y = 2x + 6$ by which it is understood that y is a function of x which is described by the given equation. It is possible to rearrange this equation in such a way that x is the subject and is thus a function of y . The rearrangement is as follows:

$$x = \frac{1}{2}y - 3$$

Although it is not always possible to exchange the roles of the dependent and independent variables unconditionally, a pair of functions which originate in this way, have properties which are of special interest in this study.

Exchanging the roles of x and y as dependent and independent variables in the exponential function $y = e^x$, yields the following pair of functions:

$$y = e^x \quad \text{and} \quad x = \ln y$$

Consider the function $y = y(x)$ and the function $x = x(y)$ which is obtained if the roles of x and y are exchanged. It is assumed that the change of the subject of the equation is possible and valid. For such a pair of functions, a simple relationship exists between their graphs of which possible examples are shown in Figure 2.8-3.

At position P_1 in sketch (a), the gradient of the function $y = y(x)$ is given by

$$\left. \frac{dy}{dx} \right|_{P_1} = \tan \alpha = \tan (90^\circ - \beta) = \cot \beta = 1 / \tan \beta$$

and at the corresponding point P_2 on the graph of $x = x(y)$ in which the roles of x and y are interchanged, the gradient is given by

$$\left. \frac{dx}{dy} \right|_{P_2} = \tan \beta$$

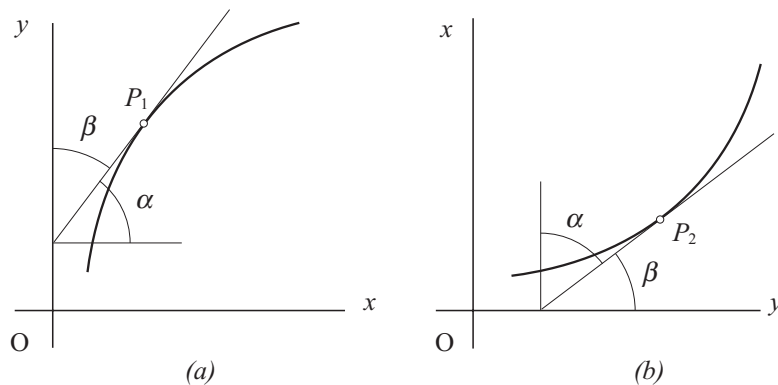


Figure 2.8-3

From this follows an important result which is often used in physics:

$$\frac{dy}{dx} = \left(\frac{dx}{dy} \right)^{-1} \quad \text{if } \frac{dy}{dx} \neq 0 \quad 2.8(6)$$

Consider once again the following pair of functions $y = y(x)$ and $x = x(y)$ in which the roles of x and y as dependent and independent variables are interchanged:

$$\begin{aligned} y &= \ln x \quad \text{and} \quad x = e^y \\ \text{From 2.8(3):} \quad \frac{dx}{dy} &= \frac{d}{dy}(e^y) = e^y = x \\ \text{but} \quad \frac{dx}{dy} &= \left(\frac{dy}{dx} \right)^{-1} \quad \text{or} \quad \frac{dy}{dx} = \left(\frac{dx}{dy} \right)^{-1} \\ \text{so that} \quad \frac{d}{dx}(y) &= \frac{d}{dx}(\ln x) = \frac{1}{x} \end{aligned}$$

This gives the fifth differentiation formula which is important enough to remember.

$$\frac{d}{dx}(\ln x) = \frac{1}{x} \quad 2.8(7)$$

This result enables one to differentiate exponential functions with base other than e .

$$\text{If} \quad y = a^x \quad \text{then} \quad \ln y = x \ln a \quad \text{or} \quad x = \ln y / \ln a$$

$$\begin{aligned} \text{now} \quad \frac{dx}{dy} &= \frac{d}{dy} \left(\frac{\ln y}{\ln a} \right) = \frac{1}{\ln a} \frac{1}{y} \\ \text{so that} \quad \frac{dy}{dx} &= (\ln a)y = (\ln a)a^x \end{aligned}$$

This result is the last of the differentiation formulas which are necessary for this course.

$$\frac{d}{dx}(a^x) = (\ln a)a^x \quad 2.8(8)$$

This result shows that the constant $k(a)$ which occurred in 2.8(2), is given by $k(a) = \ln a$.

Examples:

(a) $y = e^{2x^3}$ Calculate dy/dx .

$$\begin{aligned} \text{Let} \quad u &= 2x^3 & \Rightarrow du/dx &= 6x^2 \\ \text{now} \quad y &= e^u & \Rightarrow dy/du &= e^u = e^{2x^3} \\ \text{and} \quad \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} &= (e^{2x^3}) \times (6x^2) &= 6x^2 e^{2x^3} \end{aligned}$$

(b) $y = e^{-6x^2}$ Calculate dy/dx .

$$\frac{dy}{dx} = (e^{-6x^2}) \times \frac{d}{dx}(-6x^2) = -12xe^{-6x^2}$$

(c) $y = \ln x^2$ Calculate du/dx .

$$\begin{aligned} \text{Let} \quad u &= x^2 & \Rightarrow du/dx &= 2x \\ \text{now} \quad y &= \ln u & \Rightarrow dy/du &= 1/u = 1/x^2 \\ \text{so that} \quad \frac{dy}{dx} &= \frac{dy}{du} \frac{du}{dx} &= 2x/x^2 &= 2x^{-1} \end{aligned}$$

This result may also be calculated in the following way:

$$y = \ln x^2 = 2 \ln x \quad \Rightarrow dy/dx = (d/dx)(2 \ln x) = 2x^{-1}$$

(d) $y = \ln[x^3/(x+1)]$ Calculate dy/dx .

In the first step this function is written as the sum of two terms and then the sum rule and the chain rule are used to calculate its derivative. The substitution will not be shown as it was done in the previous problems.

$$y = \ln x^3 - \ln(x+1)$$

$$\begin{aligned}
dy/dx &= (x^3)^{-1} \times \frac{d}{dx}(x^3) - (x+1)^{-1} \times \frac{d}{dx}(x+1) \\
&= (x^{-3})(3x^2) - (x+1)^{-1} \times 1 \\
&= 3x^{-1} - (x+1)^{-1} = (2x+3)/(x^2+x)
\end{aligned}$$

2.9 Summary of the differentiation formulas and rules

$\frac{d}{dx}(ax^n) = nax^{n-1}$	$\frac{d}{dx}(e^{ax}) = ae^{ax}$
$\frac{d}{dx}(\sin ax) = a \cos ax$	$\frac{d}{dx}(a^x) = (\ln a)a^x$
$\frac{d}{dx}(\cos ax) = -a \sin ax$	$\frac{d}{dx}(\ln x) = x^{-1}$

If $u = u(x)$ and $v = v(x)$, then the following apply:

$$\begin{aligned}
\text{sum rule:} \quad (d/dx)(u \pm v) &= du/dx \pm dv/dx \\
\text{product rule:} \quad (d/dx)(u \times v) &= u(dv/dx) + v(du/dx) \\
\text{quotient rule:} \quad (d/dx)(u \div v) &= [v(du/dx) - u(dv/dx)] \times v^{-2}
\end{aligned}$$

If $y = y(p)$ and $p = p(x)$, then the following applies:

$$\text{chain rule:} \quad dy/dx = (dy/dp)(dp/dx)$$

On condition that $dx/dy \neq 0$, then

$$dy/dx = (dx/dy)^{-1}$$

It is *imperative* that the contents of this summary are *understood and remembered*. It is *vital* to the study of introductory physics.

2.10 Second and higher-order derivatives

Consider the function $y = x^2 - 4x + 3$. Its graphic representation is shown in Figure 2.10-1(a). The derivative of this function is given by $dy/dx = 2x - 4$, and its graph is shown in Figure 2.10-1(b). From the sketch it should be clear that the function $g = dy/dx$ has a constant gradient of 2. This means that $g = g(x) = [\text{the gradient of the function } y = y(x)]$ increases by 2 if x increases by 1. The gradient function also has a gradient.

The gradient of the derivative may be calculated by differentiating it to x as follows:

$$\frac{d}{dx} \frac{dy}{dx} = \frac{d}{dx} (2x - 4) = 2$$

This result is called the **second derivative** of y and is indicated by any of the following notations:

$$\frac{d^2 y}{dx^2}, \quad \frac{d^2}{dx^2}(y), \quad \frac{d}{dx} \frac{dy}{dx}, \quad \ddot{y}, \quad y'', \quad D^2(y)$$

They are pronounced *d-two-y-d-x-squared*, *d-two-d-x-squared of y*, *d-d-x of d-y-d-x*, *y-double-dot*, *y-double-prime* and *d-two of y*, respectively.

By means of the derivative of a function the rate of change of the function may be calculated at each position as the independent variable increases. In the same way the derivative of the derivative enables one to calculate the rate at which the gradient changes at a given position when the independent variable increases.

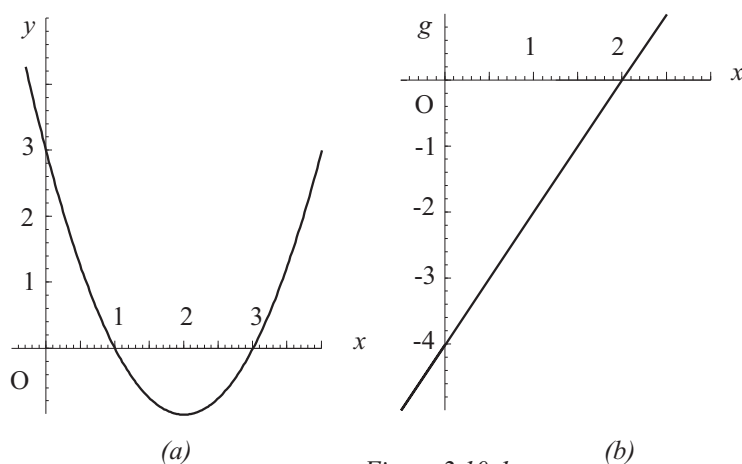


Figure 2.10-1

As it was possible to differentiate the derivative of a function, it will, in general, be possible to differentiate the derivative of the derivative of a function, etc. to

obtain the third, fourth and higher-order derivatives. The notation corresponds with that of the second-order derivative namely d^2y/dx^2 , in which n is the **order** of the derivative. The meaning of these higher-order derivatives will not be treated in this book but left to the ingenuity and intuition of the reader.

The sign of a second-order derivative has an interesting geometric interpretation as shown in Figure 2.10-2. In sketch (a) a curve is shown of which the gradient constantly increases from left to right, i.e. dy/dx increases with increase in x or d^2y/dx^2 is positive for the portion of the graph which is shown. If the value of d^2y/dx^2 had been larger than that shown in the sketch, the graph would have had a larger **curvature**.

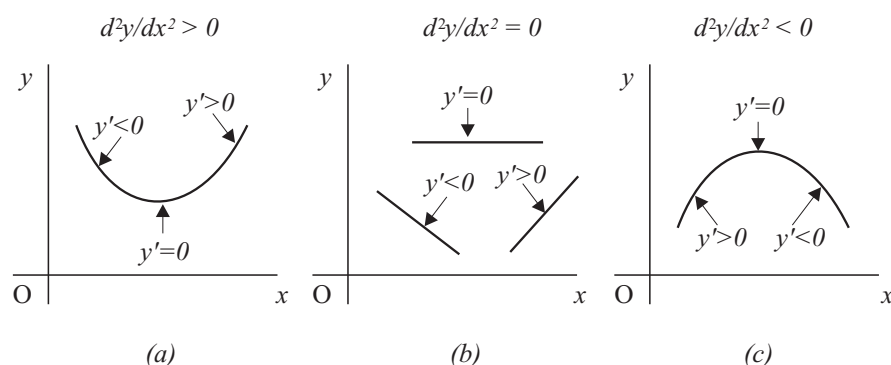


Figure 2.10-2

In sketch (b) three cases are shown in which the gradient remains constant but with a different sign in each case. The graphs are straight with no curvature. The value of the second derivative is zero in each case.

In sketch (c) the gradient decreases with increase in x . The sign of the second derivative is negative over the portion of the graph which is shown and the curve is concave when viewed in the direction in which y increases.

Examples:

(a) $y = 3x^4 - 2x^2 + 3x - 5$. Calculate d^2y/dx^2 .

$$\begin{aligned} \frac{dy}{dx} &= 12x^3 - 4x + 3 \quad \Rightarrow \quad \frac{d^2y}{dx^2} = \frac{d}{dx}(12x^3 - 4x + 3) \\ &= 36x^2 - 4 \end{aligned}$$

- (b)
- $y = 3 \sin 2x$
- . Calculate
- d^3y/dx^3
- .

$$\frac{dy}{dx} = 6 \cos 2x, \quad \frac{d^2y}{dx^2} = -12 \sin 2x, \quad \frac{d^3y}{dx^3} = -24 \cos 2x$$

- (c)
- $y = e^{2x}$
- . Calculate
- d^2y/dx^2
- .

$$\frac{dy}{dx} = 2e^{2x} \quad \Rightarrow \quad \frac{d^2y}{dx^2} = 4e^{2x}$$

- (d)
- $y = e^{2x^2}$
- . Calculate
- d^2y/dx^2
- .

$$\begin{aligned} \frac{dy}{dx} &= 4xe^{2x^2} & \frac{d^2y}{dx^2} &= \frac{d}{dx}(4xe^{2x^2}) \\ & & &= 16x^2e^{2x^2} + 4e^{2x^2} \quad (\text{product rule}) \\ & & &= (16x^2 + 4)e^{2x^2} \end{aligned}$$

2.11 Maxima and minima

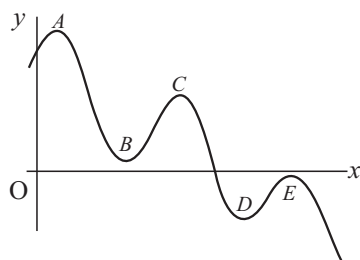


Figure 2.11-1

Consider the graph in Figure 2.11-1. Just to the left of point A , the function increases with increasing x (the gradient is positive) and just to the right of it, it decreases with increasing x (the gradient is negative.) Point A on the graph is called a **stationary point** or a **relative extreme value** of the function which the graph represents.

Just to the left of B , y decreases with increasing x and just to the right of it, it increases. B is also a stationary point or relative extreme value.

The value of the function, y , is larger at position A than at other positions in the immediate vicinity. A is called a **local maximum**. Positions C and E are also local maxima.

At B the value of the function is less than at other positions in its immediate vicinity and it is called a **local minimum**. The function also has a local minimum at position D .

The concepts of *maximum*, *minimum* and *immediate vicinity* may be confusing. Note that the maximum value at E is less than the minimum value at B . The terms *maximum* and *minimum* refer to values of the function at a *relative extreme value* in relation to positions which are very near the position. The confusion disappears if one notes the following:

1. At each relative extreme value (or stationary point), the gradient of the function is zero.
2. An extreme value is a minimum if the gradient increases on both sides of it with increasing x . This condition is met if $d^2y/dx^2 > 0$ at that position. At a relative minimum the graph is concave when viewed from above.
3. An extreme value is a maximum if the gradient decreases on both sides of it with increasing x and the condition is met if $d^2y/dx^2 < 0$ at that position. At a relative maximum, the graph is convex when viewed from above.

Positions of zero gradient may exist on a curve with no extreme value there. It is necessary to have knowledge of the properties of such conditions so that they are not erroneously confused with maxima or minima.

One such condition is when the x -axis or a line parallel to it, is an **asymptote** of the graph of a function. The exponential decay and limited exponential growth functions which are described by Equations 2.1.2(9) and (10) and of which the graphs are shown in Figure 2.1-10, are examples of such behaviour. The rectangular hyperbola which is described in Equation 2.1.2(3) and of which the graph is shown in Figure 2.1-5(b), is another example. The possibility of confusing these states with an extreme value because $dy/dx = 0$, should not happen since the value of d^2y/dx^2 is also equal to zero in each case.

Another possibility for the gradient to be equal to zero without the existence of an extreme value, is at a **point of inflexion**. A point of inflexion is the position where a graph changes from convex to concave or vice versa. Consider the graph in Figure 2.11-2(a). The gradient of the graph is zero at point a . To the left side of it, the graph is convex as viewed from above ($d^2y/dx^2 < 0$, i.e. the gradient is decreasing) and to the right side it is concave as viewed from above ($d^2y/dx^2 > 0$, i.e. the gradient is increasing). At A the value of d^2y/dx^2 is zero. Points of inflexion may also exist at positions where the gradient is not zero. The graph of such a function is shown in Figure 2.11-2(b). The value of d^2y/dx^2 is of course equal to zero at point B .

It may thus be stated that a curve has a point of inflexion at each position where $d^2y/dx^2 = 0$ and the sign of d^2y/dx^2 is different at a position immediately before and immediately after this position. It is further true that a monotonic

continuous function must have a point of inflexion between two consecutive

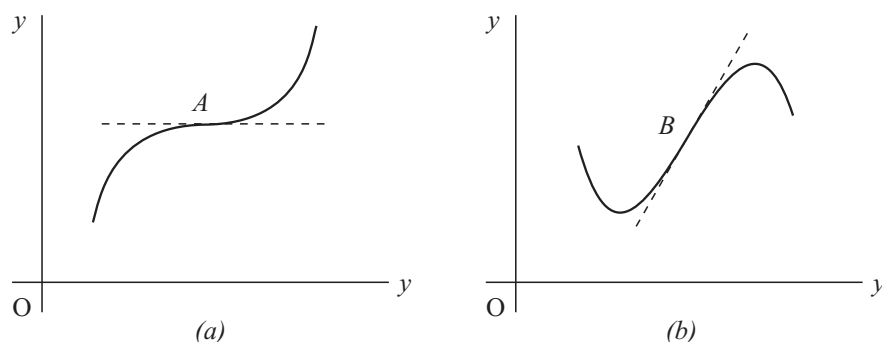


Figure 2.11-2

extreme points. A monotonic function is one which does not have more than one gradient at each point on it, and a continuous function is one which is defined at each value of the independent variable within its domain. For the purposes of this study, points of inflexion are of importance in cases such as that illustrated in Figure 2.11-2(a) where the danger exists that they might be confused with extreme values.

Summary of tests to determine the position and nature of local extreme values of a function $y = y(x)$:

1. At a local extreme value $dy/dx = 0$. At a point of inflexion dy/dx might be equal to zero, but not necessarily.
2. If $d^2y/dx^2 > 0$ at a position where $dy/dx = 0$, the function has a local extreme value at that position and it will be a local minimum.
If $d^2y/dx^2 < 0$ at a position where $dy/dx = 0$, the function has a local extreme value at that position and it will be a local maximum.
3. If both $dy/dx = 0$ and $d^2y/dx^2 = 0$ at a given position, it necessitates an examination of higher-order derivatives before a conclusion can be reached. The following is stated without proof:

If $(dy/dx)_{x=a} = 0$ and the smallest positive integer n for which $(d^n y/dx^n)_{x=a} \neq 0$ (i.e. all order derivatives are inspected from the first upwards and the first one which is *not* equal to zero at $x = a$, is considered), then

- (a) for even n and $(d^n y/dx^n)_{x=a} < 0$, $y(a)$ is a local maximum
 > 0 , $y(a)$ is a local minimum

(b) for uneven n a point of inflexion exists at $x = a$.

In order to understand the problem of extreme values, it is necessary to study the **Taylor series** expansion of a function about the point $x = a$. This theory is not complicated. The Maclaurin series which was mentioned in section 2.8.1, is a Taylor series about the point $x = 0$.

Problems:

(1) Determine the position and nature of the extreme value(s) of the function $y = x^2 - 4x + 3$.

$$y = x^2 - 4x + 3 \quad \Rightarrow \quad dy/dx = 2x - 4$$

The function possibly has an extreme value where $dy/dx = 0$, i.e. where

$$2x - 4 = 0 \quad \text{or} \quad x = 2. \quad \text{There} \quad y = y(2) = (2)^2 - 4(2) + 3 = -1$$

But $d^2y/dx^2 = (d/dx)(2x - 4) = 2$. The second derivative is always positive. The function thus has an extreme value at $x = 2$ which is a local minimum. The minimum value of the function is $y_{min} = y(2) = -1$.

Comment: In the same way it can be shown that the function $y = ax^2 + bx + c$ has an extreme value at $x = -b/2a$. The function is symmetrical about $x = -b/2a$. The extreme value is a minimum if $a > 0$ and a maximum if $a < 0$.

(2) Determine the position and nature of each extreme value of the function $y = 2x^3 - 9x^2 + 12x + 4$.

$$y = 2x^3 - 9x^2 + 12x + 4 \quad \Rightarrow \quad dy/dx = 6x^2 - 18x + 12$$

The function possibly has extreme values where $dy/dx = 0$, i.e. where

$$\begin{aligned} 6x^2 - 18x + 12 &= 0 \\ x^2 - 3x + 2 &= 0 \\ (x - 2)(x - 1) &= 0 \\ \Rightarrow \quad x = 2 \quad \text{and} \quad x &= 1 \\ d^2y/dx^2 = (d/dx)(6x^2 - 18x + 12) &= 12x - 18 \end{aligned}$$

At $x = 1$: $(d^2y/dx^2)_{x=1} = 12(1) - 18 = -6 < 0$
 $\Rightarrow$ at $x = 1$ the function has a local maximum.

At $x = 2$: $(d^2y/dx^2)_{x=2} = 12(2) - 18 = +6 > 0$
 $\Rightarrow$ at $x = 2$ the function has a local minimum.

$$\text{The local maximum is } y(1) = 2(1)^3 - 9(1)^2 + 12(1) + 4 = 9$$

$$\text{The local minimum is } y(2) = 2(2)^3 - 9(2)^2 + 12(2) + 4 = 8$$

(3) An object is projected vertically upwards and its height above the ground, h , is given by $h = 20t - 5t^2$ metre, in which the time, t , is measured in seconds. Calculate the maximum height which it will reach and also when it will be reached.

$$h = 20t - 5t^2 \quad \Rightarrow \quad dh/dt = 20 - 10t$$

The height, h , possibly has an extreme value where $dh/dt = 0$, i.e. where

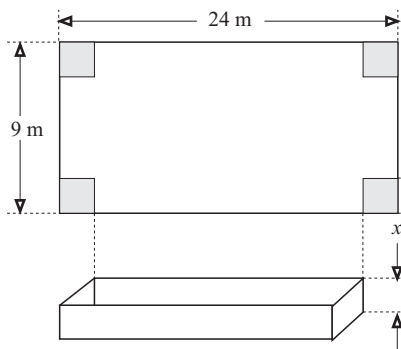
$$20 - 10t = 0 \quad \text{or} \quad t = 2 \text{ s}$$

$d^2h/dt^2 = (d/dt)(20 - 10t) = -10 \Rightarrow h$ has an extreme value which can be a maximum only. The maximum is given by

$$h(2) = 20(2) - 5(2)^2 = 20 \text{ m}$$

$\Rightarrow$ At time $t = 2$ s after it is projected upwards, the object reaches its maximum height of 20 m.

(4)



A farmer possesses a rectangular sheet of polyethylene plastic of 24 metres by 9 metres which he wishes to use as a waterproof lining for a rectangular dam with constant depth and vertical walls. To achieve this, equal squares of side length x are cut from the four corners and the edges of the cuts glued to form a hollow rectangular parallelepiped as is shown in the sketch.

First calculate the volume of the dam as a function of x and leave the answer in the form of a polynomial. Calculate

the value of x for which the volume will give a maximum value. Calculate the maximum value.

Let the volume of the dam be represented by $V = V(x)$. Then

$$\begin{aligned} V = \text{length} \times \text{breadth} \times \text{height} &= (24 - 2x)(9 - 2x)x \\ &= 216x - 66x^2 + 4x^3 \text{ m}^3 \end{aligned}$$

V possibly has an extreme value where $dV/dx = 0$, i.e. where

$$\begin{aligned} dV/dx &= (d/dx)(216x - 66x^2 + 4x^3) = 216 - 132x + 12x^2 = 0 \\ \text{i.e. where} \quad x^2 - 11x + 18 &= (x - 2)(x - 9) = 0 \\ \text{or} \quad x &= 2 \quad \text{and} \quad x = 9 \end{aligned}$$

The value $x = 9$ is not physically possible (the width of the sheet is 9 m) and therefore the only possibility left is $x = 2$ m. The sign of the second derivative is calculated at $x = 2$ m to verify the existence of a maximum.

$$d^2V/dx^2 = (d/dx)(216 - 132x + 12x^2) = 24x - 132$$

$$\text{and } (d^2V/dx^2)_{x=2} = 24(2) - 132 = -84 < 0.$$

For $x = 2$, the volume, V , has a maximum value. The maximum value of the volume is given by

$$V = V(2) = 216(2) - 66(2)^2 + 4(2)^3 = 200 \text{ m}^3$$

2.12 Differentials

For the function $y = y(x)$ the derivative was defined and represented by the notation dy/dx . The symbols dy and dx were not defined and when written separately they have no meaning at this stage. The symbols Δy and Δx were defined and their meanings are well known and they are used quite often.

It is now possible to define dx and dy which are known as **differentials** of x and y respectively, in such a way that they will have a useful meaning. They are defined in such a way that $dy \div dx$ is the same as the gradient of the function at any given position. The meaning of these symbols becomes clear if they are studied by means of the sketch in Figure 2.12-1.

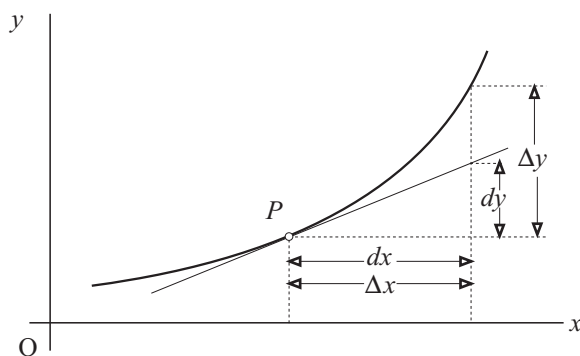


Figure 2.12-1

Consider point $P(x, y)$ on the graph of the function $y = y(x)$ at which a tangent is drawn. If x increases by an amount $\Delta x = dx$, the value of y changes along

the curve by an amount Δy . The corresponding change *along the tangent* is dy . Where Δx and Δy refer to changes along the curve, dx ($= \Delta x$) and dy refer to changes along the *tangent* to the curve.

$$\text{This allows one to write} \quad dy = (dy/dx)dx \quad 2.12(1)$$

In physics the following approximation for the change in a function is often used: If Δx is small, then

$$\Delta y \approx (dy/dx)\Delta x \quad 2.12(2)$$

The following example will illustrate the use of equation 2.12(1):

Example:

Consider a circle with radius r . Calculate the increase in area if the radius increases by Δr . It is known that Δr is much smaller than r .

(a) First method (rather cumbersome).

The change in the area is given by

$$\begin{aligned} \Delta A &= \pi(r + \Delta r)^2 - \pi r^2 \\ &= \pi r^2 + 2\pi r\Delta r + \pi(\Delta r)^2 - \pi r^2 \\ &= 2\pi r\Delta r + \pi(\Delta r)^2 \end{aligned}$$

If Δr is much smaller than r , the term which contains $(\Delta r)^2$ may be disregarded when compared to the other. For this case

$$\Delta A = 2\pi r\Delta r$$

(b) Second method (in which 2.12(2) is used and which is shorter and much more elegant than the first method).

$$A = \pi r^2 \quad \Rightarrow \quad dA/dr = 2\pi r$$

$$\text{Then } \Delta A \approx (dA/dr)\Delta r = 2\pi r\Delta r.$$

2.13 PROBLEMS: CHAPTER 2

1. Calculate the derivative of the function $y = y(x)$ in each case. Also calculate the gradient of the tangent to its curve at positions $x = -1$ and $x = 2$.

- | | |
|-------------------------------------|-----------------------------------|
| (a) $y = 3x - 7$ | (b) $y = 3x^2 - 2x + 14$ |
| (c) $y = 16x^3 - 15x + 18$ | (d) $y = 3x^{-3} + 12x^{-2}$ |
| (e) $y = 7x^3 + 7x^{-3}$ | (f) $y = 5x^5 - 15x^3 - 8/x$ |
| (g) $y = 5/x^6 + 8/x^3 - (4x)^{-2}$ | (h) $y = 6x^{-2} - 4/x + (4/x)^2$ |

2.(a) $z = 2 \sin \theta, 5\theta$. Calculate $dz/d\theta$ where $\theta = 0, \pi$ and 2π .

(c) $s = 30 + 200t - 5t^2$. Calculate ds/dt where $t = 2, 20$ and 50 .

(c) $p = \ln q$. Calculate dp/dq where $q = 0, 1$ and 2 .

(d) $y = e^x$. Calculate dy/dx where $x = 0, -1$ and 2 .

(e) $g = 3 \cos 2\phi$. Calculate $dg/d\phi$ where $\phi = 0, \pi/2$ and π .

3.(a) $s = (3t^2 - 5)(2t - 7)$. Calculate ds/dt in two different ways.

(b) $p = (5r^3 - 7)/2r$. Calculate dp/dr in two different ways.

(c) $f = (27t^3 + 8)/(3t + 2)$. Calculate df/dt in two different ways.

(d) $h = \cos^2 p$. Calculate dh/dp in two different ways. First use the product rule and then Equation 1.7(9).

(e) $y = \cot x$. Calculate dy/dx in two different ways. First express $\cot x$ in terms of $\sin x$ and $\cos x$ and apply the quotient rule. Then use the result of example (c) in section 2.7.3 and apply the quotient rule.

4. Calculate the derivatives of the following functions:

- | | |
|---|--------------------------------------|
| (a) $y = \sqrt{2x - 3}$ | (b) $y = \sin^3 x$ |
| (c) $y = \cos^4 x$ | (d) $y = \sin^2(2x - 5)$ |
| (e) $y = \cos^3(3x^2 - 2)$ | (f) $y = \sin \sqrt{3x + 4}$ |
| (g) $y = \ln(\sin x)$ | (h) $y = \ln(\sin \sqrt{2x^3 - 3x})$ |
| (i) $y = 2kqa/(a^2 + x^2)^{1/2}$ in which k, q and a are constants. | |
| (j) $y = 3e^{3x^2}$ | |

5. Calculate the first four derivatives of each of the following functions:

- | | |
|---------------------------------------|---|
| (a) $y = \sin x$ | (b) $y = \sin 2x$ |
| (c) $y = \cos x$ | (d) $y = e^{2x}$ |
| (e) $y = 3x^4 + 2x^3 + 3x^2 + 5x + 6$ | (f) $y = 3x^{-4} + 2x^{-3} + 3x^{-2} + 5x^{-1} + 6$ |

6. Calculate the values of x where the following functions have extreme values. Investigate their nature (maximum or minimum) and the value of the function at each.

- | | |
|----------------------------------|-----------------------------|
| (a) $y = 3x^2 - 18x + 12$ | (b) $y = -x^2 + 12x + 5$ |
| (c) $y = x^3 - 8x$ | (d) $y = 2x^3 - 9x^2 + 12x$ |
| (e) $y = 2x^3 + 3x^2 - 12x + 30$ | (f) $y = x^3$ |

7. In section 2.1.1 an example was considered in which an object was projected from the edge of a precipice. From the data in Table 2.1-1 it was not possible to calculate the maximum height. Use the theory of extreme values and calculate the instant at which the object reaches its maximum height. Use the function which is given in Equation 2.1(2). Calculate the maximum height.

8. A mortar shell is fired over a horizontal field in a direction which makes an angle of θ with the field. The position of the point of impact relative to the position from which it is fired, is represented by x which may be negative or positive as determined by the magnitude of θ . For this problem the domain of θ is given by $0 \leq \theta \leq \pi$. As a function of θ , the position of the point of impact is given by $x = 4000 \sin 2\theta$ metre in which the elevation angle, θ is measured in degrees. Calculate the values of the elevation angle for which x will have extreme values, investigate their nature (maximum or minimum) and then calculate the positions in question. Try to give a physical interpretation of the answers.

9. An existing stone wall on a farm is about 200 m long. The farmer has enough material to erect a fence of length 100 m. He wishes to use a portion of the existing wall as the one side of a rectangular pen for his cattle and the other three sides are to be completed by using the available fencing material. Calculate the dimensions and the area of the pen which is constructed in this way and will have a maximum area.

10. A manufacturer requires cylindrical tin cans in which 500 ml of fruit juice is to be marketed. The curved surface, the lid and the bottom of each tin is made from the same material. The radius of the tin is r cm and the height, h cm.
(a) Write h in terms of r . (b) Calculate A , the area of the material required to make such a can, as a function of r . This function must not contain h . (c) Calculate the value of r for which A will be a minimum. (d) Calculate the ratio r/h for a can of which the area of the material will be a minimum for a given volume.

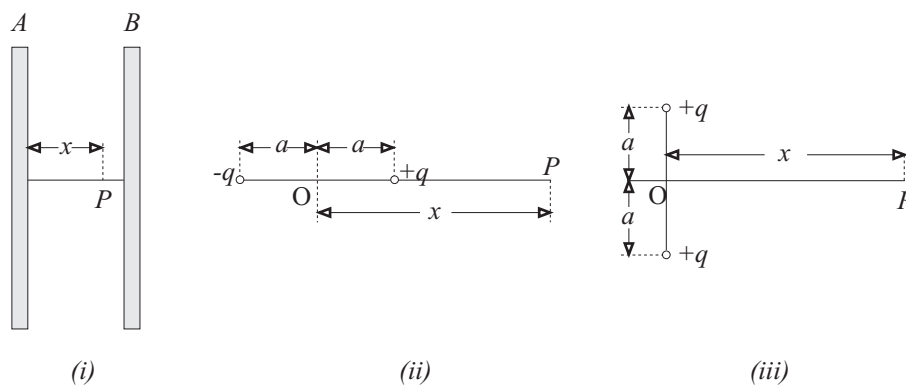
11. Repeat problem 10 for a can which does not have a lid.

12. A point moves along a straight line. A fixed point, O, on this line is used as origin from which the position, x , of the moving point is measured. To the one side of O, x is positive and negative to the other side. If x is constant, the point is at rest. If the point is in motion, x is a function of the time, t . The velocity of a moving point is defined as $v = dx/dt$. If v is not a function of t , the velocity remains constant. If, however, it is a function of t , the point is said to

accelerate and the acceleration is defined as $a = dv/dt = d^2x/dt^2$. Use the two definitions and calculate the velocity and the acceleration of a point of which the position is given by the following functions:

- (a) $x = 6t - 5$ (b) $x = 5t^2 - 3t + 5$
 (c) $x = 4t - 3t^3 - 2t^2 + 5$ (d) $x = 5 \sin 3t$
 (e) $x = 5 \sin 3t + 5 \cos 3t$

13. When the electric potential (the SI units are volts) is constant along a straight line, no electric field exists in the direction of the line. If the electric potential, V , depends upon the position along the line, an electric field exists. The intensity of this field, E , is defined by the equation $E = -dV/dx$ in which x is the position along the line. E is measured in volts per metre ($V m^{-1}$). Use the definition to answer the following questions:



(a) Figure (i) shows two flat parallel metal plates which carry electric charges of equal magnitude but opposite sign. At position P which is at distance x from plate A , the electric potential is given by $V = 2000x$ volts. Calculate the intensity of the electric field between the plates.

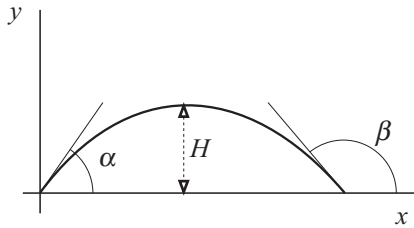
(b) At a distance of x metre from a single point charge, the electric potential is given by $V = 200x^{-1}$ volts. Calculate the intensity of the electric field at distance x from the charge.

(c) Figure (ii) shows two electric point charges of equal magnitude but opposite sign at position $x = a$ and $x = -a$ metre relative to point O . Point P is at position x ($x > a$) relative to O and the electric potential there is given by the function $V = 2kqa(x^2 - a^2)^{-1}$ volt. k , q and a are constants. Calculate the intensity of the electric field at P .

(d) Two identical electric point charges, $+q$, are on either side of the x -axis at $y = -a$ and $y = a$ metre respectively. The electric potential at a position P

on the x -axis, is given by $V = 2kq(a^2 + x^2)^{-1/2}$ volt. Calculate the intensity of the electric field at P . Investigate the possibility of local extreme values of E in the region $-5a \leq x \leq 5a$. If such values exist, determine their positions and nature.

14.



An object is projected over a horizontal field at an initial speed of 40 m s^{-1} . It follows a trajectory which resembles the curve in the sketch. The horizontal position measured from O is indicated by x and the vertical height by y . The Cartesian equation of the trajectory is

$$y = -x^2/20 + x/\sqrt{3}$$

in which x and y are measured in metre.

Use the equation for the trajectory to calculate the following: (a) The value of x for which the object reaches its maximum height. (b) At what angle, α , the object leaves the origin and at what angle, β , it strikes the ground.

15. At instant $t = 0 \text{ s}$, 10^5 radioactive atomic nuclei exist in a sample of material. Since radioactive nuclei decay, there will be less at any later instant. The number of radioactive nuclei, N , which exist at time t , is given by the function $N = N(t) = 10^5 e^{-t/50}$.

(a) Calculate dN/dt , the rate at which the radioactive nuclei decay. Explain the negative sign of the answer.

(b) Show that dN/dt is directly proportional to N , the number of radioactive nuclei which still exist at a given instant.

(c) Calculate the time when $0,5 \times 10^5$ radioactive nuclei will exist in the sample. Also calculate the time when $0,25 \times 10^5$ will exist.

16. A burette of which the cross-sectional area, A , is constant, contains water to a height $h = h_0$. When the stopcock is opened the water flows out at a rate of $A dh/dt$. The height of the water in the tube is given by the function $h = h_0 e^{-\lambda t}$ metre in which the time, t , is measured in seconds and λ is a positive constant.

(a) Calculate the rate (volume per unit time) at which the water flows from the burette and explain the negative sign of the answer.

(b) Show that the flow rate is directly proportional to the height of the water in the burette.

17. One of the many surprising results of Einstein's general theory of relativity, is that the apparent mass of an object depends on its speed. The mass, m , of an object is given by the function $m = m_0(1 - v^2/c^2)^{-1/2}$ in which m_0 is a constant which is known as the rest mass of the object. The speed of the object is v and c is the constant speed of light in free space (vacuum). Calculate dm/dv , the

rate at which the apparent mass of an object changes as its speed increases.

18. A square has a side length of x metre. Write its area, A , as a function of x . Calculate the rate of change of the area if the side length increases. Use this result to calculate the change in area if the side length increases by a small amount Δx .

19. A popular model of rain gauge is conical with base radius R and depth (height) H . The dimensions are in metre. Water is poured into it to a depth of h above the apex and the circular surface has a radius of r . Write the volume, V , of the water in terms of h and r . Use the dimensions of the conical gauge to eliminate r from this expression. Use this result to write the height h of the water in the gauge as a function of its volume, V . Calculate the rate at which the height changes as the volume increases. Use this result to calculate the increase in the height if a small amount of water, ΔV , is added to an existing volume, V .

Chapter 3

INTEGRAL CALCULUS

3.1 The indefinite integral

When a function is known and it is required to calculate its derivative, it is done according to the methods which are explained in chapter 2. The name of the process is *differentiation* and the verb is *differentiate*.

Now the opposite process will be considered. The derivative is known and the function has to be calculated. The name of the process is **integration** and the verb is **integrate**.

If one is skilled in the technique of differentiation, then simple integration can be done forthwith as the following examples illustrate:

(1) $dy/dx = x^2$. Calculate $y = y(x)$.

The answer is obtained by using one's experience in differentiation and by answering the following question: Which function has to be differentiated to give an answer of x^2 ? The answer to this question is the function which has to be determined.

$$(d/dx)(\text{of which function of } x?) = x^2. \quad \text{By inspection: } y = \frac{1}{3}x^3$$

The correctness of the answer may, and should always be tested by differentiating it.

$$(d/dx)(\frac{1}{3}x^3) = x^2. \quad \text{The integration is correct.}$$

From experience the reader should know that if *any* constant is added to $\frac{1}{3}x^3$, the derivative will still be x^2 , thus

$$(d/dx)(\frac{1}{3}x^3 + k) = x^2$$

At this stage it seems that no better answer can be given than one in which an unknown constant k appears. For this reason it is called an **indefinite integral**.

Although this undetermined constant seems to have nuisance value only, its numerical determination which will be discussed in different sections of this book, plays a most important role in physics. For this reason it should **never** be omitted when giving the answer to an indefinite integral. It is called an **integration constant**.

(2) $dp/dt = t^3 - 3$. Calculate $p = p(t)$.

$$(d/dt)(\text{which function of } t?) = t^3 - 3. \quad \text{By inspection: } p = \frac{1}{4}t^4 - 3t + k$$

$$\text{Test: } (d/dt)(\frac{1}{4}t^4 - 3t + k) = t^3 - 3. \quad \text{The integration is correct.}$$

(3) $(dv/dt) = \sin 3t$. Calculate $v = v(t)$.

$$\text{By inspection: } v = -\frac{1}{3}\cos 3t + c, \quad \text{in which } c = \text{constant.}$$

The test for the correctness of this answer is left to the reader.

The process by which the answers to the preceding three problems were found, is known as **integration**. The term *anti-differentiation* would also have been valid and quite descriptive. A notation exists by means of which calculations of this kind are indicated. Using this notation, example (1) is written as follows:

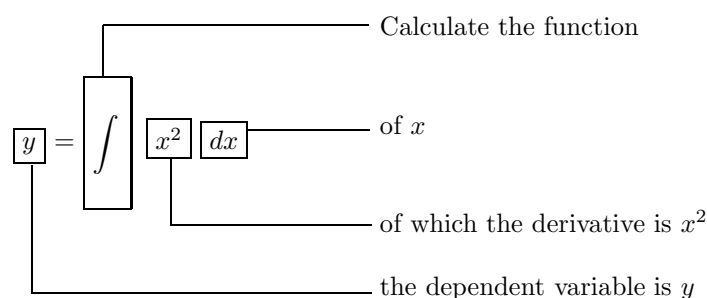
$$y = \int x^2 dx = \frac{1}{3}x^3 + k$$

The origin and actual meaning of this notation will become clear later in this chapter. At this stage it is useful to interpret it as an instruction to perform an integration as follows:

$$\int : \text{ This symbol is called an **integral sign** and may be taken to mean an instruction to calculate a function.}$$

dx : The **differential** indicates that a function of x is to be calculated.

x^2 : Whatever is between the integral sign and the differential (in this case x^2), is known as the **integrand** and it is the derivative of the function which is to be calculated.



Examples:

$$(1) \int (6x^2 - 8x + 5) dx = \frac{6}{3}x^3 - \frac{8}{2}x^2 + 5x + k = 2x^3 - 4x^2 + 5x + k$$

$$(2) \int (4 \cos 2x) dx = \frac{4}{2} \sin 2x + c = 2 \sin 2x + c$$

$$(3) \int (5 \sin 2x + 3e^{2x}) dx = -\frac{5}{2} \cos 2x + \frac{3}{2}e^{2x} + k = -2,5 \cos 2x + 1,5e^{2x} + k$$

$$(4) \int (\sin^2 x dx) = \int \frac{1}{2}(1 - \cos 2x) dx = 0,5x - 0,25 \sin 2x + k$$

Comment: Note that when the integrand consists of more than one term, it is always written between brackets.

3.2 Integration formulas

Inspection of the differentiation formulas and rules, leads to the following integration formulas which are sufficient for this course. For constant numbers b, n

and k , the following are valid:

$$\begin{aligned}
 \int bx^n dx &= (b/[n+1])x^{n+1} + k, \quad n \neq -1 \\
 \int \frac{1}{x} dx &= \int x^{-1} dx = \ln|x| + k \\
 \int (\sin bx) dx &= -(1/b) \cos bx + k \\
 \int (\cos bx) dx &= (1/b) \sin bx + k \\
 \int e^{bx} dx &= (1/b)e^{bx} + k \\
 \int b^{nx} dx &= (1/n \ln b)b^{nx} + k \\
 \int \frac{f'(x)}{f(x)} dx &= \ln|f(x)| + k \\
 \int b f(x) dx &= b \int f(x) dx \\
 \int [f(x) \pm g(x)] dx &= \int f(x) dx \pm \int g(x) dx
 \end{aligned}$$

3.3 The integration constant

Consider the function $y = 2x+3$. If its gradient is to be calculated, the procedure is simple and gives an unambiguous answer: $dy/dx = 2$. If the problem is to calculate the function of which the gradient is 2, the solution is not quite as straightforward. It is known that the answer is given by $y = \int 2 dx = 2x + k$

but the presence of the integration constant, k , makes the answer ambiguous.

The equation of a straight line is $y = mx + k$ in which m is its gradient and k the intercept on the ordinate axis. If the function is differentiated, the k disappears and confirms what is already known, namely that the intercept on the ordinate axis has no influence on the gradient. If one has to determine the function of which the gradient is 2, the best answer would be a **family** of straight lines, $y = 2x + k$ which all have a common gradient. If one member of the family is to be identified, supplementary information is needed. The co-ordinates of one position on a line will identify it unambiguously. If the co-ordinates of such a point are known, the value of k may be calculated.

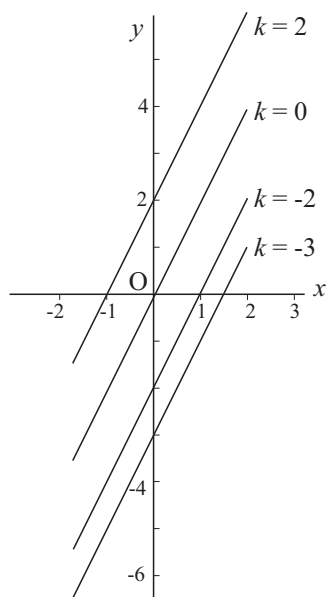


Figure 3.3-1

The graph in Figure 3.3-1 shows a number of the members of the family of straight lines of which the gradient is 2 and which are described by the equation $y = 2x + k$. If the line which contains the point (1, 4) is to be identified, these values are substituted into the equation as follows:

$$y = 2x + k$$

$$4 = 2 + k$$

$$\text{so that } k = 2$$

The required equation is

$$y = 2x + 2$$

The same procedure may be followed for any other line of the family.

Example:

Calculate the function of which the derivative is $2x - 4$ and which contains the point (2, -1).

$$\begin{aligned} dy/dx = 2x - 4 &\Rightarrow y = \int (2x - 4) dx \\ &= x^2 - 4x + k \end{aligned}$$

This answer represents a family of parabolas. It remains to calculate the intercept on the ordinate axis of the one which contains the given point.

$$y = x^2 - 4x + k$$

$$\begin{aligned}
 -1 &= 2^2 - 4(2) + k \\
 \Rightarrow \quad k &= 3 \\
 \text{so that} \quad y &= x^2 - 4x + 3
 \end{aligned}$$

The graph of this function is shown in Figure 2.10-1.

3.4 The definite integral

In this section the reader is introduced to the concept of a definite integral. It will be defined and then a number of examples of calculations will be shown. The actual meaning and use of the concept will be treated in the following sections.

If the derivative of the function $y = y(x)$ is given by $dy/dx = g(x)$, the following is true:

$$\int g(x) dx = y(x) + k, \quad \text{in which } k \text{ is constant}$$

The **definite integral** of $g(x)$ between the **limits** $x = a$ and $x = b$ (in this order) is defined by the following:

$$\int_a^b g(x) dx = [y(x)]_a^b = y(x)|_a^b = y(b) - y(a) \quad 3.4(1)$$

In this relationship, a is called the **lower limit of the integral** and b the **upper limit**. Their order is of prime importance. The answer cannot contain the independent variable since it is replaced by a and b when the answer is calculated in accordance with the definition. In the first example the integration constant will be introduced to show that it plays no role in the final answer and may thus always be omitted when a definite integral is calculated.

Examples:

$$\begin{aligned}
 \text{(a)} \quad \int_1^3 (4x - 3) dx &= [2x^2 - 3x + k]_1^3 \\
 &= [2(3)^2 - 3(3) + k] - [2(1)^2 - 3(1) + k] \\
 &= [9 + k] - [-1 + k] = 10
 \end{aligned}$$

This result shows very clearly that it is superfluous to make use of an integration constant when a definite integral is calculated.

$$\text{(b)} \quad \int_3^1 (4x - 3) dx = [2x^2 - 3x]_3^1 = [-1] - [9] = -10$$

Examples (a) and (b) illustrate a theorem which will not be proved but of which the reader should be thoroughly aware. *If the upper and lower limits of a definite integral are interchanged, the sign of the answer changes.*

$$(c) \int_0^{\pi/2} (\cos 2x) dx = [0, 5 \sin 2x]_0^{\pi/2} = [0, 5 \sin \pi] - [0, 5 \sin 0] = 0$$

$$\begin{aligned} (d) \int_0^{\pi/2} (\sin x - 3 \cos 3x) dx &= [-\cos x - \sin 3x]_0^{\pi/2} \\ &= [-\cos(\pi/2) - \sin(3\pi/2)] - [-\cos 0 - \sin 0] \\ &= [-0 - (-1)] - [-1 - 0] \\ &= 2 \end{aligned}$$

3.5 The definite integral represented by an area

Consider the function $y = y(x)$ of which the derivative is given by $dy/dx = g(x)$ so that

$$\int g(x) dx = y(x) + k \quad 3.5(1)$$

Figure 3.5-1 shows a graphic representation of a portion of the function $g = g(x)$. At position C the value of x is a , and at F , $x = b$. Consider an arbitrary value of x at D with corresponding value of the function of g . At E the x -value is $x + \Delta x$ and the corresponding value of the function, $g + \Delta g$.

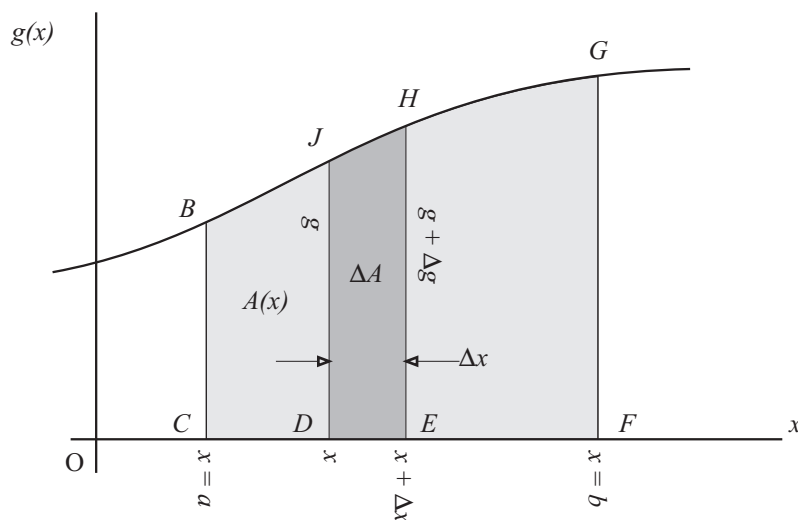


Figure 3.5-1

Let A be the area between the graph $g = g(x)$ and the x -axis from the fixed abscissa $x = a$ to the arbitrary abscissa $x = x$. In the sketch, A is the area of figure $BCDJ$. A is thus a function of x and when x increases by an amount Δx , A changes by an amount ΔA and it may be written that

$$\begin{aligned} A(x + \Delta x) &= A(x) + \Delta A \\ \text{and} \quad \Delta A &= A(x + \Delta x) - A(x) \\ &= \text{area of } JDEH \end{aligned}$$

If Δx is small, figure $JDEH$ may be approximated by a trapesium of which the area is given by

$$\begin{aligned} \Delta A &= \frac{JD + HE}{2} \times DE = \frac{1}{2}(g + [g + \Delta g]) \times \Delta x \\ \text{so that } \frac{\Delta A}{\Delta x} &= \frac{1}{2}(2g + \Delta g) \end{aligned}$$

If $\Delta x \rightarrow 0$, then $\Delta A \rightarrow 0$ and $\Delta g \rightarrow 0$ and

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta A}{\Delta x} = \frac{dA}{dx} = g(x) \quad 3.5(2)$$

From this it follows directly that

$$A = \int g(x) dx = y(x) + k \quad [\text{from 3.5(1)}] \quad 3.5(3)$$

in which k is a constant which may be calculated if the value of A is known for a given value of x . According to the way in which A was defined, $A = 0$ where $x = a$ (i.e. the area between $x = a$ and $x = a$). From Equation 3.5(3) it follows that

$$\begin{aligned} 0 &= y(a) + k \Rightarrow k = -y(a) \\ \text{and } A &= y(x) - y(a) \end{aligned}$$

$$\text{At } x = b : \quad A = y(b) - y(a) = \int_a^b g(x) dx \quad 3.5(4)$$

which is the area between the graph of $g = g(x)$ and the x -axis and which lies between $x = a$ and $x = b$, i.e. the area $BCFG$ in Figure 3.5-1.

In the study of physics this result is of great importance since the area under a given curve, often represents a physical quantity. The units of the area are determined by the units along the two axes. If, for example, g is measured in metres per second and x in seconds, the area represents metres. If g is measured in ampère and x in seconds, the area represents coulombs.

In the deduction of this result, the abscissa was represented by x and the ordinate by y . It should be noted that any symbols may be used. Similarly

$$\int_p^q y(t) dt$$

would represent the area between the graph of the function $y = y(t)$ and the t -axis from $t = p$ to $t = q$.

In Figure 3.5-1 a portion of the graph which lies above the abscissa-axis was considered. In the examples which follow, cases in which the graph lies below the axis will also be considered and an important extended interpretation of the result will have to be treated.

Examples:

(a) Calculate the area between the curve $y = t^2 - 4t + 5$ and the t -axis which lies between $t = 1$ and $t = 4$. y is the gas flow in a pipe in litres per second and t the time in seconds. The area is shown in Figure 3.5-2(a).

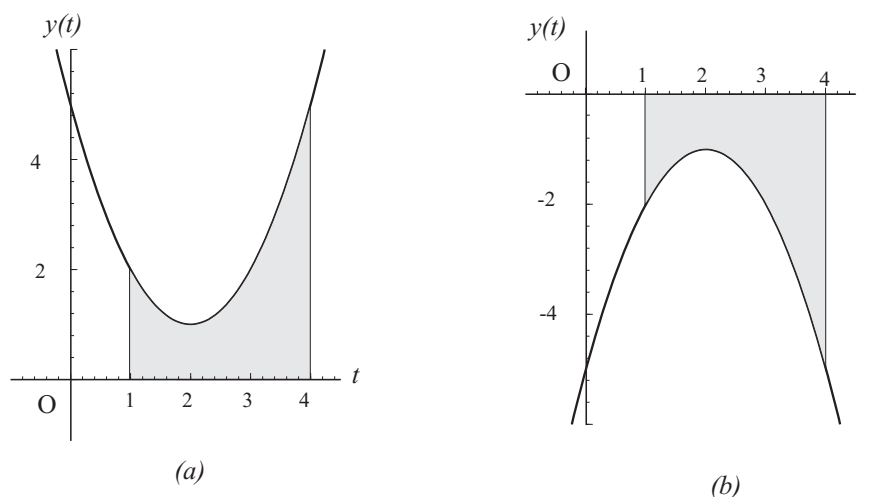


Figure 3.5-2

$$\begin{aligned} A &= \int_1^4 (t^2 - 4t + 5) dt = \left[\frac{1}{3}t^3 - 2t^2 + 5t \right]_1^4 \\ &= \left[\frac{1}{3}(4)^3 - 2(4)^2 + 5(4) \right] - \left[\frac{1}{3}(1)^3 - 2(1)^2 + 5(1) \right] \\ &= 9\frac{1}{3} - 3\frac{1}{3} \\ &= 6 \text{ litres} \end{aligned}$$

(b) Calculate the area between the graph of the function $y = -t^2 + 4t - 5$ and the t -axis between $t = 1$ and $t = 4$. The graph is the mirror image about the t -axis of that in example (a). As in the previous example, y is the gas flow in a pipe in litres per second and t the time in seconds. The area is shown in Figure 3.5-2(b).

$$\begin{aligned} A &= \int_1^4 (-t^2 + 4t - 5) dt = -\frac{1}{3}t^3 + 2t^2 - 5t \Big|_1^4 \\ &= -9\frac{1}{3} + 3\frac{1}{3} \\ &= -6 \text{ litres} \end{aligned}$$

The minus sign indicates that the area which was calculated, lies under the t -axis. If the actual *magnitude* of the area was to be calculated, the sign would have to be disregarded and the answer given as 6 litres. Negative area (as area) has no meaning. The physical interpretation of the negative answer of example (b) is that the flow was in the opposite sense.

(c) Calculate the area between the graph of $y = t^2 - 4t + 3$ and the t -axis between (i) $t = 0$ and $t = 1$, (ii) $t = 1$ and $t = 3$, (iii) $t = 0$ and $t = 3$. As in the previous questions, y is the gas flow in a pipe in litres per second and t the time in seconds. These areas are shown in Figure 3.5-3.

$$\begin{aligned} \text{(i)} \quad A &= \int_0^1 (t^2 - 4t + 3) dt \\ &= \frac{1}{3}t^3 - 2t^2 + 3t \Big|_0^1 \\ &= 1\frac{1}{3} \text{ litres} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad B &= \int_1^3 (t^2 - 4t + 3) dt \\ &= \frac{1}{3}t^3 - 2t^2 + 3t \Big|_1^3 \\ &= -1\frac{1}{3} \text{ litres} \end{aligned}$$

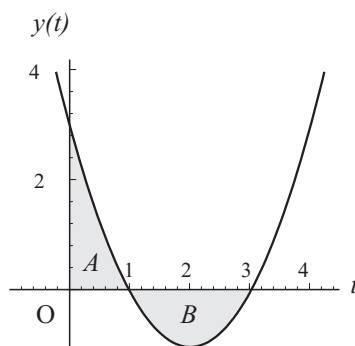


Figure 3.5-3

This answer is negative and indicates that the flow is in the opposite sense to that in the previous answer. The area representing the negative amount is under the t -axis.

(iii) The magnitude of the area between $y = y(t)$, the t -axis from $t = 0$ and $t = 3$ is $|A| + |B| = 1\frac{1}{3} + 1\frac{1}{3} = 2\frac{2}{3}$ square units. In this case it indicates that $2\frac{2}{3}$ litres of gas have flowed and the answer disregards the sense of the flow. If the

integral is calculated between the given limits, it gives the following answer:

$$\int_0^3 (t^2 - 4t + 3)dt = \frac{1}{3}t^3 - 2t^2 + 3t \Big|_0^3 = 0$$

The zero value indicates that the magnitudes of the areas above and below the t -axis are equal. The gas flow was the same in both ways, i.e. no net flow.

If the intersection points of a graph and the abscissa axis are not known and a definite integral which is calculated between given limits is positive, the answer should be interpreted as net area above the abscissa axis and when it is negative, as net area below the axis. A zero answer indicates equal areas above and below the axis.

3.6 The definite integral as the limit of a sum

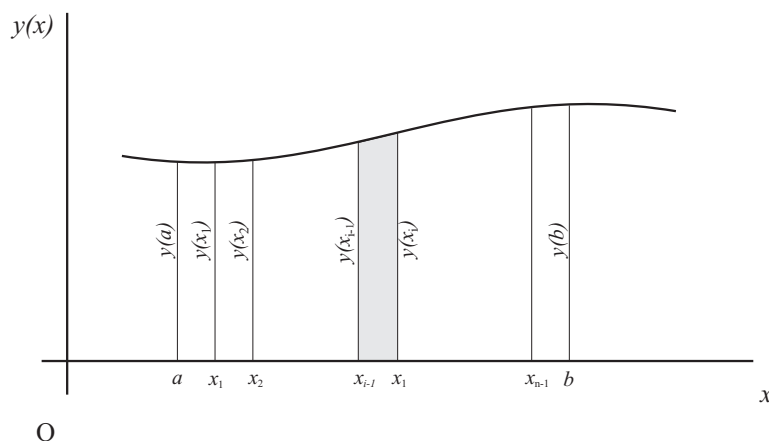


Figure 3.6-1

Figure 3.6-1 shows a portion of the graph of the function $y = y(x)$ between the limits $x = a$ and $x = b$. From 3.5(4) it is known that the area between the graph of the function and the x -axis from $x = a$ to $x = b$ is given by

$$A = \int_a^b y(x) dx \quad 3.6(1)$$

The area may be calculated in a different way. The interval between $x = a$ and $x = b$ along the x -axis, is divided into n equal intervals, each with magnitude $\Delta x = (b - a)/n$. The lower limits of these intervals are the x -values $a, x_1,$

$x_2, \dots, x_{i-1}, x_i \dots x_{n-1}$. According to the notation which is used, $x_0 = a$ and $x_n = b$. The total area is divided into n strips, which, if Δx is small enough (or n large enough), may each be approximated by a trapezium. Consider strip number i (which is limited by x_{i-1} and x_i). Its area is given by

$$\Delta A_i \approx \frac{y(x_{i-1}) + y(x_i)}{2} \times \Delta x \quad 3.6(2)$$

The entire area between the limits $x = a$ and $x = b$ is given by

$$\begin{aligned} A &\approx \Delta A_1 + \Delta A_2 + \Delta A_3 \dots \Delta A_{n-1} + \Delta A_n \\ &= \sum_{i=1}^n \Delta A_i \end{aligned} \quad 3.6(3)$$

In this notation, the Greek letter Σ (sigma) represents the instruction to calculate the sum of the terms indicated by the index i which in consecutive terms assumes the values from 1 to n . From Equations 3.6(2) and 3.6(3) it follows that

$$A \approx \sum_{i=1}^n \frac{y(x_{i-1}) + y(x_i)}{2} \times \Delta x$$

This approximation may be replaced by an equality if $\Delta x \rightarrow 0$ (i.e. $n \rightarrow \infty$). If $\Delta x \rightarrow 0$, then $y(x_{i-1}) \rightarrow y(x_i)$ so that

$$\begin{aligned} A &= \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n \frac{y(x_{i-1}) + y(x_i)}{2} \times \Delta x \\ &= \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n y(x_i) \Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n y(x_i) \Delta x \end{aligned}$$

To say that the index i assumes all the numerical values from 1 to n , is equivalent to the statement that x assumes all the values from a to b with intervals of Δx so that

$$A = \lim_{\Delta x \rightarrow 0} \sum_{x=a}^b y(x) \Delta x \quad 3.6(4)$$

This is an exact answer for the calculation of the area under consideration. The fact that an infinite number of intervals of which the magnitude tends to zero are used, makes the practical calculation of the area by this method impossible. Another exact way of calculating the area by means of an integral is known. By equating the results of Equations 3.6(1) and 3.6(4), we have

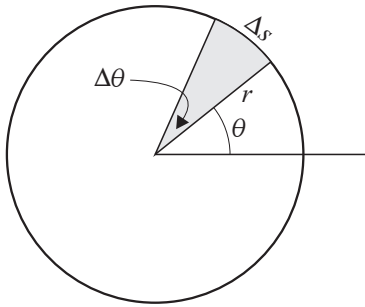
$$\lim_{\Delta x \rightarrow 0} \sum_{x=a}^b y(x) \Delta x = \int_a^b y(x) dx \quad 3.6(5)$$

This result shows that a definite integral is the limit of a sum. This interpretation explains the origins of the word *integrate* and the integral sign which is a deformed letter *S* which was introduced by Wilhelm Leibniz. The interpretation also explains the necessity for writing the differential dx after the integrand.

Few techniques are used more often by physicists than the definite integral. It is a most powerful tool without which many results would be impossible, or at the very best, extremely difficult to calculate. It requires much practice before one is skilled in its use and the reader should constantly keep in mind that it will be one of his most important mathematical tools and that time spent to become familiar with it, is a good investment. A number of examples will illustrate its use.

Examples:

(a)



In this example an integral is used to derive the formula for the area of a circle of which the radius is r metre.

Consider a small sector formed by two radii which make an angle $\Delta\theta$. The angle subtends an arc Δs as shown in the sketch. From the definition of radian measure it follows that

$$\Delta s = r \times \Delta\theta$$

Since the angle is very small, the sector may be approximated by a triangle with base Δs and height r .

When approximated by a triangle, the area of the sector is given by

$$\Delta A \approx \frac{1}{2}(\text{base}) \times (\text{height}) = \frac{1}{2} \Delta s \times r = \frac{1}{2} (r \Delta\theta) \times (r) = \frac{1}{2} r^2 \Delta\theta$$

The area of the circle is given by

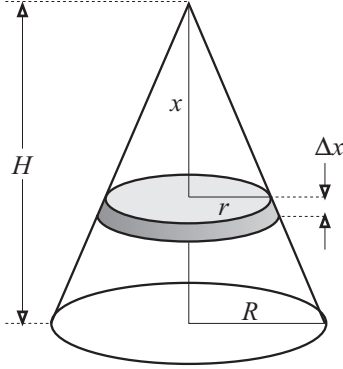
$$A = \lim_{\Delta\theta \rightarrow 0} \sum_{\theta=0}^{2\pi} \frac{1}{2} r^2 \Delta\theta = \int_0^{2\pi} \frac{1}{2} r^2 d\theta$$

$$= \frac{1}{2}r^2\theta\Big|_0^{2\pi} = \frac{1}{2}r^2(2\pi) - 0 = \pi r^2$$

The way in which this result was calculated is rather clumsy. Furthermore, the “approximation” was not necessary since the result for the area of the small sector is exactly correct. A person who is skilled in the use of definite integrals would do the calculation without referring to a sum since he always thinks of a definite integral as a sum. The calculation will be repeated by making direct use of differentials.

Consider a sector which is formed by two radii of length r which make an angle $d\theta$. The length of the arc which subtends the angle is given by $ds = r d\theta$, and the area of the sector by $dA = \frac{1}{2}r \times r d\theta = \frac{1}{2}r^2 d\theta$. The integral remains to be calculated and the answer is the same as that of the previous calculation.

(b).



A definite integral will be used to calculate the formula for the volume of a right circular cone with base radius R and height H .

Consider a circular slice with thickness Δx at distance x from the apex of the cone and parallel to its base. From the geometry of a cone it follows that

$$r = (R/H)x$$

If Δx is small, the volume of the slice may be approximated by that of a cylinder as follows:

$$\Delta V \approx \pi r^2 \Delta x = \pi(R^2/H^2)x^2 \Delta x$$

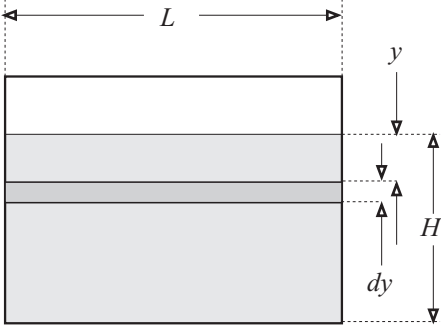
The volume of the cone is given by

$$\begin{aligned} V &= \lim_{\Delta x \rightarrow 0} \sum_{x=0}^H \pi(R^2/H^2)x^2 \Delta x = \int_0^H \pi(R^2/H^2)x^2 dx \\ &= \frac{1}{3}\pi(R^2/H^2)x^3 \Big|_0^H \\ &= \frac{1}{3}\pi R^2 H \end{aligned}$$

As in the previous example, the calculation may be done by avoiding reference to a sum. In this problem the approximation was necessary since the slice is not a true cylinder. In the limit its difference from a cylinder is of no consequence

whatsoever. Consider a circular slice of thickness dx at distance x from the apex of the cone and parallel to its base. Its volume is given by $dV = \pi r^2 dx = \pi(R^2/H^2)x^2 dx$. The volume of the cone follows by integration between the limits $x = 0$ and $x = H$.

(c)



The sketch shows a rectangular dam wall with horizontal width L which holds water of density $\rho \text{ kg m}^{-3}$ to a depth of H metres. The hydrostatic pressure of the water at depth y below the surface, is given by

$$p = \rho g y \text{ Pa (N m}^{-2}\text{)}$$

in which g = magnitude of gravitational acceleration. Calculate the total force exerted by the water on the dam wall as a result of the hydrostatic pressure.

Since the pressure varies with depth, it is not possible to calculate the force by multiplying the pressure by the area. If a narrow horizontal strip of area which spans the entire width of the wall is considered, this method of calculation would be in order since the variation of pressure over the height of the strip, may be neglected. The total force is equal to the sum (integral) of the forces on all the strips on which the water exerts pressure. The calculation is made by direct reference to differentials.

Consider a horizontal element of area of which the width is dy and which is at depth y below the surface. The area of the strip is $dA = L dy$, and the force on it is given by

$$(\text{pressure}) \times (\text{area}) = p dA = (\rho g y)(L dy)$$

The total force against the dam wall is given by the following definite integral:

$$\begin{aligned} F &= \int_0^H \rho g L y dy = \rho g L \int_0^H y dy \\ &= \rho g L \left(\frac{1}{2} y^2 \right) \Big|_0^H = \frac{1}{2} \rho g L H^2 \end{aligned}$$

3.7 The calculation of integration constants

In section 3.3 it was shown that a known pair of co-ordinates is required to calculate an integration constant. This may be done in a more elegant way by using a definite integral as will be illustrated in the following examples. In each

example the method used in 3.3 will be done first and then a definite integral will be used to perform the same task.

Examples:

(a) $dy/dx = 6x^2 - 4x + 5$. It is known that $y(1) = 2$. Calculate $y = y(x)$.

1. The method as used in section 3.3:

$$y = \int (6x^2 - 4x + 5)dx = 2x^3 - 2x^2 + 5x + k$$

It is known that $y = 2$ if $x = 1$. Substitute these values in the above equation:

$$\begin{aligned} 2 &= 2(1)^3 - 2(1)^2 + 5(1) + k = 5 + k \Rightarrow k = -3 \\ \text{so that } y &= 2x^3 - 2x^2 + 5x - 3 \end{aligned}$$

2. By means of a definite integral:

$$\text{Since } dy/dx = 6x^2 - 4x + 5 \Rightarrow dy = (6x^2 - 4x + 5)dx$$

Both sides are integrated and the following pairs of co-ordinates are used as limits: *Lower limits:* On the left side 2 (the value of y) and on the right side, 1 (the corresponding value of x). *Upper limits:* On the left side y and on the right side x .

$$\begin{aligned} \int_2^y dy &= \int_1^x (6x^2 - 4x + 5)dx \\ y|_2^y &= 2x^3 - 2x^2 + 5x|_1^x \\ y - 2 &= (2x^3 - 2x^2 + 5x) - (2 - 2 + 5) \\ \Rightarrow y &= 2x^3 - 2x^2 + 5x - 3 \end{aligned}$$

Question: Would it be admissible to exchange the upper and lower limits on both sides? If the answer is not clear, test and see what happens.

(b) A body is limited to motion along a straight line. Its position relative to a fixed point on this line which is called the origin, is represented by x . x is positive for positions to the one side of the origin and negative for positions to the opposite side. The velocity of the body is defined as $v = dx/dt$. For the body under consideration the velocity is a function of time and is given by $v = dx/dt = -6 \sin 2t \text{ m s}^{-1}$ in which the time, t , is measured in seconds. If $t = 0, x = 0 \text{ m}$ (or $x(0) = 0 \text{ m}$). Calculate the position of the body as a function of time, i.e. $x = x(t)$.

1. By the method used in section 3.3:

$$x = \int (-6 \sin 2t) dt = 3 \cos 2t + k$$

It is given that $x = 0$ when $t = 0$. Substitute these values in the above equation:

$$\begin{aligned} 0 &= 3 \cos 0 + k = 3 + k \Rightarrow k = -3 \\ \text{so that } x &= 3 \cos 2t - 3 \text{ metres} \end{aligned}$$

2. By means of a definite integral:

$$\text{Since } dx/dt = -6 \sin 2t \Rightarrow dx = (-6 \sin 2t) dt$$

Integrate on both sides, using the following pairs of co-ordinates as limits:

Lower limits: On the left side 0 (the value of x) and on the right side 0 (the corresponding value of t). *Upper limits:* On the left side x and on the right side, t .

$$\begin{aligned} \int_0^x dx &= \int_0^t (-6 \sin 2t) dt \\ x|_0^x &= 3 \cos 2t|_0^t \\ x - 0 &= 3 \cos 2t - 3 \cos 0 = 3 \cos 2t - 3 \\ \Rightarrow x &= 3 \cos 2t - 3 \text{ metres} \end{aligned}$$

3.8 PROBLEMS: CHAPTER 3

1. Calculate $y = y(x)$ in each case if it is known that dy/dx is given by the following:

- | | |
|------------------------|---|
| (a) $3x^2 - 6x + 5$ | (b) $3x^4 + 2x^3 - 5x^2 + 7x$ |
| (c) $7x^3 + 7x^{-3}$ | (d) $4x^2 + 6x^{-2}$ |
| (e) $16x^7 + 29x^{-9}$ | (f) $4x^3 + 3x^2 + 2x + 1 - x^{-1} - x^{-2} - x^{-3}$ |
| (g) $6e^{3x}$ | (h) $4 \sin 2x$ |
| (i) $6 \cos(3x)$ | (j) $(12x^3 + 5)/(3x^4 + 5x)$ |

2. Calculate the following indefinite integrals:

- | | |
|---|---|
| (a) $\int (4x^2 + 2x - 3) dx$ | (b) $\int (8x^6 - 12x^3 - 5) dx$ |
| (c) $\int (8s^3 - 3 - 4s^{-3}) ds$ | (d) $\int (20 - 10t) dt$ |
| (e) $\int -18x dx$ | (f) $\int -18 dx$ |
| (g) $\int (-18x^{-1}) dx$ | (h) $\int (v - gt) dt$ v and g are constant |
| (i) $\int \left(\int -10t dt \right) dt$ | (j) $\int \left(\int -g dt \right) dt$ g is constant |
| (k) $\int 5 \cos 4x dx$ | (l) $\int 3 \sin 2x dx$ |

2. Calculate the following definite integrals:

- | | |
|---------------------------------|---|
| (a) $\int_2^5 3x dx$ | (b) $\int_{0^\circ}^{90^\circ} \sin x dx$ |
| (c) $\int_0^{\pi/2} \sin 2x dx$ | (d) $\int_{-1}^0 (1 + 3x - 2x^2) dx$ |

(e) $\int_0^{\pi/6} \cos 3x dx$

(f) $\int_0^{\pi/2} (\cos \theta - \sin 2\theta) d\theta$

(g) $\int_1^4 \sqrt{x} dx$

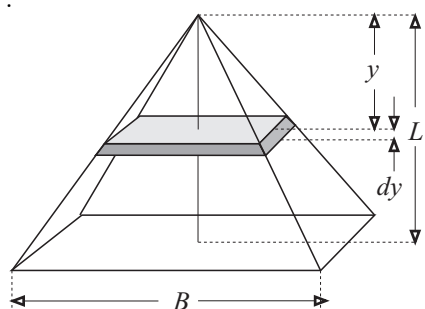
(h) $\int_0^{10} x^{-0.6} dx$

4. Calculate the area bounded by the t -axis and the graph of the function $v = 3 + 4t$, between $t = 1$ and $t = 5$. The units of v are metres per second. What are the units of the quantity which the area represents?

5. Make a rough sketch of the graph of the function $y = 4x^3 - 12x^2 + 8x$. Calculate the area between the graph of the function and the x -axis (i) from $x = 0$ to $x = 1$, (ii) $x = 1$ to $x = 2$. Use the two quantities to calculate the definite integral of the function between the limits $x = 0$ and $x = 2$.

6. Sketch the graphs of the functions $y = x^2$ and $y = 2x$ on one and the same set of axes. First calculate the co-ordinates of their points of intersection and then the area between the graphs.

7.



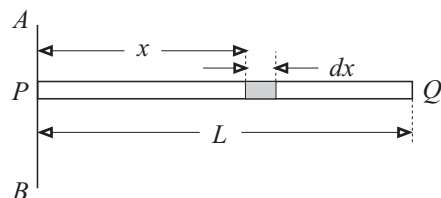
The sketch shows a right pyramid on a square base with side length B metres. The height of the pyramid is L metres. Consider a section parallel to the base at distance y from the apex and with infinitesimal thickness dy . First write down the volume, dV , of the section in terms of y , L , and B , only. (*Hint:* Let the side length of the square section be represented by b . First write dV in terms of b and dy . Then eliminate b by writing it in terms of y , L and B .)

Use the expression for dV to calculate the volume of the pyramid.

8. When a gas of volume V cubic metres is at a pressure of p pascal (newton per square metre), work is done when the volume changes. The work done when the volume changes by an infinitesimal amount dV , is given by $dA = p dV$. A change in volume generally changes the pressure and the way in which it is changed depends on the way in which the volume is changed. An ideal gas of which the volume changes **isothermally** (the temperature remains constant), obeys Boyle's law: $pV = k$ in which k is constant. If the volume changes **adiabatically** (no heat crosses the border of the system), it obeys the law

$pV^\gamma = K$ in which K and γ are constant quantities. Calculate the work done when an ideal gas changes from the initial state p_1, V_1 to the final state p_2, V_2 during an (a) isothermal, (b) adiabatic change.

9.



The sketch shows a thin straight homogeneous rod with length L metres and mass M kilograms. The mass per unit length is given by $\mu = M/L \text{ kg m}^{-1}$. AB is an axis perpendicular to the rod, PQ , and dx is an infinitesimal element of length at position x from AB . The mass of this element is given by $dm = \mu dx = (M/L)dx$. The quantity

$dI = x^2 dm$ is known as the **moment of inertia** of the element about the axis AB . (a) Calculate the moment of inertia of the entire rod about axis AB . (b) Calculate the the moment of inertia of the rod about an axis parallel to AB and through the centre of the rod.

Chapter 4

VECTOR ALGEBRA

4.1 Introductory discussion and definitions

Many quantities which are encountered in the study of physics are relatively simple in the sense that they can be described by a single number. Calculations involving such numbers are simple since they require the rules of ordinary arithmetic. Other quantities are more involved since they require the use of more than one number each. A physical quantity in which direction is involved, is an example of such a quantity. In this chapter an elegant way to describe such quantities will be developed and the rules which are required to use them in calculations, will be investigated. These rules are known as **vector algebra**. In this introductory discussion the necessary definitions will be given.

1. A **scalar** is a physical quantity which has magnitude but no direction. Examples: time, length, mass, temperature, etc. A scalar is usually represented by a Roman or Greek character as follows: $x, y, a, A, D, \alpha, \beta$, etc. (never boldface). A scalar is specified by a single number.
2. Provisionally a vector may be defined as a physical quantity which has magnitude and direction. A more stringent definition which is required for advanced work, and which depends on its so-called **transformation properties**, falls outside the scope of this book. Examples of vector quantities are: force, displacement, acceleration, electric field intensity, etc. Unfortunately many different notations are used to indicate vectors. The following are found in textbooks: $\vec{H}, \vec{H}, \mathcal{H}, \underline{H}, \mathbf{H}, \vec{H}$. Greek letters are also used: $\vec{\mu}, \vec{\mu}$, etc. In most modern physics textbooks, boldface characters are used to indicate vectors. Such a notation is not suitable for

handwriting and typewriters and for this reason the first notation ($\vec{H}$, $\vec{a}$, etc.) will be used in this book.

3. A vector may be specified by its magnitude and direction relative to a suitable frame of reference. A vector can never be specified by a single number. In the following *two-dimensional* example, two numbers are necessary: A velocity of 20 ms^{-1} in the direction N 30° E. Although this way of specifying a vector will be used initially in this discussion, it is not always the best and another will be developed as the study progresses.
4. The **magnitude**, **modulus** or **norm** of vector $\vec{A}$, is indicated by $|\vec{A}|$. Often, when no possibility of confusion exists, the following notation is used for the modulus of a vector: $A = |\vec{A}|$. It is defined as a **positive number**.
5. A vector may be represented graphically by means of a directed line segment of which the length represents the magnitude of the vector according to a suitable scale. The end which terminates in an arrowhead is called the **terminal point** of the vector and the other point, the **initial point**.

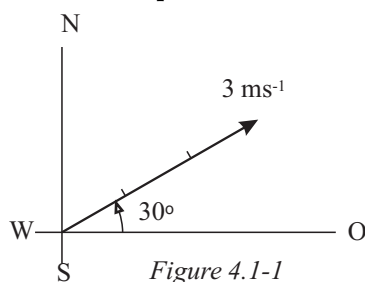


Figure 4.1-1 shows a graphical representation of a velocity of 3 ms^{-1} in the direction 30° north of east. The direction may also be indicated as 60° east of north, E 30° N or N 60° E. A navigator would specify the direction simply as 60° . According to this notation, north is taken as 0° (or 360°), and the angle is measured clockwise.

6. A **unit vector** is one of which the magnitude is unity. Unit vectors which do not have units are used to specify directions. Since they indicate direction only, they are dimensionless. The following notation will be used to indicate unit vectors: $\hat{A}$, $\hat{a}$, $\hat{\mu}$, $\hat{x}$, $\hat{y}$, etc. If a given vector is $\vec{A}$, a unit vector may be constructed which has the same direction as $\vec{A}$.

$$\hat{A} = \vec{A}/|\vec{A}| = \vec{A}/A \quad 4.1(1)$$

Example: $\vec{A} = 20 \text{ ms}^{-1}, \text{ north} \Rightarrow |\vec{A}| = 20 \text{ ms}^{-1}$
 and $\hat{A} = (20 \text{ ms}^{-1}, \text{ north})/(20 \text{ ms}^{-1}) = 1 \text{ north}$

The way in which this unit vector is specified is rather awkward (in fact, fairly useless) and a more elegant way will be developed later in the chapter.

7. The **negative vector**, $-\vec{A}$ has the same magnitude as $\vec{A}$, but it is in the opposite direction. It is said that they are **anti-parallel**.

8. A vector $\vec{A}$ multiplied by a scalar quantity k , gives a new vector $k\vec{A}$ of which the magnitude is $|k|$ times the magnitude of $\vec{A}$ and with direction parallel or antiparallel to $\vec{A}$, depending on whether k is positive or negative. If any vector with finite magnitude is multiplied by zero, the answer is the **zero vector** which is written as $\vec{0}$. The zero vector does not have direction.

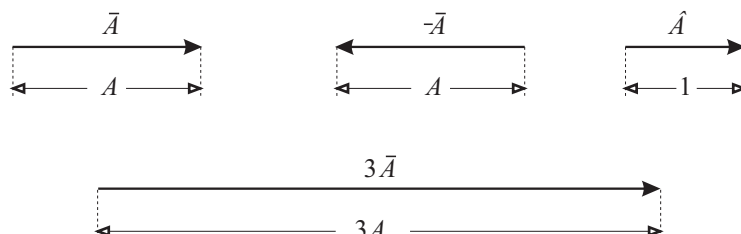


Figure 4.1-2

9. The **component of a vector in a given direction** is defined as the magnitude of the vector multiplied by the cosine of the angle between the direction of the vector and the given direction. For this purpose the smallest angle θ is used ($0 \leq \theta \leq \pi$) and it is measured as follows: Measure the smallest angle from the given direction to the direction of the vector. If it is measured anticlockwise, it is taken to be positive and when it is measured clockwise, it is negative.¹ Since $\cos \theta = \cos(-\theta)$, the effect would be the same if positive angles only between 0 and 2π are used. It is useful to remember that $\cos \theta = -\cos(\pi - \theta)$.

Other terms used for the component of a vector in a given direction, are the **projection** of the vector on the given direction and the **mapping** of the vector on the given direction.

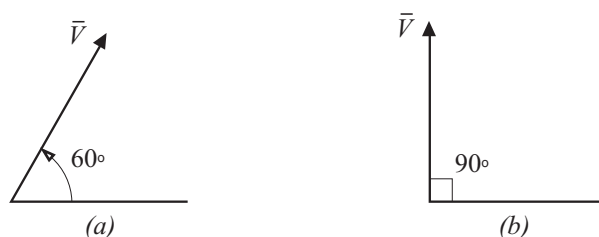


Figure 4.1-3

¹This method of measurement is introduced since it is used in calculators which are programmed for conversions from Cartesian co-ordinates to polar co-ordinates.

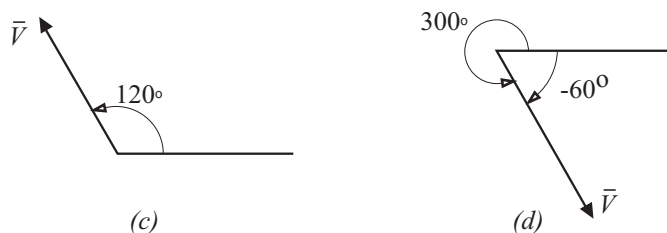


Figure 4.1-3

The components of $\bar{V}$ in the given direction OX (in this order) for the four examples illustrated in Figure 4.1-3 are given by the following:

- (a) $V \cos 60^\circ = 0,5V$
- (b) $V \cos 90^\circ = 0$
- (c) $V \cos 120^\circ = -V \cos 60^\circ = -0,5V$
- (d) $V \cos(-60^\circ) = V \cos 300^\circ = 0,5V$

The advantages of measuring the angles in the way explained above, become evident when the components of a vector are also to be calculated in a second direction, perpendicular to the first.

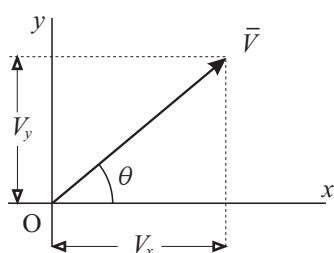


Figure 4.1-4

Consider two such directions which form a two-dimensional orthogonal frame of reference. The two straight lines which form the axes, are distinguished by naming them the x -axis and y -axis respectively. The component of V along the x -axis is called its x -component (indicated by V_x) and that along the y -axis, its y -component (indicated by V_y). The two components are calculated from V , the magnitude of the vector, and θ , the angle between $\bar{V}$ and the x -axis.

$$\begin{aligned}
 \text{Now} \quad V_x &= V \cos \theta \\
 \text{and} \quad V_y &= V \cos(\pi/2 - \theta) = V \sin \theta \\
 \text{note that } V_x^2 + V_y^2 &= V^2 \cos^2 \theta + V^2 \sin^2 \theta \\
 &= V^2 (\cos^2 \theta + \sin^2 \theta) = V^2 \\
 \text{so that} \quad V &= |\bar{V}| = (V_x^2 + V_y^2)^{\frac{1}{2}}
 \end{aligned}
 \tag{4.1(2)}$$

$$\tag{4.1(3)}$$

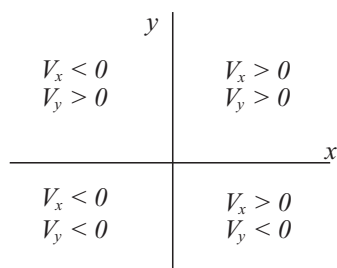


Figure 4.1-5

If the vector lies in the second quadrant, $\pi/2 \leq \theta \leq \pi$ and then $\cos \theta < 0$ and $\sin \theta > 0$ so that V_x is negative and V_y , positive.

For a vector in the third quadrant, $-\pi \leq \theta \leq -\pi/2$ so that $\cos \theta < 0$ and $\sin \theta < 0$ and both V_x and V_y are negative.

In the fourth quadrant, $-\pi/2 \leq \theta \leq 0$ so that $\cos \theta > 0$ and $\sin \theta < 0$. In this case $V_x > 0$ and $V_y < 0$. These results are summarised in Figure 4.1-5.

All calculators which are known as *scientific models* provide a function by means of which the x and y -components are obtained directly if the magnitude of the vector and the angle between the vector and the x -axis (known as the **azimuth angle**) are keyed in, in the prescribed manner. The inverse function is also available. It is of prime importance that the reader becomes skilled in the use of these functions. On some models the function is indicated by the notation $(r, \theta) \rightarrow (x, y)$ with inverse $(x, y) \rightarrow (r, \theta)$. Other models use the notations $P \rightarrow R$ and the inverse is $R \rightarrow P$. In the literature supplied with the calculator, (r, θ) and P are referred to as **polar notation** and (x, y) or R as the **rectangular notation**. If such functions are not available on a calculator, the calculations have to be made as illustrated in the example which follows.

Example: Consider a frame of reference in which the x -axis is to the east and the y -axis, north. At a given instant an aircraft has a speed of 200 km h^{-1} while flying horizontally. Calculate the x and y -components if the direction in which it flies is (a) N 40° E, (b) N 30° W, (c) W 15° S, (d) south, (e) S 30° E.

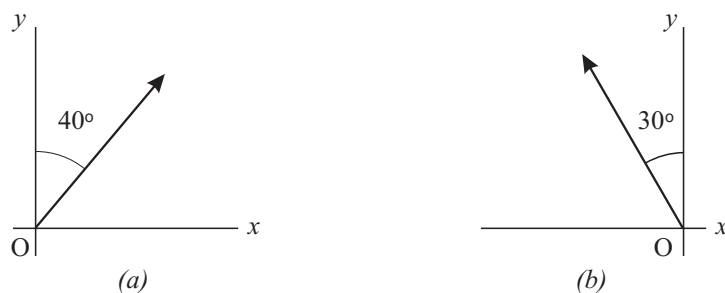


Figure 4.1-6

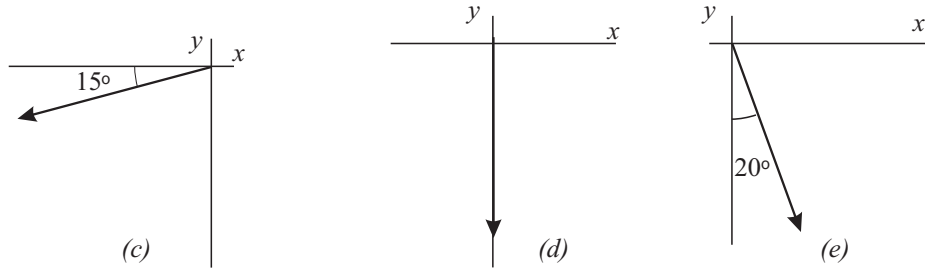


Figure 4.1-6

- (a) $V_x = 200 \cos 50^\circ = 200(0,6428) = 128,6 \text{ m s}^{-1}$
 $V_y = 200 \sin 50^\circ = 200(0,7660) = 153,2 \text{ m s}^{-1}$
- (b) $V_x = 200 \cos 120^\circ = 200(-0,5000) = -100,0 \text{ m s}^{-1}$
 $V_y = 200 \sin 120^\circ = 200(0,8660) = 173,2 \text{ m s}^{-1}$
- (c) $V_x = 200 \cos(-165^\circ) = 200(-0,9659) = -193,2 \text{ m s}^{-1}$
 $V_y = 200 \sin(-165^\circ) = 200(-0,2588) = -51,8 \text{ m s}^{-1}$
- (d) $V_x = 200 \cos(-90^\circ) = 200(0,0000) = 0,0 \text{ m s}^{-1}$
 $V_y = 200 \sin(-90^\circ) = 200(-1,0000) = -200,0 \text{ m s}^{-1}$
- (e) $V_x = 200 \cos(-70^\circ) = 200(0,3420) = 68,4 \text{ m s}^{-1}$
 $V_y = 200 \sin(-70^\circ) = 200(-0,9396) = -187,9 \text{ m s}^{-1}$

4.2 Vector addition

4.2.1 The vector parallelogram

The addition of two vectors is quite a different process from the addition of two scalar quantities and it involves a rather cumbersome process which is known as the vector parallelogram. The answer to the “addition” of two vectors is called their **resultant** and the following notation is used:

$$\vec{A} + \vec{B} = \vec{R} \quad 4.2(1)$$

It should be noted that even if the equals sign is used, it does not necessarily mean that the two vectors $\vec{A}$ and $\vec{B}$ when representing physical quantities, will have the same effect on a system as the resultant $\vec{R}$.

Consider the two vectors $\vec{A}$ and $\vec{B}$ in Figure 4.2-1 of which the resultant $\vec{R}$ is

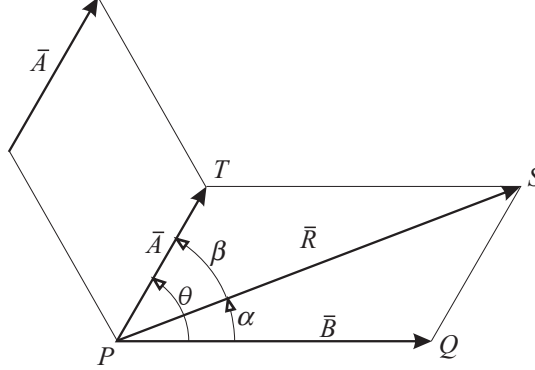


Figure 4.2-1

to be calculated. The process is as follows: One of the two vectors is given a parallel displacement so that their initial points coincide. With these two vectors as adjacent sides, a parallelogram is completed. The resultant, $\bar{R}$, is the vector represented by the diagonal from the common initial point of $\bar{A}$ and $\bar{B}$ to the opposite vertex of the parallelogram as shown in the sketch. The magnitude of $\bar{R}$ may be determined by accurate construction and measurement or it may be calculated by using the cosine rule and the sine rule as follows:

In ΔPQS ($QS = PT = A$) in Figure 4.2-1 we have:

$$\begin{aligned} R^2 &= A^2 + B^2 - 2AB \cos(\pi - \theta) = A^2 + B^2 - 2AB(-\cos \theta) \\ &= A^2 + B^2 + 2AB \cos \theta \\ \text{so that } R &= (A^2 + B^2 + 2AB \cos \theta)^{\frac{1}{2}} \end{aligned} \quad 4.2(2)$$

The direction of $\bar{R}$ can be determined by calculating either angle α or angle β . In ΔPQS

$$\begin{aligned} (\sin \alpha)/QS &= (\sin [\pi - \theta])/PS \quad \Rightarrow \quad \sin \alpha = (QS \sin [\pi - \theta])/PS \\ &= (A \sin \theta)/R \end{aligned}$$

$$\text{so that} \quad \alpha = \arcsin ([A \sin \theta]/R) \quad 4.2(3)$$

From the preceding calculation it should be clear that the construction of the parallelogram is superfluous since ΔPQS could have been constructed directly by placing the initial point of $\bar{A}$ at the terminal point of $\bar{B}$. The vector represented by the side which completes the triangle, is the resultant and its direction is from the open initial point to the open terminal point. Some prefer to speak of the **triangle of vectors** instead of the parallelogram.

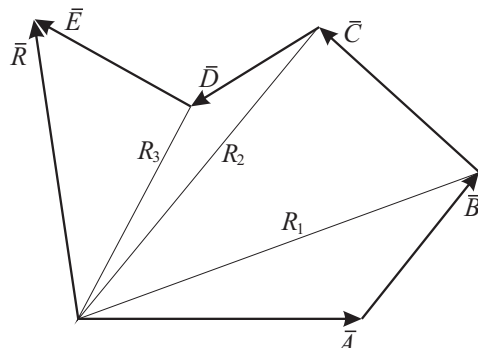


Figure 4.2-2

For the calculation of the resultant of more than two coplanar vectors, the resultant of any two may be added to a third and the new resultant to a fourth, etc. The method leads to a so-called **vector polygon** in which the vectors may be displaced in any order, parallel to their original directions to form a polygon. The terminal point of each vector must coincide with the initial point of the next. The side which closes the polygon, represents the resultant and its direction is the same as that for the triangle of vectors. This process is illustrated in Figure 4.2-2. The magnitude and direction may also be determined by construction and measurement. The use of the polygon of vectors cannot be recommended since it leads to cumbersome calculations in which calculation errors may easily occur.

For the definition of the difference between two vectors, the concept of a negative vector is used.

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B}) \quad 4.2(4)$$

4.2.2 The method of mutually perpendicular components for the addition of coplanar vectors

This method is in agreement with the parallelogram method since it is derived from it. Although the proof will not be given in this book, the method will be described in detail.

The first step in this method is the parallel displacement of all the vectors so that their initial points coincide. The common initial point is used as the origin of a

two-dimensional frame of reference. Provided they are mutually perpendicular, any coplanar orientation of the axes is admissible. For this calculation the axes will be called the x -axis and y -axis respectively. The method is illustrated by the example in Figure 4.2-3.

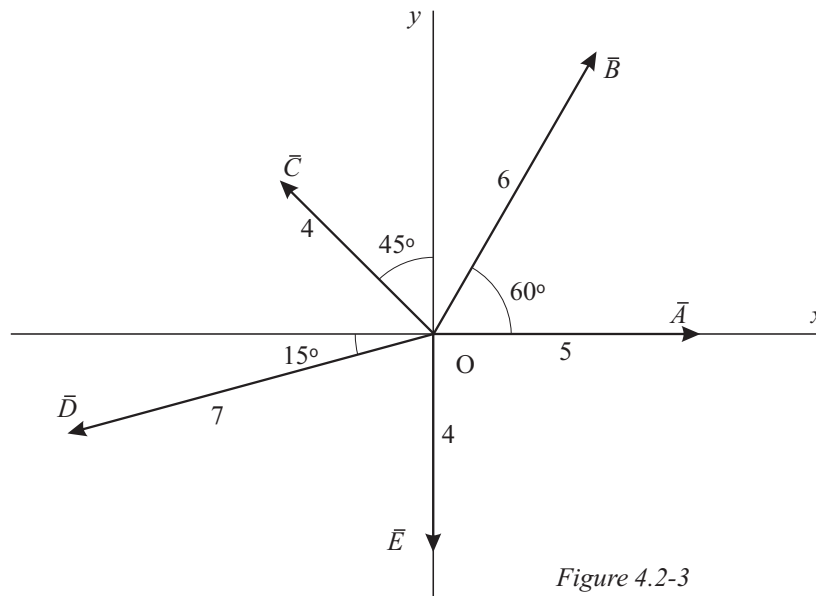


Figure 4.2-3

First the sum of all the components along the x -axis is calculated. This sum is represented by X . Then the sum of the components along the y -axis is calculated and it is denoted by Y . In the calculation which is shown below, we start with vector $\bar{A}$ and then proceed to the others in a counter-clockwise sense.

$$\begin{aligned}
 X &= 5 \cos 0^\circ + 6 \cos 60^\circ + 4 \cos 135^\circ + 7 \cos (-165^\circ) + 4 \cos (-90^\circ) \\
 &= 5(1,0000) + 6(0,5000) + 4(-0,7071) + 7(-0,9659) + 4(0,0000) \\
 &= 5,0000 + 3,0000 - 2,8284 - 6,7614 + 0,0000 \\
 &= -1,5898
 \end{aligned}$$

$$\begin{aligned}
 Y &= 5 \sin 0^\circ + 6 \sin 60^\circ + 4 \sin 135^\circ + 7 \sin (-165^\circ) + 4 \sin (-90^\circ) \\
 &= 5(0,0000) + 6(0,8660) + 4(0,7071) + 7(-0,2588) + 4(-1,0000) \\
 &= 0,0000 + 5,1961 + 2,8284 - 1,8117 - 4,0000 \\
 &= 2,2128
 \end{aligned}$$

The resultant of the vectors which are shown in Figure 4.2-4, is the vector $\bar{R}$ of which the x -component is $X = -1,5898$ and the y -component, $Y = 2,2128$. This result is shown in Figure 4.2-4.

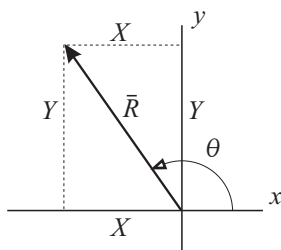


Figure 4.2-4

From this follows:

$$\begin{aligned}
 R &= (X^2 + Y^2)^{\frac{1}{2}} \\
 &= ([-1,5898]^2 + [2,2128]^2)^{\frac{1}{2}} \\
 &= (2,5274 + 4,8964)^{\frac{1}{2}} \\
 &= (7,4238)^{\frac{1}{2}} \\
 &= 2,7246
 \end{aligned}$$

$$\begin{aligned}
 \text{Further} \quad \tan \theta &= Y/X = (2,2128)/(-1,5898) = -1,3918 \\
 \text{so that} \quad \theta &= \arctan(-1,3918) \\
 &= 125,7^\circ
 \end{aligned}$$

Now all the information about the resultant is known.

If a scientific calculator is available which provides the function for the transformation between polar and rectangular notations and also two memory positions in which the sums of the x and y components can be accumulated, the above calculation may be performed without writing down any intermediate steps. The procedure is fairly simple and it is a good investment to explore these facilities on a calculator.

4.2.3 The specification of a vector in terms of its components along two mutually perpendicular axes in two dimensions

The addition of vectors as illustrated in sections 4.2.1 and 4.2.2 requires much calculation. It is surely not as simple as the addition of scalar quantities. One of the reasons for this is that a vector is a more complicated concept than a scalar. A second, and probably more important reason, is that people often prefer to specify vectors in terms of their magnitudes and directions relative to a suitable frame of reference.

From section 4.2.2 it should be clear that a vector is *fully* specified if its components along two mutually perpendicular axes are known. Vector $\vec{B}$ in Figure 4.2-3 has a magnitude of 6 units and its azimuth angle is 60° . This information inevitably means that $B_x = 3,0000$ and $B_y = 5,1961$. Similarly in Figure 4.2-4, the information that $R_x = -1,5898$ and $R_y = 2,2128$ unambiguously specifies a vector $\vec{R}$ of which the magnitude is 2,7246 units and of which the azimuth angle is $125,7^\circ$.

If the problem shown in Figure 4.2-3 could be repeated but with each vector

specified in terms of its components, it becomes very simple.

$$\begin{array}{lll} \bar{A}: & A_x = & 5,0000 \quad A_y = \quad 0,0000 \\ \bar{B}: & B_x = & 3,0000 \quad B_y = \quad 5,1961 \\ \bar{C}: & C_x = & -2,8284 \quad C_y = \quad 2,8284 \\ \bar{D}: & D_x = & -6,7614 \quad D_y = -1,8117 \\ \bar{E}: & E_x = & 0,0000 \quad E_y = -4,0000 \end{array}$$

R_x is simply the sum of all the x -components and R_y the sum of all the y -components.

$$\bar{R}: \quad R_x = -1,5898 \quad R_y = \quad 2,2128$$

which specifies $\bar{R}$ in full and is thus also a satisfactory answer to the problem.

By introducing this way of specifying a vector, a very cumbersome calculation is reduced to two simple additions. Many ways exist for the specification of a vector in terms of its components. Although it is seldom used, one way is that which is used above.

$$\bar{B}: \quad B_x = \quad 3,0000 \quad B_y = \quad 5,1961$$

In the so-called **matrix notation**, the vector is written as an ordered pair of numbers with the x -component first.

$$\bar{B}: \quad (3,0000; 5,1961)$$

Matrix notation has many advantages and is used extensively in the study of physics but it does have the problem that the frame of reference which is used, cannot be indicated. In introductory physics it is preferable to make use of **component vectors** along mutually perpendicular axes.

Consider a rectangular frame of reference. Let $\hat{x}$ be a unit vector along the x -axis and $\hat{y}$ a unit vector along the y -axis. These unit vectors are referred to as the **base vectors** of the frame of reference. For vector $\bar{V}$ in Figure 4.2-5, the two components along the axes are as follows:

$$V_x = V \cos \alpha \quad \text{and} \quad V_y = V \sin \alpha = V \cos \beta$$

The **component vectors** of vector $\bar{V}$ along the axes of the system are defined as follows:

$$V_x \hat{x} = (V \cos \alpha) \hat{x} \quad \text{and} \quad V_y \hat{y} = (V \cos \beta) \hat{y}$$

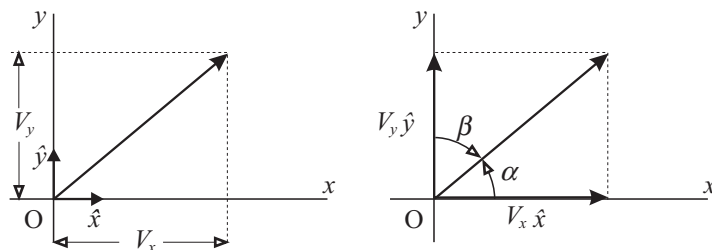


Figure 4.2-5

The component vector $V_x \hat{x}$ is in the direction of the positive x -axis (i.e. $\hat{x}$) with a magnitude of $V \cos \alpha$ and $V_y \hat{y}$ is in the direction $\hat{y}$ with magnitude $V \cos \beta$. According to the definition of vector addition, it may be written that

$$\bar{V} = V_x \hat{x} + V_y \hat{y} \quad 4.2(5)$$

which is the notation to which preference is given in the study of introductory physics. The magnitude (modulus or norm) of $\bar{V}$ may be calculated from this as follows:

$$V = |\bar{V}| = (V_x^2 + V_y^2)^{\frac{1}{2}} \quad 4.2(6)$$

The angles which it makes with the co-ordinate axes are

$$\begin{aligned} \alpha &= \arccos(V_x/V) & (0 \leq |\alpha| \leq \pi) \\ \beta &= \arccos(V_y/V) & (0 \leq |\beta| \leq \pi) \end{aligned} \quad 4.2(7)$$

Comment: A component vector is a vector and should not be confused with the component of a vector in a given direction which is a scalar quantity.

4.2.4 The specification of a vector in terms of its component vectors along mutually perpendicular axes in three dimensions

The space in which we live requires three independent numbers to specify the position of a point unambiguously. For this reason we speak of three-dimensional space. It is impossible to solve some problems in physics in two-dimensional frames of reference such as those which we have used up to know.

For three-dimensional problems we make use of a frame of reference with three mutually perpendicular axes with base vectors $\hat{x}$, $\hat{y}$ and $\hat{z}$. Such a frame of reference is called a **three-dimensional Cartesian frame of reference** (*Renè*

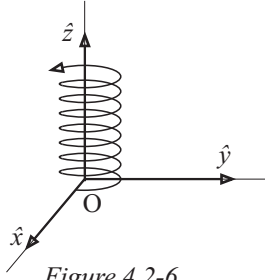


Figure 4.2-6

Descartes, 1596 – 1650, French mathematician, physicist and philosopher). In this book only **right-handed systems** will be treated. It is said that $\hat{x}$, $\hat{y}$ and $\hat{z}$ (in this order) form a right-handed system if the handle of a conventional corkscrew (right-handed screw) which is along the x -axis is rotated through 90° to coincide with $\hat{y}$ and the screw advances in the direction of $\hat{z}$. See Figure 4.2-6.

Consider vector $\bar{V}$ in such a frame of reference with base vectors $\hat{x}$, $\hat{y}$ and $\hat{z}$ along the axes Ox , Oy and Oz respectively as shown in Figure 4.2-7.

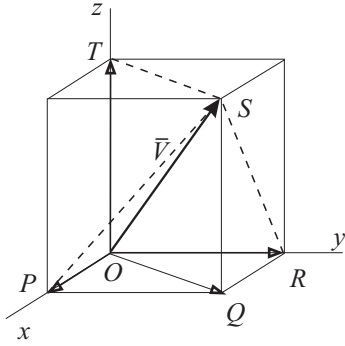


Figure 4.2-7

From the terminal point, S , of the vector, perpendiculars are dropped to the axes. The projections on the axes are the components of $\bar{V}$ in the directions of the base vectors. $V_x = OP$, $V_y = OR$ and $V_z = OT$. By multiplying each component by its corresponding base vector, the component vectors, $V_x\hat{x}$, $V_y\hat{y}$ and $V_z\hat{z}$ are obtained.

In rectangle $OPQR$ we have

$$V_x\hat{x} + V_y\hat{y} = \overline{OQ}$$

and in rectangle $OQST$,

$$\begin{aligned} \overline{OQ} + V_z\hat{z} &= \bar{V} \\ \Rightarrow \bar{V} &= V_x\hat{x} + V_y\hat{y} + V_z\hat{z} \end{aligned} \quad 4.2(8)$$

which is in agreement with the notation by which a vector was represented in two dimensions as in Equation 4.2(5). To calculate the magnitude (modulus or norm) of this vector, first consider $\triangle ORQ$. Here angle $ORQ = 90^\circ$, $RQ = OP = V_x$ and $OR = V_y$ so that

$$OQ^2 = V_x^2 + V_y^2$$

In $\triangle OQS$, angle $OQS = 90^\circ$ and $QS = OT = V_z$ so that

$$\begin{aligned} V^2 &= OQ^2 + V_z^2 = V_x^2 + V_y^2 + V_z^2 \\ \text{and } V &= |\bar{V}| = (V_x^2 + V_y^2 + V_z^2)^{\frac{1}{2}} \end{aligned} \quad 4.2(9)$$

Let $\alpha = \text{angle } POS = \text{the angle between } \bar{V} \text{ and } \hat{x}$, then we have in $\triangle POS$ (angle $OPS = 90^\circ$)

$$\cos \alpha = OP/OS = V_x/V$$

In the same manner we have for $\beta = \text{angle } ROS$ and $\gamma = \text{angle } TOS$

$$\begin{aligned} \cos \beta &= V_y/V & \text{and} & & \cos \gamma &= V_z/V \\ \text{so that} \quad \alpha &= \arccos(V_x/V) \\ \beta &= \arccos(V_y/V) \\ \gamma &= \arccos(V_z/V) \end{aligned} \quad 4.2(10)$$

The following three numbers

$$\cos \alpha = V_x/V, \quad \cos \beta = V_y/V, \quad \cos \gamma = V_z/V \quad 4.2(11)$$

are called the **direction cosines** of vector $\bar{V}$. The direction cosines of a vector have the following interesting property:

$$\begin{aligned} (V_x/V)^2 + (V_y/V)^2 + (V_z/V)^2 &= (V_x^2 + V_y^2 + V_z^2)/V^2 = 1 \\ \text{so that} \quad \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma &= 1 \end{aligned} \quad 4.2(12)$$

If a vector $\bar{V} = V_x\hat{x} + V_y\hat{y} + V_z\hat{z}$ is given, it is simple to calculate a unit vector $\hat{V}$ which has the same direction as $\bar{V}$. From Equations 4.1(1) and 4.2(9) follows

$$\hat{V} = \bar{V}/V = (V_x/V)\hat{x} + (V_y/V)\hat{y} + (V_z/V)\hat{z} \quad 4.2(13)$$

$$= (\cos \alpha)\hat{x} + (\cos \beta)\hat{y} + (\cos \gamma)\hat{z} \quad 4.2(14)$$

in which $\cos \alpha$, $\cos \beta$ and $\cos \gamma$ are the direction cosines of vector $\bar{V}$ in the given frame of reference.

4.2.5 The position vector

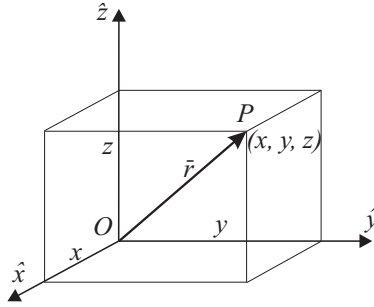


Figure 4.2-8

One of the most important and surely the most fundamental vector in the study of physics, is the **position vector** which specifies the position of a point relative to a given frame of reference. Consider a point P with coordinates (x, y, z) . If the initial point of the vector $\bar{r} = x\hat{x} + y\hat{y} + z\hat{z}$ is placed at the origin of the frame of reference, its terminal point will be at the position of P .

This vector is the position vector of P and its magnitude is given by $r = (x^2 + y^2 + z^2)^{\frac{1}{2}}$. This is the distance from the origin to point P .

The unit vector in the direction of $\bar{r}$, is given by

$$\hat{r} = \bar{r}/r = (x\hat{x} + y\hat{y} + z\hat{z})/(x^2 + y^2 + z^2)^{\frac{1}{2}}$$

Examples:

(a)

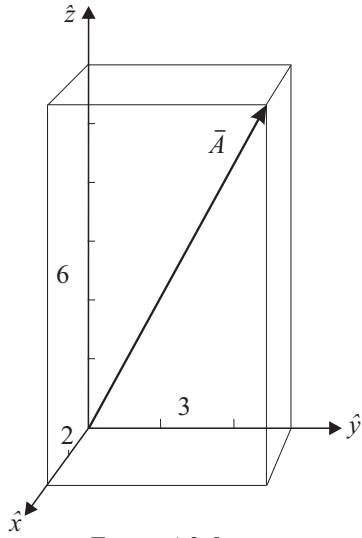


Figure 4.2-9

Consider the vector $\bar{A} = 2\hat{x} + 3\hat{y} + 6\hat{z}$. It is shown in Figure 4.2-9 in which its components are clearly visible. The following are to be calculated: The magnitude of $\bar{A}$, the unit vector $\hat{A}$, the direction cosines of $\bar{A}$ and the angles which it makes with the axes.

$$\begin{aligned} A &= (2^2 + 3^2 + 6^2)^{\frac{1}{2}} = (49)^{\frac{1}{2}} = 7 \\ \hat{A} &= \bar{A}/A \\ &= (2/7)\hat{x} + (3/7)\hat{y} + (6/7)\hat{z} \end{aligned}$$

The direction cosines are as follows:

$$\begin{aligned} \cos \alpha &= A_x/A = 2/7 \\ \cos \beta &= A_y/A = 3/7 \\ \cos \gamma &= A_z/A = 6/7 \end{aligned}$$

The angles which $\bar{A}$ makes with the base vectors are given by

$$\begin{aligned} \alpha &= \arccos(2/7) & \beta &= \arccos(3/7) & \gamma &= \arccos(6/7) \\ &= 73^\circ 24' & &= 64^\circ 37' & &= 31^\circ 0' \end{aligned}$$

(b) The calculations of the previous example are repeated for vector $\bar{B} = 4\hat{x} - 4\hat{y} - 7\hat{z}$. It is left to the reader as an exercise to sketch the vector in the same manner as that in example (a).

$$\begin{aligned} B &= (16 + 16 + 49)^{\frac{1}{2}} = (81)^{\frac{1}{2}} = 9 \\ \hat{B} &= \bar{B}/B \\ &= (4/9)\hat{x} - (4/9)\hat{y} - (7/9)\hat{z} \end{aligned}$$

The direction cosines are as follows:

$$\begin{aligned}\cos \alpha &= B_x/B = 4/9 \\ \cos \beta &= B_y/B = -4/9 \\ \cos \gamma &= B_z/B = -7/9\end{aligned}$$

The angles which it makes with the base vectors are as follows:

$$\begin{aligned}\alpha &= \arccos(4/9) = 63^\circ 37' \\ \beta &= \arccos(-4/9) = 116^\circ 23' \\ \gamma &= \arccos(-7/9) = 141^\circ 3'\end{aligned}$$

(c) The vectors $\bar{A}$ and $\bar{B}$ are given by $\bar{A} = 3\hat{x} - 2\hat{y} + 4\hat{z}$ and $\bar{B} = -2\hat{x} + 3\hat{y} - 5\hat{z}$ respectively. Calculate (i) $\bar{A} + \bar{B}$, (ii) $\bar{A} - \bar{B}$, (iii) $\bar{B} - \bar{A}$, (iv) $3\bar{A} + 2\bar{B}$, (v) $2\bar{A} - 3\bar{B}$.

$$\begin{aligned}\text{(i)} \quad \bar{A} + \bar{B} &= (3\hat{x} - 2\hat{y} + 4\hat{z}) + (-2\hat{x} + 3\hat{y} - 5\hat{z}) \\ &= (3 - 2)\hat{x} + (-2 + 3)\hat{y} + (4 - 5)\hat{z} \\ &= \hat{x} + \hat{y} - \hat{z} \\ \text{(ii)} \quad \bar{A} - \bar{B} &= (3\hat{x} - 2\hat{y} + 4\hat{z}) - (-2\hat{x} + 3\hat{y} - 5\hat{z}) \\ &= 5\hat{x} - 5\hat{y} + 9\hat{z} \\ \text{(iii)} \quad \bar{B} - \bar{A} &= (-2\hat{x} + 3\hat{y} - 5\hat{z}) - (3\hat{x} - 2\hat{y} + 4\hat{z}) \\ &= -5\hat{x} + 5\hat{y} - 9\hat{z} \\ \text{(iv)} \quad 3\bar{A} + 2\bar{B} &= 3(3\hat{x} - 2\hat{y} + 4\hat{z}) + 2(-2\hat{x} + 3\hat{y} - 5\hat{z}) \\ &= (9\hat{x} - 6\hat{y} + 12\hat{z}) + (-4\hat{x} + 6\hat{y} - 10\hat{z}) \\ &= 5\hat{x} + 2\hat{z} \\ \text{(v)} \quad 2\bar{A} - 3\bar{B} &= 2(3\hat{x} - 2\hat{y} + 4\hat{z}) - 3(-2\hat{x} + 3\hat{y} - 5\hat{z}) \\ &= (6\hat{x} - 4\hat{y} + 8\hat{z}) + (6\hat{x} - 9\hat{y} + 15\hat{z}) \\ &= 12\hat{x} - 13\hat{y} + 23\hat{z}\end{aligned}$$

(d) Calculate the resultant of the following vectors and also calculate the magnitude of the resultant: $\bar{A} = 4\hat{x} - 3\hat{y} - 2\hat{z}$, $\bar{B} = \hat{x} - 2\hat{y} + 4\hat{z}$, $\bar{C} = -5\hat{x} + \hat{y} + 3\hat{z}$ and $\bar{D} = 2\hat{x} + 3\hat{y} - 5\hat{z}$.

$$\begin{aligned}\bar{R} &= \bar{A} + \bar{B} + \bar{C} + \bar{D} = 2\hat{x} - \hat{y} \\ R &= (4 + 1)^{\frac{1}{2}} = 2, 236\end{aligned}$$

(e) The co-ordinates of point A are $(2, 2, 2)$ and those of point B , $(4, 5, 6)$. Calculate the distance AB by using the position vectors of the points.

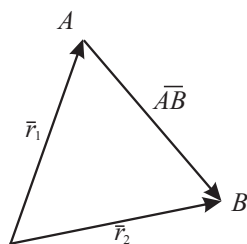


Figure 4.2-10

The position vector of point A is given by $\bar{r}_1 = 2\hat{x} + 2\hat{y} + 2\hat{z}$ and that of point B by $\bar{r}_2 = 4\hat{x} + 5\hat{y} + 6\hat{z}$. Both are measured in metres.

To calculate the required distance, the vector $\overline{AB}$ is used.

$$\begin{aligned}
 \overline{AB} &= \bar{r}_2 - \bar{r}_1 \\
 &= 2\hat{x} + 3\hat{y} + 4\hat{z} \text{ metres} \\
 AB &= |\overline{AB}| \\
 &= (4 + 9 + 16)^{\frac{1}{2}} \\
 &= (29)^{\frac{1}{2}} = 5.385 \text{ metres}
 \end{aligned}$$

4.3 The scalar product of two vectors

4.3.1 Definition and derivation

The notation by which the **scalar product** (also known as **dot product** or **inner product**) of vectors $\bar{A}$ and $\bar{B}$ is indicated is $\bar{A} \cdot \bar{B}$. It is read as A dot B . If the dot is omitted the result will not be a scalar product and will be devoid of meaning in this course. The scalar product is defined as follows:

$$\bar{A} \cdot \bar{B} = AB \cos \theta \quad (0 \leq \theta \leq \pi) \quad 4.3(1)$$

in which A and B are the magnitudes of the two vectors and θ the smallest angle between them. As the definition indicates, $\bar{A} \cdot \bar{B}$ is a scalar quantity.

$\bullet$	$\hat{x}$	$\hat{y}$	$\hat{z}$
$\hat{x}$	1	0	0
$\hat{y}$	0	1	0
$\hat{z}$	0	0	1

Table 4.3-1

From the definition follows:

$$\begin{aligned}
 \hat{x} \cdot \hat{x} &= 1 \times 1 \times \cos 0^\circ = 1 \\
 \hat{y} \cdot \hat{x} &= 1 \times 1 \times \cos 90^\circ = 0 \\
 \hat{z} \cdot \hat{y} &= 1 \times 1 \times \cos 90^\circ = 0
 \end{aligned}$$

In the same way the reader may explore all the possible combinations for the formation of scalar products between the base vectors $\hat{x}$, $\hat{y}$ and $\hat{z}$. The results are given in Table 4.3-1.

If the magnitudes of two vectors and the angle between them are known, Equation 4.3(1) may be used directly for the calculation of their scalar product. If, however, the two vectors are specified in terms of their components along the axes of a Cartesian frame of reference, that will not be possible. In such a case the information in Table 4.3-1 can be used to calculate the scalar product.

If $\bar{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z}$ and $\bar{B} = B_x\hat{x} + B_y\hat{y} + B_z\hat{z}$, then

$$\begin{aligned}\bar{A} \cdot \bar{B} &= (A_x\hat{x} + A_y\hat{y} + A_z\hat{z}) \cdot (B_x\hat{x} + B_y\hat{y} + B_z\hat{z}) \\ &= A_xB_x\hat{x} \cdot \hat{x} + A_xB_y\hat{x} \cdot \hat{y} + A_xB_z\hat{x} \cdot \hat{z} \\ &\quad + A_yB_x\hat{y} \cdot \hat{x} + A_yB_y\hat{y} \cdot \hat{y} + A_yB_z\hat{y} \cdot \hat{z} \\ &\quad + A_zB_x\hat{z} \cdot \hat{x} + A_zB_y\hat{z} \cdot \hat{y} + A_zB_z\hat{z} \cdot \hat{z} \\ &= \begin{array}{rrr} A_xB_x & + 0 & + 0 \\ + 0 & + A_yB_y & + 0 \\ + 0 & + 0 & + A_zB_z \end{array}\end{aligned}$$

This result supplies an algorithm (recipe) by which the scalar product of two vectors may be written down directly if they are known in terms of their component vectors.

$$\bar{A} \cdot \bar{B} = A_xB_x + A_yB_y + A_zB_z \quad 4.3(2)$$

Example: Calculate the scalar product of $\bar{P} = -2\hat{x} - 3\hat{y} - 7\hat{z}$ and $\bar{Q} = 5\hat{x} - 4\hat{y} + 2\hat{z}$.

$$\begin{aligned}\bar{P} \cdot \bar{Q} &= (-2\hat{x} - 3\hat{y} - 7\hat{z}) \cdot (5\hat{x} - 4\hat{y} + 2\hat{z}) \\ &= -10 + 12 - 14 \\ &= -12\end{aligned}$$

4.3.2 A number of useful applications of scalar products

If vectors $\bar{A}$ and $\bar{B}$ are known in terms of their component vectors, their scalar product may be used to calculate the angle between them.

$$\begin{aligned}\text{Since} \quad \bar{A} \cdot \bar{B} &= AB \cos \theta \\ \Rightarrow \quad \cos \theta &= \bar{A} \cdot \bar{B} / AB\end{aligned} \quad 4.3(3)$$

$$\text{and} \quad \theta = \arccos(\bar{A} \cdot \bar{B} / AB) \quad 4.3(4)$$

From the scalar product the component of $\bar{A}$ in the direction of $\bar{B}$, i.e. $A \cos \theta$ (which is also known as the projection of $\bar{A}$ on $\bar{B}$ or the mapping of $\bar{A}$ on $\bar{B}$) may also be calculated.

$$\begin{aligned}
\text{Since } \quad \bar{A} \cdot \bar{B} &= AB \cos \theta \\
\Rightarrow \quad A \cos \theta &= \bar{A} \cdot \bar{B} / B & 4.3(5) \\
&= \bar{A} \cdot \hat{B} & 4.3(6)
\end{aligned}$$

In the same way the component of $\bar{B}$ in the direction of $\bar{A}$ may be calculated.

$$B \cos \theta = \bar{A} \cdot \bar{B} / A = \hat{A} \cdot \bar{B}$$

Examples: The following two vectors are known: $\bar{A} = 7\hat{x} - 4\hat{y} + 4\hat{z}$ and $\bar{B} = -3\hat{x} + 2\hat{y} + 6\hat{z}$.

(a) Calculate the angle between them.

$$\begin{aligned}
\bar{A} \cdot \bar{B} &= AB \cos \theta \\
\Rightarrow \quad \cos \theta &= \frac{\bar{A} \cdot \bar{B}}{AB} = \frac{(7\hat{x} - 4\hat{y} + 4\hat{z}) \cdot (-3\hat{x} + 2\hat{y} + 6\hat{z})}{(49 + 16 + 16)^{\frac{1}{2}}(9 + 4 + 36)^{\frac{1}{2}}} \\
&= \frac{-21 - 8 + 24}{(81)^{\frac{1}{2}}(49)^{\frac{1}{2}}} = \frac{-5}{63} \\
\text{and } \quad \theta &= \arccos(-5/63) = 94^\circ 33'
\end{aligned}$$

(b) Calculate the component of $\bar{A}$ in the direction of $\bar{B}$.

$$A \cos \theta = \bar{A} \cdot \bar{B} / B = -5/7 = -0.7143$$

(c) Calculate the projection of $\bar{B}$ on $\bar{A}$.

$$B \cos \theta = \bar{A} \cdot \bar{B} / A = -5/9 = -0.5556$$

That the last two answers are negative stems from the fact that the angle between the vectors is larger than 90° . In each case the projection has to be made on the extension of the vector in question.

The scalar product of two vectors provides a test to determine whether they are perpendicular to each other (**orthogonal**). If $\bar{A} \cdot \bar{B} = AB \cos \theta = 0$ and both $A \neq 0$ and $B \neq 0$, then $\cos \theta = 0$ which means that $\theta = 90^\circ$.

Example:

$$\bar{P} = 3\hat{x} - 5\hat{y} + 4\hat{z} \quad \text{and} \quad \bar{Q} = 2\hat{x} + 2\hat{y} + \hat{z}$$

The magnitude of a vector can be zero only if each and every component is equal to zero. In this case neither vector has a magnitude of zero. Further we have that

$$\bar{P} \cdot \bar{Q} = (3\hat{x} - 5\hat{y} + 4\hat{z}) \cdot (2\hat{x} + 2\hat{y} + \hat{z}) = 6 - 10 + 4 = 0$$

$\Rightarrow \bar{P}$ and $\bar{Q}$ are orthogonal.

The angles which a vector makes with the co-ordinate axes, may also be calculated by means of scalar products as follows:

Consider $\bar{V} = V_x\hat{x} + V_y\hat{y} + V_z\hat{z}$. Then

$$\begin{aligned}\bar{V} \cdot \hat{x} = V \times 1 \times \cos \alpha &\Rightarrow \cos \alpha = \bar{V} \cdot \hat{x} / V = (V_x\hat{x} + V_y\hat{y} + V_z\hat{z}) \cdot \hat{x} / V \\ &= V_x / V\end{aligned}$$

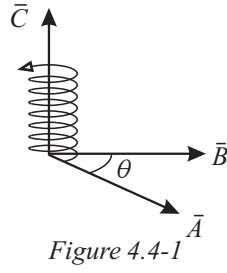
In the same way it follows that $\cos \beta = V_y / V$ and $\cos \gamma = V_z / V$ as was previously derived in a different manner.

4.3.3 Summary of useful results and remarks regarding scalar products

1. $\bar{A} \cdot \bar{B} = AB \cos \theta$ ($0 \leq \theta \leq \pi$)
2. $\bar{A} \cdot \bar{B} = \bar{B} \cdot \bar{A}$. The factors in a scalar product may be commuted without altering the product. This is known as the **commutative** law of the scalar product.
3. $A \cos \theta = \bar{A} \cdot \bar{B} / B = \bar{A} \cdot \hat{B}$ = the component of $\bar{A}$ in the direction of $\bar{B}$. Similarly $B \cos \theta = \bar{A} \cdot \bar{B} / A = \hat{A} \cdot \bar{B}$.
4. $\hat{A} \cdot \hat{B} = \bar{A} \cdot \bar{B} / AB = (AB \cos \theta) / AB = \cos \theta$
5. $A_x = \bar{A} \cdot \hat{x}$, $A_y = \bar{A} \cdot \hat{y}$, $A_z = \bar{A} \cdot \hat{z}$ so it may be written that $\bar{A} = (\bar{A} \cdot \hat{x})\hat{x} + (\bar{A} \cdot \hat{y})\hat{y} + (\bar{A} \cdot \hat{z})\hat{z}$
6. $\cos \alpha = \bar{A} \cdot \hat{x} / A$, $\cos \beta = \bar{A} \cdot \hat{y} / A$, $\cos \gamma = \bar{A} \cdot \hat{z} / A$.
7. If $\bar{A} \cdot \bar{B} = 0$ with $A \neq 0$ and $B \neq 0$, then $\cos \theta = 0$ and $\bar{A} \perp \bar{B}$.

4.4 The vector product

In the previous section the properties of two vectors were used to define a scalar quantity which was called the scalar product, dot product or inner product. In this section the properties of two vectors are used to define a *vector quantity* which is called a **vector product**, **cross product** or **outer product**. The vector product of the two vectors $\bar{A}$ and $\bar{B}$ is written as $\bar{A} \times \bar{B}$ and is read as



$\bar{A}$ cross $\bar{B}$. Unfortunately the sign between the vectors is similar to the multiplication sign for scalars but has a quite different meaning. For this reason some prefer the notation $\bar{A} \wedge \bar{B}$. Definition:

$$\bar{A} \times \bar{B} = \bar{C} \quad 4.4(1)$$

in which the magnitude of $\bar{C}$ is given by

$$C = AB \sin \theta \quad 4.4(2)$$

and θ is the *smallest* angle between the two vectors. The direction of $\bar{C}$ is perpendicular to the plane which contains $\bar{A}$ and $\bar{B}$ in such a way that $\bar{A}$, $\bar{B}$, and $\bar{C}$ in this order, form a right-handed system. These vectors are shown in Figure 4.4-1. A right-handed system is explained in section 4.2.4 by means of Figure 4.2-6. From this definition it follows that

$$\bar{A} \times \bar{B} = -\bar{B} \times \bar{A} \quad 4.4(3)$$

$\times$	$\hat{x}$	$\hat{y}$	$\hat{z}$
$\hat{x}$	$\bar{0}$	$\hat{z}$	$-\hat{y}$
$\hat{y}$	$-\hat{z}$	$\bar{0}$	$\hat{x}$
$\hat{z}$	$\hat{y}$	$-\hat{x}$	$\bar{0}$

Table 4.4-1

and it is said that two vectors are **anticommutative** in a vector product. From the definition it follows that

$$\begin{aligned} \hat{x} \times \hat{x} &= \hat{y} \times \hat{y} = \hat{z} \times \hat{z} = 0 \\ \hat{x} \times \hat{y} &= -\hat{y} \times \hat{x} = \hat{z} \\ \hat{y} \times \hat{z} &= -\hat{z} \times \hat{y} = \hat{x} \\ \hat{z} \times \hat{x} &= -\hat{x} \times \hat{z} = \hat{y} \end{aligned}$$

A summary of these results is shown in Table 4.4-1 and it gives the vector product of the unit vectors in the first column with those in the first row, *in this order*.

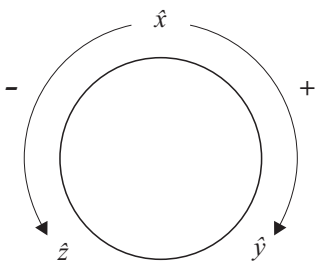


Figure 4.4-2

A simple way to remember the results in Table 4.4-1, is shown in Figure 4.4-2. The vector product of any vector with itself gives the zero vector. The three base vectors, $\hat{x}$, $\hat{y}$ and $\hat{z}$ in *this order* are written clockwise as shown in the sketch. If, in the calculation, the order is clockwise, the vector product of any two is equal to the third and the answer is positive: $\hat{x} \times \hat{y} = +\hat{z}$, $\hat{y} \times \hat{z} = +\hat{x}$ and $\hat{z} \times \hat{x} = +\hat{y}$.

If the calculation is in the counter-clockwise sense, the answer is negative: $\hat{x} \times \hat{z} = -\hat{y}$, $\hat{z} \times \hat{y} = -\hat{x}$ and $\hat{y} \times \hat{x} = -\hat{z}$.

If two vectors are specified in terms of their component vectors, the definition given in Equations 4.4(1) and 4.4(2) is not of much use in calculating their vector product. The results of Table 4.4-1, however, may be used for this purpose.

If $\bar{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z}$ and $\bar{B} = B_x\hat{x} + B_y\hat{y} + B_z\hat{z}$, then

$$\begin{aligned}
 \bar{A} \times \bar{B} &= (A_x\hat{x} + A_y\hat{y} + A_z\hat{z}) \times (B_x\hat{x} + B_y\hat{y} + B_z\hat{z}) \\
 &= A_xB_x\hat{x} \times \hat{x} + A_xB_y\hat{x} \times \hat{y} + A_xB_z\hat{x} \times \hat{z} \\
 &\quad + A_yB_x\hat{y} \times \hat{x} + A_yB_y\hat{y} \times \hat{y} + A_yB_z\hat{y} \times \hat{z} \\
 &\quad + A_zB_x\hat{z} \times \hat{x} + A_zB_y\hat{z} \times \hat{y} + A_zB_z\hat{z} \times \hat{z} \\
 &= \bar{0} \quad \quad \quad + A_xB_y\hat{z} \quad \quad - A_xB_z\hat{y} \\
 &\quad - A_yB_x\hat{z} \quad \quad + \bar{0} \quad \quad + A_yB_z\hat{x} \\
 &\quad + A_zB_x\hat{y} \quad \quad - A_zB_y\hat{x} \quad \quad + \bar{0} \\
 &= (A_yB_z - A_zB_y)\hat{x} - (A_xB_z - A_zB_x)\hat{y} + (A_xB_y - A_yB_x)\hat{z} \quad 4.4(4)
 \end{aligned}$$

This algorithm is much more difficult to remember than that for a scalar product and the calculation illustrated above is rather cumbersome and fraught with possibilities for errors. The use of a **determinant** simplifies the procedure greatly.

The notation $\begin{vmatrix} a & c \\ b & d \end{vmatrix}$ is called a determinant and it is defined as follows:

$$\begin{vmatrix} a & c \\ b & d \end{vmatrix} = ad - bc \quad 4.4(5)$$

A determinant which consists of two rows and two columns is called a **second-order determinant** and one which consists of three rows and three columns, a **third-order determinant**.

To expand a third-order determinant, it is first reduced to second-order determinants according to the following algorithm:

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \times \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \times \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \times \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

in which the values of the second-order determinants are calculated according to the definition. Note that the second term in the development is preceded by a negative sign. The development proceeds as follows: Write down a_1 and multiply it by the determinant obtained from the elements which remain after both the row and the column which contain a_1 are deleted.

$$\begin{vmatrix} \cancel{a_1} & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

The process is repeated for the other two elements in the top row.

$$\begin{vmatrix} \cancel{a_1} & \cancel{a_2} & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \quad \begin{vmatrix} a_1 & a_2 & \cancel{a_3} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Example:

$$\begin{aligned} \begin{vmatrix} 4 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} &= 4 \times \begin{vmatrix} 3 & 4 \\ 4 & 5 \end{vmatrix} - 2 \times \begin{vmatrix} 2 & 4 \\ 3 & 5 \end{vmatrix} + 3 \times \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} \\ &= 4(15 - 16) - 2(10 - 12) + 3(8 - 9) \\ &= -4 + 4 - 3 = -3 \end{aligned}$$

By calculation the reader may verify the following interesting and very useful properties of a determinant:

1. If any row or any column consists of zeros only, the value of the determinant is zero.
2. If any row is identical to another row or if one row is a multiple of another, the value of the determinant is zero. The same rule is valid for two columns.
3. If two *adjacent* rows are interchanged, the numerical value of the determinant remains unchanged but its sign changes. This also applies to two adjacent columns.

By means of a determinant, the vector product of two vectors may be calculated as follows:

$$\bar{A} \times \bar{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \quad 4.4(6)$$

If the determinant is expanded, it gives the same result as in Equation 4.4(4). Take note of property number (3) mentioned above. If the order of vectors $\bar{A}$

and $\bar{B}$ is changed, the third and second rows have to be interchanged in the corresponding determinant. This causes a change in the sign of the answer and this is in agreement with the anticommutative property of a vector product.

Examples:

(a) $\bar{A} = 2\hat{x} + \hat{y} + 3\hat{z}$ and $\bar{B} = \hat{x} + 3\hat{y} + 4\hat{z}$.

$$\begin{aligned}\bar{A} \times \bar{B} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 2 & 1 & 3 \\ 1 & 3 & 4 \end{vmatrix} \\ &= \hat{x} \times \begin{vmatrix} 1 & 3 \\ 3 & 4 \end{vmatrix} - \hat{y} \times \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} + \hat{z} \times \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} \\ &= \hat{x}(4 - 9) - \hat{y}(8 - 3) + \hat{z}(6 - 1) \\ &= -5\hat{x} - 5\hat{y} + 5\hat{z}\end{aligned}$$

(b) Use the vectors in the previous question. Let $\bar{A} \times \bar{B} = \bar{C}$. Calculate $\bar{A} \cdot \bar{C}$

$$\bar{A} \cdot \bar{C} = (2\hat{x} + \hat{y} + 3\hat{z}) \cdot (-5\hat{x} - 5\hat{y} + 5\hat{z}) = -10 - 5 + 15 = 0$$

It was unnecessary to actually calculate the scalar product since, by definition, $\bar{C} = \bar{A} \times \bar{B}$ is perpendicular to the plane which contains $\bar{A}$ and $\bar{B}$. For the same reason $\bar{B} \cdot \bar{C} = 0$.

(c) $\bar{A} = 2\hat{x} - \hat{y} + 3\hat{z}$ and $\bar{B} = -4\hat{x} + 2\hat{y} - 6\hat{z}$. Calculate $\bar{A} \times \bar{B}$

$$\begin{aligned}\bar{A} \times \bar{B} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 2 & -1 & 3 \\ -4 & 2 & -6 \end{vmatrix} \\ &= \hat{x}(6 - 6) - \hat{y}(-12 + 12) + \hat{z}(4 - 4) = \bar{0}\end{aligned}$$

Since $\bar{B} = -2\bar{A}$, the calculation of this result was also unnecessary. $\bar{A}$ and $\bar{B}$ are antiparallel ($\theta = \pi$) so that $|\bar{A} \times \bar{B}| = AB \sin \pi = 0$. Note that the last row in the determinant is a multiple of the middle row.

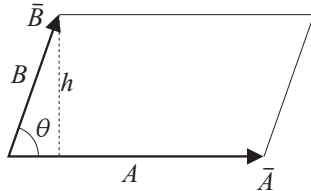
(d) Calculate a unit vector which is perpendicular to the plane which contains the vectors $\bar{A} = 2\hat{x} - 2\hat{y} + \hat{z}$ and $\bar{B} = \hat{x} + 2\hat{y} + 2\hat{z}$.

Since the vector product of two vectors is a vector which is perpendicular to the plane which contains them, the vector product of $\bar{A}$ and $\bar{B}$ has to be calculated first and then a unit vector which has the same direction.

$$\begin{aligned}
\bar{C} = \bar{A} \times \bar{B} &= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 2 & -2 & 1 \\ 1 & 2 & 2 \end{vmatrix} \\
&= \hat{x}(-4-2) - \hat{y}(4-1) + \hat{z}(4+2) \\
&= -6\hat{x} - 3\hat{y} + 6\hat{z} \\
\hat{C} = \bar{C}/C &= -\frac{2}{3}\hat{x} - \frac{1}{3}\hat{y} + \frac{2}{3}\hat{z} \\
-\hat{C} &= \frac{2}{3}\hat{x} + \frac{1}{3}\hat{y} - \frac{2}{3}\hat{z}
\end{aligned}$$

is also a unit vector perpendicular to the plane which contains $\bar{A}$ and $\bar{B}$.

(e) Show that $|\bar{A} \times \bar{B}|$ is equal to the area of the parallelogram of which $\bar{A}$ and $\bar{B}$ form the sides.

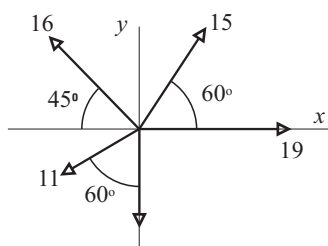


Consider the parallelogram in the sketch. Here θ is the angle between $\bar{A}$ and $\bar{B}$ and $\sin \theta = h/B$ in which h is the height of the parallelogram.

$$\begin{aligned}
\Rightarrow h &= B \sin \theta \\
|\bar{A} \times \bar{B}| &= AB \sin \theta = Ah \\
&= \text{base} \times \text{height} \\
&= \text{area of the parallelogram}
\end{aligned}$$

4.5 PROBLEMS CHAPTER 4

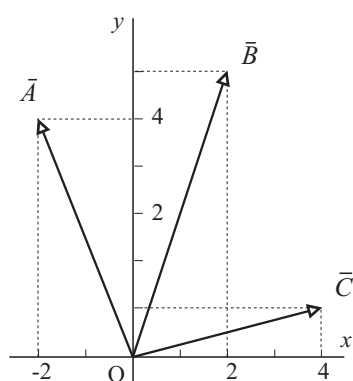
1.



The magnitudes and directions of five vectors are given in the sketch. Calculate their resultant by first using a graphical method (the vector polygon) and then by the use the method of components.

2. Construct a parallelogram of which the vectors $\vec{A}$ and $\vec{B}$ form two adjacent sides. Show that one diagonal represents their sum and the other their difference.

3.



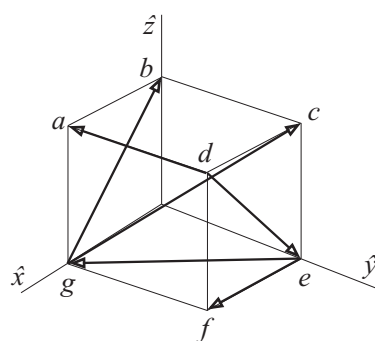
Consider vectors $\vec{A}$, $\vec{B}$, and $\vec{C}$ which are shown in the sketch. Write each in terms of their component vectors along the co-ordinate axes. Calculate the magnitude and azimuth angle of each. Show that

$$\vec{A} = \vec{B} - \vec{C}$$

Define a new frame of reference with axes parallel to those in the sketch and with origin at the terminal point of $\vec{A}$. Calculate the terminal point of $\vec{B}$ in the new frame.

4.

The cube in the sketch has a side length of 2 metre. It is placed in a Cartesian frame of reference as shown.



(a) Write down the position vectors of points g , e , b , a , f , c and d .

(b) Write down each of the following vectors in terms of their component vectors along the co-ordinate axes and calculate the magnitude of each: $\overrightarrow{ef}$, $\overrightarrow{da}$, $\overrightarrow{gb}$, $\overrightarrow{eg}$, $\overrightarrow{de}$ and $\overrightarrow{gc}$.

(c) A particle moves at a constant speed of 6 m s^{-1} . Speed is the magnitude of the velocity vector. Write down the velocity vector of the particle while moving along $\overrightarrow{ef}$, $\overrightarrow{da}$, $\overrightarrow{gb}$, $\overrightarrow{eg}$, $\overrightarrow{de}$ and $\overrightarrow{gc}$.

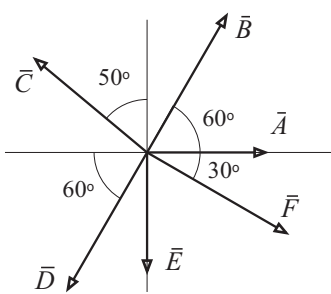
5. $\bar{A} = \hat{x} - 2\hat{y} - 3\hat{z}$, $\bar{B} = -2\hat{x} + 4\hat{y} + 3\hat{z}$ and $\bar{C} = -3\hat{x} + 2\hat{y} + \hat{z}$. Calculate (a) $|\bar{A}|$, $|\bar{B}|$, $|\bar{C}|$, (b) $\bar{A} + \bar{B} + \bar{C}$, $|\bar{A} + \bar{B} + \bar{C}|$, $|\bar{A}| + |\bar{B}| + |\bar{C}|$, (c) $5\bar{A} + 3\bar{B} - \bar{C}$ and $|5\bar{A} + 3\bar{B} - \bar{C}|$.

6. The initial points of vectors $\bar{P}$ and $\bar{Q}$ coincide to form the adjacent sides of a triangle. Show that the third side may be represented by either $\bar{P} - \bar{Q}$ or $\bar{Q} - \bar{P}$.

7. Do problem 6 first and then use the principles involved to solve the following problem: Point A has co-ordinates $(4, -2, 4)$ and point B , $(8, 2, 7)$. Write the position vectors of these two points. Use these position vectors to calculate the distance between A and B .

8. Use the method of the previous question to calculate the distance between the points (x_1, y_1, z_1) and (x_2, y_2, z_2) .

9.



The relative directions of vectors $\bar{A}$, $\bar{B}$, $\bar{C}$, $\bar{D}$, $\bar{E}$ and $\bar{F}$ are given in the sketch. Their magnitudes are: $A = 4$, $B = 6$, $C = 5$, $D = 6$, $E = 4$, $F = 5$.

Use the magnitudes and the angles which may be read off from the sketch to calculate the following scalar products: $\bar{A} \cdot \bar{B}$, $\bar{A} \cdot \bar{C}$, $\bar{A} \cdot \bar{D}$, $\bar{A} \cdot \bar{E}$, $\bar{A} \cdot \bar{F}$, $\bar{B} \cdot \bar{D}$, $\bar{B} \cdot \bar{F}$, $\bar{D} \cdot \bar{C}$, $\bar{D} \cdot \bar{F}$.

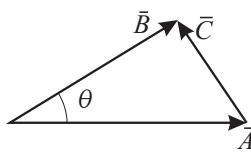
10. $\bar{A} = 6\hat{x} - 3\hat{y} + 2\hat{z}$, $\bar{B} = 2\hat{x} + 2\hat{y} - \hat{z}$. Calculate (a) A and B , (b) $\bar{A} \cdot \bar{B}$, (c) the angle between $\bar{A}$ and $\bar{B}$.

11. $\bar{V} = 4\hat{x} - 4\hat{y} + 7\hat{z}$. Calculate (a) $\hat{V}$, the unit vector parallel to $\bar{V}$, (b) the angles that $\bar{V}$ makes with the co-ordinate axes.

12. $\bar{A} = -3\hat{x} + 6\hat{y} - 2\hat{z}$, $\bar{B} = -\hat{x} + 2\hat{y} - 2\hat{z}$. Calculate (a) A and B and $\bar{A} \cdot \bar{B}$, (b) the component of $\bar{A}$ in the direction of $\bar{B}$, (c) the projection of $\bar{B}$ on $\bar{A}$, (d) the angle between $\bar{A}$ and $\bar{B}$.

13. Repeat the previous problem with $\bar{A} = -3\hat{x} + 6\hat{y} - 2\hat{z}$ and $\bar{B} = \hat{x} - 2\hat{y} + 2\hat{z}$.

14.



In the sketch, $\bar{B} = \bar{A} + \bar{C}$ and $\bar{C} = \bar{B} - \bar{A}$. From the study of scalar products, it should be clear that $C^2 = \bar{C} \cdot \bar{C}$. Use the calculation of a scalar product to show that the cosine rule for the calculation of an unknown side of a triangle, follows from it.

15. The vectors $\bar{P} = 3\hat{x} + 5\hat{y} - 2\hat{z}$ and $\bar{Q} = 3\hat{x} - \hat{y} + 2\hat{z}$ form two sides of a

triangle in such a way that their initial points coincide. Calculate a vector $\bar{C}$ that may represent the third side and also calculate the angles of the triangle.

16. $\bar{A} = \hat{x} + 2\hat{y} + 4\hat{z}$, $\bar{B} = 3\hat{x} - \hat{y} + 5\hat{z}$ and $\bar{C} = -2\hat{x} + 3\hat{y} - \hat{z}$. Show that these three vectors may represent the sides of a right-angled triangle. Calculate the other two angles of the triangle.

17. Calculate the value(s) of c for which vectors $\bar{P} = 4\hat{x} - c\hat{y} - 2c\hat{z}$ and $\bar{Q} = \hat{x} - 2\hat{y} + c\hat{z}$ will be orthogonal.

18. Calculate the following determinants:

$$(a) \begin{vmatrix} -1 & 2 & 3 \\ 1 & 0 & 4 \\ 4 & 3 & -2 \end{vmatrix} \qquad (b) \begin{vmatrix} 3 & 0 & 2 \\ 4 & 1 & -1 \\ -2 & -3 & 1 \end{vmatrix}$$

$$(c) \begin{vmatrix} -1 & -3 & 2 \\ -5 & -1 & 1 \\ 4 & 2 & -2 \end{vmatrix} \qquad (d) \begin{vmatrix} -2 & -1 & -9 \\ 2 & -3 & 4 \\ -2 & 3 & 2 \end{vmatrix}$$

19. $\bar{A} = -3\hat{x} + 2\hat{y} - 4\hat{z}$, $\bar{B} = 2\hat{x} + \hat{y} - \hat{z}$. Calculate $\bar{A} \times \bar{B}$ by the method which led to the result in Equation 4.4(4) (see page 108). Repeat the calculation by using a determinant.

20. $\bar{A} = 2\hat{x} - 3\hat{y} - \hat{z}$ and $\bar{B} = \hat{x} + 4\hat{y} - 2\hat{z}$. Calculate (a) $\bar{A} \times \bar{B}$, (b) $(\bar{A} + \bar{B}) \times (\bar{A} - \bar{B})$. Calculate the latter in two ways: (i) First calculate the sum and the difference of the vectors and then the vector product. (ii) Expand the expression to four terms and complete the calculation.

21. The quantity $\bar{A} \cdot \bar{B} \times \bar{C}$ is known as a **box product** or a **scalar triple product**. Is the answer a scalar or a vector? Is the prescription for the calculation unambiguous or are brackets required to make it unique? In order to answer this question it will be helpful to determine whether the order in which the scalar product and the vector product are calculated may be reversed. Now show that

$$(a) \quad \bar{A} \cdot \bar{B} \times \bar{C} = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix}$$

$$(b) \quad \bar{A} \cdot \bar{B} \times \bar{C} = \bar{C} \cdot \bar{A} \times \bar{B} = \bar{B} \cdot \bar{C} \times \bar{A}$$

(c) $|\bar{A} \cdot \bar{B} \times \bar{C}|$ = the volume of a parallelepiped of which vectors $\bar{A}$, $\bar{B}$ and $\bar{C}$ are the edges.

22. The torque, $\bar{\tau}$, of a force $\bar{F}$ about an axis through the origin of a frame of reference and which is perpendicular to the plane which contains the force vector and the origin, is given by $\bar{\tau} = \bar{r} \times \bar{F}$, in which $\bar{r}$ is the position of the point of application of the force. In a given frame of reference, the position vector of the point of application of a force is $\bar{r} = 2\hat{x} - 6\hat{y} - 3\hat{z}$ metres. The force is $\bar{F} = 4\hat{x} + 3\hat{y} - \hat{z}$ newton. Calculate (a) the torque of the force about the axis which is described above, (b) the magnitude of the torque vector, (c) the direction of the axis around which the torque tends to cause rotation. (*Hint:* The torque vector is along the said axis and a unit vector along this axis is a sufficient specification of its direction.)

23. When an electric point charge of q coulomb moves at velocity $\bar{v}$ metres per second in a magnetic field of which the induction vector is $\bar{B}$ tesla, it experiences a force which is given by $\bar{F} = q\bar{v} \times \bar{B}$. A proton with charge $q = 1.6 \times 10^{-19}$ C, moves at velocity $\bar{v} = 3 \times 10^7\hat{x} - 2 \times 10^7\hat{y}$ ms⁻¹ through a magnetic field of $\bar{B} = 500\hat{y} - 500\hat{z}$ tesla. Calculate the force on the proton.

Chapter 5

VECTOR DIFFERENTIATION

5.1 Introductory discussion and definitions

If a vector $\bar{A} = \bar{A}(u)$ depends on a single scalar quantity, u , in general it means that each component is a function of u .

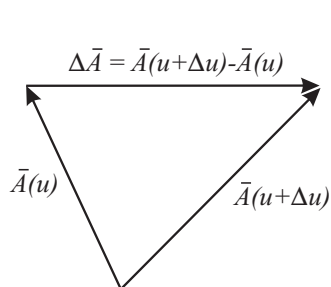


Figure 5.1-1

$$\bar{A}(u) = A_x(u)\hat{x} + A_y(u)\hat{y} + A_z(u)\hat{z}$$

Examples:

$$\bar{A} = (2u^2 - 5)\hat{x} + (3u + 2)\hat{y} + (u^3 - 8)\hat{z}$$

$$\bar{B} = (\sin 2t)\hat{x} + (\cos 2t)\hat{y} + (3t)\hat{z}$$

If u increases by an amount Δu , then, in general, $\bar{A}$ will change by $\Delta\bar{A}$ as shown in Figure 5.1-1.

It is important to note that while it is quite in order to speak of a *decrease* or *increase* of a scalar quantity, these two

terms can never apply to a vector quantity. One may speak only of a *change* in a vector. It is, however, permissible to refer to an increase or decrease in the magnitude of a vector since it is a scalar quantity.

The derivative of vector $\bar{A}$ with respect to the scalar variable q , is defined as

$$\begin{aligned}
\frac{d\bar{A}}{du} &= \lim_{\Delta u \rightarrow 0} \frac{\Delta \bar{A}}{\Delta u} = \lim_{\Delta u \rightarrow 0} \left(\frac{\bar{A}(u + \Delta u) - \bar{A}(u)}{\Delta u} \right) \\
&= \lim_{\Delta u \rightarrow 0} \left(\frac{A_x(u + \Delta u)\hat{x} + A_y(u + \Delta u)\hat{y} + A_z(u + \Delta u)\hat{z} -}{\Delta u} \right. \\
&\quad \left. \frac{A_x(u)\hat{x} + A_y(u)\hat{y} + A_z(u)\hat{z}}{\Delta u} \right) \\
&= \lim_{\Delta u \rightarrow 0} \left(\frac{A_x(u + \Delta u) - A_x(u)}{\Delta u} \hat{x} + \frac{A_y(u + \Delta u) - A_y(u)}{\Delta u} \hat{y} + \right. \\
&\quad \left. \frac{A_z(u + \Delta u) - A_z(u)}{\Delta u} \hat{z} \right) \\
&= \frac{dA_x}{du} \hat{x} + \frac{dA_y}{du} \hat{y} + \frac{dA_z}{du} \hat{z} \tag{5.1(1)}
\end{aligned}$$

Although it was not mentioned, the above derivation is based on the assumption that the base vectors, $\hat{x}$, $\hat{y}$ and $\hat{z}$ are constant, i.e. they are not functions of u . For a Cartesian frame of reference this assumption is correct. In systems which rotate the base vectors do not remain constant. Such systems are not discussed in this book but it is necessary that the reader take note of their existence and note that the differentiation of such vectors is more complicated.

In general, it is not possible to make a graphical representation of a variable vector $\bar{A} = \bar{A}(u)$ on a single graph. For this three graphs are necessary; one for each component. In such a set of graphs, the gradient of the $A_x = A_x(u)$ -graph is dA_x/du , the gradient of the $A_y = A_y(u)$ -graph, dA_y/du and the gradient of the $A_z = A_z(u)$ -graph, dA_z/du .

From the definition of the derivative of a vector function, the derivative is also a vector. From the result given in Equation 5.1(1), it should be clear that the differentiation of a vector function which is specified in a frame of reference with constant base vectors, is simply the independent differentiation of its three components which, in general, are each a function of the independent variable.

Examples:

(a) $\bar{A} = (2u^2 - 5)\hat{x} + (3u + 2)\hat{y} + (u^3 - 8)\hat{z}$. Calculate $d\bar{A}/du$.

$$\begin{aligned}
d\bar{A}/du &= (d/du)(2u^2 - 5)\hat{x} + (d/du)(3u + 2)\hat{y} + (d/du)(u^3 - 8)\hat{z} \\
&= (4u)\hat{x} + (3)\hat{y} + (3u^2)\hat{z}
\end{aligned}$$

(b) $\bar{B} = (\sin 2t)\hat{x} + (\cos 2t)\hat{y} + (3t)\hat{z}$. Calculate $d\bar{B}/dt$.

$$d\bar{B}/dt = (2 \cos 2t)\hat{x} + (-2 \sin 2t)\hat{y} + (3)\hat{z}$$

5.2 Space curves

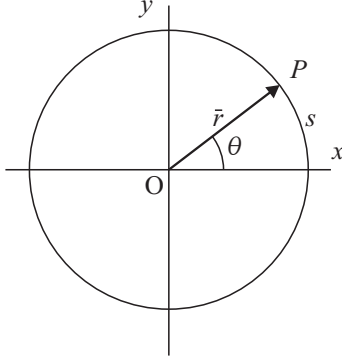


Figure 5.2-1

Consider the position vector of point P :

$$\bar{r} = (r \cos \theta)\hat{x} + (r \sin \theta)\hat{y}$$

which is the same as the statement that

$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned} \quad 5.2(1)$$

By assigning different values to θ in the domain $0 \leq \theta \leq 2\pi$, pairs of co-ordinates are generated which describe the positions of points which lie on the circle shown in Figure 5.2-1.

The two Equations 5.2(1) which jointly describe the circle in Figure 5.2-1, are known as the **parametric equations** of the circle. The independent variable θ is called a **parameter** (i.e. auxiliary variable. Greek: $\pi\alpha\rho\acute{\alpha}$ $\equiv$ alongside or conjugate, and $\mu\acute{\epsilon}\tau\rho\omicron\nu$ $\equiv$ dimension or gauge). By eliminating the parameter θ , the Cartesian equation for a circle is obtained.

$$\begin{aligned} \text{From 5.2(1)} \quad x^2 &= r^2 \cos^2 \theta \quad \text{and} \quad y^2 = r^2 \sin^2 \theta \\ \Rightarrow x^2 + y^2 &= r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2 (\sin^2 \theta + \cos^2 \theta) \\ \Rightarrow x^2 + y^2 &= r^2 \end{aligned} \quad 5.2(2)$$

Since $\theta = s/r$ (see Figure 5.2-1) with r constant, the arc length, s , could have been used as parameter in the description of a circle.

The intention of this short exposition is to introduce the reader to the concepts *parameter* and *parametric equations*.

Consider a point of which the position vector has three components, thus:

$$\bar{r} = \bar{r}(u) = x(u)\hat{x} + y(u)\hat{y} + z(u)\hat{z} \quad 5.2(3)$$

which is the same as the statement that each position co-ordinate is a function of u .

$$x = x(u), \quad y = y(u), \quad z = z(u) \quad 5.2(4)$$

The three Equations 5.2(4) are the parametric equations of a **space curve**. As the parameter u assumes all admissible values, sets of numerical values (x, y, z) are generated for the co-ordinates of the terminal point of $\bar{r}$ as it moves along a

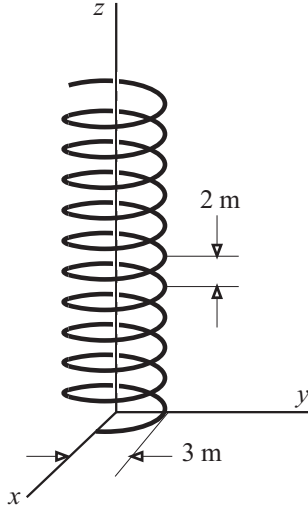


Figure 5.2-2

three-dimensional space curve. Contrary to the case of a two-dimensional space curve (see the example for a circle which is treated above), it is not possible to eliminate the parameter from three parametric equations. The use of three parametric equations is the simplest way to describe a three-dimensional space curve.

Example: Consider the following position vector which is a function of time, t :

$$\bar{r} = (3 \cos 2\pi t)\hat{x} + (3 \sin 2\pi t)\hat{y} + 2t\hat{z}$$

in which r is measured in metres.

It should be possible for the reader to see that this represents a right circular helix with a radius of 3 m and a **pitch** of 2 metres per revolution.

5.3 Rules for the differentiation of vectors

The following rules for the differentiation of vectors, follow directly from the rules for the differentiation of scalars. The reader is advised to try to deduce some of them. For the vector functions $\bar{P} = \bar{P}(u)$ and $\bar{Q} = \bar{Q}(u)$ and the scalar function $\phi = \phi(u)$, the following are valid:

$$(1) (d/du)(\bar{P} \pm \bar{Q}) = d\bar{P}/du \pm d\bar{Q}/du$$

$$(2) (d/du)(\bar{P} \cdot \bar{Q}) = (d\bar{P}/du) \cdot \bar{Q} + \bar{P} \cdot (d\bar{Q}/du)$$

$$(3) (d/du)(\bar{P} \times \bar{Q}) = (d\bar{P}/du) \times \bar{Q} + \bar{P} \times (d\bar{Q}/du)$$

$$(4) (d/du)(\phi \bar{P}) = (d\phi/du)\bar{P} + \phi(d\bar{P}/du)$$

$$(5) \text{ If } \bar{P} = P_x\hat{x} + P_y\hat{y} + P_z\hat{z}, \text{ then}$$

$$d\bar{P} = dP_x\hat{x} + dP_y\hat{y} + dP_z\hat{z}$$

Specifically

$$d\bar{r} = dx\hat{x} + dy\hat{y} + dz\hat{z}$$

$$(6) d(\bar{P} \cdot \bar{Q}) = (d\bar{P}) \cdot \bar{Q} + \bar{P} \cdot d\bar{Q}$$

$$(7) d(\bar{P} \times \bar{Q}) = (d\bar{P}) \times \bar{Q} + \bar{P} \times d\bar{Q}$$

If the order of any two factors on the right-hand side of equations (3) and (7) is reversed, the sign of the corresponding term must be changed.

5.4 Applications to kinematics

Kinematics is that branch of mechanics which deals with the measurement of motion. In this study the motion of points only, will be considered.

The most fundamental concept in the kinematics of a point, is its position. The position of a point is specified by its position vector.

$$\bar{r} = x\hat{x} + y\hat{y} + z\hat{z} \quad 5.4(1)$$

from which the distance between the point and the origin of the frame of reference may be calculated directly as follows:

$$r = |\bar{r}| = (x^2 + y^2 + z^2)^{\frac{1}{2}} \quad 5.4(2)$$

If the point is in motion, its position vector must be a function of time, which implies that at least one of its components should be a function of time. In general, all the components will be functions of time as follows:

$$\bar{r} = x(t)\hat{x} + y(t)\hat{y} + z(t)\hat{z} \quad 5.4(3)$$

As time increases by an amount Δt , the position vector changes by the vector $\Delta\bar{r}$ which is called the **displacement** of the point.

$$\begin{aligned} \Delta\bar{r} &= \bar{r}(t + \Delta t) - \bar{r}(t) \\ &= [x(t + \Delta t) - x(t)]\hat{x} + [y(t + \Delta t) - y(t)]\hat{y} + [z(t + \Delta t) - z(t)]\hat{z} \end{aligned} \quad 5.4(4)$$

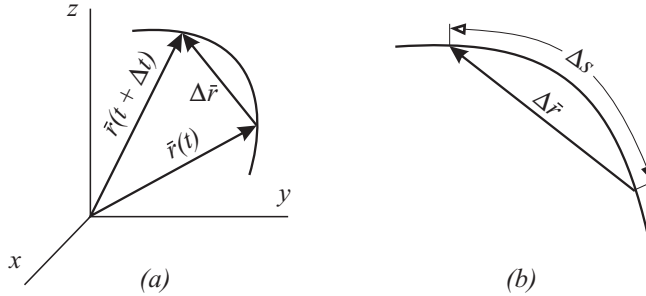


Figure 5.4-1

As time increases, the point moves along a space curve with parametric equations $x = x(t)$, $y = y(t)$ and $z = z(t)$. In Figure 5.4-1(a) the path of the point between instants t and $t + \Delta t$ is shown. As can clearly be seen, the magnitude

of the displacement is not equal to the distance that the point moves along the space curve. If the displacement vector tends to zero (i.e. the time interval tends to zero) the path length will be equal to the magnitude of the displacement vector.

$$\lim_{\Delta t \rightarrow 0} \Delta s = \lim_{\Delta t \rightarrow 0} |\Delta \vec{r}|$$

which is equivalent to $ds = |d\vec{r}|$ 5.4(5)

That necessarily means that $d\vec{r}$ is a vector that is tangential to the space curve (trajectory) of the particle.

For a moving point the position vector is always a function of time and its **velocity**, $\vec{v}$, is defined as the time derivative of its position vector.

$$\begin{aligned} \vec{v} = d\vec{r}/dt &= (d/dt)(x\hat{x} + y\hat{y} + z\hat{z}) \\ &= (dx/dt)\hat{x} + (dy/dt)\hat{y} + (dz/dt)\hat{z} \end{aligned} \quad 5.4(6)$$

As $d\vec{r}$, $\vec{v}$ is also tangential to the space curve at each position on the space curve. The **speed** of the point is the magnitude of the velocity vector.

$$v = |\vec{v}| = [(dx/dt)^2 + (dy/dt)^2 + (dz/dt)^2]^{\frac{1}{2}} \quad 5.4(7)$$

If the velocity is a function of time, it means that the velocity changes with time and the point is said to **accelerate**. The **acceleration** of the point is defined as follows:

$$\begin{aligned} \vec{a} &= d\vec{v}/dt = d^2\vec{r}/dt^2 \\ &= (dv_x/dt)\hat{x} + (dv_y/dt)\hat{y} + (dv_z/dt)\hat{z} \\ &= (d^2x/dt^2)\hat{x} + (d^2y/dt^2)\hat{y} + (d^2z/dt^2)\hat{z} \end{aligned} \quad 5.4(8)$$

If the position vector is measured in metres and the time in seconds, the units of velocity are metres per second (m s^{-1}) and that of acceleration, metres per second per second (m s^{-2}).

Example:

The position of a point is given by the following position vector:

$$\vec{r} = (3t^2 - 2t + 3)\hat{x} + (1, 5t^2 - 2t + 2)\hat{y} + (-t^2 - t + 1)\hat{z} \text{ m}$$

in which the time, t , is measured in seconds. Calculate the following: (a) The initial position. (b) The distance between the point and the origin at time $t = 0$. (c) The displacement between instants $t = 0$ and $t = 1$ s. (d) The velocity of the point as a function of time. (e) The initial velocity. (f) The initial speed. (g) The speed as a function of time. (h) The acceleration.

(a) By *initial* position is meant the position at time $t = 0$.

$$\bar{r}(0) = 3\hat{x} + 2\hat{y} + \hat{z} \text{ m}$$

$$(b) \quad r(0) = |\bar{r}(0)| = (9 + 4 + 1)^{\frac{1}{2}} = 3,742 \text{ m}$$

$$\begin{aligned} (c) \quad \Delta \bar{r} = \bar{r}(1) - \bar{r}(0) &= (4\hat{x} + 1, 5\hat{y} - \hat{z}) - (3\hat{x} + 2\hat{y} + \hat{z}) \\ &= \hat{x} - 0, 5\hat{y} - 2\hat{z} \text{ m} \end{aligned}$$

$$\begin{aligned} (d) \quad \bar{v} &= d\bar{r}/dt \\ &= (d/dt)[(3t^2 - 2t + 3)\hat{x} + (1, 5t^2 - 2t + 2)\hat{y} + (-t^2 - t + 1)\hat{z}] \\ &= (6t - 2)\hat{x} + (3t - 2)\hat{y} + (-2t - 1)\hat{z} \text{ m s}^{-1} \end{aligned}$$

$$(e) \quad \bar{v}(0) = -2\hat{x} - 2\hat{y} - \hat{z} \text{ m s}^{-1}$$

$$(f) \quad v(0) = |\bar{v}(0)| = (4 + 4 + 1)^{\frac{1}{2}} = 3 \text{ m s}^{-1}$$

$$\begin{aligned} (g) \quad v = |\bar{v}| &= [(6t - 2)^2 + (3t - 2)^2 + (-2t - 1)^2]^{\frac{1}{2}} \\ &= [36t^2 - 24t + 4 + 9t^2 - 12t + 4 + 4t^2 + 4t + 1]^{\frac{1}{2}} \\ &= [49t^2 - 32t + 9]^{\frac{1}{2}} \text{ m s}^{-1} \end{aligned}$$

$$(h) \quad \bar{a} = d\bar{v}/dt = 6\hat{x} + 3\hat{y} - 2\hat{z} \text{ m s}^{-2}$$

5.5 PROBLEMS: CHAPTER 5

1. $\bar{A} = \bar{A}(u) = (2u^3 - 5u)\hat{x} + (\sin 2u)\hat{y} + 12\hat{z}$. Calculate (a) $d\bar{A}/du$, (b) $d\bar{A}/du$ when $u = 0$, (c) $d^2\bar{A}/du^2$, (d) The angle between $\bar{A}$ and $d\bar{A}/du$ when $u = 0$.

2. The position vector of a point is given by $\bar{r} = (4 \sin 2t)\hat{x}$ metres in which the time, t is measured in seconds. Calculate (a) The velocity as a function of time. (b) The acceleration as a function of time. (c) The acceleration as a function of the position vector.

3. The velocity of a point is given by the following vector function of time:

$$\bar{v} = (4t - 3)\hat{x} + (7t + 6)\hat{y} + (-4t + 2)\hat{z} \text{ m s}^{-1}$$

in which the time, t , is measured in seconds. Calculate (a) The acceleration. (b) The magnitude of the acceleration. (c) The angle between the velocity and the acceleration at time $t = 0$.

4. The position vector of a particle is given by $\bar{r} = (2 \cos 4\pi t)\hat{x} + (2 \sin 4\pi t)\hat{y}$ metres in which the time, t , is measured in seconds. (a) Describe the path of the particle. (b) Calculate the velocity, $\bar{v}$, of the particle. (c) Calculate the speed of the particle and show that it is constant. (d) Calculate the acceleration of the particle. (e) Show that $\bar{v} \perp \bar{r}$, and that $\bar{a} \perp \bar{v}$. (f) Show that $\bar{a}$ is always directed towards the origin. (g) Show that $\bar{r} \times \bar{v}$ is a constant vector.

5. When an electron moves in a magnetic field with induction vector $\bar{B} = B\hat{z}$ tesla, its space curve has the following parametric equations:

$$x = 2 \times 10^5 \cos t; \quad y = 2 \times 10^5 \sin t \quad z = 10^5 t$$

in which x , y and z are measured in metres and the time, t , in seconds. (a) Write down the position vector of the electron and describe the curve which it follows. (b) Calculate the speed of the electron. (c) Calculate the acceleration of the electron. (d) Calculate the magnitude of the acceleration.

6. The parametric equations of a space curve are as follows: $x = 1, 5u^2 - 3$, $y = 3 + 11u - 2u^2$, $z = 1 - 2u$. Calculate a unit vector tangential to the curve at the position where $u = 2$. (*Hint*: The vector $d\bar{r}/du$ is tangential to the curve. Read the remarks following Equations 5.4(5) and 5.4(6).)

Chapter 6

VECTOR INTEGRATION

6.1 “Ordinary” vector integration

If a vector $\bar{V}$ is a function of a single scalar variable, u , then its derivative is $d\bar{V}/du = (dV_x/du)\hat{x} + (dV_y/du)\hat{y} + (dV_z/du)\hat{z}$, which, in general, is also a function of u . Let $d\bar{V}/du = \bar{G}$. Then

$$\bar{V} = \int \bar{G} du \quad 6.1(1)$$

This vector equation represents three different independent integrals as follows:

$$V_x = \int G_x du \quad V_y = \int G_y du \quad V_z = \int G_z du \quad 6.1(2)$$

Equation 6.1(1) may be rewritten as follows:

$$\bar{V} = \left(\int G_x du \right) \hat{x} + \left(\int G_y du \right) \hat{y} + \left(\int G_z du \right) \hat{z}$$

Each of these integrals supplies an integration constant independently of the other two. These three constants form a constant vector which will remain undetermined unless the value of $\bar{V}$ is known for one value of u .

As is the case with a scalar function, a definite integral of a vector function between two given limits may be calculated. The definite integral of $\bar{G}(u)$ between the limits $u = a$ and $u = b$ means

$$\int_a^b \bar{G} du = \bar{V}(b) - \bar{V}(a) \quad 6.1(3)$$

Examples:

(1) $d\bar{B}/du = \bar{A}(u) = (2u-3)\hat{x} + (6u^2-1)\hat{y} + (4u+7)\hat{z}$. Calculate $\bar{B}$.

$$\begin{aligned}\bar{B} &= \int \bar{A} du = \int [(2u-3)\hat{x} + (6u^2-1)\hat{y} + (4u+7)\hat{z}] du \\ &= (u^2-3u+k_x)\hat{x} + (2u^3-u+k_y)\hat{y} + (2u^2+7u+k_z)\hat{z}\end{aligned}$$

in which k_x , k_y and k_z are scalar constants.

It would be perfectly in order to write the answer as

$$\bar{B} = (u^2-3u)\hat{x} + (2u^3-u)\hat{y} + (2u^2+7u)\hat{z} + \bar{k}$$

in which $\bar{k}$ is a constant vector with components k_x , k_y and k_z .

(2) $d\bar{B}/dr = \bar{A}(r) = (8r^3-2r)\hat{x} + (3r^2+2)\hat{y} + (2r-7)\hat{z}$ and $\bar{B}(0) = 3\hat{x} - 4\hat{y} + 2\hat{z}$. Calculate $\bar{B} = \bar{B}(r)$.

$$\begin{aligned}\bar{B} &= \int \bar{A} dr = \int [(8r^3-2r)\hat{x} + (3r^2+2)\hat{y} + (2r-7)\hat{z}] dr \\ &= (2r^4-r^2+k_x)\hat{x} + (r^3+2r+k_y)\hat{y} + (r^2-7r+k_z)\hat{z} \\ \Rightarrow \bar{B}(0) &= k_x\hat{x} + k_y\hat{y} + k_z\hat{z} \quad (\text{calculated from the above}) \\ \text{but } \bar{B}(0) &= 3\hat{x} - 4\hat{y} + 2\hat{z} \quad (\text{given}) \\ \Rightarrow k_x &= 3, \quad k_y = -4, \quad k_z = 2 \\ \text{so that } \bar{B} &= (2r^4-r^2+3)\hat{x} + (r^3+2r-4)\hat{y} + (r^2-7r+2)\hat{z}\end{aligned}$$

(3) $\bar{P} = (e^{0.5t})\hat{x} + (\sin 0.5t)\hat{y} + (3t^2)\hat{z}$. Calculate $\bar{Q} = \int \bar{P} dt$. It is known that $\bar{Q}(0) = 2\hat{x} + 2\hat{y} + 3\hat{z}$.

$$\begin{aligned}\bar{Q} &= \int \bar{P} dt = \int [(e^{0.5t})\hat{x} + (\sin 0.5t)\hat{y} + (3t^2)\hat{z}] dt \\ &= (2e^{0.5t}+k_x)\hat{x} + (-2\cos 0.5t+k_y)\hat{y} + (t^3+k_z)\hat{z} \\ \Rightarrow \bar{Q}(0) &= (2+k_x)\hat{x} + (-2+k_y)\hat{y} + (k_z)\hat{z} \\ \text{but } \bar{Q}(0) &= 2\hat{x} + 2\hat{y} + 3\hat{z} \\ \Rightarrow k_x &= 0, \quad k_y = 4, \quad k_z = 3 \\ \text{so that } \bar{Q} &= (2e^{0.5t})\hat{x} + (-2\cos 0.5t+4)\hat{y} + (t^3+3)\hat{z}\end{aligned}$$

(4) $\bar{A} = (4t-2)\hat{x} + (6t^2-1)\hat{y} + (8t)\hat{z}$. Calculate $\int_1^3 \bar{A} dt$.

$$\int_1^3 \bar{A} dt = \int_1^3 [(4t-2)\hat{x} + (6t^2-1)\hat{y} + (8t)\hat{z}] dt$$

$$\begin{aligned}
&= (2t^2 - 2t)\hat{x} + (2t^3 - t)\hat{y} + (4t^2)\hat{z}|_1^3 \\
&= (12\hat{x} + 51\hat{y} + 36\hat{z}) - (0\hat{x} + \hat{y} + 4\hat{z}) \\
&= 12\hat{x} + 50\hat{y} + 32\hat{z}
\end{aligned}$$

6.2 Applications to kinematics

With the definitions of kinematic quantities and a knowledge of the necessary mathematics, it is possible to solve problems in kinematics *in which time is the independent variable*. There are, in fact, only three different problems of this type and they are discussed in sections 6.2.1 to 6.2.3. The problems are treated by means of examples. A schematic summary is given in section 6.2.5 and the reader is advised to make a brief study of this summary before reading the following sections.

6.2.1 First kind: The position vector, $\bar{r} = \bar{r}(t)$, is known

Actually the motion of a point is fully known if its position vector is known as a function of time. The calculation of each of the other quantities which may be required, follows directly from its definition:

$$\text{displacement:} \quad \Delta \bar{r} = \bar{r}(t_2) - \bar{r}(t_1) \quad 6.2(1)$$

$$\text{velocity:} \quad \bar{v} = d\bar{r}/dt \quad 6.2(2)$$

$$\text{acceleration:} \quad \bar{a} = d\bar{v}/dt = d^2\bar{r}/dt^2 \quad 6.2(3)$$

The example in section 5.4 is such a problem and no vector integration is required for its solution.

6.2.2 Second kind: The velocity vector, $\bar{v} = \bar{v}(t)$, is known

By definition $\bar{v} = \frac{d\bar{r}}{dt}$ and therefore we may write $\bar{r} = \int \bar{v} dt$ 6.2(4)

which is equivalent to the following three equations:

$$x = \int v_x dt, \quad y = \int v_y dt, \quad z = \int v_z dt$$

In order to calculate the integration constants which appear in these, it is necessary to know the vector $\bar{r}$ completely at one given instant.

If the displacement between instants $t = t_1$ and $t = t_2$ needs to be calculated, it may be done directly by means of the following definite integral:

$$\Delta \bar{r} = \bar{r}(t_2) - \bar{r}(t_1) = \int_{t_1}^{t_2} \bar{v} dt \quad 6.2(5)$$

The vector equation represents **three** equations, one for each component. That for the x -component is as follows:

$$\Delta x = x(t_2) - x(t_1) = \int_{t_1}^{t_2} v_x dt$$

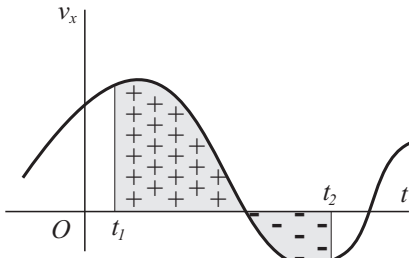


Fig. 6.2-1

As is known from the theory of definite integrals, the quantity Δx is represented by the area between the curve $v_x = v_x(t)$ and the t -axis from the value $t = t_1$ to the value $t = t_2$. In this representation, the area above the t -axis is taken as positive and that below, negative. This is shown in Figure 6.2-1. The two opposite signs indicate displacements in opposite directions.

Similar relationships exist for Δy and Δz and they may also be represented by areas on the graphs for $v_y = v_y(t)$ and $v_z = v_z(t)$ respectively.

Example: The velocity of a point is known as a function of time and is given by the following vector equation:

$$\bar{v} = \bar{v}(t) = (2t - 3)\hat{x} + (4t + 2)\hat{y} + (6t - 1)\hat{z} \text{ m s}^{-1}$$

in which the time, t , is measured in seconds. The position of the point is known at $t = 3$ s, and is given by $\bar{r}(3) = 2\hat{x} + 25\hat{y} + 21\hat{z}$ m.

(a) Calculate the position of the point as a function of time.

$$\bar{r} = \int \bar{v} dt = \left(\int (2t - 3) dt \right) \hat{x} + \left(\int (4t + 2) dt \right) \hat{y} + \left(\int (6t - 1) dt \right) \hat{z}$$

$$\begin{aligned}
&= (t^2 - 3t + k_x)\hat{x} + (2t^2 + 2t + k_y)\hat{y} + (3t^2 - t + k_z)\hat{z} \\
\text{but } \bar{r}(3) &= (k_x)\hat{x} + (24 + k_y)\hat{y} + (24 + k_z)\hat{z} \quad (\text{calculated from above}) \\
\text{and } \bar{r}(3) &= 2\hat{x} + 25\hat{y} + 21\hat{z} \quad (\text{specified in problem})
\end{aligned}$$

from which the numerical values of k_x, k_y and k_z may be calculated as follows: $k_x = 2, k_y = 1$ and $k_z = -3$, and $\bar{r}$ is known in full.

$$\bar{r} = (t^2 - 3t + 2)\hat{x} + (2t^2 + 2t + 1)\hat{y} + (3t^2 - t - 3)\hat{z} \text{ m}$$

(b) Calculate the displacement between $t = 1$ s and $t = 2$ s.

This may be calculated in two different ways. From the answer of question (a) it follows directly that

$$\Delta \bar{r} = \bar{r}(2) - \bar{r}(1) = (0\hat{x} + 13\hat{y} + 7\hat{z}) - (0\hat{x} + 5\hat{y} - \hat{z}) = 8\hat{y} + 8\hat{z} \text{ m.}$$

It may also be calculated directly by means of a definite integral as was explained in Equation 6.2(5).

$$\begin{aligned}
\Delta \bar{r} &= \int_1^2 [(2t - 3)\hat{x} + (4t + 2)\hat{y} + (6t - 1)\hat{z}] dt \\
&= (t^2 - 3t)\hat{x} + (2t^2 + 2t)\hat{y} + (3t^2 - t)\hat{z} \Big|_1^2 \\
&= (-2\hat{x} + 12\hat{y} + 10\hat{z}) - (-2\hat{x} + 4\hat{y} + 2\hat{z}) \\
&= 8\hat{y} + 8\hat{z} \text{ m}
\end{aligned}$$

(c) Calculate the acceleration of the point.

To calculate the acceleration, the same procedure is followed as that in 5.4(8). The reader should verify the correctness of the following answer:

$$\bar{a} = d\bar{v}/dt = 2\hat{x} + 4\hat{y} + 6\hat{z} \text{ m s}^{-2}$$

6.2.3 Third kind: The acceleration vector, $\bar{a} = \bar{a}(t)$, is known

By definition

$$\bar{a} = \frac{d\bar{v}}{dt} \quad \text{from which it follows that} \quad \bar{v} = \int \bar{a} dt \quad 6.2(6)$$

which is equivalent to the following three equations:

$$v_x = \int a_x dt, \quad v_y = \int a_y dt, \quad v_z = \int a_z dt$$

In order to calculate the integration constants which appear in these integrals, it will be necessary to know $\bar{v}$ completely at a given instant.

The change in velocity between $t = t_1$ and $t = t_2$ may be calculated directly by means of a definite integral as follows:

$$\Delta \bar{v} = \bar{v}(t_2) - \bar{v}(t_1) = \int_{t_1}^{t_2} \bar{a} dt \quad 6.2(7)$$

This vector equation represents three relationships, one for each component. That for the x -component is as follows:

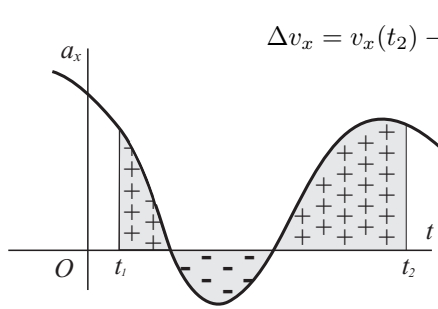


Fig. 6.2-2

$$\Delta v_x = v_x(t_2) - v_x(t_1) = \int_{t_1}^{t_2} a_x dt$$

From the theory of definite integrals it is known that the quantity Δv_x may be represented as the area between the curve $a_x = a_x(t)$ and the t -axis from $t = t_1$ to $t = t_2$. This is shown in Figure 6.2-2. The area above the t -axis is positive and that below it, negative. The opposite signs represent changes in opposite directions.

Similar expressions exist for changes in the other two components of the velocity and they may also be represented by areas as is shown in Figure 6.2-2. The reader should take note of the perfect analogy that exists between the calculations in 6.2.3 and 6.2.2.

After the velocity vector has been calculated from the acceleration, the position vector may also be calculated, but it will be necessary to know the position vector, $\bar{r}$, completely at any given instant.

Example: The acceleration of a point is constant, and is given by $\bar{a} = 4\hat{x} + 7\hat{y} - 4\hat{z}$ ms^{-2} . At time $t = 0$ the position is $\bar{r}(0) = 2\hat{x} - 2\hat{y} + \hat{z}$ m and at time $t = 1$ s, the velocity is $\bar{v}(1) = \hat{x} + 13\hat{y} - 2\hat{z}$ ms^{-1} .

(a) Calculate the velocity as a function of time.

$$\begin{aligned}
 \bar{v} &= \int \bar{a} \, dt = \int (4\hat{x} + 7\hat{y} - 4\hat{z}) \, dt \\
 &= (4t + k_x)\hat{x} + (7t + k_y)\hat{y} + (-4t + k_z)\hat{z} \\
 \text{but } \bar{v}(1) &= (4 + k_x)\hat{x} + (7 + k_y)\hat{y} + (-4 + k_z)\hat{z} \quad (\text{calculated}) \\
 \text{and } \bar{v}(1) &= \hat{x} + 13\hat{y} - 2\hat{z} \quad (\text{given})
 \end{aligned}$$

from which the integration constants may be calculated. The values are as follows: $k_x = -3$, $k_y = 6$ and $k_z = 2$. The velocity vector, $\bar{v}(t)$, is now completely known.

$$\bar{v}(t) = (4t - 3)\hat{x} + (7t + 6)\hat{y} + (-4t + 2)\hat{z} \quad \text{m s}^{-1}$$

(b) Calculate the change in velocity between $t = 1$ s and $t = 2$ s.

$$\begin{aligned}
 \Delta \bar{v} = \bar{v}(2) - \bar{v}(1) &= (5\hat{x} + 20\hat{y} - 6\hat{z}) - (\hat{x} + 13\hat{y} - 2\hat{z}) \\
 &= 4\hat{x} + 7\hat{y} - 4\hat{z} \quad \text{m s}^{-1}
 \end{aligned}$$

It may also be calculated directly by using a definite integral as explained in 6.2(7).

$$\begin{aligned}
 \Delta \bar{v} &= \int_1^2 (4\hat{x} + 7\hat{y} - 4\hat{z}) \, dt = (4t)\hat{x} + (7t)\hat{y} + (-4t)\hat{z} \Big|_1^2 \\
 &= (8\hat{x} + 14\hat{y} - 8\hat{z}) - (4\hat{x} + 7\hat{y} - 4\hat{z}) \\
 &= 4\hat{x} + 7\hat{y} - 4\hat{z} \quad \text{m s}^{-1}
 \end{aligned}$$

(c) Calculate the position of the point as a function of time.

$$\begin{aligned}
 \bar{r} &= \int \bar{v} \, dt = \int [(4t - 3)\hat{x} + (7t + 6)\hat{y} + (-4t + 2)\hat{z}] \, dt \\
 &= (2t^2 - 3t + 2)\hat{x} + (3.5t^2 + 6t - 2)\hat{y} + (-2t^2 + 2t + 1)\hat{z} \quad \text{m} \\
 \text{since } \bar{r}(0) &= 2\hat{x} - 2\hat{y} + \hat{z} \quad \text{m}
 \end{aligned}$$

The complete solution of a problem of the kind shown in the above example, involves the calculation of the velocity and position vectors from the acceleration. For the calculation of the integration constants additional information was required. In this example the velocity at a given instant and also the position at a given instant were used. It would also be possible to make the calculations if the positions were known at two different instants and this would make knowledge of the velocity vector at a given time superfluous. The following example shows how this is accomplished.

Example: The acceleration of a point is constant and is given by $\bar{a} = 4\hat{x} - 6\hat{y} + 2\hat{z}$ m s^{-2} . Its position at time $t = 1$ s is $\bar{r}(1) = 3\hat{x} - 2\hat{y} + 2\hat{z}$ m, and that at time $t = 2$ s, $\bar{r}(2) = 6\hat{x} - 9\hat{y} + 3\hat{z}$ m. Calculate its velocity and position as functions of time.

$$\begin{aligned}\bar{v} &= \int \bar{a} dt = \int (4\hat{x} - 6\hat{y} + 2\hat{z}) dt \\ &= (4t + k_x)\hat{x} + (-6t + k_y)\hat{y} + (2t + k_z)\hat{z}\end{aligned}$$

It is impossible to calculate the integration constants k_x , k_y and k_z at this stage because no value for the velocity is known at a given time. The calculation may, however, proceed as if these constants were known and they will be calculated at a later stage.

$$\begin{aligned}\bar{r} &= \int \bar{v} dt = \int [(4t + k_x)\hat{x} + (-6t + k_y)\hat{y} + (2t + k_z)\hat{z}] dt \\ &= (2t^2 + k_x t + c_x)\hat{x} + (-3t^2 + k_y t + c_y)\hat{y} + (t^2 + k_z t + c_z)\hat{z} \\ \text{but } \bar{r}(1) &= (2 + k_x + c_x)\hat{x} + (-3 + k_y + c_y)\hat{y} + (1 + k_z + c_z)\hat{z} \quad (\text{calculated}) \\ \text{and } \bar{r}(1) &= 3\hat{x} - 2\hat{y} + 2\hat{z} \quad (\text{given})\end{aligned}$$

By equating the corresponding components in the last two expressions, the following three equations are obtained:

$$k_x + c_x = 1 \cdots (1) \quad k_y + c_y = 1 \cdots (2) \quad k_z + c_z = 1 \cdots (3)$$

$$\begin{aligned}\text{also } \bar{r}(2) &= (8 + 2k_x + c_x)\hat{x} + (-12 + 2k_y + c_y)\hat{y} + (4 + 2k_z + c_z)\hat{z} \\ &\quad (\text{calculated}) \\ \text{and } \bar{r}(2) &= 6\hat{x} - 9\hat{y} + 3\hat{z} \quad (\text{given})\end{aligned}$$

This leads to the following three equations:

$$2k_x + c_x = -2 \cdots (4) \quad 2k_y + c_y = 3 \cdots (5) \quad 2k_z + c_z = -1 \cdots (6)$$

From these six equations the unknown constants may be calculated. Their values are as follows:

$$\begin{array}{lll} k_x &= & -3, \\ c_x &= & 4, \end{array} \quad \begin{array}{lll} k_y &= & 2, \\ c_y &= & -1, \end{array} \quad \begin{array}{lll} k_z &= & -2 \\ c_z &= & 3 \end{array}$$

The position and velocity vectors are now fully known as functions of the time.

$$\begin{aligned}\bar{r} &= (2t^2 - 3t + 4)\hat{x} + (-3t^2 + 2t - 1)\hat{y} + (t^2 - 2t + 3)\hat{z} \text{ m} \\ \bar{v} &= (4t - 3)\hat{x} + (-6t + 2)\hat{y} + (2t - 2)\hat{z} \text{ m s}^{-1}\end{aligned}$$

With these two vector functions known, all questions of the kind treated in 5.4 may be answered.

6.2.4 The use of position as independent variable

Many problems in kinematics involve questions such as, “What is the position, velocity or speed of an object *at a given instant*?” Some problems do not involve the time at all and questions such as, “What is the velocity or speed of an object when it is at a given position?” “Where will an object be when its speed has a given value?” or “What is the speed or velocity of an object as a function of a given position co-ordinate?” will have to be answered. In answering questions like these, it is often possible to make use of time, but the calculations are usually long and involved. For this reason a technique is developed to deal with problems in which time is not used.

Consider the x -component of a motion.

$$\begin{aligned}
 \text{By definition} \quad a_x &= \frac{dv_x}{dt} \\
 &= \frac{dv_x}{dx} \frac{dx}{dt} && \text{chain rule} \\
 &= \frac{dv_x}{dx} v_x && v_x = dx/dt \\
 \text{from which it follows} \quad a_x dx &= v_x dv_x && 6.2(8)
 \end{aligned}$$

The equation $a_x = (dv_x/dx)v_x$ is called a **differential equation** of the type **first order** (no higher-order derivatives than the first occur) and **first degree** (no derivative appears to a higher power than one). The variables in such an equation are always **separable**, i.e. they may be separated to appear on opposite sides of the equals sign. Equation 6.2(8) shows this equation after the separation of the variables has been completed. It is now ready to be solved by integration on both sides.

Before integration can take place, subsidiary information is required to calculate the integration constants. The supplementary information is known as the **boundary conditions** or **initial conditions**.

Suppose the following are the boundary conditions of a problem under consideration: $v_x = v_1$ where $x = x_1$ and $v_x = v_2$ where $x = x_2$. The solution is obtained by using these values as limits and integrating Equation 6.2(8) on both sides.

$$\int_{x_1}^{x_2} a_x dx = \int_{v_1}^{v_2} v_x dv_x \quad 6.2(9)$$

in which any one of x_1, x_2, v_1, v_2 or a_x may be an unknown quantity, or the integral may be used to calculate the function $v_x = v_x(x)$, i.e. the velocity component as a function of the position along a given axis. The use of this integral will be illustrated in the following three examples.

Examples:

1. The velocity of a motor-car is $20\hat{x} \text{ m s}^{-1}$ when the driver applies the brakes to bring it to rest over a distance of 50 m along a straight line. Assume that the acceleration is constant while the brakes are in action. Choose the origin where the brakes are first applied. Calculate the acceleration.

$$\begin{aligned} \text{From 6.2(9) follows} \quad \int_0^{50} a_x dx &= \int_{20}^0 v_x dv_x \\ a_x x|_0^{50} &= \frac{1}{2} v_x^2|_{20}^0 \\ 50 a_x &= \frac{1}{2}(0)^2 - \frac{1}{2}(20)^2 \\ \text{from which follows} \quad a_x &= -4 \\ \text{and} \quad \bar{a} &= -4\hat{x} \text{ m s}^{-2} \end{aligned}$$

2. The magnitude of the acceleration due to gravitation is equal to 10 m s^{-2} . An object at rest falls from a height of 20 m above ground level. Calculate the speed at which it hits the ground. Disregard frictional drag.

Choose a frame of reference with $\hat{z}$ vertically upwards and origin on the ground. Then $\bar{a} = -10\hat{z} \text{ m s}^{-2}$ and $a_z = -10 \text{ m s}^{-2}$. As before

$$\begin{aligned} \int_{20}^0 -10 dz &= \int_0^{v_z} v_z dv_z \\ -10 z|_{20}^0 &= \frac{1}{2} v_z^2|_0^{v_z} \\ 200 &= \frac{1}{2} v_z^2 \\ \text{and} \quad v = |v_z| &= 20 \text{ m s}^{-1} \end{aligned}$$

Since it is required to calculate the *speed*, only the positive root is used.

3. An object is projected vertically upwards at an initial velocity of $\bar{v}(0) = 40\hat{z} \text{ m s}^{-1}$. Gravitational acceleration is $\bar{a} = -10\hat{z} \text{ m s}^{-2}$. Calculate the velocity of the object as a function of its height above the ground.

$$\int_0^z -10 dz = \int_{40}^{v_z} v_z dv_z$$

$$\begin{aligned}
 -10z|_0^z &= \frac{1}{2}v_z^2|_{40}^{v_z} \\
 \text{from which follows } v_z &= \pm(1600 - 20z)^{\frac{1}{2}} \\
 \text{and } \bar{v} = v_z\hat{z} &= \pm(1600 - 20z)^{\frac{1}{2}}\hat{z} \text{ m s}^{-1}
 \end{aligned}$$

For each admissible value of the height ($20z \leq 1600$) two values exist for the velocity, one whilst it is ascending and the other when it is descending.

6.2.5 A summary of the kinematics of a point.

Definitions:

Position: $\bar{r} = x(t)\hat{x} + y(t)\hat{y} + z(t)\hat{z}$

Displacement: $\Delta\bar{r} = \bar{r}(t_2) - \bar{r}(t_1)$

Velocity: $\bar{v} = (dx/dt)\hat{x} + (dy/dt)\hat{y} + (dz/dt)\hat{z}$

Speed: $v = |\bar{v}| = (v_x^2 + v_y^2 + v_z^2)^{\frac{1}{2}}$

Acceleration: $\bar{a} = d\bar{v}/dt = d^2\bar{r}/dt^2$

$$\begin{aligned}
 &= (dv_x/dt)\hat{x} + (dv_y/dt)\hat{y} + (dv_z/dt)\hat{z} \\
 &= (d^2x/dt^2)\hat{x} + (d^2y/dt^2)\hat{y} + (d^2z/dt^2)\hat{z}
 \end{aligned}$$

If the quantities shown in boxes are known, the solution of a problem proceeds as indicated in the following schematic representation.

$$\begin{aligned}
 1. \quad & \boxed{\bar{r} = \bar{r}(t)} \xrightarrow{d\bar{r}/dt} \bar{v} = \bar{v}(t) \xrightarrow{d\bar{v}/dt} \bar{a} = \bar{a}(t) \\
 2. \quad & \bar{r} = \bar{r}(t) \xleftarrow[\boxed{\bar{r}(0)}]{\int \bar{v} dt} \boxed{\bar{v} = \bar{v}(t)} \xrightarrow{d\bar{v}/dt} \bar{a} = \bar{a}(t) \\
 3. \quad & \bar{r} = \bar{r}(t) \xleftarrow[\boxed{\bar{r}(0)}]{\int \bar{v} dt} \bar{v} = \bar{v}(t) \xleftarrow[\boxed{\bar{v}(0)}]{\int \bar{a} dt} \boxed{\bar{a} = \bar{a}(t)}
 \end{aligned}$$

Although it is shown in the scheme that $\bar{r}(0)$ and $\bar{v}(0)$ should be known for the calculation of the integration constants, known values for $\bar{r}$ and $\bar{v}$ at any instant will satisfy this need as was indicated in the examples.

In problems in which time is not specified and a position co-ordinate is used as an independent variable, integrals are used which follow from one of the following differential equations: $a_x dx = v_x dv_x$, $a_y dy = v_y dv_y$, $a_z dz = v_z dv_z$.

6.3 Line integrals

6.3.1 The calculation of the lengths of curves

In section 5.2 it was shown that the vector function $\bar{r} = x(u)\hat{x} + y(u)\hat{y} + z(u)\hat{z}$ represents a space curve. If u increases by an infinitesimal amount du , the position vector $\bar{r}$ changes by the following infinitesimal vector:

$$d\bar{r} = dx\hat{x} + dy\hat{y} + dz\hat{z} \quad 6.3(1)$$

which is always tangential to the space curve. The path length which corresponds to the displacement $d\bar{r}$, is given by

$$ds = |d\bar{r}| = (d\bar{r} \cdot d\bar{r})^{\frac{1}{2}} = [(dx)^2 + (dy)^2 + (dz)^2]^{\frac{1}{2}} \quad 6.3(2)$$

The position of a point on the space curve is specified unambiguously by either its co-ordinates (x, y, z) or the corresponding numerical value of the parameter u . The possibility to use a single parameter u instead of three co-ordinates, allows one to change a three-dimensional problem to a one-dimensional problem.

If one is required to calculate the distance *along a given space curve* between points P (with position vector $\bar{r}_P = x_1\hat{x} + y_1\hat{y} + z_1\hat{z}$ which corresponds to $u = u_1$) and Q (with position vector $\bar{r}_Q = x_2\hat{x} + y_2\hat{y} + z_2\hat{z}$ which corresponds to $u = u_2$) on the curve, it may be done by using the fact that a line integral is the limit of a sum (see section 3.6). The required distance is represented by s . Then

$$\begin{aligned} s &= \lim_{\Delta s \rightarrow 0} \sum_{i=1}^n \Delta s_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta s_i = \int_P^Q ds \\ &= \int_P^Q [(dx)^2 + (dy)^2 + (dz)^2]^{\frac{1}{2}} \end{aligned} \quad 6.3(3)$$

The P and the Q which are written in the positions of the limits of the definite integral, are not limits but a symbolic notation to indicate that the integral is to be calculated from P to Q along the space curve. In the examples it will be shown how the actual limits are to be treated. The integral shown in Equation 6.3(3) still contains three variables namely x , y , and z . In the examples it will also be shown how to transform it to a one-dimensional problem.

Examples:

(1) Consider the space curve which has the following parametric equations: $x = u$, $y = 2u$, $z = -2u$, in which x , y and z are measured in metres. This description is equivalent in all respects to the specification of the position vector,

$\bar{r} = u\hat{x} + 2u\hat{y} - 2u\hat{z}$, of an arbitrary point on the space curve. Calculate the length between the following pairs of points: (a) (0, 0, 0), and (1, 2, -2), (b) (0, 0, 0) and (2, 4, -4).

At each position on the space curve, the following applies:

$$\begin{aligned} x &= u & y &= 2u & z &= -2u \\ dx &= du & dy &= 2du & dz &= -2du \\ (dx)^2 &= (du)^2 & (dy)^2 &= 4(du)^2 & (dz)^2 &= 4(du)^2 \end{aligned}$$

(a) By equating the values of x , y and z which are known in terms of u to the corresponding numerical values of the co-ordinates of the points, it is simple to show that the point (0, 0, 0,) corresponds to the value $u = 0$ and that the point (1, 2,-2) corresponds to $u = 1$. From Equation 6.3(3), it follows that

$$\begin{aligned} s &= \int_{(0,0,0)}^{(1,2,-2)} [(dx)^2 + (dy)^2 + (dz)^2]^{\frac{1}{2}} \quad \text{along the space curve} \\ &= \int_{u=0}^{u=1} [(du)^2 + 4(du)^2 + 4(du)^2]^{\frac{1}{2}} \\ &= \int_0^1 3du = 3u|_0^1 = 3 \text{ metres} \end{aligned}$$

(b) The position (2,4,-4) corresponds to the value $u = 2$. In exactly the same manner which was used in (a), it follows that

$$s = \int_0^2 3du = 3u|_0^2 = 6 \text{ metres}$$

(2) The position vector of a particle in motion is given by

$$\bar{r} = (3t^2)\hat{x} + (1, 5t^2)\hat{y} + (-t^2)\hat{z} \text{ metres}$$

in which the time, t , is measured in seconds. Calculate the distance which it travels along the space curve on which it moves during the following time intervals: (a) From $t = 0$ to $t = 2$ s, (b) from $t = 1$ s to $t = 5$ s.

At each position on the space curve along which the particle moves, the following applies:

$$\begin{aligned} x &= 3t^2 & y &= 1, 5t^2 & z &= -t^2 \\ dx &= 6t dt & dy &= 3t dt & dz &= -2t dt \\ (dx)^2 &= 36t^2(dt)^2 & (dy)^2 &= 9t^2(dt)^2 & (dz)^2 &= 4t^2(dt)^2 \end{aligned}$$

$$\begin{aligned}
s = \int_{t=0}^{t=2} ds &= \int_{t=0}^{t=2} [(dx)^2 + (dy)^2 + (dz)^2]^{\frac{1}{2}} \\
&= \int_0^2 [(6t \, dt)^2 + (3t \, dt)^2 + (-2t \, dt)^2]^{\frac{1}{2}} \\
&= \int_0^2 [49t^2(dt)^2]^{\frac{1}{2}} = \int_0^2 7t \, dt = 3,5t^2 \Big|_0^2 = 14 \text{ metres}
\end{aligned}$$

(b) Exactly as in problem (a), it follows that

$$s = \int_1^5 7t \, dt = 3,5t^2 \Big|_1^5 = 84 \text{ metres}$$

6.3.2 Vector fields

If one unique vector $\bar{V} = \bar{V}(\bar{r}) = \bar{V}(x, y, z)$ exists at each position $\bar{r} = x\hat{x} + y\hat{y} + z\hat{z}$ within a given region, then $\bar{V}$ is called a **vector function of position** or a **vector field**. This means that each component of the vector field is, in general, a function of x , y , and z as follows:

$$\bar{V} = V_x(x, y, z)\hat{x} + V_y(x, y, z)\hat{y} + V_z(x, y, z)\hat{z} \quad 6.3(4)$$

It also means that if the co-ordinates of a point in the given region are specified, the components of $\bar{V}$ at that position, can be calculated.

If a vector field does not depend on time, it is called a **steady** or **stationary** field. Otherwise it is known as a **time-dependent** field. In this book steady fields only will be treated.

Example:

(1)

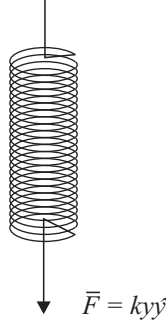


Figure 6.3-1

A light helical spring hangs parallel to the y -axis of a Cartesian frame of reference. The upper end is fixed and the lower end is at the origin of the frame of reference when the spring is not extended. A force $\bar{F} = ky\hat{y}$ newton is required to extend the spring so that its lower point is at position $\bar{r} = y\hat{y}$ metres. The constant, k , which appears in the formula for the force, is measured in N m^{-1} and is called the **force constant** or **stiffness** of the spring.

The force vector, which is given by

$$\bar{F} = 0\hat{x} + ky\hat{y} + 0\hat{z}$$

is an example of a vector field. It is a very simple vector field since it has one non-trivial component which depends on y only. The following example is more complicated.

(2) An electric point charge causes an electric field in the space surrounding it. Consider a point charge of q coulomb at the origin of a Cartesian frame of reference. At position $\bar{r}$, the intensity of the electric field, $\bar{E}$, is given by

$$\bar{E} = (kq/r^2)\hat{r}$$

in which k is a constant and $\hat{r} = \bar{r}/r$ is a unit vector parallel to $\bar{r}$. This intensity may be written in terms of Cartesian co-ordinates as follows:

$$\begin{aligned}\bar{E} &= (kq/r^2)\hat{r} = (kq/r^2)\bar{r}/r = (kq/r^3)\bar{r} \\ &= \frac{kq}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}(x\hat{x} + y\hat{y} + z\hat{z}) \\ &= \frac{kqx}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}\hat{x} + \frac{kqy}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}\hat{y} + \frac{kqz}{(x^2 + y^2 + z^2)^{\frac{3}{2}}}\hat{z}\end{aligned}$$

from which it can be seen clearly that each component of $\bar{E}$ is a function of x , y , and z .

6.3.3 Definition and properties of line integrals

Consider a space curve $\bar{r} = \bar{r}(u)$. Points P and Q on this curve correspond to the values $u = u_P$ and $u = u_Q$ respectively. If $\bar{V} = \bar{V}(x, y, z)$ is a vector function of position (vector field), the **line integral** of $\bar{V}$ along the space curve between the given two positions, is defined as follows:

$$\lim_{\Delta r_i \rightarrow 0} \sum_{i=1}^n \bar{V}_i \cdot \Delta \bar{r}_i = \int_P^Q \bar{V} \cdot d\bar{r} = \int_{u=u_P}^{u=u_Q} \bar{V} \cdot d\bar{r} \quad 6.3(5)$$

For the calculation of the result, the space curve is subdivided into n equal intervals between the two given positions and the limit of the sum calculated for the case where n tends to infinity (or the size of the intervals tends to zero). The answer is, of course, a scalar (note the point which indicates a scalar product). In general, the answer will not depend only on the positions of the two points, but also on the route or space curve which is followed from the one to the other.

As is the case with all definite integrals, the sign of the answer will change if the limits are interchanged. On condition that the space curve between the positions remains the same, the following will be valid:

$$\int_P^Q \bar{V} \cdot d\bar{r} = - \int_Q^P \bar{V} \cdot d\bar{r} \quad 6.3(6)$$

For any *steady* vector field, the following equation must apply:

$$\int_P^Q \bar{V} \cdot d\bar{r} + \int_Q^P \bar{V} \cdot d\bar{r} = 0 \quad 6.3(7)$$

if the same path is followed from P to Q as that which is followed in the opposite sense, i.e. on the same space curve. If the paths differ, the sum of the two integrals given in Equation 6.3(7) will, in general, not be equal to zero.

A line integral along a closed circuit (beginning and end-point the same) is indicated by the following notation:

$$\oint \bar{V} \cdot d\bar{r}$$

The symbol $\oint$ which is used in the above expression indicates that the line integral is calculated along a *closed circuit*.

Some vector fields have the property that the value of the line integral between two points is independent of the path which is followed from the one to the other. Such fields are known as **conservative fields**. For such fields, the sum of the two integrals given by Equation 6.3(7) is identically zero, independent of the paths followed. For a conservative field the following is always valid:

$$\oint \bar{V} \cdot d\bar{r} \equiv 0 \quad \bar{V} \text{ conservative} \quad 6.3(8)$$

A conservative field may always be described in terms of a scalar function which is known as a **potential function**. Since some physical phenomena may be described in a much more elegant (and simpler) way when these potential functions are used, it is of prime importance in the study of physics to know whether a vector field is conservative or not. This mathematical property forms the base of what is known as the principle of energy conservation.

Examples:

- (1) $\bar{F} = (x^2 - y^2)\hat{x} + 2xy\hat{y}$. Calculate the line integral of $\bar{F}$ from point P (0,0) to point Q (2,4) along the following three paths: (a) The straight line $y = 2x$, (b) the parabola $y = x^2$, (c) the parabola with parametric equations $x = 2t^2$,

$y = 4t$. (d) Also calculate the line integral along the straight line between points $P(0,0)$ and $R(2,0)$ and then along the straight line from $R(2,0)$ to $Q(2,4)$.

For all the paths specified in the problem, the following applies:

$$\begin{aligned}\int_P^Q \bar{F} \cdot d\bar{r} &= \int_P^Q [(x^2 - y^2)\hat{x} + (2xy)\hat{y} + (0)\hat{z}] \cdot [dx\hat{x} + dy\hat{y} + dz\hat{z}] \\ &= \int_P^Q [(x^2 - y^2)dx + (2xy)dy]\end{aligned}\quad 6.3(9)$$

(a) Along the given straight line the following are valid:

$$\begin{array}{llll} y & = & 2x & \text{or} & x & = & 0,5y \\ \text{and } dy & = & 2dx & & dx & = & 0,5dy \end{array}$$

The calculation may be done by eliminating either x or y . To show that it makes no difference, the answer is calculated in both ways.

(i) With $y = 2x$ and $dy = 2dx$, eliminate y from equation 6.3(9). Furthermore $x = 0$ at P and $x = 2$ at Q . Therefore

$$\begin{aligned}\int_P^Q \bar{F} \cdot d\bar{r} &= \int_0^2 [(x^2 - 4x^2)dx + (2x \times 2x)2dx] = \int_0^2 5x^2 dx \\ &= \frac{5}{3}x^3 \Big|_0^2 = 13,33\end{aligned}$$

(ii) With $x = 0,5y$ and $dx = 0,5dy$, eliminate x from equation 6.3(9). Furthermore $y = 0$ at P and $y = 4$ at Q . Therefore

$$\begin{aligned}\int_P^Q \bar{F} \cdot d\bar{r} &= \int_0^4 [(0,25y^2 - y^2)0,5dy + 2(0,5y)ydy] = \int_0^4 \frac{5}{8}y^2 dy \\ &= \frac{5}{24}y^3 \Big|_0^4 = 13,33\end{aligned}$$

(2) Along the given parabola $y = x^2$ and $dy = 2x dx$. Therefore

$$\begin{aligned}\int_P^Q \bar{F} \cdot d\bar{r} &= \int_0^2 [(x^2 - x^4)dx + 2x(x^2)2x dx] = \int_0^2 (x^2 + 3x^4)dx \\ &= \frac{1}{3}x^3 + \frac{3}{5}x^5 \Big|_0^2 = 21,87\end{aligned}$$

The same answer will be obtained if y is used as integration variable instead of x .

(c) Along the specified parabola the following are valid:

$$\begin{array}{rclcl} x & = & 2t^2 & \text{and} & y & = & 4t \\ \text{so that } dx & = & 4t \, dt & & dy & = & 4 \, dt \end{array}$$

The point $P(0,0)$ corresponds to $t = 0$ and $Q(2,4)$ to $t = 1$. Therefore

$$\begin{aligned} \int_P^Q \vec{F} \cdot d\vec{r} &= \int_0^1 [(4t^4 - 16t^2)4t \, dt + 2(2t^2)(4t)4t \, dt] = \int_0^1 (16t^5) \, dt \\ &= \left. \frac{8}{3} t^6 \right|_0^1 = 2,67 \end{aligned}$$

(d) All along the straight line between $P(0,0)$ and $R(2,0)$, $y = 0$ so that $dy = 0$. Therefore

$$\int_P^R \vec{F} \cdot d\vec{r} = \int_0^2 [(x^2 - 0)dx + 2x(0)(0)] = \left. \frac{1}{3} x^3 \right|_0^2 = 2,67$$

All along the straight line between $R(2,0)$ and $Q(2,4)$, $x = 2$ so that $dx = 0$. Therefore

$$\int_R^Q \vec{F} \cdot d\vec{r} = \int_0^4 [(2^2 - y^2)(0) + 2(2)y \, dy] = \left. 2y^2 \right|_0^4 = 32,00$$

Therefore the answer for this path is

$$\int_P^Q \vec{F} \cdot d\vec{r} = 2,67 + 32,00 = 34,67$$

Comments on this problem:

1. The elimination of either x or y was possible because the problem involved two dimensions only. The solution of three-dimensional problems is possible only by the use of a parameter as in problem (c).
2. All four line integrals were between the same points but along different paths. Since the answers differ, it is correct to conclude that $\vec{F}$ is not a conservative vector field.

(2) Calculate $\oint \bar{F} \cdot d\bar{r}$ for the vector field in question 1 along the path which begins at P (0,0) and then along the straight line $y = 2x$ to Q (2,4) and back to P along the parabola with parametric equations $y = 4t$, $x = 2t^2$.

The answers of 1(a) and 1(c) may be used to calculate this integral.

$$\begin{aligned}\oint \bar{F} \cdot d\bar{r} &= \left(\int_P^Q \bar{F} \cdot d\bar{r} \right)_{\text{along straight line}} + \left(\int_Q^P \bar{F} \cdot d\bar{r} \right)_{\text{along parabola}} \\ &= \left(\int_P^Q \bar{F} \cdot d\bar{r} \right)_{\text{along straight line}} - \left(\int_P^Q \bar{F} \cdot d\bar{r} \right)_{\text{along parabola}} \\ &= 13,33 - 2,67 = 10,66\end{aligned}$$

If the same path had been followed in the opposite sense (“direction”), the answer would have been -10,66. The fact that the answer is not equal to zero, confirms that the field is not conservative.

(3) Calculate the line integral of the vector field $\bar{F} = 2xy\hat{x} + 3x\hat{y} - 5y\hat{z}$ along the space curve with parametric equations $x = 2t$, $y = t^2$, $z = t^3$ from point P (2,1,1) to point Q (4,4,8).

It is important that the reader verifies that point P corresponds to the value $t = 1$ and point Q to $t = 2$. Along the space curve the following are valid:

$$\begin{array}{llll} \text{so that} & x & = & 2t, \\ & dx & = & 2dt, \end{array} \quad \begin{array}{llll} & y & = & t^2, \\ & dy & = & 2tdt, \end{array} \quad \begin{array}{llll} & z & = & t^3 \\ & dz & = & 3t^2dt \end{array}$$

$$\begin{aligned}\int_P^Q \bar{F} \cdot d\bar{r} &= \int_P^Q (2xy\hat{x} + 3x\hat{y} - 5y\hat{z}) \cdot (dx\hat{x} + dy\hat{y} + dz\hat{z}) \\ &= \int_P^Q (2xydx + 3xdy - 5ydz) \\ &= \int_1^2 (2(2t)(t^2)(2dt) + 3(2t)(2tdt) - 5(t^2)(3t^2dt)) \\ &= \int_1^2 (8t^3 + 12t^2 - 15t^4)dt = 2t^4 + 4t^3 - 3t^5 \Big|_1^2 \\ &= (32 + 32 - 96) - (2 + 4 - 3) \\ &= -35\end{aligned}$$

(4) When a force $\vec{F}$ acts on a body while it is displaced from point P to point Q , the line integral of the force vector along the given space curve, is called the **work** which the force does.

A helical spring in the unstretched condition, hangs vertically and parallel to the x -axis of a frame of reference. Its upper point is fixed and the lower point is at the origin of the frame. In order to displace the lower point to position $\vec{r} = x\hat{x}$ metres by stretching the spring, a force of $\vec{F} = 20x\hat{x}$ newton is required. Calculate the work done by the applied force if the lower end of the spring is displaced as follows: (a) From the origin to $\vec{r} = 2\hat{x}$ m, (b) from $\vec{r} = 2\hat{x}$ m to $\vec{r} = 4\hat{x}$ m.

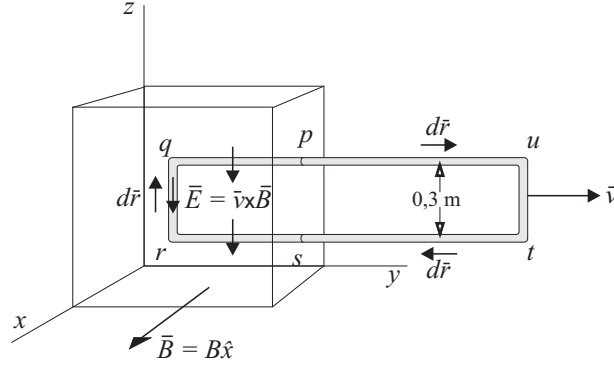
$$\begin{aligned} \text{(a)} \quad A &= \int_O^P \vec{F} \cdot d\vec{r} = \int_O^P (20x\hat{x}) \cdot (dx\hat{x} + dy\hat{y} + dz\hat{z}) \\ &= \int_0^2 20x \, dx = 10x^2 \Big|_0^2 = 40 \text{ joule} \end{aligned}$$

$$\text{(b)} \quad A = \int_2^4 20x \, dx = 10x^2 \Big|_2^4 = 120 \text{ joule}$$

(5) An electric point charge of q coulomb is at the origin of a Cartesian frame of reference. At the position $\vec{r} = x\hat{x}$ metres, this charge causes an electrostatic field of intensity $\vec{E} = (kqx^{-2})\hat{x}$ volt metre⁻¹, in which k is a constant. The **electric potential difference** between position Q ($\vec{r}_Q = a\hat{x}$ metres) and position S ($\vec{r}_S = b\hat{x}$ metres) is defined as the line integral of $\vec{E}$ from Q to S along any path. It can be shown that electrostatic fields are conservative. Calculate the potential difference, V_{QS} .

$$\begin{aligned} V_{QS} &= \int_Q^S \vec{E} \cdot d\vec{r} = \int_Q^S (kqx^{-2})\hat{x} \cdot (dx\hat{x} + dy\hat{y} + dz\hat{z}) \\ &= \int_a^b kqx^{-2} \, dx = -kqx^{-1} \Big|_a^b \\ &= kq[(1/a) - (1/b)] \text{ volt} \end{aligned}$$

(6) A magnetic field of which the induction vector is $\vec{B} = 2\hat{x}$ tesla, exists within the cubic space which is shown in the sketch. Outside the cube no magnetic field exists. The plane of a rectangular wire conductor which is partially within the cube, is parallel to the plane $x = 0$. The loop moves at a velocity of $\vec{v} = 7\hat{y}$ ms⁻¹. When a conductor moves at velocity $\vec{v}$ through a magnetic field with induction vector $\vec{B}$ tesla, a non-electrostatic field which is given by $\vec{E} = \vec{v} \times \vec{B}$ volt meter⁻¹ is generated within the conductor. The line integral of this non-conservative vector field around the conductor, is known as the **electromotive force** of **emf**, $\mathcal{E}$ in the conductor. Calculate (a) the non-conservative vector field, $\vec{E}$, in the conductor, (b) the emf in the conductor.



(a) $\bar{E} = \bar{v} \times \bar{B} = (7\hat{y} \times 2\hat{x}) = -14\hat{z}$ volt metre⁻¹.

(b) For the calculation of the line integral of $\bar{E}$ around the closed circuit, the different parts of the circuit are considered separately. The integration will be made in the clockwise sense as seen in the sketch.

In portions rs and pq , $\bar{E}$ is perpendicular to $d\bar{r}$ so that $\bar{E} \cdot d\bar{r} = 0$. In these portions of the circuit the line integral is everywhere equal to zero. Therefore

$$\int_s^r \bar{E} \cdot d\bar{r} = \int_q^p \bar{E} \cdot d\bar{r} = 0$$

In the portion of the loop which is outside the magnetic field, $\bar{B} = \bar{0}$ and thus $\bar{E} = \bar{0}$ there. Therefore

$$\int_p^u \bar{E} \cdot d\bar{r} = \int_u^t \bar{E} \cdot d\bar{r} = \int_t^s \bar{E} \cdot d\bar{r} = 0$$

The only contribution which is not equal to zero, is from the portion rq and for this we have

$$\int_r^q \bar{E} \cdot d\bar{r} = \int_0^{0,3} -14 dz = -14z \Big|_0^{0,3} = -4,2 \text{ volts}$$

The minus sign indicates that the emf is positive in the counter-clockwise sense, opposite to that in which the integration took place.

6.4 Surface integrals

6.4.1 The representation of area by a vector

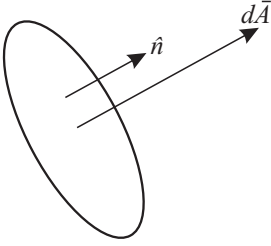


Figure 6.4-1

Any infinitesimal surface is flat and may be represented by a vector as follows:

$$d\vec{A} = dA\hat{n} \quad 6.4(1)$$

In this representation dA is the magnitude of the area and $\hat{n}$ a unit vector perpendicular to the surface as shown in Figure 6.4-1.

Such a representation of a *finite* surface with area A , will be possible only if it is flat. If a finite surface is not flat, an

infinitesimal element only may be represented by a vector because, in general, $\hat{n}$ will differ in direction from one position to the next.

One and the same surface may be represented by two different vectors in opposite directions. In many cases it will be of no consequence which of the two options is chosen, providing $d\vec{A}$ is perpendicular to the surface. An area which entirely encloses a volume (like the unbroken shell of an egg), is called a **closed surface**. If it does not enclose a volume it is known as a **bounded surface**. With a closed surface, $d\vec{A}$ is chosen to point outwards. This convention is in accordance with flow problems in which quantities which cross the surface from the inside to the outside, are taken as positive.

6.4.2 Solid angles

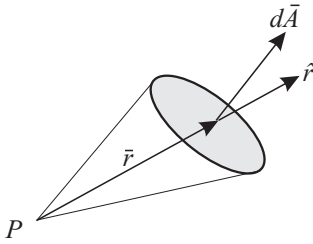


Figure 6.4-2

The vector representation of an infinitesimal area may be used to define the **solid angle**, $d\Omega$, which such a surface subtends at a point P .

$$d\Omega = \frac{d\vec{A} \cdot \hat{r}}{r^2} \quad 6.4(2)$$

in which $\vec{r}$ is the position vector of the infinitesimal surface relative to P . From the definition it should be clear

that $d\Omega$ is a scalar quantity. The units in which solid angles are measured are **steradians (sr)**.

In order to measure the solid angle which a finite surface subtends at a point, it must be taken into account that $\hat{r}$ differs from one point to the next on the surface. The finite surface may be subdivided into n elements, each with magnitude ΔA_i which should be small enough so that $\bar{r}_i$ and r_i may be taken to be constant over each element. The finite solid angle is given by

$$\Omega = \lim_{\Delta A \rightarrow 0} \sum_{i=1}^n \frac{\Delta \bar{A}_i \cdot \hat{r}_i}{r_i^2} = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{\Delta \bar{A}_i \cdot \hat{r}_i}{r_i^2}$$

This sum, as shown in section 3.5, represents an integral, in this case, over the entire surface. Therefore

$$\Omega = \iint_A \frac{d\bar{A} \cdot \hat{r}}{r^2} \quad 6.4(3)$$

The double integral sign indicates that a surface integral is to be calculated. In the calculation of many surface integrals, it actually means that two integrations will be performed. The subscript A indicates that the integral is to be calculated over the entire surface A . If A is a closed surface, the integral is indicated by the following notation:

$$\Omega = \oiint \frac{d\bar{A} \cdot \hat{r}}{r^2} \quad 6.4(4)$$

Equations 6.4(3) and 6.4(4) are only two examples of surface integrals.

Examples:

Consider a sphere with radius r with its centre at the origin of a Cartesian frame of reference. Calculate the the solid angle which (a) the portion of the spherical surface in the first octant, (b) the entire spherical surface, subtends at the centre.

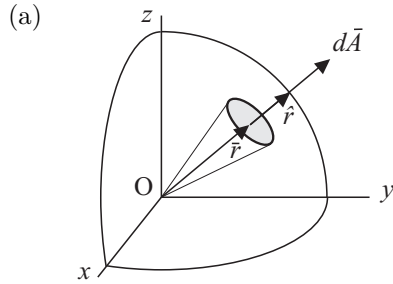


Figure 6.4-3

Consider an element of area, $d\bar{A}$, on the surface of the sphere as shown in Figure 6.4-3. The solid angle, $d\Omega$ which it subtends at the centre, is given by

$$d\Omega = (d\bar{A} \cdot \hat{r})/r^2$$

and for the spherical surface, $d\bar{A} \parallel \hat{r}$ so that

$$d\bar{A} \cdot \hat{r} = dA \times 1 \times \cos 0^\circ = dA$$

$$\begin{aligned}
\Rightarrow \quad \Omega &= \iint dA/r^2 = (1/r^2) \iint dA \\
&= (1/r^2) \left(\frac{1}{8} \times 4\pi r^2 \right) = \pi/2 \text{ steradians}
\end{aligned}$$

(b) For the entire surface of the sphere the procedure is the same.

$$\begin{aligned}
\Omega &= \oiint (d\bar{A} \cdot \hat{r})/r^2 = (1/r^2) \oiint dA \quad r \text{ is constant} \\
&= (1/r^2)(4\pi r^2) = 4\pi \text{ steradians}
\end{aligned}$$

It is worthwhile to remember this result.

Comments on this problem

1. The magnitude of $\hat{r}$ is equal to unity at each position on the spherical surface but its direction is different at each position. $\hat{r}$ is thus a vector function of position (a vector field) and it may be written as $\hat{r} = \hat{r}(\vec{r}) = \hat{r}(x, y, z)$.
2. This calculation was relatively simple because $\hat{r}$ and $d\bar{A}$ are parallel at each position on the surface.

6.4.3 More examples of surface integrals

The surface integral of a vector field $\vec{F} = \vec{F}(\vec{r}) = \vec{F}(x, y, z)$ over a surface with area A , is called the **flux** of $\vec{F}$ through the surface. If $\vec{F}$ is an electrostatic field vector, the corresponding flux will be the number of field lines crossing the surface. If the surface is closed, this flux supplies information about the total electric charge enclosed by the surface. A problem of this kind follows below.

If $\vec{F}$ is the velocity vector of a fluid in a current, the flux is the volume of fluid which passes through the surface per unit time. In more advanced theory, such surface integrals are used to localise the presence of **sources** and **sinks**.

Example:

An electric point charge of q coulomb is at the origin of a Cartesian frame of reference. The electric field intensity which this charge causes at position $\vec{r}$, is given by $\vec{E} = (kq/r^2)\hat{r} \text{ V m}^{-1}$, in which k is a constant. Calculate the surface integral of $\vec{E}$ over the entire surface of a sphere with radius r metre and with centre at the origin.

Since the surface is spherical, $\bar{r} \parallel d\bar{A}$. Therefore

$$\begin{aligned}\bar{E} \cdot d\bar{A} &= \oiint (kq/r^2) \hat{r} \cdot d\bar{A} = \oiint (kq/r^2) dA \\ &= (kq/r^2) \oiint dA = (kq/r^2)(4\pi r^2) = 4\pi kq\end{aligned}$$

This result is a special case of the **Gauss law in electrostatics** and may be interpreted as follows: Since a positive flux exists towards the outside of the sphere ($d\bar{A}$ was chosen to point outwards), a so-called source (in this case the point charge) is located within the sphere.

6.5 PROBLEMS: CHAPTER 6

1. $\bar{A} = (2p^3)\hat{x} + (p - p^2)\hat{y} + (-3)\hat{z}$. Calculate (a) $\int \bar{A}(p) dp$; (b) $\int_1^2 \bar{A}(p) dp$.
2. It is given that $\bar{A} = \bar{A}(u)$. (a) Calculate $(d/du)(\bar{A} \times [d\bar{A}/du])$. (b) Use the result of (a) to calculate $\int (\bar{A} \times d^2\bar{A}/du^2) du$.
3. The motion of a point is limited to the z -axis of a Cartesian frame of reference. The initial velocity and initial position of a point are given by $\bar{v}(0) = -3\hat{z} \text{ m s}^{-1}$ and $\bar{r}(0) = -10\hat{z} \text{ m}$ respectively. The acceleration of the point is independent of time and is given by $\bar{a} = 2\hat{z} \text{ m s}^{-2}$. Calculate (a) $\bar{v} = \bar{v}(t)$, (b) $\bar{r} = \bar{r}(t)$, (c) $v_z = v_z(z)$ directly from the data, not using the time as a variable.
4. The acceleration of a point is given by $\bar{a} = 4\hat{x} + 7\hat{y} - 4\hat{z} \text{ m s}^{-2}$. Its initial velocity is $\bar{v}(0) = -3\hat{x} + 6\hat{y} + 2\hat{z} \text{ m s}^{-1}$ and its initial position, $\bar{r}(0) = 2\hat{x} - 2\hat{y} + \hat{z} \text{ m}$. (a) Calculate the magnitude of the acceleration. (b) Calculate $\bar{v} = \bar{v}(t)$. (c) Calculate the initial speed of the point. (d) Calculate the angle between the velocity and the acceleration at instant $t = 0$. (e) Calculate the displacement vector between $t = 0$ and $t = 2 \text{ s}$. (f) Calculate $\bar{r} = \bar{r}(t)$. (g) Calculate the angle between the velocity and the position vector at $t = 0$.
5. Gravitational acceleration has a magnitude of 10 m s^{-2} . At $t = 0$ an object is projected at 40 m s^{-1} in a direction of 30° above the horizontal. Choose a Cartesian frame of reference with origin at the position from which the object was projected (i.e. $\bar{r}(0) = 0\hat{x} + 0\hat{y}$), $\hat{x}$ horizontally forwards and $\hat{y}$ vertically upwards. (a) Write down the acceleration vector and the initial velocity vector in this frame of reference. (b) Calculate $\bar{v} = \bar{v}(t)$. (c) Calculate $\bar{r} = \bar{r}(t)$. (d) Calculate $v_y = v_y(y)$. (e) Write the parametric equations of the trajectory (space curve) which the object follows. (f) Eliminate the time, t , from these equations and calculate the equation of the trajectory in the form $y = y(x)$. Describe the trajectory in words. (g) What is the maximum height that the object reaches above the x -axis? (h) What is the range (the maximum horizontal distance from the origin) over a horizontal field?
6. The position vector of a particle is given by $\bar{r} = (2 \sin 2t)\hat{x} + (2 \cos 2t)\hat{y} + (2t)\hat{z}$ metres in which the time, t , is measured in seconds. Calculate the distance traversed by the particle in the following time intervals: (a) From $t = 0$ to $t = 5 \text{ s}$. (b) From $t = 2 \text{ s}$ to $t = 12 \text{ s}$.
7. Given: A vector field $\bar{V} = (x^2 - y^2)\hat{x} - (2xy)\hat{y}$. Calculate $\int \bar{V} \cdot d\bar{r}$ from point (0,0) to point (1,2) along the following paths: (a) $x = t^2$, $y = 2t$, (b) $x = u$, $y = 2u^2$, (c) from (0,0) along a straight line to (2,0) and then along the straight line from (2,0) to (1,2).
8. Consider a cube of side length 1 metre and with vertices at the origin,

$(0,0,1)$, $(0,1,0)$ and $(1,0,0)$. $d\vec{A}$ is an infinitesimal element of area perpendicular to the surfaces of the cube and it points outwards. Write $d\vec{A}$ in terms of the base vectors for each of the six surfaces. (The magnitude is dA in each case.) Calculate the surface integral of the vector field $\vec{V} = x^2\hat{x} + y^2\hat{y} + z^2\hat{z}$ over the entire surface of the cube.

9. An infinitely long thin straight wire has a homogeneous linear charge density of λ coulomb per metre. Assume that the intensity of the electric field, $\vec{E}$, which it causes at a distance r from the wire, is radial and has cylindrical symmetry. It means that the magnitude of the field is the same at all positions on the curved surface of a cylinder which is coaxial with the wire and the direction is perpendicular to it. The field may thus be written in the form $\vec{E} = E(r)\hat{r}$.

The information supplied above, follows from Coulomb's law and a study of the symmetry. Use a surface integral to calculate the unknown intensity of the electric field, $\vec{E}$, at distance r from the wire. Describe a cylinder with radius r metres and length L metres coaxial with the wire. Calculate the surface integral of $\vec{E}$ over the entire cylinder. First calculate it over the curved surface (where E is unknown but constant over the entire curved surface) and then over the two circular end surfaces (where E is also unknown but parallel to the surface). Add these values for the final answer.

According to the Gauss law in electrostatics, this surface integral is equal to $4\pi k \times$ (the electric charge enclosed in the cylinder). Now show that $\vec{E} = (2k\lambda/r)\hat{r}$ at the curved surface of the cylinder. k is a constant which appears in Coulomb's law and $\hat{r}$ is a unit vector perpendicular to the curved surface and which points outwards.

Chapter 7

DIFFERENTIAL EQUATIONS

7.1 Introduction

The setting up and solution of differential equations form a most important part of the mathematical skills which a physicist needs to solve problems. To master the complete contents of the subject requires a relatively long and difficult study. Among others the student is required to be acquainted with a large number of standard forms and special techniques for their solution.

It is not an aim of this book to present a formal course in differential equations. The intention is to alter the reader's approach to existing knowledge. This alteration has as aim the development of a more elegant and streamlined way of thinking in the solution of problems.

In this chapter the reader will be introduced to a few differential equations which often occur in the study of introductory physics and related subjects. The differential equations which describe periodic phenomena, exponential decay and exponential growth, are applicable to a vast number of physical phenomena and some of these will be used as examples in the study of the subject.

7.1.1 Differential equations and their solutions

Consider the following problem in kinematics: The motion of a particle is limited to the x -axis of a Cartesian frame of reference. The acceleration of the particle is $\bar{a} = 6\hat{x} \text{ m s}^{-2}$, its initial velocity, $\bar{v}(0) = 7\hat{x} \text{ m s}^{-1}$ and the initial position, $\bar{r}(0) = -5\hat{x} \text{ m}$. In order to be able to answer questions about the motion, it is necessary, as in most kinematics problems, to calculate the position as a function of time.

Since the problem is one-dimensional, the vector notation may be omitted in order that the relevant mathematical procedures may be stressed. From the definitions of acceleration, velocity and position, and the knowledge acquired in chapter 6, the position of the particle may be calculated as follows:

$$v = \int a \, dt = \int 6 \, dt = 6t + 7 \quad \text{because } v(0) = 7$$

$$x = \int v \, dt = \int (6t + 7) \, dt = 3t^2 + 7t - 5 \quad \text{because } x(0) = -5$$

The solution of the problem will be repeated but this time in a different manner.

$$\text{Given:} \quad a = \frac{dv}{dt} = \frac{d}{dt} \left(\frac{dx}{dt} \right) = \frac{d^2x}{dt^2} = 6$$

The equation $d^2x/dt^2 = 6$ is known as a **differential equation** for the simple reason that it contains a derivative. This specific differential equation is called a **second-order differential equation** because the *highest order* derivative which it contains, is a second-order derivative. If it contained a first-order derivative as well, it would still be a second-order differential equation.

The problem is the calculation of the function $x = x(t)$ which has the property that its second-order derivative is 6. This function must also conform to two requirements which are known as **initial conditions**, **boundary conditions**, **accessory conditions** or **constraints**. In this problem they are $x(0) = -5$, and $(dx/dt)_{t=0} = 7$. As a mathematical problem it is set in the following way: Solve the differential equation $d^2x/dt^2 = 6$ for the boundary conditions (or initial conditions) $x(0) = -5$ and $(dx/dt)_{t=0} = 7$. The solution is the function $x = x(t)$.

This specific differential equation may be solved by simply integrating twice and using the boundary conditions to calculate the integration constants.

$$\frac{d^2x}{dt^2} = 6 \quad \Rightarrow \quad \frac{dx}{dt} = \int \left(\frac{d^2x}{dt^2} \right) dt = \int 6 \, dt = 6t + 7$$

$$\text{and} \quad x = \int \left(\frac{dx}{dt} \right) dt = \int (6t + 7) dt = 3t^2 + 7t - 5$$

The fact that this differential equation could be solved by two successive integrations is due to its fairly simple properties. On the left is a derivative and on the right a constant or a function of the independent variable. Any order differential equation which has this property can be solved if enough boundary conditions are known for the calculation of the integration constants.

7.1.2 Summary of the definitions

In the preceding problem the following new concepts were introduced:

$d^2x/dt^2 = 6$	is called	a differential equation
$x(0) = -5$ & $(dx/dt)_{t=0} = 7$	are called	initial conditions, boundary conditions, accessory conditions or constraints
$x = x(t) = 3t^2 + 7t - 5$	is called	the solution

7.2 The differential equation for periodic phenomena

7.2.1 The differential equation and its general solutions

A **periodic phenomenon** is one which is repeated in the same manner as time progresses. Examples: The motion of the balance wheel of a watch, the motion of a pendulum, the vertical motion of a mass hanging from the end of a helical spring, the tides at sea, the rotation of a wheel, etc.

In the study of these phenomena a differential equation of the following form occurs frequently:

$$\frac{d^2x}{dt^2} = -\omega^2 x \quad 7.2(1)$$

in which ω^2 is a constant. At this stage it might seem strange that one would prefer to use ω -squared to indicate a constant. The advantages of this choice will be evident when the solutions of the equation are studied and a physical interpretation of ω is given.

As was the case with the differential equation in section 7.1.1, we may attempt to solve this one by integration. As before

$$d^2x/dt^2 = -\omega^2x \quad \text{so that} \quad dx/dt = \int (d^2x/dt^2)dt = \int (-\omega^2x)dt$$

The last term shows that this will be impossible since the solution, $x = x(t)$, will have to be known before integration can be attempted. (The answer is needed to calculate the answer!) The only method remaining is to make use of knowledge about functions which have the property that if they are differentiated twice, the answer will be the original function multiplied by $-\omega^2$.

From experience with trigonometric and exponential functions, it is known that

$$\begin{aligned} (d^2/dt^2)(\sin \omega t) &= (d/dt)(\omega \cos \omega t) = -\omega^2 \sin \omega t \\ (d^2/dt^2)(\cos \omega t) &= (d/dt)(-\omega \sin \omega t) = -\omega^2 \cos \omega t \\ (d^2/dt^2)(e^{i\omega t}) &= (d/dt)(i\omega e^{i\omega t}) = i^2\omega^2 e^{i\omega t} = -\omega^2 e^{i\omega t} \end{aligned}$$

In the last function, the **imaginary quantity** $i = \sqrt{-1}$ which has the property $i^2 = -1$, was used.

By simply differentiating twice in each case, the reader can show that each of the following functions are solutions of the differential equation given in Equation 7.2(1):

$$x = A \sin(\omega t + \alpha) \quad A \text{ and } \alpha \text{ constant} \quad 7.2(2)$$

$$x = B \cos(\omega t + \beta) \quad B \text{ and } \beta \text{ constant} \quad 7.2(3)$$

$$x = A \sin(\omega t + \alpha) + B \cos(\omega t + \beta) \quad A, B, \alpha \text{ and } \beta \text{ constant} \quad 7.2(4)$$

$$x = Ae^{\pm i(\omega t + \phi)} \quad A \text{ and } \phi \text{ constant} \quad 7.2(5)$$

The quantities A , B , α , β and ϕ in these solutions are equivalent to integration constants and each has to be calculated from the given boundary conditions.

Comments:

1. The solutions of this differential equation are all **oscillatory** with a **period** of $T = 2\pi/\omega$. This means that the function value at any value of t will be the same if t increases by an amount of $\Delta t = T = 2\pi/\omega$. During this period, x is said to complete **one cycle**. The relationship between T and ω is of prime importance.

$$T = 2\pi/\omega \quad \text{or} \quad \omega = 2\pi/T \quad 7.2(6)$$

2. From the second equation it can be seen that ω has the same dimensions as angular velocity, i.e. radians per second. This fact is of importance in the physical interpretation of ω .
3. The number of cycles completed per second, is known as the **frequency**. Frequency is measured in **cycles per second** which, in short, is called **hertz (Hz)**. Frequency is usually indicated by either f or the Greek letter ν . In this book, ν will be used. It should be clear that the period and the frequency are reciprocals of each other.

$$T = 1/\nu = 2\pi/\omega \quad \text{or} \quad \nu = 1/T = \omega/2\pi \quad 7.2(7)$$

The quantity ω is simply the frequency multiplied by 2π .

$$\omega = 2\pi\nu \quad 7.2(8)$$

For this reason ω is called the **angular frequency**.

4. The constants A and B in the solutions shown in Equations 7.2(2), 7.2(3) and 7.2(5), are known as the **amplitude** of the functions and each constant is the maximum value of $|x|$. The differential equation 7.2(1) contains no information about the value of the amplitude. For its calculation suitable boundary conditions are required.
5. The quantity which appears between brackets in all the solutions, e.g. $(\omega t + \phi)$, is called the **phase angle** or, in short, **phase**. The constant portion of the phase, e.g. ϕ , is known as the **phase constant** or **initial phase**. The phase constant is the value of the phase angle when $t = 0$. As is the case with the amplitude, the differential equation contains no information about the phase constant and a suitable boundary condition is necessary to calculate it. The specification of $x(0)$ and the initial value of dx/dt are sufficient to calculate both the amplitude and the phase constant.

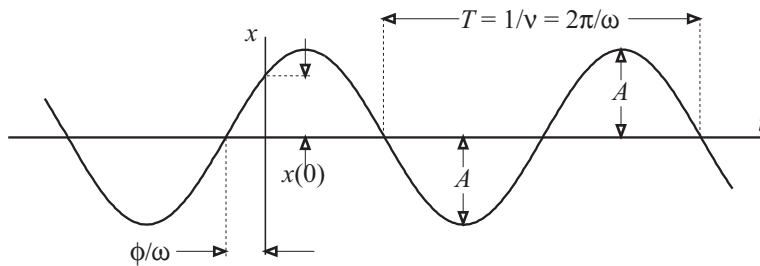


Figure 7.2-1

6. Figure 7.2-1 shows a graphical representation of the solution $x = A \sin(\omega t + \phi)$. In this figure, most of the quantities which are defined in Comments (1) to (4) are indicated. Note that $x(0) = A \sin \phi$ and that the same graph may just as well have been described by means of a cosine function with a different phase constant.

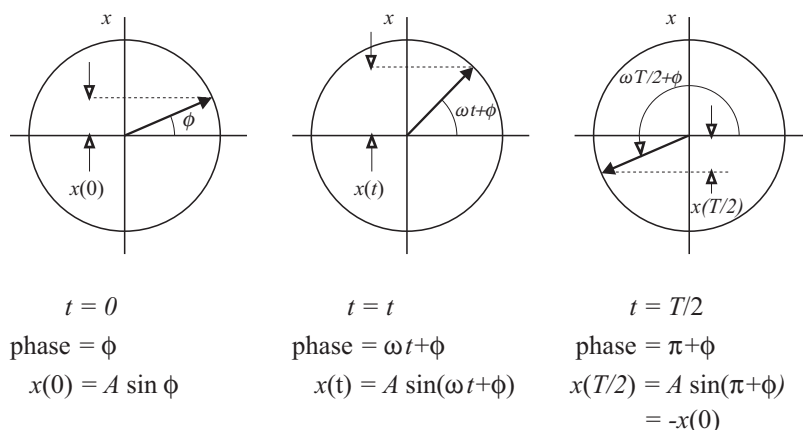


Figure 7.2-2

7. The use of a **phasor diagram** provides a simple graphical way to generate numerical values of the oscillatory solution. A **phasor** is a vector of which the length is equal to the amplitude of the function (A in Figure 7.2-1) and with its initial point at the zero point of a scale along which x is indicated. At time $t = 0$, the phasor makes an angle which is equal to the phase constant (ϕ in Figure 7.2-2) with the direction perpendicular to the axis along which the function is plotted. The phasor rotates at a constant angular velocity ω in a positive sense (counter-clockwise). The projection of the phasor on the function-axis gives the numerical value of x at any instant t . It is said that the circular motion of the phasor and the oscillation of the function are **in harmony**. For each revolution completed by the phasor, the function completes one cycle. The physical interpretation of ω should be clear. It is the angular velocity of that phasor which is in harmony with the function.

Figure 7.2-2 shows a phasor at time $t = 0$, at time t seconds later and again at time $t = T/2 = \pi/\omega$.

The technique of phasor diagrams may be used for the composition (“addition”) of two or more oscillatory phenomena of the same kind. In the study of physical optics, waves in general, and especially the theory of alternating electric currents and voltages, the use of phasors simplifies the calculations to a great extent. Phasors also form the base of the descrip-

tion of alternating electrical currents by means of complex numbers, i.e. numbers which contain real and imaginary portions.

7.2.2 The construction of solutions for the differential equation $d^2x/dt^2 = -\omega^2x$

By means of an example, it will be shown how solutions for this differential equation may be constructed for a variety of boundary conditions. In this example the quantity x which is dependent on the time, t , will be considered. To keep it general, no units will be assigned to x . It should, however, be kept in mind that x could represent a position of which the units could be metres, or a velocity, an acceleration, the intensity of an electric field, an electric potential difference, or an angle, etc., in each case with the appropriate units.

Example A physical quantity, x , depends on the time, t . The magnitude of x is determined by means of the following differential equation:

$$d^2x/dt^2 = -16\pi^2x$$

in which the time is measured in seconds. The amplitude of the solution is 100 units. (a) Calculate the period and the frequency of the solution. (b) Construct solutions for the following initial values: (i) $x(0) = 100$, (ii) $x(0) = -100$, (iii) $x(0) = 0$ and $dx/dt > 0$, i.e. x is increasing at the instant when $x = 0$, (iv) $x(0) = 0$ and $dx/dt < 0$, i.e. x is decreasing when $x = 0$, (v) $x = 50$ and $dx/dt > 0$, i.e. x is increasing when $x = 50$, (vi) $x(0) = 50$ and $dx/dt < 0$, i.e. x is decreasing when $x = 50$.

(a) It is not necessary to construct a solution for the differential equation to answer this portion of the question. From the differential equation it follows directly that

$$\begin{aligned} \omega^2 = 16\pi^2 \quad \text{from which follows:} \quad \omega &= 4\pi \text{ rad s}^{-1} \\ \text{so that the period is given by} \quad T &= 2\pi/\omega = 0,5 \text{ seconds} \\ \text{and the frequency by} \quad \nu &= 1/T = 2,0 \text{ hertz} \end{aligned}$$

This calculation emphasises the fact that the period and the frequency have no connection with the amplitude and that they are not influenced by the boundary conditions which are used for the construction of special solutions.

(b) From the discussion of the theory it is known that the solutions are oscillatory, i.e. the phenomenon is repetitive in a way shown in Figure 7.2-1. An observer who wishes to describe this phenomenon as a function of time, will have to start a stop-watch in order to define the instant $t = 0$, i.e. the zero-point on the time scale. The construction of a solution which is in accordance

with given boundary conditions (initial conditions) is an exercise to allow for the possibility that an observer may choose $t = 0$ at any stage of a cycle. Different functions which correspond to different initial conditions, describe the same physical phenomenon in spite of differences in the mathematical expressions.

In order to construct a solution for the given differential equation, any of the functions 7.2(2) to 7.2(5) may be used. We simply *choose* the following solution:

$$x = A \sin(\omega t + \phi)$$

which will be adapted to the given differential equation and in each case to the given initial conditions.

The value of ω follows from the differential equation as shown in the answer to question (a). $\omega = 4\pi \text{ rad s}^{-1}$. The differential equation does not provide any information about the amplitude, A . In the absence of supplementary information, its value is not known. In this problem it is known that the amplitude is 100 units. Using this information, it may be written that

$$x = 100 \sin(4\pi t + \phi) \quad \dots\dots(1)$$

From this the value of $x(0)$ may be calculated in terms of ϕ .

$$x(0) = 100 \sin(4\pi \times 0 + \phi) = 100 \sin \phi \quad \dots\dots(2)$$

From the solution it can also be seen that

$$\begin{aligned} dx/dt &= 4\pi \times 100 \cos(4\pi t + \phi) \\ \text{so that } (dx/dt)_{t=0} &= 400\pi \cos \phi \quad \dots\dots(3) \end{aligned}$$

In each of the six examples which follow, the observer starts his stop-watch at a different stage of the cycle. It will be shown how the value of ϕ is calculated in each case.

(i) $x(0) = 100$. This means that the stop-watch is started when $x = 100$. This value is equated to the value of $x(0)$ expressed in terms of ϕ , i.e. the value in Equation(2) above.

$$\begin{aligned} 100 &= 100 \sin \phi && \text{from which follows} && \sin \phi = 1 \\ \text{so that } \phi &= \pi/2 \pm n \times 2\pi && && n = 0, 1, 2, 3, \dots \end{aligned}$$

Any of the above values for ϕ would be an acceptable phase constant. *It is, however, customary (but not necessary) to assign the smallest possible value to ϕ when the problem involves a single oscillatory phenomenon.* In accordance

with this custom (which is introduced to avoid the unnecessary use of large numbers) the value $\phi = \pi/2$ is accepted. By substituting this value in Equation (1), the required solution is complete.

$$x = x(t) = 100 \sin(4\pi t + \pi/2) = 100 \cos 4\pi t$$

If the choice was made at the beginning rather to use a cosine function, the appropriate phase constant would have been equal to zero and the solution exactly the same. A graph of the solution is shown in Figure 7.2-3(a).

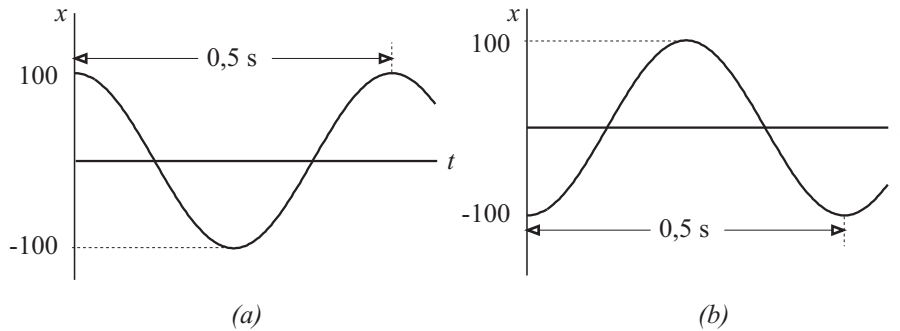


Figure 7.2-3

(ii) $x(0) = -100$. This means that the observer starts the stop-watch when $x = -100$. If this value is substituted in Equation (2), the phase constant may be calculated in exactly the same way as in the problem (i). The complete solution is as follows:

$$x = x(t) = 100 \sin(4\pi t + 3\pi/2) = 100 \cos(4\pi t + \pi)$$

The graph representing this solution is shown in Figure 7.2-3(b).

(iii) $x(0) = 0$ and $dx/dt > 0$. A function oscillating about $x = 0$ can be zero in two different ways – while it is increasing and while it is decreasing. In Figure 7.2-1 these two possibilities correspond to positions where the function intersects the t -axis with a positive and a negative gradient, respectively. In the previous two solutions it was not necessary to specify dx/dt since the specified initial values of x can be either a maximum (100) or a minimum (-100) in a unique way.

$$\begin{aligned} \text{As before:} \quad x(0) &= 0 = 100 \sin \phi \\ \text{so that} \quad \sin \phi &= 0 \\ \text{and} \quad \phi &= \pm n\pi \quad n = 0, 1, 2, 3, \dots \end{aligned}$$

The requirement that $dx/dt \geq 0$ excludes some of the values of n which are mentioned above. Only the even values of n , give positive values of dx/dt and

therefore the odd values are excluded. ($dx/dt = 400\pi \cos(\text{even}n \times \pi) > 0$, and $dx/dt = 400\pi \cos(\text{odd}n \times \pi) < 0$.)

If, as before, the smallest value of ϕ is chosen, the complete solution is as follows:

$$x = x(t) = 100 \sin 4\pi t$$

A graphical representation of this solution is shown in Figure 7.2-4(a). This solution is one of the simplest that can be used for a periodic phenomenon. Excluding cases in which the boundary conditions are of such a nature that the phase constant cannot be zero, one simply chooses the initial conditions to give $\phi = 0$. In practice the zero-point on the time scale is chosen when the function is equal to zero while it is increasing.

(iv) $x(0) = 0$ with $dx/dt < 0$. In this case the observer starts the stop-watch when $x = 0$ while x is decreasing. By using the same argument as in example (iii), it follows that the solution is the same but this time with the exclusion of the even values of n . Choosing the smallest possible value of ϕ , the complete solution is as follows:

$$\begin{aligned} x = x(t) &= 100 \sin(4\pi t + \pi) \\ &= -100 \sin(4\pi t) \\ &= 100 \cos(4\pi t + \pi/2) \end{aligned}$$

The graph which represents this function, is the mirror image about the t -axis of that shown in Figure 7.2-4(a).

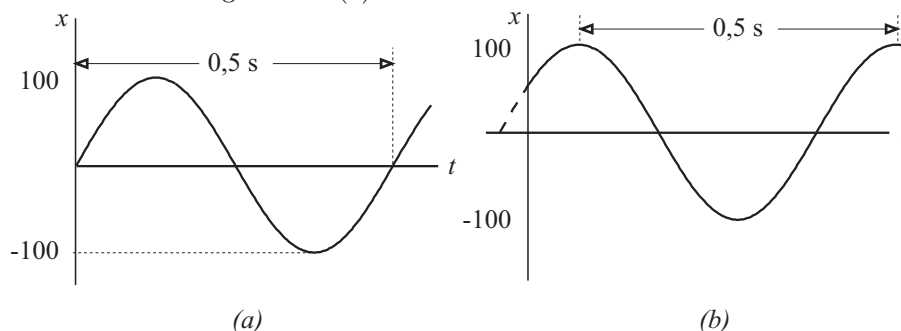


Figure 7.2-4

(v) $x(0) = 50$ with $dx/dt > 0$. This means that the observer starts the stop-watch when $x = 50$ and increasing.

$$\begin{aligned} \text{As before:} \quad x(0) &= 50 = 100 \sin \phi \\ \text{so that} \quad \sin \phi &= 0,5 \\ \text{and} \quad \phi &= (-1)^n(\pi/6) \pm n\pi \quad n = 0, 1, 2, 3, \dots \end{aligned}$$

On condition that dx/dt is positive, only even values of n are admissible. The simplest complete solution is as follows:

$$x = x(t) = 100 \sin(4\pi t + \pi/6)$$

The observer starts the stop-watch when the phase is equal to $\pi/6$. That means that one-twelfth of a period elapses after the function is equal to zero and increasing.

A graph which represents this solution is shown in Figure 7.2-4(b)

(vi) $x(0) = 50$ with $dx/dt \leq 0$. The procedure is the same as that in the previous problem. The answer is the same in all respects but for the fact that only odd values of n are admissible. With the smallest positive value of ϕ , the solution is as follows:

$$x = x(t) = 100 \sin(4\pi t + 5\pi/6)$$

The observer sets the stop-watch in motion at the instant when the phase angle is equal to $5\pi/6$. When t is chosen as zero, five twelfths of a period has elapsed since $x = 0$ and decreasing.

A graph which represents this solution, is shown in Figure 7.2-5.

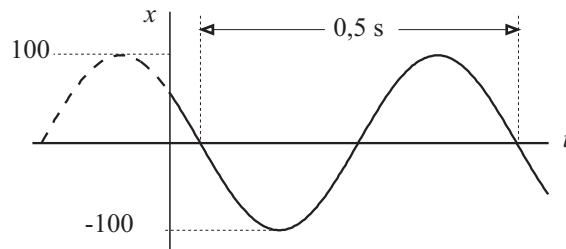


Figure 7.2-5

7.3 Exponential decay

7.3.1 The differential equation and its solution

In this study we encounter a function $y = y(x)$ whose behaviour may be described by the following differential equation:

$$\frac{dy}{dx} = -ky \quad 7.3(1)$$

in which k is a positive constant. With the differential equation as the only guide-line the behaviour of the function may be analysed as follows:

1. If y is positive, it will *decrease* as x increases. This property is caused by the minus sign which appears in the differential equation. (The gradient is always negative for positive values of y .)
2. If y is large and positive, the *rate* at which it decreases with increasing x , will be large. As the value of y decreases, the rate of further decrease will become less.
3. The gradient of the graph of the function is directly proportional to the value of y .

Since the differential equation is of the first order, only one boundary condition will be needed to calculate the the integration constant. The initial value for this differential equation is usually of the form $y(0) = Y$ in which Y is a constant number.

The differential equation has a property which makes its solution a simple process. Since it is a first-order differential equation, it is **separable** which means that it may be rewritten in such a way that the dependent variable (y in this case) will appear on one side only of the equals sign, and the independent variable (x in this case) on the other side only. Once this separation has been accomplished, the solution is quite simple. The integration constants will be calculated in two ways.

With the variables separated, the equation is as follows:

$$\frac{dy}{y} = -k \, dx \quad 7.3(2)$$

The solution follows by integrating both sides of the equation above.

First method:

$$\begin{aligned} \int \frac{dy}{y} &= \int y^{-1} dy &= \int -k \, dx \\ \text{from which follows} \quad \ln y + c_1 &= -kx + c_2 \end{aligned}$$

in which c_1 and c_2 are two unknown integration constants. Since both c_1 and c_2 are unknown, c_2 may be subtracted from both sides of the equation and the difference of the two, $c_1 - c_2$, equated to a single constant, c as follows:

$$\ln y + c = -kx \quad 7.3(3)$$

The boundary condition, $y(0) = Y$, is known and it may be used to calculate the value of c . Substitute $x = 0$ and $y = Y$ in Equation 7.3(3).

$$\begin{aligned} \ln Y + c &= 0 \\ \text{so that } c &= -\ln Y \end{aligned}$$

Substitute this value for c in Equation 7.3(3). From this follows:

$$\begin{aligned} \ln y - \ln Y &= -kx \\ \text{or } \ln y/Y &= -kx \\ \text{so that } y/Y &= e^{-kx} \\ \text{i.e. } y &= Y e^{-kx} = Y \exp(-kx) \end{aligned} \quad 7.3(4)$$

which is the required solution. The second notation, $\exp(-kx)$, is used when the exponent is an unwieldy expression which cannot easily be written in the conventional way.

Second method:

In section 3.7 it was shown that a definite integral may be used to calculate the integration constants in an elegant way. This will now be used as an alternative (and preferred) method to calculate the solution. The values $y = Y$ and $x = 0$ are used as the lower limits of the definite integrals respectively, and y and x as their upper limits as follows:

$$\begin{aligned} \int_Y^y y^{-1} dy &= \int_0^x -k dx \\ \ln y|_Y^y &= -kx|_0^x \\ \ln y - \ln Y &= -kx \\ \ln(y/Y) &= -kx \\ \Rightarrow y &= Y e^{-kx} = Y \exp(-kx) \end{aligned}$$

7.3.2 Properties of the function $y = Y e^{-kx}$

When a solution to a differential equation is calculated, it is of the utmost importance to determine whether it satisfies the differential equation and the boundary conditions. The reader should become accustomed to performing

these tests as a matter of routine. The first test is to determine if the calculated function satisfies the differential equation.

$$\text{Test 1: } dy/dx = (d/dx)(Ye^{-kx}) = -kYe^{-kx} = -ky$$

The function satisfies the differential equation.

The second test is to determine whether the function satisfies the boundary condition.

$$\text{Test 2: } y(0) = Ye^{-k \times 0} = Ye^0 = Y \times 1 = Y$$

The function also satisfies the boundary condition.

To investigate further properties of the function, it is useful to rewrite it as follows:

$$y = y(x) = Ye^{-kx} = Y/e^{kx} \quad 7.3(5)$$

If only values of x which are not negative, are investigated, the minimum value of the factor e^{kx} is unity which corresponds to the value $x = 0$. The corresponding value of y is Y . As x increases, the denominator of the fraction increases and the numerical value of y decreases. Figure 7.3-1 shows graphs of this function for different values of k .

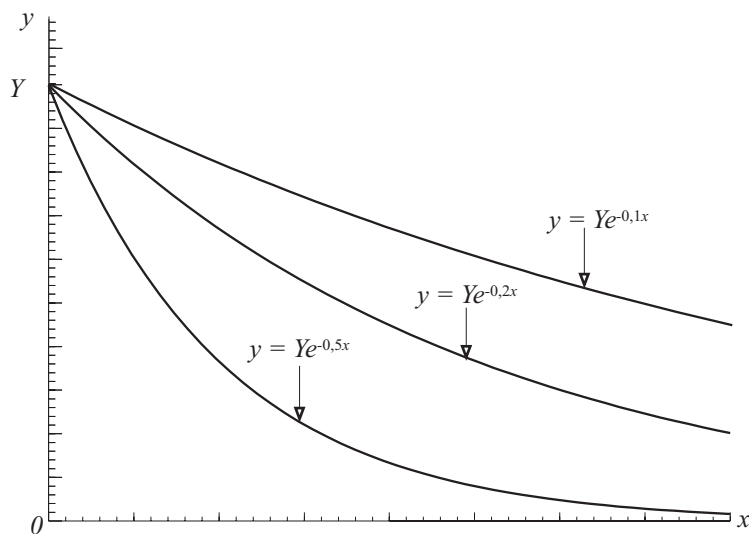


Figure 7.3-1

From Figure 7.3-1 it can be seen that a large value of k , which is called the **decay constant**, causes a greater rate of decrease than one which is less. All the functions in the graph begin at $y = Y$ where $x = 0$ and Equation 7.3(5)

shows that each function tends to zero when x tends to infinity. This is true for all values of k , and therefore this information is not of much value.

More useful information is gained if a graph of one of these functions is inspected in greater detail. The graph in Figure 7.3-2 shows that an interval Δx exists for which the value of y will be halved. A good term for this interval would be the **halving interval**, but this term is not used by English-speaking physicists. If x is time, the term which is used, is **half-life** and if x is a distance, it is called a **half-thickness**, **half-value thickness** or **half-layer value**. Some of these terms are misnomers and could lead to confusion. (The half-life of a collection of radioactive atomic nuclei of a given kind, is not half of the life of the collection but the time in which one half of the total number will decay.)

This interval is usually indicated by $\Delta x = x_h$ (some prefer $x_{\frac{1}{2}}$). If $x = 0$, then $y = Y$ and if $x = x_h$, $y = Y/2$. Half-life, which is an interval of time, is usually indicated by t_h , $t_{\frac{1}{2}}$ or τ .

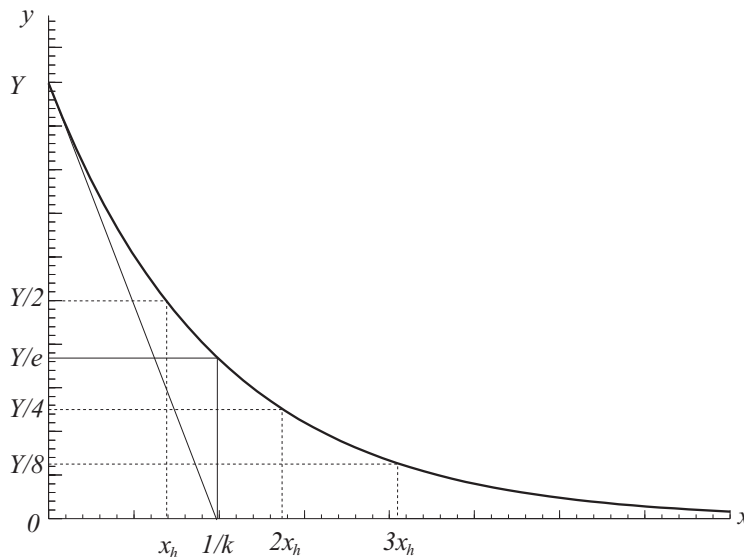


Figure 7.3-2

A processes which is described by this function and which exhibits these interesting properties, is known as **exponential decay**. The halving interval is a very useful concept since it allows one to calculate the progress of such a process by simple arithmetic. In Table 7.3-1, y is given as a percentage of its initial value, Y , after intervals which are multiples of the halving interval.

The halving interval is calculated by substituting $y = Y/2$ and $x = x_h$ in

Halving intervals	y as % of Y
0	100,000
1	50,000
2	25,000
3	12,500
4	6,250
5	3,125
6	1,563
7	0,781
8	0,391

Table 7.3-1

Intervals of $1/k$	y as % of Y
0	100,000
1	36,788
2	13,533
3	4,979
4	1,832
5	0,674
6	0,248
7	0,091
8	0,034

Table 7.3-2

Equation 7.3(5) as follows:

$$\begin{aligned} Y/2 &= Y/e^{kx_h} \\ \text{so that } e^{kx_h} &= 2 \end{aligned}$$

If logarithms to the base e are taken on both sides, it follows that

$$\begin{aligned} kx_h(\ln e) &= kx_h \times 1 = kx_h = \ln 2 \\ \text{so that } x_h &= (\ln 2)/k = 0,6931/k \end{aligned} \quad 7.3(6)$$

Some prefer not to use the halving interval and rather work with the interval of $1/k$ which differs from the former by a factor of $\ln 2 = 0,6931$. If x measures time, the factor $1/k$ is called the **time constant**. After an interval of $1/k$ as measured from $x = 0$, y will decrease to Y/e and after two such intervals to Y/e^2 , etc. In Table 7.3-2, y is shown as a percentage of its initial value, Y , after intervals of $1/k$. It is of interest to note that if the function continues decreasing at its *initial rate*, it would be zero after an interval of $\Delta x = 1/k$. See the graph in Figure 7.3-2.

7.3.3 The use of logarithmic graph paper

Consider the function $y = 5e^{-0,75x}$. Two different graphical representations will be made of this function on graph paper of which both scales are **linear**. In this sense, linear means that the quantity represented on the scale, is directly proportional to the length along the scale as measured from a suitable zero-point. Linear scales are usually subdivided into intervals of ten equal units. The groups of ten units are called **decades**.

First y will be represented as a function of x and then $\ln y$ as a function of x . In order to do that, it is necessary to calculate the following pairs of co-ordinates.

x	0	1	2	3	4	5	6	7
y	5,00	2,36	1,12	0,53	0,25	0,12	0,06	0,03
$\ln y$	1,61	0,86	0,11	-0,64	-0,139	-2,14	-2,89	-3,64

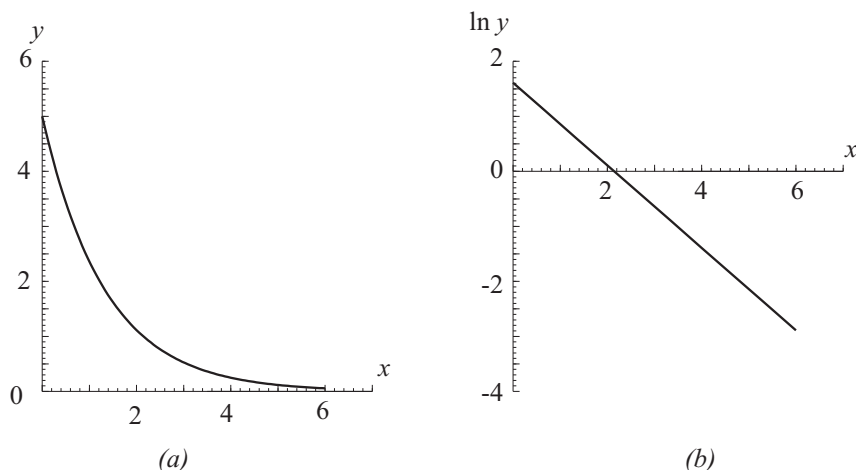


Figure 7.3-3

While the graph in Figure 7.3-3(a) might seem the most acceptable for the reason that it gives the relationship between x and y very simply and directly, it has two disadvantages: (i) If only a portion of the graph is available, **graphical extrapolation** is difficult. Graphical extrapolation is the extension of the graph in a region beyond that in which pairs of co-ordinates are known. The existing shape of the known portion is the only clue for its extension. (ii) The poor sensitivity of the y -scale for small values of y .

Although a plot of the natural logarithm of y against x eliminates these disadvantages, it introduces a new one. One needs a calculator to find the value of y when x is given or to find the value of x if y is specified.

In order to eliminate the disadvantages, **logarithmic graph paper** may be used. On a logarithmic scale the intervals representing different numbers (as measured from $\log 1$ or $\ln 1$) are directly proportional to the *logarithms* of the numbers involved. From this it follows that a logarithmic scale cannot have a zero point. (The logarithm of $0 \rightarrow -\infty$.) A logarithmic scale is subdivided into **cycles**, each of which represents a factor ten as shown in Figure 7.3-4

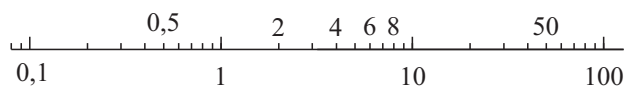


Figure 7.3-4

The number of cycles which will be used for a given graph, depends on the range in which the function is to be studied. A study of Figure 7.3-3(b) shows that the independent variable is still plotted against a linear scale. Graph paper which combines a linear and a logarithmic scale, is known as **semi-logarithmic paper**, **linear-logarithmic paper** or **ratio paper**. Graph paper on which both scales are logarithmic, is also available and is known as **double logarithmic paper** or **log-log paper**. The latter kind is not of importance to this study and is used where the graphs of power functions are studied or where the graph of a function is to be read off quite accurately over many orders of magnitude.

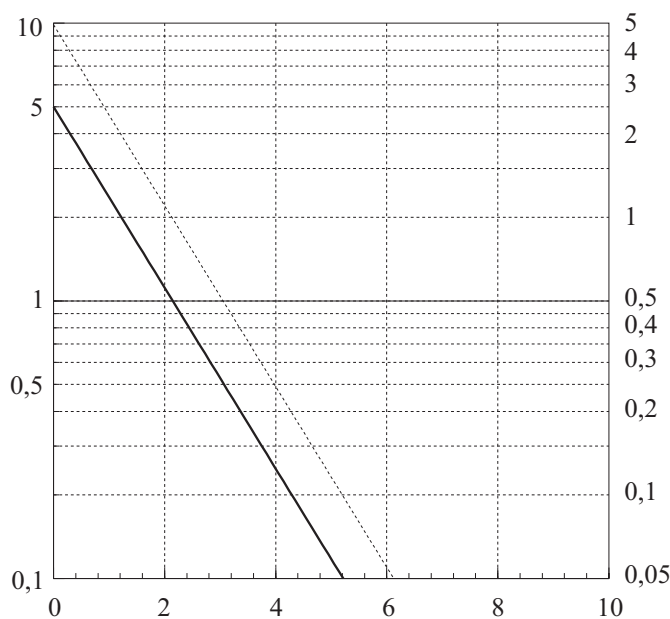


Figure 7.3-5

On linear-logarithmic paper the graph of $y = 5e^{-0,75x}$ has the same shape as that in Figure 7.3-3(b) but the scale on the ordinate-axis is different. In Figure

7.3-3(b) the graph of $\ln y$ was plotted as a function of x on double linear graph paper. In Figure 7.3-5 the graph of y as a function of x is plotted on log-linear paper. The solid line corresponds to the scale on the left-hand side. If the scale on the right-hand side is used, the graph will be that shown as a dotted line parallel to the first.

The use of linear-log graph paper is of great help when routine recordings of exponential-decay processes are studied. If it is known that this is the nature of the process, three or four points only are needed and the best straight line is fitted on linear-log paper. (Question: Why are *three* points the absolute minimum required even though the graph is a straight line?) The halving interval can be read off directly from the graph and, if necessary, the decay constant may be calculated from Equation 7.3(6). It is suggested that the reader try to read the halving interval from the graph in Figure 7.3-5. (Answer: Approximately 0.9)

7.3.4 Problems on exponential decay

(1) **Radioactive decay:** The radioactive decay of atomic nuclei is a discrete process. This means that an individual nucleus has either decayed or not and the number of nuclei in question is always an integer number. The behaviour of a large number of such nuclei may be described very well by the exponential-decay function, but it should be kept in mind that this description is a mathematical model and that uncertainty always exists regarding the exact number of radioactive nuclei in a given sample of material at any instant after the measurement has commenced, even if the number at $t = 0$ is known exactly.

A knowledge of statistics enables one to describe the limits of the uncertainty in terms of a number which is called the standard deviation. If N_0 radioactive nuclei of a certain kind exist in a given sample of material at instant $t = 0$, the number will be less at a later instant. If the number at the later instant is N , it should be indicated as $N \pm \sqrt{N_0 - N}$. This specific standard deviation (the number after the $\pm$ -sign) is characteristic of a so-called **Poisson distribution**. The meaning of the terms *standard deviation* and *Poisson distribution* is explained in textbooks on elementary mathematical statistics.

The principle which governs the decay of radioactive atomic nuclei of a given kind, is

$$dN/dt = -\lambda N \quad 7.3(7)$$

in which $N = N(t)$ is the number of nuclei which still exist at instant t . The decay constant, λ may be described as the probability per unit time that one of the nuclei of this kind will decay. The decay constant is characteristic of a given kind of radioactive nuclei and it differs vastly from one kind to the next.

In words it may be put as follows: The rate at which the radioactive nuclei decay at a given instant, dN/dt , is directly proportional to the number which still exist at that instant, namely N . The principle may be rewritten in the following form: $dN/N = -\lambda dt$ and the interpretation is as follows: In equal infinitesimal time intervals, the same fraction of nuclei decays.

Consider the following numerical example: The rate at which radioactive nuclei of a given kind decay, is given by $dN/dt = -0,05N$, in which the time, t is measured in seconds. At instant $t = 0$, 10^6 of these nuclei exist in a given sample of material. Calculate the following: (a) $N = N(t)$. (b) The half-life (halving time) of the decay. (c) At which time will $6,25 \times 10^4$ radioactive nuclei still exist? (d) Calculate the number of nuclei and the standard deviation at time $t = 30$ seconds. (e) The instant at which the sample may be expected to be free of radioactivity.

(a) As before the variables are separated and the equation integrated.

$$\int_{10^6}^N N^{-1} dN = \int_0^t -0,05 dt$$

from which follows $N = 10^6 e^{-0,05t}$

(b) As before: $\tau = \ln 2 / 0,05 = 13,86$ seconds.

$$\begin{aligned} \text{(c)} \quad 6,25 \times 10^4 &= 10^6 e^{-0,05t} \\ \Rightarrow e^{-0,05t} &= 1/16 = 2^{-4} \end{aligned}$$

from which follows $0,05t = 4 \ln 2$ so that $t = 55,452$ s (4 half-life periods)

$$\begin{aligned} \text{(d)} \quad N(30) &= 10^6 \exp(-0,05 \times 30) = 2,231 \times 10^5 \\ \text{standard deviation: } \sigma &= \sqrt{N_0 - N} \\ &= (10^6 - 2,231 \times 10^5)^{1/2} = 881 \end{aligned}$$

(e) In the theory it was shown that each and every function which describes exponential decay will be zero only when the time tends to infinity and that suggests that no sample of material which contains radioactive atomic nuclei, will ever be free of radioactivity. This misconception arises from the fact that a continuous function is used to describe a discrete process.

To answer the question when the sample will be free of radioactivity, the instant when N will probably be unity, is calculated. The best answer will be to say that after that time the probability that the sample is free of radioactivity, will increase as the time increases. On the other hand, even if several multiples of ten or hundred of these radioactive nuclei are present, the rate at which they

decay will be so low, that the sample may, for practical purposes, be taken to be free of radioactivity.

$$\begin{aligned} 1 &= 10^6 e^{-0,05t} \\ \Rightarrow 0,05t &= \ln 10^6 = 6 \ln 10 = 13,82 \\ \text{so that } t &= 276,4 \text{ s} \end{aligned}$$

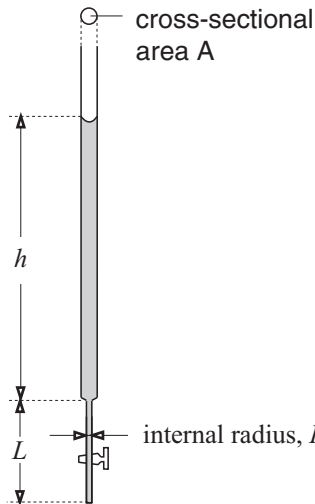
After this time the sample may be considered free of radioactivity.

(2) **The effluence of a liquid.** The term **viscosity** refers to the internal resistance or opposition to flow in a liquid. It is measured by a number which is known as the **coefficient of viscosity** of the liquid (usually indicated by the symbol η). If a constant current of liquid is to be maintained in a tube, a **pressure gradient** (change of pressure per unit length) is required.

The quantity $\mathcal{R} = 2\rho Rv/\eta$, in which ρ is the density of the liquid, R the radius of the cylindrical tube in which it flows and v the average speed at which it flows, is called the **Reynolds number**. If this number is less than about 2000, the flow is **steady, ordered** or **non-turbulent**.

The stream (volume of flow per unit time) which emerges from the end of a cylindrical tube when the flow is steady, is given by **Poiseuille's law** which is as follows:

$$\frac{dV}{dt} = \frac{\pi R^4}{8\eta} \frac{p_2 - p_1}{L}$$



In this expression, $p_2 - p_1$ is the difference in pressure at the two extremities of the tube with length L .

Now this law is applied to a burette which is in a vertical position and which is initially filled with a liquid to a height h_0 when the cock is opened. The height of the liquid as a function of time is to be calculated. The assumption is made that the radius of the tube in the vicinity of the stop-cock is much less than that of the main tube and that steady flow occurs in that portion.

Since the tube diameter in the lower portion of the burette is much less than that of the upper portion, the effect of viscosity may be disregarded in the latter. The rate at which liquid flows from the burette, is determined mainly by what happens in the portion of the tube of which the radius is small. If h is the height of the liquid

in the wide portion of the burette, the pressure at the upper portion of the narrow tube will be equal to (atmospheric pressure + ρgh). Where the liquid leaves the tube, the pressure is equal to atmospheric pressure. The pressure gradient is thus equal to $\rho gh/L$.

The volume of liquid which flows from the tube per unit time is equal to the decrease per unit time of the volume remaining in the tube. Thus

$$\frac{dV}{dt} = -\frac{d}{dt}(Ah) = -A \left(\frac{dh}{dt} \right)$$

in which A is the cross-sectional area of the wider portion of the burette. Poiseuille's law applied to this problem gives the following differential equation:

$$\begin{aligned} \frac{dh}{dt} &= -\frac{\pi R^4}{8\eta A} \frac{\rho gh}{L} = -kh \\ \text{in which } k &= (\pi R^4 \rho g)/(8\eta AL) \end{aligned}$$

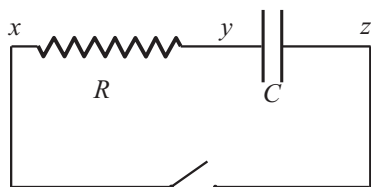
The solution of the differential equation for the given initial condition is

$$h = h_0 e^{-kt}$$

and the halving time of the function is given by

$$\tau = (\ln 2)/k$$

(3) The discharge of a capacitor through a resistor



The circuit in the sketch shows a capacitor with capacitance C farad which is electrically charged and then connected in series with a resistor with resistance R ohm and a switch which is initially open. At time $t = 0$, when the switch is closed, the charge on the capacitor is $q(0) = Q$ coulomb. The charge on the capacitor,

$q = q(t)$ and the current in the resistor, $i = i(t)$, are to be calculated as functions of time.

If plate z of the capacitor initially carries a charge of $+Q$ coulomb, the other one has a charge of $-Q$. When the switch is closed, a positive current exists from x to y in the resistor. Since there are no electrical sources in the circuit, the potential difference between x and z is zero (electrically they are the same point). If i is the current in R , it follows that

$$iR - q/C = 0 \quad \text{or} \quad i = (1/RC)q \quad \dots\dots (1)$$

The initial value of the current, I , may be calculated by substituting the initial value of the charge in Equation (1).

$$i(0) = I = Q/RC$$

If it is kept in mind that q is the charge on the capacitor and that it will decrease after the switch has been closed, the substitution $i = -dq/dt$ can be made in Equation (1) as follows:

$$dq/dt = -(1/RC)q$$

For the given initial condition, the solution of this differential equation is as follows:

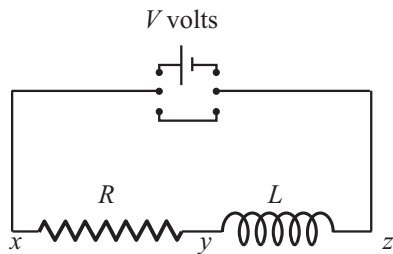
$$q = Qe^{-t/RC}$$

By using the relationship $i = -dq/dt$, the electric current in the conductor may be calculated.

$$i = (Q/RC)e^{-t/RC}$$

Both solutions are exponential decays with time constants equal to RC and halving times (half-life periods) given by $\tau = RC \ln 2$. During the discharge of the capacitor through the resistor, the energy which was initially stored in the electric field of the capacitor, was converted to heat in the resistor. This process is irreversible.

(4) The decay of an electric current in a circuit which contains resistance and inductance



If an electric current I exists in an inductor of which the inductance is L , the energy stored in the magnetic field which is caused by the current, is given by $E = \frac{1}{2}LI^2$. Consider the circuit shown in the sketch. A potential source of V volts maintains a current in the conductor of which the resistance is R ohm and an inductor which is connected in series. In the resistor, electrical energy is irreversibly converted to

heat. At time $t = 0$ a double pole switch is switched so that the potential source is removed and the points x and z short-circuited. The energy stored in the magnetic field will maintain the current until all the energy is dissipated in the resistor. Since no potential source exists in the circuit when $t = 0$, it may be written that

$$L(di/dt) + iR = 0 \quad \text{or} \quad di/dt = -(R/L)i$$

If the initial value of the electrical current is I , the solution of the differential equation is

$$i = Ie^{-(R/L)t}$$

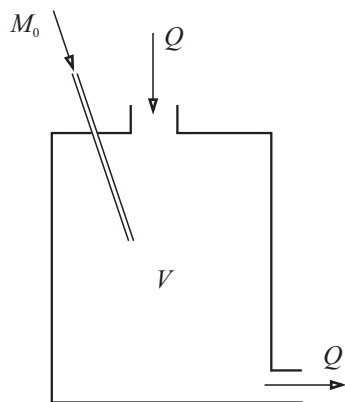
This is an exponential decay with time constant L/R and half-life period given by

$$\tau = (L/R)\ln 2$$

(5) Dilution, wash-out or clearance problems

The following are all known as dilution or wash-out or clearance problems: The removal of a detergent from clothing by rinsing in running water, the removal of fixing agent from a photographic emulsion by washing in running water, the removal of an anaesthetic from the body of a patient, the removal of a gas from an enclosed space such as the interior of a room by letting in fresh air through a door or a window, etc.

A dilution process which is of importance in the study of human physiology enables one to measure the **cardiac output** (the number of litres blood pumped by the heart per minute into the arterial system of the body). This process involves the injection of a dye into the right-hand side of the heart. The addition of blood which does not contain dye during each pulse of the heart, diminishes the concentration of the dye in the right-hand side of the heart. The measurement of the concentration of the dye at a number of different times, enables one to calculate the half-life period of the process from which the cardiac output of the heart can be calculated.



Consider a patient of whom the average cardiac output is Q litres per minute. The volume of blood in the right heart and pulmonary vessels is V litres. At time $t = 0$ a mass of M_0 mg of a suitable dye is injected into the right heart. It is assumed that the injection process and the resulting mixing of the dye and the blood, are rapid so that a homogeneous concentration exists in the heart. As blood is pumped through the aorta, the mass of dye in the heart decreases. Let the total mass of dye left in the heart after time t be $M = M(t)$.

At any given instant after $t = 0$, the average rate at which dye leaves the heart, is given by the product of the average concentration and the cardiac output.

$$dM/dt = -(M/V)Q = -(Q/V)M$$

which is a differential equation for exponential decay. The minus sign indicates that the rate at which dye leaves the heart, diminishes as the concentration diminishes.

Primarily the problem is to describe the amount of dye in the heart as a function of time, i.e. to calculate $M = M(t)$. The mathematical procedure is that described in section 7.3.1 and the answer is as follows:

$$M = M_0 e^{-(Q/V)t}$$

The halving period is calculated as one would for any other exponential decay process.

$$\tau = (V \ln 2)/Q$$

A photo-spectrometer may be used to measure the concentration of the dye. A number of measurements plotted on semi-logarithmic paper enables one to read the halving time of the process¹. The volume of blood in the heart and pulmonary vessels is fairly accurately known as a function of the mass and age of the patient and the cardiac output may be calculated as follows:

$$Q = (V \ln 2)/\tau$$

(6) Sterilisation processes

In many sterilisation processes the rate at which organisms are destroyed, is directly proportional to the number existing at a given instant. A numerical example of such a process is treated below.

To sterilise a suspension of anthrax spores, it is treated with a 5% solution of phenol. The rate at which the concentration of the spores diminishes, is directly proportional to the existing concentration at a given instant. Thus $dN/dt = -kN$, in which N is the number of spores per millilitre in the suspension, and t the time in hours. In this relationship, k is the decay constant. There are N_1 spores per ml present at time $t = t_1$ and N_2 at time $t = t_2$. Calculate the value of k in terms of the data.

As before, the variables are separated and the differential equation is solved by integration on both sides. The boundary conditions supply the limits.

$$\begin{aligned} \int_{N_1}^{N_2} N^{-1} dN &= \int_{t_1}^{t_2} -k dt \\ \text{from which follows } \ln N|_{N_1}^{N_2} &= -kt|_{t_1}^{t_2} \\ \text{so that } \ln N_2 - \ln N_1 &= -kt_2 - (-kt_1) \end{aligned}$$

¹See Hill D. W.: Physics applied to anaesthesia - Butterworths. On pages 13 to 21 this problem and also other washing-out problems are discussed

$$\begin{aligned} \text{or} \quad \ln(N_1/N_2) &= k(t_2 - t_1) \\ \text{and} \quad k &= \frac{1}{t_2 - t_1} \ln(N_1/N_2) \end{aligned}$$

For a given sample of spore suspension which is treated with a 5% phenol solution it is measured at time $t = 0$ h, that 440 spores ml^{-1} exist and at time $t = 2$ h, 60 spores ml^{-1} . Calculate the following: (a) The decay constant, k , and the halving time of the process. (b) The function $N = N(t)$. (c) The concentration of spores at time $t = 3$ h. (d) The time at which the liquid may be considered to be sterile.

The value of k may be calculated from the result of the previous calculation.

$$\begin{aligned} \text{(a)} \quad k &= \frac{1}{2 - 0} \ln 440/60 \\ &= 0,9962 \text{ h}^{-1} \\ \text{as before} \quad \tau &= (\ln 2)/k = 0,6958 \text{ h} \end{aligned}$$

(b) With $N(0)$ and the value of k known, the required function may be written down.

$$N = 440e^{-0,9962t}$$

(c) From the function above follows that

$$\begin{aligned} N(3) &= 440 \exp(-0,9962 \times 3) \\ &= 440 \times 0,0504 = 22,2 \text{ spores ml}^{-1} \end{aligned}$$

Question: Should the last answer not be rounded off to an integer? Explain.

(d) As is the case with radioactive decay, the sterilisation process deals with discrete entities and it is described by means of a continuous function. In order to determine when the sample will be sterile, one calculates the time when 1 spore only is expected to survive. This gives an indication of when the sample may be regarded sterile.

$$\begin{aligned} 1 &= 440e^{-0,9962t} \\ \text{so that} \quad -0,9962t &= \ln 1 - \ln 440 = -6,0868 \\ \text{and} \quad t &= 6,11 \text{ h} \end{aligned}$$

It should be kept in mind that, similar to radioactive decay, an uncertainty about the correct value of N , will always exist. The time which was calculated is only an indication of the time needed for the liquid to be sterile. With a knowledge of statistics, the probability can be calculated that a given number of spores may still exist. If the probability for the existence of an arbitrarily small amount of spores may be disregarded, it is assumed that the sample is sterile.

7.4 Bounded exponential growth

7.4.1 The differential equation and its solution

The differential equation for bounded exponential growth and that for exponential decay, differ by a constant term only. It may be written in the following form:

$$dy/dx = K - ky, \quad (y \leq K/k) \quad 7.4(1)$$

in which K and k are constants, and the latter is known as the **growth constant**. The initial condition is always in the form $y(0) = 0$, i.e. $y = 0$ when $x = 0$.

From equation 7.4(1) may be seen that the rate of change of y , i.e. the value of dy/dx , has its largest value when $y = 0$ and therefore $x = 0$. When the right-hand side of the equation becomes zero, then $dy/dx = 0$. This state is reached when

$$y = K/k = Y$$

in which Y is the maximum value that y can reach. Since the variables may be separated, the solution may be calculated by integration. With the given initial condition, the solution is calculated as follows:

$$\begin{aligned} \int_0^y \frac{dy}{K - ky} &= \int_0^x dx \\ (-1/k) \ln(K - ky) \Big|_0^y &= x \Big|_0^x \\ \ln(K - ky) - \ln K &= -kx \\ \ln\left(\frac{K - ky}{K}\right) &= -kx \\ \text{so that } \frac{K - ky}{K} &= e^{-kx} \\ \text{and } y &= (K/k)(1 - e^{-kx}) \\ &= Y(1 - e^{-kx}) \end{aligned} \quad 7.4(2)$$

7.4.2 The properties of the solution

If $x = 0$, the two terms on the right-hand side are equal in magnitude but opposite in sign so that $y(0) = 0$ which is in accordance with the initial condition.

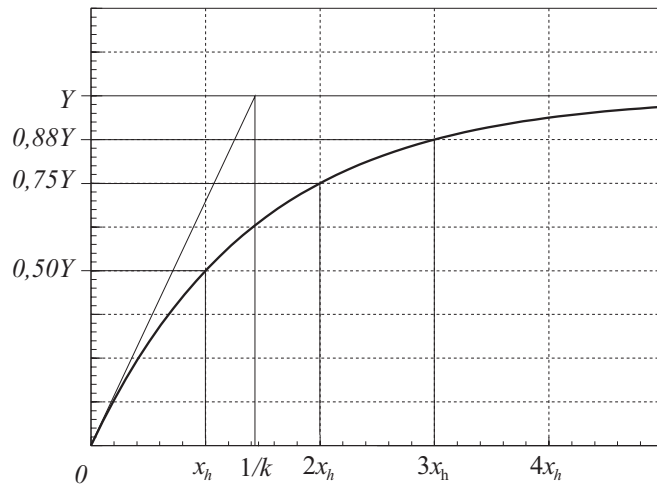


Figure 7.4-1

By differentiation the reader may show that the function satisfies the differential equation.

If x tends to infinity, the second term on the right tends to zero and y reaches its maximum value, namely Y . This behaviour is characteristic of all bounded exponential growth. From the graphical representation which is shown in Figure 7.4-1, it may be seen that this function has characteristics which are similar to exponential decay. An interval $\Delta x = x_h$ exists which will be called the **half-growth interval**. This has the property that the growth of the function during any interval $x + x_h$ to $x + 2x_h$, is half the growth which occurs during the equal and directly preceding interval x to $x + x_h$.

Half-growth intervals	y as % of Y
0	0,000
1	50,000
2	75,000
3	87,500
4	93,750
5	96,875
6	98,435
7	99,219
8	99,609

Table 7.4-1

Table 7.4-1 shows the value of y as a percentage of its final value, Y , which it will reach when $x \rightarrow \infty$. From the values in the table it is clear that y reaches more than 99% of its final value after seven half-growth intervals.

The relationship between the half-growth interval and the growth constant may be calculated in the same manner as the relationship between the halving interval and the decay constant of an exponential decay.

The relationship is exactly the same

$$x_h = (\ln 2)/k \quad 7.4(3)$$

With bounded exponential growth it is also possible to work with the interval $1/k$, rather than x_h . The study of this aspect is left to the reader as an exercise. It is suggested that a table be compiled similar to Table 7.4-1 in which multiples of $1/k$ are used rather than multiples of x_h . It is interesting to note that if the rate at which a bounded growth function grows initially could be maintained, it would reach its maximum value of Y after an interval of $1/k$, the time constant. (See Figure 7.4-1)

7.4.3 Examples of bounded exponential growth

(1) The growth of an electric current in a circuit which contains resistance and inductance

Consider the circuit as shown in example (4) of section 7.3.4 but with the switch initially in the position which short-circuits x and z . No current exists in the circuit. At instant $t = 0$ the switch is switched to the position where the short-circuit is broken and the potential source V connected between x and z . The following differential equation follows from the circuit equation:

$$\begin{aligned} L(di/dt) + iR &= V \\ \text{or} \quad di/dt &= (V - iR)/L \end{aligned}$$

Since the current is initially equal to zero, the rate at which it increases is then a maximum. The growth rate decreases as i increases and becomes zero when $V = iR$ or $i = I = V/R$, the final magnitude of the current.

With the initial condition $i(0) = 0$, the differential equation may be solved as shown in 7.4.1. The solution is as follows:

$$i = I(1 - e^{-(R/L)t})$$

and the half-growth interval is given by

$$t_h = (L \ln 2)/R$$

which is exactly the same as the halving time for the decay process when points x and z are short-circuited while a steady current exists.

(2) The growth of daughter nuclei during radioactive decay.

When a radioactive nucleus decays, a **daughter nucleus** which might be radioactive as well, is formed. If the daughter nucleus is radioactive one or more further daughter elements and even a related series of elements might be formed. Each such series ends in a **stable nucleus**. In this section the case will be studied where radioactive nuclei of one kind decay to form stable daughter nuclei, of one specific kind.

Consider a sample of material which contains N_0 radioactive nuclei with decay constant λ at instant $t = 0$. The daughter nuclei are stable and none exists when the measurement of time is commenced. Let N be the number of daughter nuclei and N' the number of radioactive nuclei at instant t . Then

$$N_0 = N' + N \quad \text{or} \quad N = N_0 - N' \quad \dots\dots(1)$$

The rate at which daughter nuclei are formed is equal to the rate at which radioactive nuclei decay. The time derivative of Equation (1) gives

$$dN/dt = -dN'/dt \quad \dots\dots(2)$$

From the principle which governs the decay of radioactive nuclei, it is known that

$$dN'/dt = -\lambda N' = \lambda(N - N_0) \quad \dots\dots(3)$$

From Equations (2) and (3) follows

$$dN/dt = \lambda N_0 - \lambda N \quad \dots\dots(4)$$

With the initial condition $N(0) = 0$, the solution of differential equation (4) is

$$N = N_0(1 - e^{-\lambda t})$$

with half-growth interval

$$t_h = (\ln 2)/\lambda$$

which is the same as the half-life period of the radioactive material from which it was formed.

The problem is interesting when a **radioactive series** is studied. If the original parent nucleus has a very long half-life period, the series becomes relatively steady as regards the numbers of the radioactive nuclei of each daughter element. The relative abundance of daughter elements and also of the final stable elements, provides a method for the determination of the age of minerals which contain them. For further information, the reader is referred to the textbook: *Wehr and Richards: Physics of the atom (Addison-Wesley)* or any other text which treats radioactive decay.

7.5 Unbounded exponential growth

7.5.1 The differential equation and its solution

Except for the difference in sign, the differential equation for unbounded exponential growth is the same as that for exponential decay.

$$dy/dx = ky \quad 7.5(1)$$

in which k is called the **growth constant**. It is of importance to realise that if y is equal to zero, the rate at which it will change will be zero also, and thus it cannot change. The initial condition for each non-trivial case must be that a non-zero value of y exists at $x = 0$, e.g. $y(0) = Y$. In this study only cases in which y is positive, will be considered.

With the given initial conditions, the solution is simple because the variables are separable as before.

$$\begin{aligned} \int_Y^y y^{-1} dy &= \int_0^x k dx \\ \text{from which follows } \ln(y/Y) &= kx \\ \text{or } y &= Ye^{kx} \end{aligned} \quad 7.5(2)$$

7.5.2 Properties of the function $y = Ye^{kx}$

In the same way as that which was used in previous examples the reader may prove that the function satisfies both the differential equation and the initial condition.

One of the important properties has been mentioned: If y is zero, no growth can occur. A graphical representation unbounded growth is shown in Figure 7.5-1. It should be clear that the function will grow at an increasing rate for any value of the growth constant and that it tends to infinity when $x \rightarrow \infty$. Hence the name *unbounded growth*. This information is not of much practical use.

From the graph another interesting property is obvious. An interval of x exists for which the magnitude of the function is doubled. This is called the **doubling interval**. If x represents time, it is called the **doubling time**. The relationship between the growth constant and the doubling interval, $\Delta x = x_2$, is derived in the same manner as the relationship between the decay constant and the halving interval of an exponential decay and is given by

$$x_2 = (\ln 2)/k \quad 7.5(3)$$

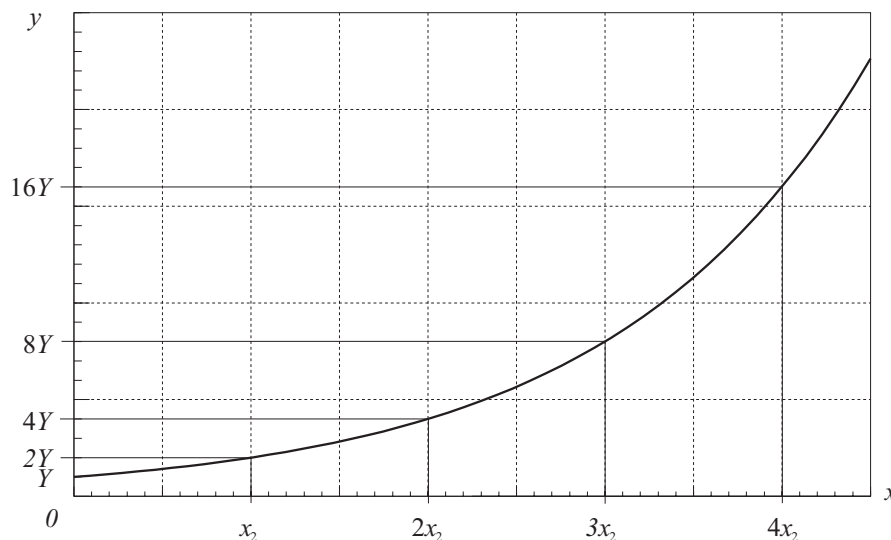


Figure 7.5-1

This kind of growth is observed on a limited scale in nature in population explosions and chain reactions. In such processes, limiting factors are present which were not taken into account in the above calculations. Each population explosion of some or other organism will be limited by the availability of nutrition and/or space and the process might lead to limited stability or even revert to a decay in numbers. Until such factors play a significant role, unbounded growth is a good model for the description of the growth of the population. In a nuclear chain reaction, the growth of the process is limited by the availability of a finite amount of fissionable material.

7.5.3 Examples of unbounded exponential growth

(1) **The growth of a colony of bacteria.** On condition that no destruction takes place, an isolated colony of bacteria with sufficient nutrition and space will grow at a rate which is directly proportional to the size of the colony. The differential equation which describes the growth is as follows:

$$dN/dt = kN$$

in which $N = N(t)$ is the number of bacteria as a function of the time, t . If N_0 bacteria are present at time $t = 0$, the solution of the differential equation is

$$N = N_0 e^{kt}$$

with a doubling time of $t_2 = (\ln 2)/k$.

(2) **The growth of kefir culture.** Kefir is a slightly alcoholic and refreshing drink which is prepared by fermenting kefir granules in milk. About 100 grams of granules are sufficient for the treatment of 1 litre of milk. During this process, the kefir grows and its doubling time is 6 weeks. Calculate the time required for 100 grams of kefir granules to grow to (a) 400 grams, (b) 150 grams.

(a) The initial 100 g has to double twice in order to produce a total of 400 g. Since the doubling time is 6 weeks the time required is 12 weeks.

(b) The growth constant may be calculated from the doubling time.

$$k = (\ln 2)/6 = 0,11552 \text{ week}^{-1}$$

The function which describes the growth may be written as follows:

$$M = 100e^{0,11552t} \text{ g}$$

If $M = 150$ g is substituted in this function and it is solved for t , the answer is

$$t = 3,51 \text{ weeks}$$

7.6 PROBLEMS: CHAPTER 7

1. Given: The differential equation $d^2x/dt^2 = -9x$. Show that each of the following functions, $x = x(t)$, is a solution of the equation: (a) $x = 12 \cos 3t$, (b) $x = -20 \sin(3t - \pi)$, (c) $x = R \cos(3t + 5)$, (d) $x = 3 \sin(3t - 5) - 8 \cos(3t + 7)$, (e) $x = 6 \sin 3t + 3 \cos 3t$, (f) $x = 8 \exp(3it)$, (g) $x = 15 \exp(-i[3t + 5])$. In the last two functions, $i = \sqrt{-1}$ and $\exp n = e^n$.

2. Consider the solutions (a), (b), (c), (f) and (g) of the previous question. Give the amplitude, frequency, period and the phase constant of the function in each case.

3. Show that the function $x = A \sin(\omega t + \phi)$ may be written in the form $x = B \sin \omega t + C \cos \omega t$ and calculate the values of B and C in terms of the amplitude, A , and the phase constant, ϕ . Consider solution (e) of question (1) and write it in the form $x = A \sin(\omega t + \phi)$. What are the amplitude and phase constant of solution 1(e)?

4. $\bar{F} = m\bar{a}$ is Newton's second law of motion for a force of $\bar{F}$ newton which acts on a body of mass m kilogram and which experiences an acceleration of $\bar{a}$ m s^{-2} . Consider the following example: A mass of 2 kg is in motion under the action of a force which is given by $\bar{F} = -3x\hat{x}$ in which F is measured in newton and x in metre. Set up the differential equation for the motion and calculate the solutions for an amplitude of 4 m and the following boundary conditions: (a) $x(0) = 4$, (b) $x(0) = -4$, (c) $x(0) = 0$ and $(dx/dt)_{t=0} > 0$, (d) $x(0) = 0$ and $(dx/dt)_{t=0} < 0$, (e) $x(0) = -2$ and $(dx/dt)_{t=0} > 0$, (f) $x(2) = 4$, (g) $x(1) = 2$ and $(dx/dt)_{t=1} < 0$. Give the smallest possible positive value of the phase constant in each case.

5. Monochromatic (one frequency or one wave length only) parallel X-rays are absorbed by a homogeneous (the same throughout the volume) isotropic (the same properties in all directions) medium according to the function $I = I_0 e^{-\mu x}$, in which I is the intensity of the rays which is a function of the thickness of the absorbing material, x . The intensity where the rays penetrate the material is $I(0) = I_0$ and μ is called the **linear coefficient of absorption** of the material for the specific rays. A monochromatic parallel beam of X-rays is incident on an absorbing material for which $\mu = 2 \text{ mm}^{-1}$. Calculate the percentage of the rays that 3 mm of the absorber will transmit. Calculate also the half value-thickness of this material for the X-rays.

6. It is found that 5 mm of polystyrene will decrease the intensity of a given sound by half. Assume that the absorption process may be described by the exponential decay equation. Calculate the thickness of polystyrene which will diminish the intensity of a sound of 1 watt per square metre to $10^{-12} \text{ W m}^{-2}$.

7. The flow of water from a burette is described by $dh/dt = -0,002h$ in which h is the height of the water in metres and t the time in seconds. At instant $t = 0$ the burette is filled to a height of 0,8 m. (a) Calculate the height as a function of time. (b) Calculate the half-life period of the height of the water. (c) When will the height of the water be 0,35 m? (d) What will the height of the water be at $t = 58$ s?
8. In the figure at example (3) in section 7.3.4, the resistance of the conductor is $5\text{ M}\Omega$ and the capacitance of the capacitor, $4\text{ }\mu\text{F}$. At instant $t = 0$, the capacitor carries a charge of $800\text{ }\mu\text{C}$ while the switch is open. The switch is closed at $t = 0$ s. (a) Calculate the charge on the capacitor and the current in the resistor as functions of time. (b) Calculate when the charge on the capacitor will be $157\text{ }\mu\text{C}$. (c) Calculate the half-life period of the process.
9. When an object of mass m kg falls from rest under the influence of gravity, it experiences drag (frictional force due to the atmosphere) which is directly proportional to its speed, v : $F_w = -kv$, in which k is a constant for the specific object. The following law governs the motion: $F = ma = m(dv/dt) = mg - kv$, in which g is the magnitude of gravitational acceleration, and a the actual acceleration of the object. From this relationship it can be seen that the acceleration, $a = dv/dt$, decreases as the speed, v , increases. This means that v will reach an upper limit which is called the **terminal speed**. (a) Solve the differential equation for the motion if the object starts falling from rest at $t = 0$. (b) Calculate the terminal speed without solving the differential equation. (c) A body of mass 50 kg falls from a helicopter which is hovering at rest and it acquires a terminal speed of 360 km h^{-1} . $g = 10\text{ ms}^{-2}$. Calculate k for this body. (d) How long will it fall to acquire a speed of 240 km h^{-1} ? (e) Calculate the half-growth period for the speed of the body.

ANSWERS TO PROBLEMS

CHAPTER 1

1. (a) 2 (b) 4 (c) $1/9$ (d) 72 (e) 2 (f) $1/4$
 (g) 48 (h) 10^{10} (i) 100 (j) 10^{10} (k) $3^5 = 243$
 (l) $4^3 = 64$ (m) $1/4$ (n) 20 (o) 8
2. (a) 5 (b) 2 (c) -5 (d) 0 (e) undefined
 (f) -2 (g) 2 (h) undefined (i) - 5
3. (a) 152,28 (b) 1879,1 (c) 621,97
 (d) $6,5670 \times 10^{-3}$ (e) 3,5128 (f) 2,3108
 (g) 0,4327 (h) 4,5164 (i) 0,4327
4. (a) 2,1826 (b) 1,1910 (c) 0,5457
 (d) 1,2084 (e) 3,6253 (f) 1,5613
 (g) 0,9163 (h) 0,0913 (i) 0,1091
 (j) -0,9087 (k) 2
5. (a) $a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$
 (b) $x^4 + 8x^3 + 24x^2 + 32x + 16$
 (c) $16y^4 - 96y^3 + 216y^2 - 216y + 81$
 (d) $1 + v^2/2c^2 + 3v^4/8c^4 + 5v^6/16c^6 + \dots$
 (e) $1 + 2(0,002) + (0,002)^2 \approx 1 + 2(0,002) = 1,004$
 (f) $(1 - 0,005)^{1/2} \approx 1 + (1/2)(-0,005) = 0,9975$
 (g) $(1,003)^{1/2} = (1 + 0,003)^{1/2} \approx 1 + (1/2)(0,003) = 1,0015$
 (h) $(0,995)^{1/2} = (1 - 0,005)^{1/2} \approx 1 + (1/2)(-0,005) = 0,9975$
 (i) $(0,995)^{-1/2} = (1 - 0,005)^{-1/2} \approx 1 + (-1/2)(-0,005) = 1,0025$
6. (a) $\pi/6$ (b) $\pi/3$ (c) $\pi/4$ (d) $3\pi/2$ (e) $6\pi/5$
 (f) 4π (g) $5\pi/2$ (h) $7\pi/2$ (i) $2\pi/5$
7. (a) 1,4895 rad (b) 2,1537 rad (c) 14,2349 rad
 (d) 1,5649 rad (e) 3,8381 rad (f) 6,2136 rad
8. (a) 270° (b) 90° (c) $51,43^\circ$ (d) 144° (e) 450°
 (f) 720° (g) 540° (h) 30° (i) 240°
9. (a) $420,84^\circ$ (b) $7,07^\circ$ (c) $32,53^\circ$
 (d) $-319,02^\circ$ (e) $102,20^\circ$ (f) $2620,80^\circ$

10. (a) 0,8366 (b) 0,5479 (c) 1,5270 (d) 0,6549 (e) 1,1954
 (f) 1,8253 (g) 0,9888 (h) -0,5804 (i) -0,9738 (j) -0,5354
 (k) 1,8618 (l) 0,4679 (m) -1,9549 (n) 0,6613 (o) -5,3367
11. (a) $50,9796^\circ$ or $0,8898$ rad (b) $-50,9796^\circ$ or $-0,8898$ rad
 (c) $149,9971^\circ$ or $2,6179$ rad (d) does not exist
 (e) $54,1324^\circ$ or $0,9448$ rad (f) $-39,0204^\circ$ or $-0,6810$ rad
 (g) $0,0688^\circ$ or $0,0012$ rad (h) $0,0688^\circ$ or $0,0012$ rad
 (i) $64,7651^\circ$ or $1,1304$ rad (j) does not exist
 (k) does not exist (l) $59,9971^\circ$ or $1,0471$ rad
12. See Table 1.7-1
13. (a) $b = 43,59$ mm, $\alpha = 23,42^\circ$, $\gamma = 36,58^\circ$
 (b) $\gamma = 80^\circ$, $a = 8,16$ mm, $b = 10,99$ mm
 (c) $\beta = 63^\circ$, $a = 67,91$ m, $b = 68,53$ m
 (d) $\alpha = 33,56^\circ$, $\beta = 50,70^\circ$, $\gamma = 95,74^\circ$
 (e) Two possibilities: $\beta = 125,25^\circ$, $\gamma = 34,75^\circ$, $b = 71,63$ m
 $\beta = 14,75^\circ$, $\gamma = 145,25^\circ$, $b = 22,33$ m
14. (a) $259,8$ mm² (b) $44,17$ mm² (c) 1906 m²
 (d) $435,3$ m² (e) $612,5$ m² or $190,9$ m²
15. (i) $1,732$ m² (ii) $1,732$ m²
16. $A \approx 2\pi r \Delta r = 12000$ m² (Exact: $A = 12113$ m²)
17. $3,031$; $2,456$ m² 18. $119,6$ m² 19. $2,257$ m
20. 7540 mm² 21. 215984 mm³; 173220 mm²; 981748 mm³ water.

CHAPTER 2

1. (a) 3; 3 for all values of x (b) $6x - 2$; -8 ; 10
 (c) $48x^2 - 15$; 33 ; 177 (d) $-9x^{-4} - 24x^{-3}$; 15 ; $-3,5625$
 (e) $21x^2 - 21x^{-4}$; 0 ; $82,6875$ (f) $25x^4 - 45x^2 + 8x^{-2}$; -12 ; 222
 (g) $-30x^{-7} - 24x^{-4} + x^{-3}/8$; $5,875$; $-1,719$
 (h) $-44x^{-3} + 4x^{-2}$; 48 ; $-4,5$
2. (a) $\cos 0, 5\theta$; 1 ; 0 ; -1 (b) $200 - 10t$; 180 ; 0 ; -300
 (c) $1/q$; undefined; 1 ; $1/2$ (d) e^x ; 1 ; $0,3679$; $7,3891$
 (e) $-6 \sin 2\phi$; 0 ; 0 ; 0
3. (a) $18t^2 - 42t - 10$ (b) $5r + 3, 5r - 2$ or $(10r^3 + 7)/2r^2$
 (c) $18t - 6$ (d) $-2 \cos p \sin p = -\sin 2p$
 (e) $-\csc 2x$
4. (a) $(2x - 3)^{-1/2}$ (b) $3 \sin^2 x \cos x$
 (c) $-4 \cos^3 x \sin x$
 (d) $4 \sin(2x - 5) \cos(2x - 5) = 2 \sin(4x - 10)$

- (e) $-18x \cos^2(3x^2 - 2) \sin(3x^2 - 2)$ (f) $1, 5(3x + 4)^{-1/2} \cos(3x + 4)^{1/2}$
 (g) $\cot x$ (h) $(3x^2 - 1, 5)(2x^3 - 3x)^{-1/2} \cot(2x^3 - 3x)^{1/2}$
 (i) $-2kqax(a^2 + x^2)^{-3/2}$ (j) $18xe^{3x^2}$
5. (a) $\cos x; -\sin x; -\cos x; \sin x$
 (b) $2 \cos 2x; -4 \sin 2x; -8 \cos 2x; 16 \sin 2x$
 (c) $-\sin x; -\cos x; \sin x; \cos x$
 (d) $2e^{2x}; 4e^{2x}; 8e^{2x}; 16e^{2x}$
 (e) $12x^3 + 8x^2 + 6x + 5; 36x^2 + 12x + 6; 72x + 12; 72$
 (f) $-12x^{-5} - 6x^{-4} - 6x^{-3} - 5x^{-2}; 60x^{-6} + 24x^{-5} + 18x^{-4} + 10x^{-3};$
 $-360x^{-7} - 120x^{-6} - 72x^{-5} - 30x^{-4}; 2520x^{-8} + 720x^{-7} + 360x^{-6} +$
 $120x^{-5}$
6. (a) At $x = 3$ y has a minimum value of -15
 (b) At $x = 6$ y has a maximum value of 41
 (c) At $x = 1,633$ y has a local minimum value of -8,709
 At $x = -1,633$ y has a local maximum value of 8,709
 (d) At $x = 2$ y has a local minimum value of 4
 At $x = 1$ y has a local maximum value of 5
 (e) At $x = 1$ y has a local minimum value of 23
 At $x = -2$ y has a local maximum value of 50
 (f) No extreme values but a point of inflexion at $x = 0$
7. $t = 2,05$ seconds. The maximum value of x is 20,61 metre
8. At $\theta = \pi/4$ x has a maximum value of 4000 metre.
 At $\theta = 3\pi/4$ x has a minimum value of -4000 metre
9. 50 m at 25 m
10. (a) $h = 500/\pi r^2$ cm (b) $A = 2\pi r^2 + 1000/r$ cm²
 (c) $r = 4,30$ cm; $A_{\min} = 348,7$ cm (d) $h = 2r = \text{diameter}$
11. (a) $h = 500/\pi r^2$ cm (b) $A = \pi r^2 + 1000/r$ cm²
 (c) $r = 5,42$ cm; $A_{\min} = 276,8$ cm² (d) $h = r$
12. (a) 6; 0 (b) $10t - 3; 10$
 (c) $-9t^2 - 4t + 4; -18t - 4$ (d) $15 \cos 3t; -45 \sin 3t$
 (e) $15 \cos 3t - 15 \sin 3t; -45 \sin 3t - 45 \cos 3t$
13. (a) -2000 Vm^{-1} (b) $200x^{-2} - 7 \text{ Vm}^{-1}$
 (c) $4kqax/(x^2 - a^2)^2 \text{ Vm}^{-1}$
 (d) $2kqx/(a^2 + x^2)^{3/2} \text{ Vm}^{-1}$. Maximum at $a/\sqrt{2}$, Magnitude: $4kq/3a^2\sqrt{3}$
 Minimum at $-a/\sqrt{2}$, Magnitude: $-4kq/3a^2\sqrt{3}$
14. (a) At $x = 5,773$ m $y = 1,667$ m, a maximum (b) $\alpha = 30^\circ, \beta = 150^\circ$
15. (a) $dN/dt = -2 \times 10^3 e^{-t/50}$, N decreases with time.
 (c) 34,66 s; 69,31 s
16. (a) $dV/dt = -\lambda Ah_0 e^{-\lambda t}$, h and V decrease with time.
17. $m_0 v/c^2 (1 - v^2/c^2)^{3/2}$
18. $x^2; 2x; 2x\Delta x$
19. $V = \pi r^2 h/3; V = \pi R^2 h^3/3H^2; h = (3H^2 V/\pi R^2)^{1/3}; dh/dV = (H^2/9\pi R^2 V^2)^{1/3}$
 or $H^2/\pi R^2 h^2; \Delta h \approx (H^2/\pi R^2 V^2)^{1/3} \Delta V$

CHAPTER 3

1. (a) $x^3 - 3x^2 + 5x + k$ (b) $\frac{3}{5}x^5 + \frac{1}{2}x^4 - \frac{5}{3}x^3 + \frac{7}{2}x^2 + k$
 (c) $\frac{7}{4}x^4 - \frac{7}{2}x^{-2} + k$ (d) $\frac{4}{3}x^3 - 6x^{-1} + k$
 (e) $2x^8 - \frac{29}{8}x^{-8} + k$ (f) $x^4 + x^3 + x^2 + x - \ln x + x^{-1} + \frac{1}{2}x^{-2} + k$
 (g) $2e^{3x} + k$ (h) $-2\cos 2x + k$
 (i) $2\sin 3x + k$ (j) $\ln(3x^4 + 5x) + k$
2. (a) $\frac{4}{3}x^3 + x^2 - 3x + c$ (b) $\frac{8}{7}x^7 - 3x^4 - 5x + c$
 (c) $2s^4 - 3s + 2s^{-2} + c$ (d) $20t - 5t^2 + c$
 (e) $-9x^2 + c$ (f) $-18x + c$
 (g) $-18\ln x + c$ (h) $vt - \frac{1}{2}gt^2 + c$
 (i) $-\frac{5}{3}t^3 + at + b$; a, b constant (j) $-\frac{1}{2}gt^2 + at + b$; a, b constant
 (k) $\frac{5}{4}\sin 4x + c$ (l) $\frac{3}{2}\cos 2x + c$
3. (a) 31,5 (b) 1 (c) 1 (d) -7/6 (e) 1/3 (f) 0
 (g) 14/3 (h) 6,28
4. 60 metre
5. (i) 1 (ii) 1 (under x -axis) (iii) 0 (real area is 2)
6. (0, 0) and (2, 4); $4/3$
7. $dV = (B^2/L^2)y^2dy$; $V = \frac{1}{3}B^2L$
8. (a) $k \ln(V_2/V_1)$, $k = p_1V_1 = p_2V_2$
 (b) $(k/[1 - \gamma])(V_2^{1-\gamma} - V_1^{1-\gamma}) = (p_2V_2 - p_1V_1)(1 - \gamma)^{-1}$
9. $ML^2/3$; $ML^2/12$

CHAPTER 4

1. $R = 8,85$; $\theta = 50,23^\circ$; $\bar{R} = 5,66\hat{x} + 6,80\hat{y}$
3. $\bar{A} = -2\hat{x} + 4\hat{y}$; $A = 4,472$; $\theta = 116,57^\circ$
 $\bar{B} = 2\hat{x} + 5\hat{y}$; $B = 5,385$; $\theta = 68,20^\circ$
 $\bar{C} = 4\hat{x} + 9$; $C = 4,123$; $\theta = 14,04^\circ$
 $\bar{r}_B = 4\hat{x} + \hat{y}$; $r_B = 4,123$; $\theta = 14,04^\circ$
4. (a) $\bar{r}_g = 2\hat{x}$ m, $\bar{r}_e = 2\hat{y}$ m, $\bar{r}_b = 2\hat{z}$ m; $\bar{r}_a = 2\hat{x} + 2\hat{z}$ m; $\bar{r}_f = 2\hat{x} + 2\hat{y}$ m
 $\bar{r}_c = 2\hat{y} + 2\hat{z}$ m; $\bar{r}_d = 2\hat{x} + 2\hat{y} + 2\hat{z}$ m
 (b) $\overline{ef} = 2\hat{x}$ m; $ef = 2$ m; $\overline{da} = -2\hat{y}$ m; $da = 2$ m
 $\overline{gb} = -2\hat{x} + 2\hat{z}$ m; $gb = 2,828$ m; $\overline{eg} = 2\hat{x} - 2\hat{y}$ m; $eg = 2,828$ m
 $\overline{de} = -2\hat{x} - 2\hat{z}$ m; $de = 2,828$ m; $\overline{gc} = -2\hat{x} + 2\hat{y} + 2\hat{z}$ m; $gc = 3,464$ m
 (c) $\bar{v}_{ef} = 6\hat{x}$ m s $^{-1}$; $\bar{v}_{da} = -6\hat{y}$ m s $^{-1}$; $\bar{v}_{gb} = -4,243\hat{x} + 4,243\hat{z}$ m s $^{-1}$
 $\bar{v}_{eg} = 4,243\hat{x} - 4,243\hat{y}$ m s $^{-1}$; $\bar{v}_{de} = -4,243\hat{x} - 4,243\hat{z}$ m s $^{-1}$

- $\bar{v}_{gc} = -3,464\hat{x} + 3,464\hat{y} + 3,464\hat{z} \text{ m s}^{-1}$
5. (a) 3,742; 5,385; 3,742 (b) $-4\hat{x} + 4\hat{y} + \hat{z}$; 5,744; 12,869
(c) $2\hat{x} - 7\hat{z}$; 7,280
 7. $4\hat{x} - 2\hat{y} + 4\hat{z}$; $8\hat{x} + 2\hat{y} + 7\hat{z}$; 6,403
 8. $[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2}$
 9. 12,00; -15,32; -12,00; 0; 19,05; -36,00; 0; -5,21; 0
 10. (a) 7 en 3; (b) 4 (c) $79,02^\circ$
 11. (a) $\frac{4}{9}\hat{x} - \frac{4}{9}\hat{y} + \frac{7}{9}\hat{z}$ (b) $\alpha = 63,61^\circ$; $\beta = 116,39^\circ$; $\gamma = 38,94^\circ$
 12. (a) 7; 3; 19 (b) 6,333 (c) 2,714 (d) $25,21^\circ$
 13. (a) 7; 3; -19 (b) -6,333 (c) -2,714 (d) $154,79^\circ$
 15. $6\hat{y} - 4\hat{z}$ or $-6\hat{y} + 4\hat{z}$; 90° ; $58,75^\circ$; $31,25^\circ$
 - 8 $16\bar{A} \cdot \bar{C} = 0 \Rightarrow$ orthogonal; $B^2 = A^2 + C^2$; $50,77^\circ$; $39,23^\circ$
 17. -1 or 2
 18. (a) 57; (b) -26; (c) 6; (d) 48 19. $2\hat{x} - 11\hat{y} - 7\hat{z}$
 20. (a) $10\hat{x} + 3\hat{y} + 11\hat{z}$ (b) $-20\hat{x} - 6\hat{y} - 22\hat{z}$
 22. (a) $15\hat{x} - 10\hat{y} + 30\hat{z}$; (b) 35 N m; (c) $0,429\hat{x} - 0,286\hat{y} + 0,857\hat{z}$
 23. $(1,6\hat{x} + 2,4\hat{y} + 2,4\hat{z}) \times 10^{-9} \text{ N}$

CHAPTER 5

1. (a) $(6u^2 - 5)\hat{x} + (2 \cos 2u)\hat{y}$ (b) $-5\hat{x} + 2\hat{y}$
(c) $12u\hat{x} - (4 \sin 2u)\hat{y}$ (d) 90°
2. (a) $(8 \cos 2t)\hat{x} \text{ m s}^{-1}$; (b) $-(16 \sin 2t)\hat{x} \text{ m s}^{-2}$; (c) $\bar{a} = -4\bar{r} \text{ m s}^{-2}$
3. (a) $4\hat{x} + 7\hat{y} - 4\hat{z} \text{ m s}^{-2}$; (b) 9 m s^{-2} ; (c) $69,56^\circ$
4. (a) circular orbit, centre at origin, radius 2 metre
(b) $-(8\pi \sin 4\pi t)\hat{x} + (8\pi \cos 4\pi t)\hat{y} \text{ m s}^{-1}$; (c) $8\pi \text{ m s}^{-1}$
(d) $-(32\pi^2 \cos 4\pi t)\hat{x} - (32\pi^2 \sin 4\pi t)\hat{y} \text{ m s}^{-2}$; (e) $\bar{v} \cdot \bar{r} = \bar{a} \cdot \bar{v} = 0$
(f) $\bar{a} = -16\pi^2 \bar{r} \Rightarrow \bar{a}$ directed towards the centre; (g) $\bar{r} \times \bar{v} = 16\pi \hat{z}$
5. (a) $\bar{r} = [(2 \cos t)\hat{x} + (2 \sin t)\hat{y} + t\hat{z}] \times 10^5 \text{ m}$; a right circular helix with radius $2 \times 10^5 \text{ metre}$ and a pitch of 10^5 m s^{-1}
(b) $\bar{v} = [(-2 \sin t)\hat{x} + (2 \cos t)\hat{y} + \hat{z}] \times 10^5 \text{ m s}^{-1}$; $|\bar{v}| = 3 \times 10^5 \text{ m s}^{-1}$
(c) $\bar{a} = [(-2 \cos t)\hat{x} + (-2 \sin t)\hat{y}] \times 10^5 \text{ m s}^{-2}$ (d) $a = 2,828 \times 10^5 \text{ m s}^{-2}$
6. $\frac{6}{7}\hat{x} + \frac{3}{7}\hat{y} - \frac{2}{7}\hat{z}$

CHAPTER 6

1. (a) $(0,5p^4 + k_x)\hat{x} + (0,5p^2 - 0,33p^3 + k_y)\hat{y} + (-3p + k_z)\hat{z}$
(b) $7,5\hat{x} - 0,833\hat{y} - 3\hat{z}$
2. (a) $\bar{A} \times (d^2 \bar{A}/du^2)$ (b) $\bar{A} \times (d\bar{A}/du)$

3. (a) $(2t - 3)\hat{z} \text{ m s}^{-1}$ (b) $(t^2 - 3t - 10)\hat{z} \text{ m}$ (c) $\pm(4z + 49)^{1/2} \text{ m s}^{-1}$
4. (a) 9 m s^{-2} (b) $(4t - 3)\hat{x} + (7t + 6)\hat{y} + (-4t + 2)\hat{z} \text{ m s}^{-2}$
(c) 7 m s^{-1} (d) $69, 56^\circ$ (e) $2\hat{x} + 26\hat{y} - 4\hat{z} \text{ m}$
(f) $(2t^2 - 3t + 2)\hat{x} + (3, 5t^2 + 6t - 2)\hat{y} + (-2t^2 + 2t + 1)\hat{z} \text{ m}$ (g) $139, 63^\circ$
5. (a) $-10\hat{y} \text{ m s}^{-2}$; $34, 64\hat{x} + 20\hat{y} \text{ m s}^{-1}$ (b) $34, 64t\hat{x} + (20 - 10t)\hat{y} \text{ m s}^{-1}$
(c) $34, 64t\hat{x} + (20t - 5t^2)\hat{y} \text{ m}$ (d) $\pm(400 - 20y)^{1/2}$
(e) $x = 34, 64t \text{ m}$; $y = 20t - 5t^2 \text{ m}$ (f) $y = 0, 5774x - 0, 0042x^2$
(g) 20 m (h) $138, 6 \text{ m}$
6. (a) $22, 36 \text{ m}$ (b) $44, 72 \text{ m}$
7. (a) $-3, 667$ (b) $-3, 667$ (c) $-3, 667$
8. $dA\hat{x} = dydz\hat{x}$; $dA\hat{y} = dx dz\hat{y}$; $dA\hat{z} = dx dy\hat{z}$
Others the negatives of those above. Surface integral = 3

CHAPTER 7

2. The angular frequencies, frequencies and periods are the same for all and equal to 3 rad s^{-1} , $3/2\pi \text{ Hz}$, $2\pi/3 \text{ s}$. Their amplitudes and phase constants are as follows: (a) 12; 0; (b) 20; $-\pi$; (c) R ; 5; (f) 8; 0; (g) 15; -5
3. $B = A \cos \phi$; $C = A \sin \phi$; $x = 6, 708 \sin(3t + 0, 4636)$
4. All solutions have the form $x = 4 \sin(1, 225t + \phi)$ with the following values for ϕ : (a) $\pi/2$, (b) $3\pi/2$, (c) 0, (d) π , (e) $11\pi/6$, (f) $\pi/2$, (g) 0, 8859
5. 0,25 percent; 0,3466 mm
6. 199,4 mm
7. (a) $h = 0, 8 \exp(-0, 002t) \text{ metre}$ (b) 346,6 seconds
(c) 413,3 seconds (d) 0,7124 metre
8. (a) $q = 800 \exp(-0, 05t)$ (b) 32,56 seconds (c) 13,86 seconds
9. (a) $v = (mg/k)(1 - \exp[-kt/m]) \text{ m s}^{-1}$ (b) terminal speed = mg/k
(c) $k = 5$ (d) 10,99 seconds (e) 6,932 seconds

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